

A Monologue on Conversation: Characterising Dialogue Across Interaction Types with a Focus on Topical Structure

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*An ocean of knowledge is apt to
drown you long before it educates you.*

– Mark Lawrence, *The Book That Wouldn't Burn*

Abstract

Dialogue constitutes a fundamental mode of human interaction and has become increasingly central in natural language processing with the growing use of conversational AI systems both in research and in practical applications. However, dialogue is a heterogeneous phenomenon encompassing a wide range of forms, arising from differences in participant configurations, interaction contexts, or communication media. This diversity creates challenges for the study of dialogue data, particularly in selecting suitable resources, defining the scope and limitations of our studies, and establishing meaningful evaluation methodologies. A principled characterisation of dialogue is therefore necessary to account for its variations and to support the design of robust computational approaches.

Among the phenomena that cut across different forms of dialogue, topical organisation plays a central role. The structure and evolution of topics shape conversational dynamics and are themselves influenced by the type of interaction being considered. Studying such phenomena requires moving beyond local conversational patterns and adopting a broader, macro-level perspective on dialogue, allowing different interacting processes and structures to be examined jointly.

This thesis investigates three main research questions: (1) how dialogue can be characterised in a way that accounts for its variability while informing the selection of suitable resources and evaluation methods; (2) whether topical structure can be automatically extracted from dialogue data, with a particular focus on open-ended multi-party conversations, that are more complex but also very common in real-world interactions; and (3) whether topical structure can be leveraged as a discriminative signal for differentiating dialogue types. Through these questions, the thesis aims to contribute both analytical and methodological perspectives toward a broader understanding of dialogue structure and variation.

Résumé Court

Le dialogue constitue un mode fondamental d'interaction humaine et occupe une place de plus en plus centrale dans le traitement du langage naturel, avec l'utilisation croissante des systèmes d'IA conversationnelle tant dans la recherche que dans les applications pratiques. Cependant, c'est aussi un phénomène hétérogène qui englobe un large éventail de formes, découlant des différences dans la configuration des participants, le contexte d'interaction ou le support de communication. Cette diversité pose des défis pour l'étude des données dialogiques, notamment en ce qui concerne la sélection de ressources adaptées, la définition de la portée et des limites de nos études, ainsi que la mise en place de méthodologies d'évaluation pertinentes. Une caractérisation fondée sur des principes du dialogue est donc nécessaire pour rendre compte de ses variations et pour soutenir la conception d'approches computationnelles robustes.

Parmi les phénomènes qui transcendent les différentes formes de dialogue, l'organisation thématique joue un rôle central. La structure et l'évolution des thèmes façonnent la dynamique conversationnelle et sont elles-mêmes influencées par le type d'interaction considéré. L'étude de ces phénomènes nécessite d'aller au-delà des schémas conversationnels locaux et d'adopter une perspective plus large, à un niveau macro, permettant ainsi d'examiner conjointement les différents processus et structures en interaction.

Cette thèse explore trois questions de recherche principales : (1) comment caractériser le dialogue de manière à rendre compte de sa variabilité tout en orientant le choix des ressources et des méthodes d'évaluation appropriées ; (2) si la structure thématique peut être extraite automatiquement à partir de données de dialogue, en mettant particulièrement l'accent sur les conversations ouvertes à plusieurs participants, qui sont plus complexes mais également très courantes dans les interactions du monde réel ; et (3) si la structure thématique peut être exploitée comme un signal discriminant pour différencier les types de dialogue. À travers ces questions, cette thèse vise à apporter des perspectives tant analytiques que méthodologiques en vue d'une compréhension plus large de la structure et de la variation du dialogue.

Résumé Long

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Prologue

(0.1) **A**₁ i love bunnies! i wish i would open a bunny daycare or hotel.
B₂ that is a cool idea. how are you today?
A₃ a little tired. worked all day today at the shipyard.
B₄ i just got back from school, not a big fan of it though.
A₅ you should join the military. i am ex military. girls love uniforms;
B₆ i have never considered it. maybe it would help me make some more friends.
A₇ i joined the navy because i like navy blue, but infantry has sexy uniforms also.
B₈ i will definitely keep the in mind if i join.
A₉ i donate part of my salary to my bunny rescue. you should too.
B₁₀ do you have any bunnies?
A₁₁ not anymore. i ate them all.
B₁₂ wow! you have quite the appetite.
A₁₃ yeah, i like to eat . what do you do for a living?
B₁₄ i am a receptionist at a dentist office

[MSC Corpus (J. Xu et al., 2022) based on PersonaChat (S. Zhang et al., 2018)]

(0.1) is a conversation between two human beings who were asked to interact while displaying a given persona, described to them as a list of sentences like ‘I used to own bunnies, but I ate them.’. This example illustrates some of the challenges that arise when studying dialogue. In order to study a specific phenomenon, we often try to gather data that focuses on it. As a result, the conversations we study may not be representative of naturally occurring dialogue. Moreover, making people interact in a too constrained way can lead to unnatural behaviours that worsen this lack of representativeness. In this example, asking the participants to display their given persona led them to build all their interaction around it, while we typically do not infuse our conversations so explicitly with our personalities and interests.

Yet natural language is the base resource in Computational Linguistics (CL) and Natural Language Processing (NLP), the core fields this thesis is situated in. CL is an interdisciplinary field that leverages computational tools to analyse and model linguistic data. NLP is a more applied field that focuses on the development of computational systems that can process and generate natural language. Both domains rely on data to study and model language, and more recently NLP has used increasingly large amounts of data to develop, first, machine learning and deep learning models, and then large language models (LLMs), that can solve a wide range of tasks in natural language understanding and generation (Karanikolas et al., 2024), though they lack in explainability and interpretability. LLMs have been used to build conversational agents that can interact with humans in a dialogical fashion. These new technologies have led to a surge of studies in NLP meant to improve the performance of LLMs on dialogue tasks through various angles, one of which being the addition of a persona to the model in order to make them more personalised. (0.1) is an example from the MSC corpus (J. Xu et al., 2022), which is itself a multi-session

extension of the PersonaChat corpus (S. Zhang et al., 2018). PersonaChat was built to train models that would be ‘more consistent and engaging’ thanks to their ‘endow[ment] with a persona’. Such practices question what a conversation *is*. While (0.1) is a conversation between two human beings, it does not feel very natural. Conversely, numerous exchanges with conversational agents now occur every day and dismissing them as not being ‘real’ conversations also seems problematic. This fuels the need to better understand what a conversation is and with what parameters it can be characterised.

This question is of particular significance to us as one of the motivations that led to this thesis is the analysis of pathological conversations, i.e. conversations that involve a patient whose pathology affects their ability to communicate. One type of disfluency we observed in real interactions with patients with schizophrenia was linked to a peculiar way to change topics without any explicit signal.

Include the idea that patients with schizophrenia produce more word associations, hence more peculiar topic shifts (Maher et al., 2005). [Amandine]

In (0.2), the patient (A) changes the interpretation of the concept of death which is used figuratively in the first turns (124 and 125) before it becomes literal in turn 126. The lack of explicit signals makes it hard for the psychologist (B) to identify the topic shift and they must explicitly ask for a confirmation.

(0.2) **I**₁₂₄ Oh yeah (↑) and complicated (↓) and it’s really very very complicated (→) politics, it’s really something when you get into it, have to win or else when you lose, well, you’re finished (↓) (Fr. *Oh ouais* (↑) *et pis compliqué* (↓) *et c’est vraiment très très compliqué* (→) *la politique c’est quelque chose quand on s’en occupe faut être gagnant parce qu’autrement quand on est perdant c’est fini quoi* (↓))

B₁₂₅ Yes (Fr. *Oui*)

A₁₂₆ JCD is dead, L is dead, P is dead uh (...) (Fr. *J. C. D. est mort, L. est mort, P. est mort euh* (...))

B₁₂₇ So you think they’re dead because they lost (↑) (Fr. *Ils sont morts parce qu’ils ont perdu à votre avis* (↑))

A₁₂₈ No they won but if they’re dead, it’s their disease well it’s it’s (→) (Fr. *Non ils gagnaient mais si ils sont morts, c’est la maladie quoi c’est c’est* (→))

[SLAM Data]

Determining whether this kind of topic shift can be considered a cue to identify a pathology or whether it normally happens in conversation requires a general theory of topical organisation in conversation. But this in turn requires a way to characterise the different types of conversations, as topical structure is likely to vary depending on the type of conversation. This thesis thus focuses on two complementary axis: first, we aim to define a characterisation of dialogue that can be used to describe, compare and select dialogue resources, but also define the scope of a study or application; second, we focus on topical structure as a transversal level of analysis. We discuss the core issues of both axes in Sections 0.1 and 0.2 respectively, and present our research questions in Section 0.3. The outline of this thesis is then given in Section 0.4.

1 Studying Conversation

Conversation is our primary mode of communication. We engage in dialogue in our everyday lives for social, practical, and professional purposes, often several times a day, sometimes in different languages and increasingly through a variety of media. While dialogue generally feels natural and effortless, it is in fact a unique and constantly renewed experience: one can feel entirely at ease in one interaction and, only moments later, experience difficulty maintaining a conversation with a different interlocutor who feels constantly slightly off-topic.

These variations are not incidental. Dialogue mobilises multiple capabilities simultaneously and requires participants to align, negotiate, and continuously adjust them in cooperation with one another. Conversational success depends not only on linguistic competence, but also on shared assumptions, attentional coordination, sensitivity to interactional cues, and ability to repair disruptions and misunderstandings. The resulting complexity makes dialogue a rich and challenging object of study, and one that has been approached from many different angles across a wide range of disciplines.

Studying it is all the more complicated because due to its situated and ephemeral nature. Capturing its essence desperately implies to destroy a part of it due to various constraints. The most direct one is that making people aware that they are being observed and recorded can alter their behaviour. This is known as the observer's paradox (Labov, 1973). In addition, the very act of recording a conversation can be disruptive, as the presence of recording equipment may alter the natural flow of interaction. Even when these issues are mitigated, open-domain conversations create additional issues related to privacy concerns. Ideally, we would be able to share all the data we collect, as it would allow other researchers to replicate our studies and build upon our work. This ideal, known as Open Science, is a growing movement in the scientific community that aims to make research more transparent and its results more openly accessible. However, sharing data from conversations can be problematic, as it may contain sensitive information about the participants. Open-domain dialogues are particularly challenging in this regard, as the freedom of the participants to discuss any topic can lead to the disclosure of personal information. Amblard et al. (2014) have shown that long exchanges are impossible to fully anonymise as long as a large enough context remains.

Since naturally occurring conversation is so complicated to collect, studies are often done on <people who were asked to have a conversation in a given setting>. Tasks that require interaction have been designed in order to elicit conversation and help participants abstract away from the experimental setting but also to analyse phenomena specific to certain conditions. (0.3) is an example of dialogue from a widely used corpus in the dialogue system community, the MultiWOZ corpus (Zang et al., 2020). In this corpus, participants are asked to play the role of a customer and a clerk in a booking scenario. The task consists in booking a reservation for a restaurant, trip, or other similar domains based on some requirements. The conversation is thus constrained by the task, which can lead to behaviours that differ from more free-form interactions.

- (0.3) **USR₁** I need train reservations from Norwich to Cambridge
SYS₂ I have 133 trains matching your request. Is there a specific day and time you would like to travel?
USR₃ I'd like to leave on Monday and arrive by 18:00.
SYS₄ There are 12 trains for the day and time you request. Would you like to book it now?
USR₅ Before booking, I would also like to know the travel time, price, and departure time please.
SYS₆ There are 12 trains meeting your needs with the first leaving at 05:16 and the last one leaving at 16:16. Do you want to book one of these?
USR₇ No hold off on booking for now. Can you help me find an attraction called cineworld cinema?
SYS₈ Yes it is a cinema located in the south part of town what information would you like on it?
USR₉ Yes, that was all I needed. Thank you very much!
SYS₁₀ Thank you for using our system.

[MultiWOZ 2.2]

In other cases dialogue resources are built by asking people to have a conversation with few constraints as in (0.1). Manually written dialogues are also used in numerous studies as they mitigate all the issues related to the collection of naturally occurring dialogues. However, they are also problematic as they may

appear artificial and do not necessarily reflect the dynamics of a real conversation. (0.4) and (0.5) are two examples from the manually written corpus DailyDialog (Y. Li et al., 2017), which is also widely used in NLP, in particular for emotion recognition tasks thanks to the emotion labels that are provided for each speech turn. While (0.4) may pass as a somewhat natural conversation, though the basic transcription scheme does not capture any interactional cues, (0.5) is clearly artificial.

- (0.4) **A**₁ Hey. Why did you take that money? You are such a cheater! I should send you to jail!
B₂ I am not cheating. When you pass go, you collect \$ 200, Everyone knows that!
A₃ Well you can't just take the money. You have to ask the bank for money. And I' m the banker.
B₄ Banker?
A₅ Yes...
B₆ Can I have my \$ 200 please?
A₇ Sure. Here you are, \$ 200, Thank you, please come again! Now it's my turn to roll the dice.
 [DailyDialog]

- (0.5) **A**₁ Are you going to the annual party ? I can give you a ride if you need one
B₂ Thanks a lot. That's the favor I was going to ask you for.
A₃ The pleasure is mine.
 [DailyDialog]

Finally the rise of LLMs has increased the use of fully synthetic conversations in NLP. These dialogues are generated by a model, often in a zero-shot setting, i.e. by simply providing a prompt to the model without any additional training. More sophisticated methods have also been developed to enhance some specific phenomena like having two models 'interact' with one another, or have one produce a 'dialogue template', i.e. a frame of a conversation with some slots to fill, that another model, or the same one in a second phase, must fill in order to produce a complete dialogue.

All these methods and examples clearly differ from each other and from face-to-face interactions in some manner. Yet they are all used to study dialogue and to build models that are meant to process or generate it. The central question that arises is thus what do these differences entail for the study of dialogue and how can we characterise them in order to select the most appropriate resource for a given study or application. In particular, topical structure intuitively seems to be affected by the type of conversation, with task-oriented dialogues being more focused and constrained than open-domain ones for instance. We discuss issues in the study of topics and topical structure in Section 0.2.

2 Issues with studying topics

Topics are not a directly observable feature of language use, but rather an inferred one. Some aspects of conversation are completely linguistic, such as the vocabulary used, while others are more interactional and require more complete transcription formats to be captured, such as overlaps between dialogue turns. Topics however are more abstract and require a more holistic view of the conversation to be identified. They can be defined at several levels or granularities, where we can classify whole conversations based on their global topic, or identify what more or less long sequences of utterances are about, or even label the focus of individual turns.

These multiple levels of granularity imply that clearly defining the goal of a study and the type of data that is analysed are crucial aspects to consider. Identifying topic shifts in a short exchange of a few turns is fundamentally different from doing so in a long conversation. Similarly, working with naturally

occurring data versus synthetic data, or with organised dialogues versus more free-form ones, is likely to yield different results. Yet most studies focus on monologic discourse or organised dialogues such as meetings. For instance PersonaChat, the corpus from which (0.1) was taken, has been used to build a benchmark for topic shift aware dialogue modelling (Xie et al., 2021). The topic transitions in (0.1) are rather clear due to the unnaturalness of the dialogue that stems from the constraints imposed on the participants. In contrast, we are interested in naturalistic data where complex topical structures can appear.

STAC (Asher et al., 2016) is a corpus of multi-party chat-based English conversations occurring between players of the game (Settlers of) Catan. In this game, players impersonate settlers building each their own settlement. They must acquire and trade resources to develop them which leads to many exchanges where one asks for a resource, like clay, and the other participant may decline or inquire for another resource in exchange, like sheep or wheat. (0.6) is a piece of dialogue from this corpus where several conversational threads are intertwined.

- (0.6) **LJ₁₆₅** anyone want sheep for clay?
GWFS₁₆₆ got none, sorry :(
GWFS₁₆₇ so how do people know about the league?
WM₁₆₈ no
LJ₁₇₀ i did the trails
LJ₁₇₃ trials even
TK₁₇₄ i know about it from my gf
GWFS₁₇₅ [yeah me too,]_a [are you an Informatics student then, lj?]_b
TK₁₇₆ did not do the trials
WM₁₇₇ has anyone got wood for me?
GWFS₁₇₈ [I did them]_a [because a friend did]_b
GWFS₁₇₉ lol wm, you cad
GWFS₁₈₀ afraid not :(
LJ₁₈₁ [no, I'm about to start math.]_a [I just hang around appleton a lot]_b
TK₁₈₂ sry no
GWFS₁₈₃ my single wood is precious
WM₁₈₄ what's a cad?

[STAC corpus (Asher et al., 2016, Ex.1)]

This example shows that topical structure can be more complex than a basic linear sequence, and that only identifying topic shifts is not always sufficient. Here the multitude of participants and the written modality allow players to hold several sub-conversations at the same time. Some participants talk about “the league” while others have an interaction on resources – typical for this game – and one participant (GWS) takes part in all of them. In this dialogue, while associating a dialogue turn with its topic is not always straightforward, the vocabulary makes it quite easy to disentangle the threads when looking further than isolated sentences. But even this is not always a guaranteed strategy to interpret a conversation.

(0.7) is a piece of naturally occurring dialogue from the British National Corpus (Consortium, 1994) described as an ‘Air UK sales meeting’. While we can understand from the vocabulary used and this description that the global topic is about flights, the conversation is hard to interpret and more specific sub-topics are not easy to identify.

- (0.7) **JN7PSUNK₁** Right, you you want thirty seats on that flight?
PS4BS₂ No no no can't be done <unclear> haven't got a clue <laugh>oh dear.

- PS4BS₃ Anyway
- PS4BP₄ Just thought I'd mention the Stansted leading forward
- PS4BP₅ That's that's the same thing as says, it's reducing the catchment all the time
- PS4BS₆ What was it before?
- PS4BP₇ Seven 0 five
- PS4BS₈ Yeah well the thing is that something's gotta shift because we've got <unclear> we've got all these bloody flights going out the same time
- PS4BP₉ Well I know but
- PS4BS₁₀ And they can't cope
- PS4BP₁₁ Every time you lead it forward like that those early hours will just reduce
- PS4BP₁₂ Well that's what I said, it's nearly half an hour later
- PS4BS₁₃ Well actually it depends on the weather don't it?
- PS4BS₁₄ It depends on the
- PS4BS₁₅ Alright well if you've got any little any little points like that then you ought to er say like, we won't actually be much longer.
- PS4BS₁₆ We'll be er five ten minutes or so
- PS4BR₁₇ Oh, is that all?
- PS4BS₁₈ Yeah.
- PS4BS₁₉ You can sit and wait if you want to
- PS4BR₂₀ Erm, okay thanks, as long as I don't to er put you off or anything
- PS4BS₂₁ Yeah, yeah you can sit and wait if you like oh no <unclear> go away and come back again.

Dialogue JN7 [BNC]

These examples show that topical structures can be complex in naturally occurring dialogue, and that even identifying the topics can be challenging. Gruenstein et al. (2005) have shown that annotators struggle to reach an agreement when asked to identify topic shifts in the ICSI Meeting Corpus (Janin et al., 2003), both in a coarse-grained and a fine-grained setting. This corpus contains naturally occurring meetings from the International Computer Science Institute (ICSI), which form open-domain conversations with a pretty loose structure though the overall range of topics is somewhat restricted by the recording location. The high-quality recording setup allowed for the production of great written transcriptions. Compared to (0.7) where some turns are not fully transcribed and contain unclear tokens, and some speakers are labelled as unknown, conversations from the ICSI corpus are probably slightly easier to understand. The two main difficulties, that also exist when studying dialogues from the BNC and many other corpora, are that annotators tend to be unfamiliar with the topic, and that they are external to the conversation. These difficulties are probably largely responsible for the low agreement between annotators when marking topic shifts.

On the same corpus, Galley et al. (2003) used Cochran's Q test (Cochran, 1950) to determine whether 'the number of subjects assigning a boundary at any position was randomly distributed'. However, they did not provide any details on how they actually performed the test. In its original version, Cochran's Q test was intended to compare the number of successes of an event between different groups. In terms of annotations, it means that it would estimate whether all annotators assigned a similar number of boundaries, without accounting for *where* they were assigned. While this estimation may provide useful insights in some biomedical settings, which is the context in which Cochran's Q test was originally developed, it cannot be considered as an inter-annotator agreement measure in the context of topic segmentation.

Check if
there is an
Agenda
[Aman-
line]

Banerjee and Rudnicky (2007) have shown that when asked to segment meetings into segments that each correspond to a single agenda item and provided with the agenda, annotators could produce very similar segmentations. This shows that text segmentation is not an impossible task but that the definition of what a topic is and the context in which the task is performed are crucial.

For these reasons, building corpora of naturally occurring dialogues annotated with topics is a challenging task as the openness of the conversations makes it difficult to formally define the notion of topic. We thus had to find indirect ways to extract and analyse topical structures as we discuss when detailing the outline of this work in Section 0.4. Before that, we present our research questions in Section 0.3.

3 Focus and Contributions

Our work is situated at the crossroads of Computational Linguistics and Natural Language Processing. These two fields, though initially closely related, have diverged in the past few years. The rise of machine learning, deep learning and large language models has led to a focus on performance in NLP that is detrimental to the study of data and language. We are firmly committed to a data-driven approach, and we try to reconcile CL and NLP by operationalising CL theories in a way that can be used in NLP. In particular, we consider conversation in all its richness and explore the limitations of current approaches to topical structure analysis when applied to complex data.

RQ1 How can dialogue be characterised in a way that accounts for its variability while informing the selection of suitable resources and evaluation methods?

RQ2 Can topical structure be automatically extracted from dialogue data, and in particular from open-ended multi-party conversations?

RQ3 Can topical structure be leveraged as a discriminative signal for differentiating dialogue types?

4 Thesis outline

This thesis is organised in two parts. The first one, Part I, focuses on the characterisation of dialogue that we envision as a tool to describe, compare and select dialogue resources, but also define the scope of a study or application. Chapter 1 introduces the basic concepts around conversation that are required to understand the rest of the thesis. Chapter 2 presents a theoretical characterisation of dialogue that is based on the literature from various linguistics and cognitive science fields. This chapter also proposes a set of practical definition parameters for conversational data that can be used in a descriptive way but also included in computational models. Chapter 3 discusses recent trends in the NLP and CL communities through a survey of recent literature and shows how the characterisation can be used to alleviate some of the issues that arise when studying dialogue. This chapter shows our positioning in the field currently increasingly dominated by studies on conversational agents, and suggests directions for a responsible and explainable research on dialogue at the era of LLMs. Together, these three chapters answer RQ1.

The second part, Part II, focuses on topical structure in conversation as a transversal level of analysis. It relates three applied studies that explore topical structure in different types of dialogue and show the complexity of open-domain multi-party conversations. Chapter 4 introduces the core concepts and issues around topical structure that we need to define the scope of our study. Chapter 5 examines the limitations of existing approaches to linear topic segmentation in dialogue. Due to the scarcity of annotated data, we work on transcriptions of the Friends TV show in English, and show that even scripted dialogues reveal weaknesses in current models. This chapter also highlights the importance of really investigating the data we work on. Chapter 6 probes a more flexible approach to topical structure investigation. Chapters 5 and 6

both participate in answering RQ2. Then, Chapter 7 contributes to answering RQ2 and RQ3 by pondering the predictability of topics in varied interactional settings. This chapter highlights the difference between monologic data and dialogues in terms of organisation but also the singularity of multi-party open conversations. This difference is crucial in NLP as it shows that models trained on monologic data are not necessarily suitable for dialogue tasks, and that not all types of dialogues are equal.

Finally, Chapter 8 concludes the thesis by summarising our contributions and discussing the limitations of our work. Moreover, it discusses the increasing space dedicated to LLMs in our community and the impact it has on the study of dialogue. It also suggests directions for future research on dialogue and topical structure.

Part I

Linguistic and Cognitive Perspectives on Dialogue: Toward an operational descriptive framework

1

Dialogue as Coordinated Interaction: Foundational Concepts and Perspectives

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Conversation is a deceptively simple phenomenon. It is our primary mode of communication and we partake in it daily, in various forms and contexts. Yet it is also remarkably complex, involving the coordination of multiple cognitive, social, linguistic, and interactional processes. The following example illustrates this complexity:

(1.1) **PASSENGER** star sign (Sw. *stjärntecken*)

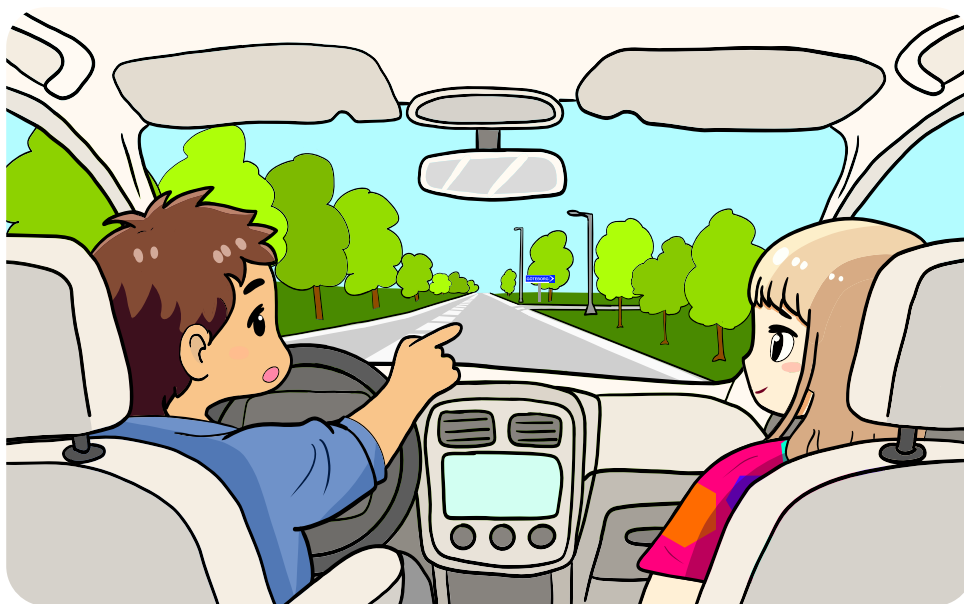


Figure 1.1: Possible illustration of the *Star sign* example (1.1)

DRIVER (*interrupting to clarify a navigation instruction*)

it's not this one (Sw. *det är inte den här det*)

PASSENGER nope (Sw. *nä*)

DRIVER nope (Sw. *nä*)

PASSENGER star sign (Sw. *stjärntecken*)

DRIVER Scorpio (Sw. *skorpion*)

[Villing (2015, p.63)]

This exchange takes place in a driving experiment where the passenger provides navigation instructions while simultaneously conducting an interview where they must gather as many trivia information as possible based on a question sheet. The navigation instructions must be delivered shortly before they are needed and thus cannot be much anticipated, making the navigation task time-sensitive. Hence the interaction is shaped by competing tasks, time pressure – because the passenger must gather as many answers as possible for the interview task – and asymmetries in access to information. Despite its apparent simplicity for the participants, this piece of conversation involves contextual interpretation, fine-grained temporal coordination, as well as syntactically incomplete sentences. Fig. 1.1 provides a possible illustration of this exchange, showing the participants and the physical environment in which the interaction takes place. The driver is pointing towards the next intersection, thereby clarifying the ‘this one’ in ‘it’s not this one’. Since we do not have access to a detailed description of the interaction, this is only a possible representation of the exchange, but it serves to illustrate the different dimensions of dialogue that we will discuss in this chapter.

In the following we introduce the foundational concepts and perspectives required to investigate the notion of dialogue and its variety of forms. We will use the above example to illustrate its different dimensions and to introduce key concepts. We begin by discussing the multimodal nature of dialogue and the role of these different modes of expression in understanding each other and building a conversation (Section 1.1). We will then discuss how participants strive to maintain coherence in their interaction and how it affects their interpretation of each other’s contributions (Section 1.2). In Section 1.3 we will see

how language is used to perform actions and how this shapes possibilities and constraints in interaction. In Section 1.4 we will discuss more formal approaches that aim at representing the different states of interaction and the coordination across participants. Finally, in Section 1.5 we will discuss the multidisciplinary nature of dialogue and how we can integrate these different perspectives in a descriptive framework to characterise dialogue. All of this serves an epistemological purpose where we aim to understand the nature of dialogue and how it can be studied, but it also has a direct impact on the way dialogical data should be studied and on how tools should be developed.

1.1 Signals Beyond Words: Multimodality, Redundancy, and Monitoring

Dialogue cannot be reduced to a mere sequence of verbal exchanges. Rather, it is a complex, situated activity that mobilises multiple channels of communication. In addition to linguistic content, interlocutors rely on non-verbal information conveyed through vocal features such as intonation and prosody, as well as through embodied cues including gestures, posture, and facial expressions. These diverse signals jointly contribute to the construction and interpretation of meaning in interaction. In this section we will discuss the different modes of expression that are involved in dialogue and how they contribute to the emergence of understanding (Section 1.1.1). We will also discuss the role of redundancy in making communication more robust (Section 1.1.2).

1.1.1 Modes of expression in interaction

The different channels through which information is conveyed in interaction are commonly referred to as *modalities*. Human communication is inherently *multimodal*, in that it simultaneously exploits several such channels. These modalities play a crucial role in the interpretation of dialogue, as they enable interlocutors to convey information that is not always explicitly encoded in linguistic form.

In the example discussed above, the participants are engaged in a face-to-face interaction – albeit in a side-by-side configuration – which affords both auditory and partial visual access to one another. Crucially, it also provides them with a shared perceptual environment: the road. This shared context is essential for the intelligibility of the exchange. Without these additional communicative resources, the sequence of utterances from (1.1) would be considerably more difficult, if not impossible, to interpret, and would likely not arise in the same form. For instance, the driver’s interruption relies on the shared visual scene to be understood as pertaining to the navigation task, despite the passenger’s expectation of a response related to the interview task. The interpretation of the utterance thus depends on the integration of linguistic input with contextual and perceptual information.

Several modalities are typically involved in dialogue. Speech conveys not only lexical and grammatical information, but also paralinguistic features such as tone and prosody. Tone may contribute to the disambiguation of lexical, grammatical, or morphological distinctions, depending on the language, while prosody plays a central role in expressing paralinguistic meanings, such as emphasis, affect, or irony (Y. Xu, 2019). Facial expressions and gaze behaviour provide cues about interlocutors’ attention, engagement, and understanding, as well as their emotional stance. Gestures can serve referential functions, for instance by pointing to entities in the environment, or descriptive functions, by visually representing aspects of the content being communicated. Finally, the physical environment itself constitutes a crucial resource: although external to the participants, it is routinely incorporated into the interaction, whether by grounding references, enabling joint attention, or introducing new topics and referents.

Fig. 1.2 provides a schematic representation of the different modalities involved in (1.1), in particular when the driver says ‘it’s not this one’. The auditory input includes the linguistic content of the utterance which is ambiguous on its own because the referent ‘this one’ must be identified. However, the visual input

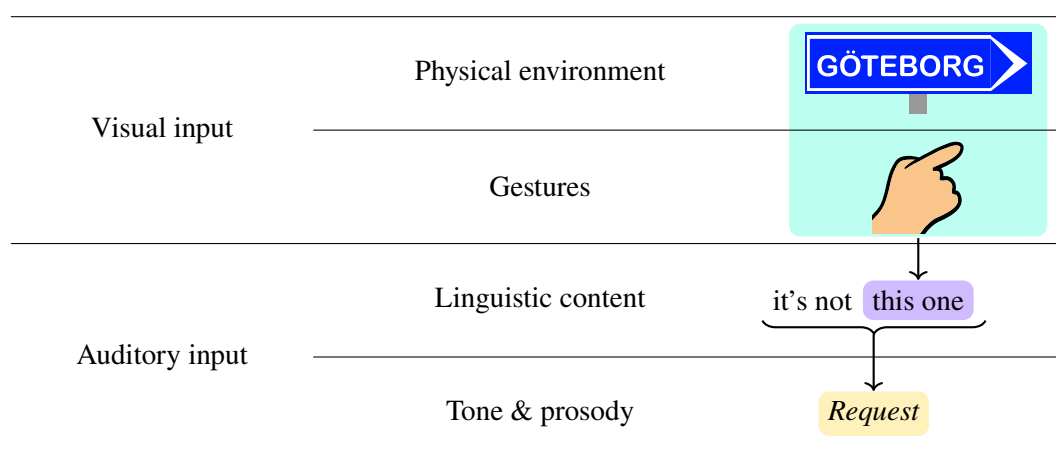


Figure 1.2: Different modalities involved in the *Star sign* example

that includes both the shared physical environment and the driver’s potential gestures, help explicit the referent of the driver’s utterance ‘it’s not this one’. The auditory input also includes the tone and prosody of the driver, which may indicate that their contribution is a confirmation request, and thus further guide the interpretation of the utterance. These different modalities thus provide complementary information that jointly contribute to the interpretation of the driver’s contribution and to the emergence of mutual understanding in this exchange.

From an NLP perspective, multimodality presents challenges for the development of computational models that can effectively process and integrate information from multiple channels. It requires to either encode several modalities into a single representation or to develop mechanisms to account for the missing ones, but also model their potential interactions. Most of the data we have access to is unimodal, and in particular textual, so enriched transcription schemes are necessary to capture the multimodal nature of dialogue. Yet such schemes require more expertise from the annotators and are more time-consuming to use.

Include refs of NLP work on multimodal models.

[Amandine]

1.1.2 Redundancy and robustness in communication

In multimodal interaction, different modalities often convey partially overlapping, or *redundant*, information. That is, content expressed through one channel may also be signalled through one or several others. This redundancy constitutes a central resource for the organisation and stability of dialogue.

A key aspect of conversational interaction is the participants’ capacity to attend to one another and to regulate the interaction accordingly. Speakers monitor their addressees to assess whether they are following, while addressees continuously provide feedback regarding their level of attention, interest, and understanding. This process can be described as *mutual monitoring*, whereby interlocutors track both their own communicative behaviour and that of others in order to maintain coordination and adapt when necessary.

Crucially, signals of attention and understanding can be conveyed through multiple modalities. For instance, addressees may produce visible cues such as head nods, as well as verbal feedback in the form of *backchannels*, i.e. short responses such as ‘uh-huh’ or ‘yeah’. These signals often co-occur or can substitute for one another, thereby ensuring that relevant information is available even if one modality is degraded or overlooked.

Redundancy thus contributes to the robustness of communication. By distributing information across

several channels, it allows participants to compensate for potential perceptual or interpretative failures. Moreover, it enables interlocutors to cross-check their interpretations by integrating cues from different modalities.

This can be illustrated by the example discussed above. The driver relies on the shared visual environment to signal that their utterance pertains to the navigation task, despite the passenger's expectation of a response related to the interview task. In addition to this contextual cue, the driver could further disambiguate their intention through gestures, such as pointing towards the road as represented on Fig. 1.1, or through head movements orienting the addressee's attention. This redundancy is represented on Fig. 1.2 by two aspects of visual input that both contribute to the disambiguation of 'this one'. This increases the likelihood that the intended interpretation is successfully achieved.

An example of NLP task that can benefit from multimodality and redundancy is the task of coreference resolution. In this task, the goal is to identify which expressions in a discourse refer to the same entity. Multimodal information can provide additional cues for resolving coreference, such as visual context or prosodic features. For instance, if a speaker refers to an object while pointing at it, the visual cue can help disambiguate which object is being referred to. By leveraging multimodal information and redundancy, NLP models can improve their performance on coreference resolution tasks. Conversely, ignoring multimodal cues can lead to misinterpretations, especially in cases where linguistic information alone is ambiguous or insufficient.

1.2 Coherence and Coordination in Dialogue

Conversation is a fundamentally collaborative activity in which participants coordinate their contributions in order to achieve mutual understanding and exchange information, ideas, and intentions. This coordination relies on a set of implicit assumptions that allow interlocutors to interpret utterances without requiring all relevant information to be made explicit. These assumptions were formalised by Grice (1975) under the *Cooperative Principle*. In this section, we first present the conversational maxims that derive from this principle (Section 1.2.1), then discuss dialogue as a joint activity following Clark's framework (Section 1.2.2), and finally examine the mechanisms through which participants repair breakdowns in interaction (Section 1.2.3).

1.2.1 Coherence and the cooperative nature of conversation

Grice's Cooperative Principle states that participants in a conversation typically orient towards making contributions that are appropriate to the ongoing interaction. This general principle is articulated through four conversational maxims: Quantity, Quality, Relation, and Manner.

The maxim of *Quantity* requires speakers to provide as much information as is needed for the current purposes of the exchange, but no more. When an utterance appears to contain either too little or too much information, interlocutors tend to infer additional meaning in order to reconcile this apparent mismatch. In particular, excessive detail may prompt the addressee to search for a reason why such information is relevant, which partly explains why over-explanation can be cognitively taxing in dialogue (Sperber & Wilson, 1986; Kalyuga, 2009).

The maxim of *Quality* concerns the truthfulness of contributions. Speakers are expected to assert only what they believe to be true or for which they have adequate evidence. Apparent violations of this maxim, such as obviously false statements, are often interpreted as signalling non-literal meanings, for instance irony or sarcasm.

The maxim of *Relation* – summarised as 'Be relevant' – requires contributions to be pertinent to the ongoing interaction. Although relevance is notoriously difficult to define formally – even in Grice's

own account – participants generally possess strong intuitions about what counts as relevant in a given context. Utterances that appear off-topic are typically interpreted by attempting to reconstruct a context in which they would make sense. In practice, relevance is dynamic and context-dependent: in highly structured exchanges, such as question–answer sequences, strong expectations constrain what counts as an appropriate response, whereas in more open or casual interactions, abrupt topic shifts may be acceptable. In the example discussed above, the driver’s interruption constitutes a marked shift from the expected answer to a question. However, it remains interpretable as relevant due to the time-sensitive nature of the navigation task, which justifies the prioritisation of this contribution.

The maxim of *Manner* pertains to *how* something is said rather than to *what* is said. It can be broadly summarised as a requirement to be clear, orderly, and unambiguous. As with the other maxims, this requirement is context-sensitive. In naturally occurring dialogue, utterances are often fragmentary or syntactically incomplete, yet they remain interpretable due to contextual support and multimodal cues. For instance, interrogative force may be conveyed through intonation rather than explicit syntactic marking, or inferred from the interactional context.

Importantly, Grice emphasises that the maxims are not strict rules but rather guiding principles. Their apparent violation is often meaningful: when a maxim is not observed, interlocutors typically assume that this deviation is purposeful and attempt to infer the underlying intention. However, not all deviations are deliberate. Speakers may inadvertently produce over-informative, irrelevant, or unclear contributions. Moreover, children, non-native speakers, or neurodivergent individuals may not consistently adhere to the maxims, either due to developmental factors or differences in communicative norms ([Amandine: Read Surian, 1996 and Asada et al., 2022]). Interpreting such cases requires balancing the assumption of cooperativeness with sensitivity to individual differences and situational factors. This interpretative process highlights the inherently collaborative nature of dialogue, where meaning is co-constructed by participants in context.

The following example illustrates the maxims and shows how they can be exploited to create humour. This excerpt is taken from the French comedy television series *Kaamelott*, set in the Arthurian legend and well known for its use of language as a comedic resource. In this scene, the two knights Perceval and Karadoc have just discovered the entrance gate of a lost temple they had been searching for, but they are unsure how to proceed. Indeed, while they typically never complete their missions successfully, they have located the entrance to this temple with surprising ease, which they find suspicious. However, the temple is sealed by a gate, which they expect to be locked and therefore impossible to open. This expectation is reflected in Perceval’s remark: ‘there’s a door to open. We never manage that. So we’ll be back at Granny’s in no time.’

(1.2) **PERCEVAL** Right, let’s go! Now that we’re here, we’re not just going to stand around outside! (Fr. *Bon on y va ! Maintenant qu’on y est on va pas rester devant !*)

KARADOC Sorry, but it stinks. (Fr. *Je m’excuse, mais ça pue.*)

PERCEVAL Do you think so? (Fr. *Vous croyez ?*)

KARADOC Hang on! We always struggle to find things, and now we’ve found it first time, as if by chance? (Fr. *Attendez ! On rame à chaque fois et là on trouve du premier coup comme par hasard ?*)

PERCEVAL Blimey! What do we do now, then? (Fr. *Merde ! Qu’est-ce qu’on fait alors ?*)

KARADOC No way, I’m not keen on that. (Fr. *Sans blague, j’suis pas chaud.*)

PERCEVAL Mind you, there’s a door to open. We never manage that. So we’ll be back at Granny’s in no time. (Fr. *Remarquez y’a une porte à ouvrir. On y arrive jamais ça. Du coup on s’ra vite rentrés chez mémé.*)

KARADOC Yeah, OK. Do you have any tools? (Fr. *Ouais, ok. Vous avez des outils ?*)

PERCEVAL We don't know any tools that open doors. (Fr. *On les connaît pas les outils qui ouvrent les portes.*)

KARADOC Yeah, but do you have any tools? Even ones that don't open doors? (Fr. *Ouais mais vous avez des outils ? Même qui ouvrent pas les portes ?*)

PERCEVAL No. I've got my shirt, but it's flimsy. I'm afraid it might be a bit too flimsy. (Fr. *Non. J'ai ma chemise mais c'est mou. J'ai peur que ce soit un peu trop mou.*)

KARADOC Well, I've got a spoon. (Fr. *Bah moi j'ai une cuillère.*)

PERCEVAL Yeah, that's not bad; it looks like a key. (Fr. *Ouais c'est pas mal, ça ressemble à une clé.*)

[Kaamelott, S03E61]

At this point, the characters begin discussing tools they could use to open the door. Karadoc first asks Perceval whether he has any tools, to which Perceval replies that they do not *know* any tools that open doors. This response can be interpreted as a violation of the maxim of Relation, as it does not directly answer the question. However, if one assumes that identifying tools capable of opening doors is a prerequisite for answering the question meaningfully, the utterance can be reinterpreted as relevant. This illustrates a case in which the relevance of a contribution is not immediately transparent and depends on assumptions about the speaker's knowledge and reasoning.

Karadoc's subsequent contribution further complicates the interaction. He asks again if Perceval has tools, specifying 'even ones that don't open doors', which also violates the maxim of Relation, since such tools would be useless to the task. Perceval then responds with a straightforward 'No.', which directly answers the question but retrospectively reveals that his previous contribution was unhelpful. In this sense, the earlier utterance may also be seen as violating the maxim of Manner, as it introduces unnecessary complexity without contributing useful information. Indeed, knowledge about which tools open doors would only have been relevant if Perceval had possessed some tools but was uncertain about their utility.

After his 'No.', Perceval continues by adding that he has his 'shirt, but it's flimsy'. This can be seen as a violation of the maxim of Quantity since it provides more information than necessary to answer the question.

Finally, Karadoc mentions that he has a spoon, and Perceval evaluates this suggestion positively, stating that it is 'not bad' because 'it looks like a key'. This final exchange illustrates how humour can arise from a discrepancy between the characters' perspective and that of the audience. While Perceval appears to genuinely consider the spoon a plausible tool, the audience recognises the inadequacy of this reasoning. From the audience's perspective, the utterance thus constitutes a violation of the maxim of Quality, as it involves an obviously implausible claim.

This example is very dense in terms of violations of the conversational maxims, and it illustrates how these violations can be exploited to create humour. While the dialogue was manually crafted for comedic effect, similar phenomena can be observed in naturally occurring interaction, though often more sparsely. Being able to distinguish *voluntary* violations of the maxims from *involuntary* ones, and understanding the *underlying reasoning* behind a violation, is crucial for interpreting the meaning of contributions in dialogue. Recovering unsaid assumptions and reasoning is a core human capacity in interaction as we often rely on common sense knowledge and world knowledge to produce and interpret contributions. Yet it remains a challenge for NLP models that often lack such knowledge and reasoning capabilities.

1.2.2 Dialogue as a joint activity

Building on this perspective, Clark conceptualises dialogue as a *joint activity* in which participants collaboratively work towards shared goals. On this view, communication is not merely the transmission

of information from a speaker to an addressee, but a coordinated process in which both parties actively contribute to the establishment of mutual understanding.

A central notion in Clark’s framework is that of *common ground*, defined as the set of knowledge, beliefs, and assumptions that participants take to be shared. Contributions to a dialogue are designed relative to this common ground, and their interpretation depends on it. Establishing and updating common ground is therefore a core function of interaction.

Clark further introduces the concept of *grounding*, the process through which interlocutors provide evidence that an utterance has been understood sufficiently for current purposes. Grounding is achieved through various forms of feedback, including acknowledgements, repetitions, reformulations, or relevant next actions. Crucially, grounding is incremental and adaptive: participants continuously monitor each other’s behaviour and adjust their contributions accordingly.

From this perspective, dialogue can be understood as a joint project in which participants coordinate their actions, manage shared resources, and negotiate meaning in real time. This account complements the Gricean framework by emphasising the interactive and procedural dimensions of communication. It has consequences for NLP research, as it highlights the importance of modelling not only the content of utterances but also the dynamics of interaction, including turn-taking, feedback, and the evolution of common ground over time. With regards to dialogue systems, determining the right degree of initiative is crucial to alleviate the asymmetry between the user and the system and to ensure that the user does not need to compensate for poor interactional capabilities. Conversely, the tendency of LLMs to always go along with the user and display sycophantic behaviour (Cheng, Lee, et al., 2026) makes them deceptive conversational partners that only reinforce the user’s own biases and beliefs. Finding the right balance in the interactive dynamics of dialogue systems is therefore a key challenge for the development of effective and trustworthy conversational AI. Understanding the principles of dialogue as a joint activity can inform the design of more natural and cooperative interactions, but more crucially it can serve to ensure that users perceive the systems for what they are and do not overestimate their capabilities.

1.2.3 Repair and the maintenance of coherence

While the preceding discussion may have presented dialogue as a smooth and predictable process, every-day experience shows that dialogue is frequently subject to misunderstandings, ambiguities, and disruptions. Rather than being exceptional, such phenomena are an integral part of interaction. The mechanisms that address them are collectively referred to as *repairs*.

Repairs encompass the practices through which participants identify and resolve problems in speaking, hearing, or understanding. It may be initiated by the speaker or by the addressee (E. A. Schegloff et al., 1977), and it can take a variety of forms (Colman & Healey, 2011; Hough & Purver, 2013). For example, an addressee may request clarification or repetition when an utterance is unclear, while a speaker may spontaneously reformulate or correct their own contribution. Similarly, interlocutors may challenge the relevance or accuracy of a statement, prompting further elaboration or correction.

These repair strategies play a crucial role in maintaining coherence in dialogue. They allow participants to restore mutual understanding, realign their interpretations, and ensure the continuity of the interaction despite local breakdowns.

The example discussed above can be interpreted through this lens. The driver’s interruption may be analysed as a form of repair, insofar as it serves to clarify a previously given navigation instruction that has become uncertain in light of the evolving situation. By momentarily suspending the expected progression of the interaction, the speaker prioritises the resolution of a potential misunderstanding, thereby contributing to the overall coherence and success of the joint activity.

In NLP, identifying repairs is a downstream task that can support various applications, such as automatic speech recognition, dialogue state tracking, and conversational agents in general. A challenge in

this task is that repairs are often considered as disfluencies and are therefore filtered out in the transcription phase or the preprocessing of speech data. However, they are crucial for understanding the dynamics of dialogue and for developing systems that can effectively manage conversational breakdowns.

Expand on the fact that understanding speaker meaning is not the only important part, being able to identify and repair breakdowns is crucial for a successful interaction. [Amandine]

1.3 Acting through Language in Interaction

Language is not merely a medium for conveying information or expressing ideas. In everyday interaction, it is also a means of performing actions and effecting changes in the social and material world. This section examines how language functions as a form of action, and how this action-oriented perspective introduces both possibilities and constraints on interaction.

1.3.1 Utterances as Actions

Language has not always been approached as a vehicle for action. In early twentieth-century philosophy, particularly within logical positivism, language was primarily conceived as a tool for making statements that could be evaluated as true or false (Ayer, 1936). According to this view, a sentence was considered meaningful only if it could, in principle, be empirically verified or falsified. As a consequence, many forms of everyday language – such as ethical, evaluative, or metaphysical statements – were deemed meaningless, insofar as they did not straightforwardly describe states of affairs subject to verification (Carnap, 1931).

This perspective was grounded in the assumption that ordinary language is too vague and ambiguous for rigorous scientific inquiry, and that formal, artificially constructed languages were required to achieve clarity and precision (Frege et al., 1879). For example, the sentence ‘Bob loves stories on faeries and dragons’ can be understood as (1) Bob loving dragons and also stories on faeries, or (2) Bob loving stories on faeries and also stories on dragons, or (3) Bob loving stories that are about both faeries and dragons. There are indeed two ambiguities in this sentence: the first one is on the scope of the verb *loves* and the second one is on the scope of the conjunction *and*. However, if we represent this sentence in a formal language, we can make these readings explicit:

- (1) $\text{love}(\text{Bob}, \text{dragons} \wedge \text{stories}(\text{faeries}))$
- (2) $\text{love}(\text{Bob}, \text{stories}(\text{faeries}) \wedge \text{stories}(\text{dragons}))$
- (3) $\text{love}(\text{Bob}, \text{stories}(\text{faeries} \wedge \text{dragons}))$

In this example we use $\text{love}(x, y)$ to represent the relation of x loving y , and $\text{stories}(x)$ to represent the set of stories about x .

This position on language gave rise to traditions of formal representation, which will be discussed in a later section (Section 1.4). However, it was also met with criticism from philosophers who observed that ordinary language, despite its apparent imperfections, functions remarkably effectively in everyday communication. Rather than replacing it, they sought to understand the principles underlying its use.

Within this alternative perspective, philosophers such as Austin (1975) and Wittgenstein (1953) emphasised the central role of context in determining meaning. Austin, in particular, argued that utterances should be understood not only in terms of what they describe, but also in terms of what they *do*. This insight led to the development of speech act theory, which conceptualises language use as a form of action.

Austin distinguishes three interrelated dimensions of any utterance. The *locutionary act* refers to the production of a linguistic expression with a particular semantic and syntactic structure. The *illocutionary act* corresponds to the action performed in saying something, such as making a request, asking a question, or issuing a promise. The *perlocutionary act* concerns the effects that the utterance has on the addressee, such as persuading, convincing, or prompting them to act. While these three dimensions co-occur in many instances of language use, the illocutionary level is often central to the speaker's communicative intention.

Importantly, the illocutionary force of an utterance is not always directly encoded in its literal form. A classical example is a sentence of the type 'Can you pass me the salt?' where the speaker is not asking about the listener's ability to pass the salt but is making a request to make the listener pass the salt. Here the locutionary act is the production of the sentence itself, the illocutionary act is the *request* to pass the salt, and the perlocutionary act is the listener actually passing the salt. Such cases illustrate that the interpretation of utterances relies on contextual and pragmatic inference, rather than solely on their literal meaning.

1.3.2 Possibilities and constraints in interaction

If utterances are understood as actions, then dialogue can be seen as a space in which participants continuously shape and respond to opportunities for action. Each contribution both opens up and constrains the range of possible subsequent moves. In this sense, interaction is structured by a dynamic set of *affordances* and constraints that emerge from the evolving context.

Certain utterances project specific types of responses. For instance, questions make answers relevant, requests make compliance or refusal relevant, and offers make acceptance or rejection relevant. These expectations, often referred to as *adjacency pairs*, contribute to the overall coherence of dialogue by narrowing the space of appropriate next actions. At the same time, these constraints are not absolute: participants may delay, suspend, or transform expected responses, for example through embedded sequences or topic shifts, as discussed earlier.

More generally, the possibilities for action in interaction are shaped by multiple factors, including the current state of the dialogue, the shared context, and the participants' goals. Multimodal resources also play a role in this process, as gestures, gaze, and the physical environment can introduce new affordances or redirect attention. Conversely, constraints arise from social norms, task demands, and the need to maintain mutual understanding.

From this perspective, dialogue is not only a process of exchanging information, but also a coordinated activity in which participants navigate a structured yet flexible space of possible actions. Understanding how these possibilities and constraints are managed is essential for analysing how interaction unfolds and how meaning is jointly constructed. This action-oriented view of dialogue has clear implications for the design of conversational systems, which must be capable of recognising and responding to the affordances and constraints present in human interaction. It also highlights the importance of modelling not only the content of utterances but also their functional role in shaping the trajectory of the conversation. This idea is at the root of dialogue state tracking (Jacqmin et al., 2022), which is a core component of many dialogue systems. By maintaining an updated representation of the interactional state, such systems can anticipate user needs, manage turn-taking, and provide contextually appropriate responses.

1.4 Representing and Coordinating Understanding

As discussed above, early twentieth-century movements in philosophy and the theory of science advocated the development of formal languages governed by explicit and unambiguous rules, with the aim

of supporting rigorous reasoning. This perspective gave rise to a range of formal representation traditions. While initially designed to capture the semantic content of utterances, these approaches have since been extended to model broader aspects of interaction, including the evolving state of dialogue and the coordination between participants.

In this section, we examine how such formalisms can be used to represent dialogue. We first consider how the state of an interaction can be modelled (Section 1.4.1), and then turn to the question of how coordination across participants can be captured within these frameworks (Section 1.4.2).

1.4.1 Representing the state of interaction

Dialogue is a complex phenomenon involving multiple interdependent dimensions, including linguistic content, contextual information, participants' goals, and the evolving structure of the interaction. A central objective of formal approaches to dialogue is to provide representations that capture these dimensions in a systematic and tractable way.

One important motivation for such representations is the need to account for the incremental nature of dialogue. Interactions unfold over time, with each contribution updating the current state and shaping the range of possible subsequent moves. Formal models therefore aim not only to describe static configurations, but also to characterise how the state of the interaction evolves in response to new utterances. In particular, they seek to make explicit how a given conversational move transforms the current state into a subsequent one.

This dynamic perspective is especially relevant for computational applications, where it enables the implementation of dialogue systems capable of tracking and updating interactional states. By representing dialogue in a structured way, it becomes possible to model expected transitions, detect deviations, and analyse the conditions under which interaction succeeds or fails.

Frameworks such as dialogue scorekeeping models (**Lewis1979Scores**) or gameboards (Ginzburg, 2012) exemplify this approach. They provide structured representations of the interaction at any given point, with varying degrees of granularity depending on the phenomena of interest. Such representations typically include information about the current and previous utterances, the commitments, and other relevant aspects of the conversational context. By formalising these elements, they allow for the computation of state changes as the dialogue progresses.

Figure 1.3 is an example of formal representation of the state of the conversation after the driver's utterance 'it's not this one' in (1.1). This representation is based on the Type Theory with Records (TTR) framework developed by Cooper (2005, 2023), which provides a flexible and expressive formalism for modelling dialogue. In this representation, we use records, modelled by brackets, that contain fields representing different aspects of the interaction. In our example, we focus on two main fields: *physC* for the physical context and *l-m* for the linguistic moves.

In the *physC* field, we represent the shared physical context, including the variable x that denotes an entity¹, and the condition c_1 that captures the fact that x is an intersection of the road.

In the *l-m* field, we separate the linguistic moves between the current one (*curr*) and the previous ones (*prev*). For the sake of simplicity, we do not explicitly represent the content of the *prev* field and instead only specify its type as *RecType*, which stands for 'Record Type' and denotes a collection of labelled fields. In practice, this field should be seen as a recursive memory of the previous moves, which would be updated after each new utterance. It could contain a field for the latest move and one for the previous ones, similarly to the *l-m* field, which would allow us to keep track of the entire history of the interaction. Evaluating the degree of accessibility of this history is a separate question related to memory management.

¹In TTR entities in the Montagovian sense are denoted by the type 'Individual' (*Ind*) (Cooper, 2023, p.14).

$$(4.i) \left[\begin{array}{l} \text{physC} : \left[\begin{array}{l} x : \text{Ind} \\ c_1 : \text{intersection}(x) \end{array} \right] \\ 1-m : \left[\begin{array}{l} \text{prev} : \text{RecType} \\ \text{curr} : \left[\begin{array}{l} \text{spkr} = \text{DRIVER} : \text{Ind} \\ \text{addr} = \text{PASSENGER} : \text{Ind} \\ \text{ctnt} : [c_2 : \neg P(\text{physC}.x)] \end{array} \right] \end{array} \right] \end{array} \right]$$

Figure 1.3: Example of a TTR representation of the driver’s utterance ‘it’s not this one’ when both participants pay attention to the road

The *curr* field contains a record that represents the current move, including the speaker (*spkr*), the addressee (*addr*), and the logical content of the utterance (*ctnt*). The content field itself is a record that includes the condition c_2 representing the meaning of the driver’s utterance. Here we represent the content as a negation of the proposition $P(\text{physC}.x)$, which can be read as ‘it is not the case that the intersection validates a certain property P ’, where P is unspecified here but clearly exists in the participants’ minds either due to a previous utterance – most likely a navigation instruction – or due to the shared context of the navigation task. We refer to the intersection as $\text{physC}.x$ to indicate that we are referring to the x defined in the field *physC*.

While the representation in Fig. 1.3 focuses on a limited set of dimensions for the sake of clarity, the TTR framework is in principle capable of modelling a much wider range of aspects of interaction. In particular, it allows for the distinction between *shared* and *private* components of the dialogue state. Shared components typically include publicly accessible information, such as established commitments or the history of accepted utterances, whereas private components capture participant-specific information, such as individual goals, beliefs, or agendas. This distinction is central to modelling the asymmetries that may arise in interaction. Extensions of TTR, such as the KoS (a toponym for Conversation Oriented Semantics) framework (Ginzburg, 2012; Maraev et al., 2018), further develop these ideas by providing more specialised tools for representing dialogue structure and dynamics.

Crucially, such formalisms make it possible to represent not only a single, global state of the interaction, but also multiple, potentially divergent representations corresponding to different participants. This perspective shifts the focus from dialogue as a monolithic object to dialogue as a coordinated activity grounded in the alignment of individual representations. The question of how these representations relate to one another, and how coherence is maintained across participants, leads directly to the issue of coordination, which we address in Section 1.4.2.

1.4.2 Coordination across participants

Traditional approaches to language often treated discourse as an abstract object, with relatively little distinction between monological production and interactive dialogue. Although the term *discourse* can encompass both, this perspective tends to obscure the fundamental differences between dialogue and monological discourse, in particular the interactive nature of dialogue as a social practice (Couper-Kuhlen & Selting, 2017; Brennan et al., 2018).

More recent approaches emphasise that dialogue must be studied as a situated interaction between participants. From this perspective, representing a conversation requires accounting not only for a global interactional state, but also for the potentially differing representations maintained by each participant (missing reference).

In formal terms, this implies that each interlocutor may hold their own representation of the dialogue state, including their beliefs about the shared context and about the other participants’ knowledge and

$$(4.ii) \left[\begin{array}{l} \text{physC} : [] \\ 1-m : \left[\begin{array}{l} \text{prev} : \text{RecType} \\ \text{curr} : \left[\begin{array}{l} \text{spkr} = \text{DRIVER} : \text{Ind} \\ \text{addr} = \text{PASSENGER} : \text{Ind} \\ \text{ctnt} : \left[\begin{array}{l} x = ? : \text{Ind} \\ c_2 : \neg P(x) \end{array} \right] \end{array} \right] \end{array} \right] \end{array} \right]$$

Figure 1.4: Example of a TTR representation of the driver’s utterance ‘it’s not this one’ interpreted by the passenger when she is not paying attention to the road

intentions. These representations may themselves include higher-order beliefs, giving rise to recursive structures of the form ‘I know that you know that I know’, and so on. While such recursive embedding is, in principle, unbounded, in practice it is constrained by cognitive limitations and task demands. The finite or infinite nature of these representations has been a subject of debate in the literature and mostly depends on the approach chosen to model common ground (Geurts, 2024).

These participant-specific representations provide a useful framework for analysing coordination in dialogue. Successful interaction can be characterised, at least in part, as a form of alignment between participants’ representations of the current state. Conversely, misalignment – where participants maintain incompatible or divergent representations – offers a natural account of many forms of misunderstanding and breakdown.

For instance in the *Star Sign* example (1.1), if the passenger is not paying attention to the road, she may fail to update her representation of the physical context in a way that aligns with the driver’s. As a result, when the driver utters ‘it’s not this one’, the passenger would not manage to resolve the anaphora as the driver expects, leading to a misinterpretation of the utterance and a potential breakdown in the interaction. Fig. 1.4 illustrates the passenger’s representation of the interaction in this case. The field **physC** contains an empty record because the passenger is not attending to the road. The content of the driver’s utterance is still represented in the 1-m field, but the passenger’s failure to update the physical context prevents her from correctly interpreting it. Instead of disambiguating ‘this one’ with **physC.x**, we introduce an entity x that we leave unspecified, and the content of the utterance is represented as $\neg P(x)$. This representation cannot be aligned with the one in Fig. 1.3 because the passenger’s representation of the physical context does not contain the relevant information to resolve the anaphora, which leads to a misinterpretation of the utterance.

From an analytical perspective, such misalignments can often be reconstructed *a posteriori*, based on observable disruptions in the interaction. However, because these representations involve internal cognitive states, they are not directly accessible and must be inferred. This limits the extent to which breakdowns can be predicted in advance, although formal models can still provide valuable insights into their structure and dynamics.

From an applied perspective, these representations are particularly relevant for the design of interactive systems. A system that maintains an explicit representation of the dialogue state, along with a model of the user’s likely representation, can in principle detect emerging misalignments and either prevent or repair them. This highlights the practical importance of formalising coordination in dialogue.

(1.1) illustrates this point. A potential misalignment would arise if the passenger were to interpret the driver’s utterance ‘it’s not this one’ as relating to the interview task rather than to the navigation task. In such a case, the anaphoric expressions ‘it’ and ‘this one’ would be resolved within an inappropriate context, leading to misunderstanding and necessitating repair. This type of divergence can be naturally captured within a framework that models participant-specific representations of the interaction.

1.5 Dialogue as a Multidisciplinary Object of Study

The preceding sections have highlighted the multifaceted nature of dialogue and the wide range of processes involved in its production and interpretation. This complexity is reflected in the diversity of disciplines that have sought to study dialogue, each bringing their own theoretical frameworks, methodologies, and research objectives. In this section, we briefly outline the domains most relevant to the present work. This overview is not intended to be exhaustive; rather, it situates our approach within a broader intellectual landscape and emphasises complementary perspectives on dialogue.

1.5.1 Pragmatics

Pragmatics studies language use in context, with a particular focus on how meaning is constructed beyond the literal content of utterances (Grice, 1975; Levinson, 1983). Central notions include speaker intention, inference, implicature, and presupposition (*missing reference*). A key insight of this field is that a substantial portion of communicated meaning is not explicitly encoded in linguistic form, but must be inferred on the basis of a combination of linguistic form, shared assumptions and context.

In the example discussed above, the utterance ‘star sign’ does not constitute a complete sentence. Its interpretation as a request for information depends on the shared understanding of the interview task, as well as on culturally grounded knowledge about astrological signs. Pragmatics thus foregrounds the role of inference and context, while also interfacing with other domains that address how such inferences are processed and coordinated in real time.

1.5.2 Formal Semantics and Dialogue Modelling

Formal semantics and dialogue modelling approaches aim to represent meaning and interaction using logical and computational frameworks (Kamp & Uwe, 1993; Ginzburg, 2012). These approaches typically conceptualise dialogue in terms of information states, which track what has been said, what is mutually accepted, and how the conversational context evolves over time.

Such models are particularly well suited to computational applications, including dialogue systems, where explicit and manipulable representations of conversational structure are required. They offer a high degree of precision and abstraction, enabling the formalisation of dialogue dynamics independently of specific linguistic realisations. However, this level of abstraction may come at the cost of simplifying or under-representing the situated, embodied, and temporally constrained aspects of interaction. As a result, formal approaches capture certain dimensions of dialogue while leaving others underspecified.

1.5.3 Psycholinguistics

Psycholinguistics investigates the cognitive processes underlying language production and comprehension (Levett, 1983; Pickering & Garrod, 2004). Dialogue constitutes a particularly demanding setting, as production and comprehension typically occur simultaneously and must be tightly coordinated.

Although the present work does not directly model cognitive mechanisms, psycholinguistic findings remain highly relevant for characterising dialogue. Factors such as cognitive load, attention, and processing limitations can significantly influence the form and timing of contributions (*missing reference*). In the example discussed above, the dual-task context likely affects how participants manage turn-taking, formulate utterances, and allocate attention. Psycholinguistics thus provides an important perspective on the constraints that shape conversational behaviour.

1.5.4 Sociolinguistics

Sociolinguistics and linguistic anthropology approach language as a social and cultural practice (missing reference). They examine how linguistic variation both reflects and constructs social identities, roles, and norms. From this perspective, dialogue is not only a means of exchanging information, but also a way of maintaining social relationships and enacting social structures.

Conversational norms – such as expectations regarding turn-taking, politeness, or topic management – play a central role in shaping interaction. For example, interruptions may be evaluated differently depending on participants' roles and the situational context. In task-oriented settings, deviations from normative patterns may be tolerated or even required due to competing priorities.

More broadly, some accounts propose that language evolved, at least in part, as a mechanism for social bonding in large groups (R. Dunbar, 1996). From this perspective, conversation can be understood not only as informational exchange but also as a form of social coordination. This highlights the importance of considering the social embedding of dialogue, which is often less central in formal approaches.

1.5.5 Conversation Analysis

Conversation Analysis (CA) is an empirically grounded approach, originating in ethnomethodology, that examines the fine-grained organisation of naturally occurring interaction (Sacks et al., 1974; E. A. Schegloff & Sacks, 1973; E. A. Schegloff, 2007). It focuses on the sequential structure of talk, including turn-taking mechanisms, adjacency pairs, and repair practices.

CA demonstrates that dialogue is highly structured at the micro-interactional level. Phenomena such as interruptions, repetitions, and self-repairs – often treated as disfluencies – are in fact integral to the organisation of interaction. The example discussed earlier illustrates several such mechanisms, including the interruption of an ongoing sequence and the use of repetition to re-establish coherence.

1.5.6 Discourse Analysis

Discourse analysis extends the study of language beyond individual turns to examine larger structures such as coherence, topic development, and rhetorical organisation (Mann & Thompson, 1988; Asher & Lascarides, 2003). It provides tools for analysing how meaning unfolds over extended stretches of interaction and how discourse is segmented into meaningful units.

This perspective complements conversation analysis by addressing higher-level organisation. In the example, the alternation between navigation instructions and interview questions reflects a form of discourse-level structuring, where multiple activities are interleaved within the same interaction.

1.5.7 Clinical and Developmental Linguistics

Clinical and developmental linguistics focus respectively on language acquisition and on language disorders (missing reference). Both domains provide valuable insights into the mechanisms underlying dialogue by examining how they emerge, develop, or break down.

Interaction plays a central role in language acquisition, as communicative abilities are developed through participation in dialogue. Conversely, disruptions observed in clinical populations can reveal the structures and processes that underpin typical interaction. These perspectives are particularly relevant for applications concerned with identifying, modelling, or supporting atypical conversational behaviour.

1.6 Toward an integrative descriptive framework

The diversity of approaches to dialogue reflects its complexity, but it also creates important challenges. Existing frameworks are typically tailored to specific research goals and levels of analysis. As a result, it is often difficult to compare findings across studies or to generalise insights across different types of dialogue.

This issue becomes particularly salient in contemporary contexts, where dialogue data vary widely: controlled experimental settings, naturally occurring conversations, human–machine interactions, and multimodal exchanges. Without a shared descriptive framework, it becomes difficult to assess how differences in data collection, annotation, or modelling assumptions influence the conclusions drawn.

Rather than converging toward a unified account, existing fields approach dialogue from distinct perspectives, each foregrounding particular aspects of interaction while leaving others in the background. Conversation thus does not admit a single, stable definition, but emerges as a composite object shaped by multiple theoretical traditions. This multiplicity motivates the need for an integrative perspective.

Figure 1.5 provides a schematic representation of the main domains of dialogue research and their relationships. It is not intended to be exhaustive, but rather to illustrate the diversity of approaches and the potential for integration. Each domain captures a particular dimension of dialogue, and the connections between them indicate areas of conceptual overlap and potential synergy.

Two main clusters can be identified in this representation. The first cluster, circled in pink, includes Conversation Analysis, Sociolinguistics, Clinical and Developmental Linguistics, and Psycholinguistics. These fields share a focus on the situated, social, and cognitive dimensions of dialogue. They often rely on empirical data from naturalistic interactions and emphasise the importance of context and participant perspectives. The second cluster includes two pairs, respectively coloured in blue on the left and teal on the right, that are weakly connected between them but strongly connected within themselves: Formal Semantics and Pragmatics on the one hand, and Psycholinguistics and Clinical and Developmental Linguistics on the other hand. The first pair shares a focus on the formal representation of meaning and interaction, while the second pair is concerned with the cognitive processes underlying language use and development. These two pairs are weakly connected because although they operate at different levels of analysis, they can inform each other. For instance, formal models can provide analytical tools that help highlight differences in the way dialogue is structured across different contexts and populations, while psycholinguistic findings can constrain and refine formal representations.

The lack of dense connections between the two clusters reflects the fact that there is no unified account of dialogue that integrates insights from all these domains. Each field has developed its own theoretical frameworks and methodologies, which are often not directly compatible with those of other fields. This fragmentation can lead to difficulties in comparing findings across studies and in synthesising insights into a coherent picture of dialogue.

While attempting to produce a single unified theory of dialogue may be unrealistic, and would likely fail to capture the richness of the phenomenon, we can still aim to develop a descriptive framework that integrates insights from different traditions and allows us to characterise resources in a way that is both comprehensive and flexible. The idea is to identify key dimensions of dialogue by considering face-to-face conversation as a reference point and then to describe how different types of dialogue data and models relate to these dimensions. Indeed, according to Fillmore (1981, p. 152), ‘The language of face-to-face conversation is the basic and primary use of language, all others being best described in terms of their manner of deviation from that base’. Such a framework would not only help us understand the different dimensions of dialogue but also provide a common tool to describe and compare different types of dialogue data and models.

The first part of this thesis is dedicated to the presentation of such a framework. In Chapter 2 we present a theoretical characterisation of dialogue that integrates insights from different disciplines and

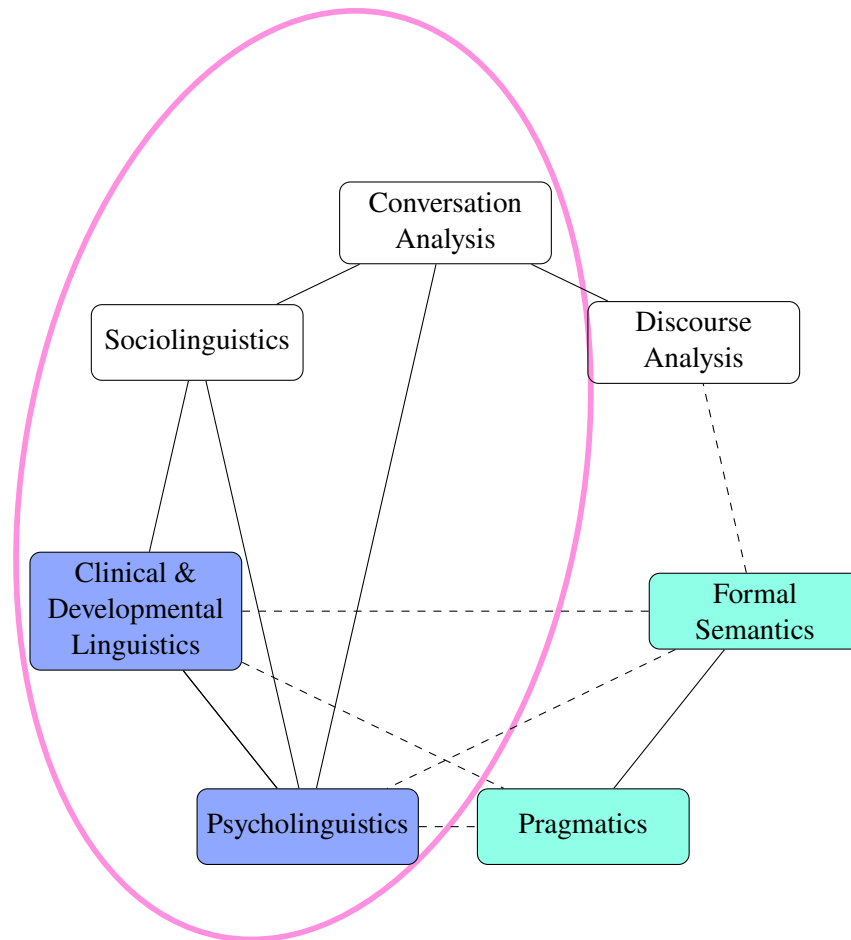


Figure 1.5: Schematic representation of the main domains of dialogue research and their relationships. Full connections indicate strong conceptual interactions while dashed ones suggest weaker or more indirect relationships.

traditions. In Chapter 3 we apply this framework to several theoretical scenarios to explain how it could help our community get out of some current issues we have in the field.

2

Parameters for Characterising Dialogue

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As semanticists, pragmaticists, and scholars of language more broadly, our ultimate goal is to understand people's linguistic behaviour in its full complexity and diversity. Conversation is one of the most fundamental human activities, and face-to-face interaction constitutes the primary environment in which language is learned, practised, and culturally shaped. It is therefore natural to treat co-present dialogue as the prototypical case against which other forms of communication can be compared (Fillmore, 1981; Levinson, 1983; Clark, 1996). Yet this 'naturalness' should not obscure the fact that conversation is a cognitively demanding activity. Speakers deploy a range of multimodal and interactional strategies – redundancy across modalities, rapid repairs, prosodic cues, gaze, gesture – to sustain mutual understanding, align perspectives, and manage the flow of interaction.

This diversity implies that tools developed to study, model, or simulate conversation must be sensitive to the specific features of the type of dialogue they are applied to. Yet conversation is complex to characterise in a systematic way that would allow us to compare different types of dialogue, and to identify the resources suitable for a given task and the relevant phenomena to model and evaluate.

The goal of this chapter is therefore to bring together insights from interactional linguistics, pragmatics, psycholinguistics, communication studies, and computational linguistics to outline a set of core parameters for characterising dialogue. We describe the features that make face-to-face interaction the prototypical conversational environment, examine how alternative forms of communication vary along these dimensions, and discuss what these differences imply for the study and modelling of dialogue. Our ultimate aim goes beyond theoretical considerations: it is to provide a descriptive tool that would enable researchers to compare different types of dialogue, assess the suitability of resources for a given task, and identify the phenomena that are relevant to model and evaluate.

The following piece of conversation illustrates the main categories of parameters we will discuss in this chapter. This dialogue is part of a corpus of French conversations between a psychologist and a patient with schizophrenia – the pathological aspect being irrelevant for this particular example.

- (2.1) **G234_1** mm hmm (0.5s) do you have a cellphone? (Fr. *humhum (0,5s) vous avez un téléphone vous ?*)
R235_1 I stopped the cellphone (Fr. *j'ai arrêté le téléphone*)
G236_1 stop (Fr. *arrêter*)
R237_1 stop (Fr. *arrêter*)
G238_1 completely (Fr. *complètement*)
R239_1 complete oulac (xxx) (1s) (Fr. *complet oulac (xxx) (1s)*)
air conditioner noises
R239_2 wouallou but (0.5s) it's (0,5s) (Fr. *wouallou mais (0,5s) c'est (0,5s)*)
G240_1 wouallou (Fr. *wouallou*)
R241_1 it's okay it's a (1.5s) (xxx) (Fr. *c'est bon c'est un (1,5s) (xxx)*)
G242_1 (xxx)
G242_2 you see the (tech-) (Fr. *vous voyez la (tech-)*)
R243_1 an old air conditioner (Fr. *un ancien climatiseur*)
G244_1 ah (Fr. *ah*)

[SLAM Data]

In this dialogue, two *participants* interact in a face-to-face setting. Their conversation revolves around technology – specifically, around owning a cellphone – when a loud noise from the air conditioner interrupts the exchange. This disruption prompts a shift in topic, initiated by the utterance « wouallou mais c'est » (En. 'wouallou but it's'). This utterance combines French and Arabic, reflecting the speaker's bilingual background as an Algerian participant. It illustrates that *language* is a non-trivial parameter in conversational analysis: even in corpora labelled as monolingual, code-switching may naturally occur. The interruption caused by the air conditioner highlights another key dimension, namely the *environment* in which the interaction takes place. External factors can shape the trajectory of a conversation, sometimes triggering unplanned topic shifts. This aspect is closely related to the *medium* of communication, which determines the extent to which environmental influences are perceptible. For instance, the shift prompted by the air conditioner would likely not have arisen in a text-based exchange. The management of this topic shift further depends on *verbal interaction*, as participants must coordinate to accommodate the new focus while maintaining conversational coherence. Finally, the likelihood and relevance of such shifts are also conditioned by the *function* underlying the interaction and the *structure* it entails.

This chapter is therefore structured around five core parameters. It begins by examining the **PARTICIPANTS** involved in conversation (Section 2.1), before turning to the **LANGUAGE** employed (Section 2.2). It then addresses the **ENVIRONMENT AND MEDIUM** of interaction (Section 2.3), followed by the dynamics

of (VERBAL) INTERACTION itself (Section 2.4). Finally, we discuss the FUNCTION AND STRUCTURE that frame conversational exchanges (Section 2.5). The chapter concludes with a summary of the parameters articulated in an operational characterisation table (Section 2.6).

2.1 Participants

Any conversation is necessarily grounded in the individuals who participate in it. Although dialogue is often described with a simple emitter-receiver model, this view obscures the complex and dynamic roles that participants simultaneously inhabit. Speakers monitor both their own production and that of their interlocutors, listeners continuously signal alignment and understanding, and all participants coordinate through a range of multimodal cues. Moreover, conversations are rarely limited to two participants: additional hearers, shifting roles, and varying degrees of involvement contribute to a layered participation structure that warrants careful analytical attention.

Recognising that dialogue unfolds between socially situated individuals brings further dimensions into view. Participants are embedded in interactional histories, asymmetries of power or expertise, and differing degrees of identifiability or anonymity, each of which shapes what can be said, how it is expressed, and how it is interpreted.

This section outlines key parameters related to the presence and identity of participants in dialogue, with the aim of clarifying how these social and structural factors constrain, enable, and diversify conversational behaviour. We begin by examining participation structure (Section 2.1.1), that is, the number of participants and their roles as speakers, addressees, and more peripheral or shifting forms of involvement. We then turn to enunciative status (Section 2.1.2), focusing on how participants are positioned and identified within the interaction. Finally, we discuss the influence of social dynamics on conversational processes (Section 2.1.3).

2.1.1 Participation Structures

Participation structure can be thought of as the ‘shell’ of a conversation – the basic configuration that defines who is present and in what capacity. Just as a player in a game first selects a class and a battlefield, the number of participants and the function they occupy establish the arena in which interaction will take place. This outer shell constrains turn-taking, attention, and coordination, providing the framework within which conversational behaviour occurs.

2.1.1.1 Number of speakers

A prototypical face-to-face conversation is held between two speakers, sometimes more but at most four. Indeed, when more than four people interact, smaller groups tend to appear following what has been called the ‘dinner party problem’ (R. I. M. Dunbar et al., 1995; Dezechache & Dunbar, 2012; Krems et al., 2016; Krems & Wilkes, 2019) or schism(ing) (Sacks et al., 1974). This name stems from the fact that when many people are sitting around a dinner table, it is difficult to have a single conversation between all of them. This is due to the complexity of coordinating with many participants but also more practically because of the physical configuration of the table which makes it difficult to talk to people sitting too far away. Thus, smaller groups tend to form and have their own conversation while still being part of the larger group.

Dialogues involving strictly more than two participants are commonly referred to as *multi-party* dialogues or *multilogues*, in contrast to *duologues* involving two people. Such interactions are generally more complex in terms of turn management (Section 2.4.1.2) and topical organisation (Traum, 2003) because each participant can bring their own ideas in the conversation and derail it from the matter at hand.

Establishing and maintaining common ground is also more demanding, as coordination must occur across a larger set of interlocutors, potentially requiring multiple repair sequences (Section 2.4.1.1). In addition, long distance attachments, i.e. producing utterances rhetorically related to a far previous one, are also more frequent in multi-party settings (Ginzburg & Fernández, 2005; Asher et al., 2016) further increasing structural complexity.

As noted above, conversations involving more than four participants often subdivide into parallel threads, with participants dynamically entering and leaving these sub-conversations, which makes them complicated to study independently (Egbert, 1997). Furthermore, the number of participants is not necessarily fixed throughout the interaction. It may vary dynamically depending on the environment and communication medium (Section 2.3), requiring participants to continuously adapt. The arrival of new participants can introduce individuals who do not share the same common ground or can shift the relevance of ongoing topics, particularly when these are tied to a specific subgroup. Conversely, participants may also leave the interaction, further altering its structure and coherence.

As explained in the Prologue, (2.2) is an example of multilogue with four participants where three sub-conversations are taking place at the same time. This example is taken from the STAC corpus (Asher et al., 2016) which contains dialogues between people playing the boardgame *Settlers of Catan* online. They interact via a chat interface. The three parallel threads are indicated with different fonts. The first one (in normal font) initiated by LJ is directly related to the game where they ask if someone would trade sheep for clay. The second one (in **bold font**) is initiated by GWFS after they answer LJ's question and is unrelated to the game as they ask whether people 'know about the league' which has to do with the real world². The third one (in *italics*) is also initiated by GWFS and can be considered a sub-thread of the second one since it arose from GWFS asking for some background on LJ after they answered the question about the league.

(2.2) LJ anyone want sheep for clay?
 GWFS got none, sorry :(
 GWFS **so how do people know about the league?**
 WM no
 LJ **i did the trials**
 TK **i know about it from my gf**
 GWFS **yeah me too, are you an Informatics student then, lj?**
 TK **did not do the trials**
 WM has anyone got wood for me?
 GWFS **I did them because a friend did**
 GWFS lol wm, you cad
 GWFS afraid not :(
 LJ *no, I'm about to start math. I just hang around appleton a lot*
 TK sry no
 GWFS my single wood is precious
 WM what's a cad?

[STAC corpus (Asher et al., 2016, Ex.1)]

This example shows that a dialogue can branch into several sub-conversations. The written format of this specific dialogue allowed the participants to simultaneously follow the different threads but similar

²The 'league' they are talking about may be a way to refer to the dialogue collection experiment they are taking part in. Asher et al. (2016) explain that 'players were grouped into several leagues'. The 'trials' could refer to the pilot of the experiment.

dynamics can also exist in face-to-face conversations where several threads can co-exist and the participants follow the one they are most interested in or most able to take part in depending on the spatial configuration. In some cases, however, the genre of the interaction can impose additional structure. For example task-oriented conversations such as meetings or tourist information conversations can be subdivided in sub-dialogues which are less complex to analyse (Fernández et al., 2008; Frampton et al., 2009; Sun et al., 2016). Nevertheless, as the number of participants increases, conversational organisation generally becomes less stable, particularly in informal settings.

Importantly, not all interactions involving multiple individuals qualify as dialogues. Indeed, the degree of interactivity plays a crucial role in determining how closely a discourse approximates face-to-face conversation. For instance, a lecture may involve both a teacher and students, yet remain highly asymmetric, with limited opportunities for student participation, or even become entirely one-sided in recorded formats. Physical configuration further modulates interactivity: greater spatial distance reduces access to visual feedback (Section 2.3.2.2). In large lecture halls, participants in the front rows typically engage more directly with the speaker than those seated further away, effectively occupying different interactional roles. At the extreme, both interactivity and co-presence are minimised in asynchronous exchanges (Section 2.4.2.1).

Conversely, even monologues can rarely be seen as involving a single participant. When someone produces a monologue, they typically do so with some kind of representation of who they are talking to in mind, even if this addressee is not explicitly defined. This representation influences the way they produce their utterances. For example, when someone is talking to themselves, they are actually talking to an imagined addressee which can be themselves or a third party. When someone is writing a book, they are writing it for an audience which can be more or less defined but is still there. Y. Wu et al. (2024) showed that we can predict the questions that will be answered in the rest of a news article based on the beginning of it, which suggests that the writer indeed has a representation of the readers in mind and produces the article accordingly.

In face-to-face interaction, a participant is typically equated with a single individual. However, this assumption does not always hold in the data used for dialogue studies. Scripted dialogues, such as those found in film or television, may involve a limited number of real contributors – the script writers – while representing a larger set of fictional participants. Conversely, some corpora attribute utterances to a single speaker that are in fact produced by multiple contributors, as in collaborative writing contexts or shared social media accounts. Comparable issues arise in interactions involving language models, where a system may evolve across versions while maintaining the appearance of a single, stable interlocutor.

Overall, the number of participants constitutes a fundamental parameter for characterising dialogue. It strongly influences both the structure and the degree of organisation of the interaction. The assumption that each participant corresponds to a single, stable individual also has implications for the establishment of common ground and the degree of spontaneity (Section 2.4.2.2). While a simple count of participants provides a useful initial description, more fine-grained distinctions are often necessary. In particular, it is important to differentiate between the number of apparent (or simulated) participants and the number of actual contributors, as well as to specify the mapping between them. These sub-parameters are summarised in Table 2.1.

2.1.1.2 Functions

In conversation, participants usually take up the speaker function one at a time. While there can be overlaps for feedback for example, in most cases one person at a time can be considered as main speaker. The other participants are listeners until the floor is taken by one of them who then becomes main speaker (see Section 2.4.1.2 for a discussion on turn-taking). However, not all listeners should be considered the same. Goffman (1976), Clark and Carlson (1982), and Clark and Schaefer (1989, 1992) suggest that there

Parameter	Minimal	Extended description
Number of participants	Number	Number of simulated participants Number of workers Workers - Simulated participants mapping

Table 2.1: Operational characterisation of dialogue resources and tasks – Participation structures: number of participants

exists a hierarchy of listeners functions where they could be separated between *participants*, i.e. people who are allowed to take the floor in the conversation, and *overhearers*.

In a multilogue, not all listeners are always considered as main targets by the speaker. Depending on the matter at hand, some of them can be *addressees* whom we expect to be more likely to participate actively and answer questions for example. The other listeners can be considered *side listeners*: the speaker takes them into account when producing their utterances but they do not necessarily expect them to take part into this section of the conversation. Both speakers and listeners behave differently depending on the different functions of the participants. Speakers must make sure their utterances are understandable by the listeners and in particular the addressees but they have no reason to care about the overhearers unless they have additional intentions related to them such as deceiving potential eavesdroppers. Listeners on the other hand expect the speaker to make themselves understandable by them while overhearers would typically not. More details on this aspect can be found under the *Dogma of Undifferentiated Hearers* in Clark (1997).

Fig. 2.1 illustrates those functions in a real-life situation. In this example, a group of three friends is waiting in the queue of a café with other people standing in the queue. At this specific time, two of the friends are talking to each other about a topic while the third friend is focused on her phone. The two conversation participants are thus alternating between being the speaker and the addressee of the conversation while the third person is a side listener. The other people in the queue are bystanders who are not part of the conversation but can still hear it and be influenced by it. For example, the lady at the counter could decide to change her order based on what they heard from the exchange, while the man behind the three friends could be tempted to barge in the conversation. Finally, we can imagine that there is an eavesdropper behind the ‘staff only’ door who can hear the conversation and may even be listening on purpose and pay more attention than the other bystanders. This person however cannot see the speakers and thus does not have access to the same modalities as the other participants which can influence their interpretation of the conversation.

As we discussed in Section 2.1.1.1, conversations involving more than four people tend to subdivide into sub-conversations. While the different threads can be studied independently, from the point of view of the global conversation there is more than one speaker at the same time and a listener could try to follow both sub-conversations at the same time. This could happen for example at a large dinner table where the people sitting in the centre can take part in dialogues on both of their sides. Studying such conversations requires to keep track of all the sub-threads and the interaction between them since the different sub-conversations can influence each other. Taking into account the function of each participant at a given time also plays a role in understanding the representation they have of the conversation.

The function a participant sees themselves in at a given time probably influences the way they will remember the corresponding part of the conversation. An addressee is presumably more involved than a side listener and even more than a bystander. The representation of the conversation they would keep in mind most likely differs depending on the function they took, and their views of the conversation in retrospect would be different as well. Similarly an eavesdropper who chooses to listen to a conversation

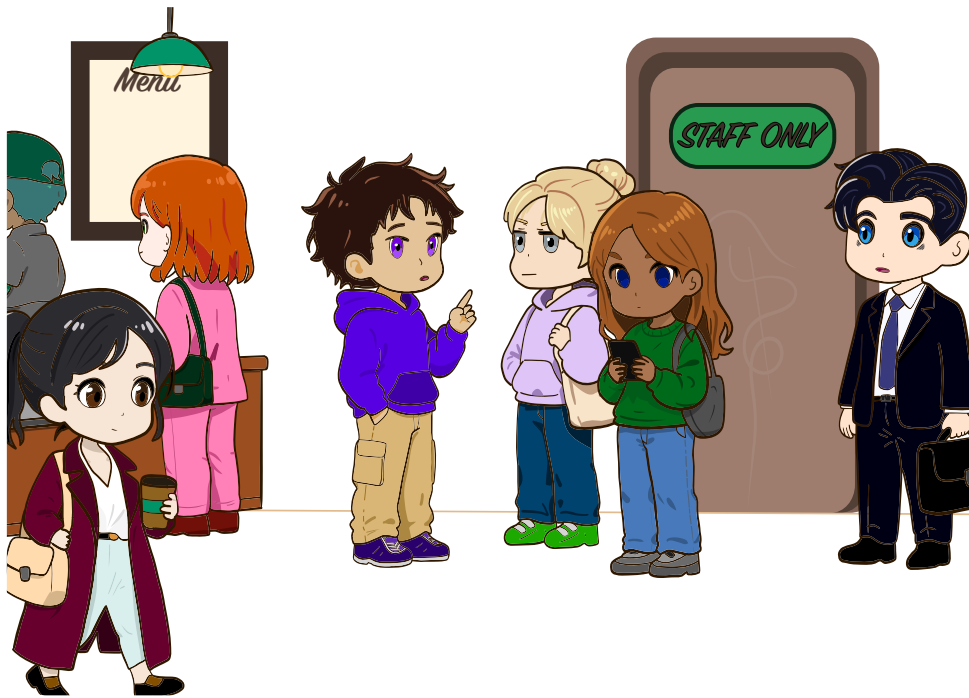


Figure 2.1: Representation of participants' functions in the queue of a café

they do not take part in has an external point of view and can notice things that the active participants would not be conscious about.

As researchers studying language data, the function we are the closest to is the one of an eavesdropper as we have no influence on the conversation but choose to pay attention to it. Several differences exist and will be discussed in following sections such as the fact that we study the conversation after it took place through a medium that only *represents* this conversation. As eavesdropper we must rebuild the context of the conversation without actually belonging to this context and it is possible to imagine a wrong context which leads to a false interpretation of the dialogue while seeming coherent from our perspective. In a research setting it could be possible to ask the participants to detail in retrospect some parts of the conversation that we do not manage to analyse. However, the spontaneity of dialogue implies that their answers would not necessarily reflect the exact mindset they had during the exchange.

The functional configuration of a conversation therefore constitutes an important parameter in the description and analysis of dialogue data. At a basic level, it can be represented as the set of functions attributed to each participant. However, more fine-grained characterisations may prove useful in capturing the dynamics of interaction. For instance, in multi-party settings, one may consider measuring the entropy of transitions between functions for each participant, thereby quantifying the extent to which individuals shift between roles and their degree of involvement across different phases of the interaction. Such a parametrisation can also accommodate the presence of implicit or 'hidden' participants, such as the audience of a TV show, who are not directly addressed by the characters but nonetheless shape the production of dialogue. These sub-parameters are summarised in Table 2.2.

2.1.2 Enunciative Status

The enunciative status of a participant in a conversation refers to the identity they present and how they are perceived by others. In face-to-face interactions, this is typically straightforward as participants can see and interact with each other directly. However, in mediated conversations, especially on social media, the

Parameter	Minimal	Extended description
Functions	List	Time-aligned list Entropy over function transitions

Table 2.2: Operational characterisation of dialogue resources and tasks – Participation structures: functions

link between identity and content can be more tenuous. Anonymity allows individuals to communicate without revealing their true identities, which can lead to behaviours that differ from those in face-to-face contexts. Moreover, the rise of synthetic agents such as language models introduces new questions about the nature of interaction and the assumptions participants make about their conversational partners. The enunciative status of participants is therefore a crucial parameter for characterising dialogue data, as it shapes the dynamics of interaction, the establishment of common ground, and the interpretation of utterances. In this section, we begin by discussing the implications of anonymity in dialogue (Section 2.1.2.1), then we examine the influence of role and persona instantiation (Section 2.1.2.2), and finally we consider the impact of commitment level on conversational dynamics (Section 2.1.2.3).

2.1.2.1 Anonymity

A conversation typically involves participants who are aware of each other’s identities. In face-to-face conversations, this is immediately evident as participants can see and interact with each other directly but it is also the case in most forms of conversation such as text messages and phone calls. This allows them to adjust their communication styles such as tone, register, detail level, and word choice, according to the social context and the people they are addressing. This principle is not limited to dialogue; even in texts that appear monologic, communication is never produced in isolation: it is oriented toward an interlocutor – real, implied, or projected – whose presence guides the organisation, style, and content of the message. As mentioned in Section 2.1.1.1, Y. Wu et al. (2024) have shown that ‘reader and writer expectations [seem to] align to a certain degree’. Similarly, participants in conversation project particular personas, consciously or unconsciously signalling aspects of their identity, social role, or expertise.

Social media and other anonymous platforms complicate this dynamic by decoupling identity from content. Participants can communicate with unknown individuals, adopt pseudonyms, or even share accounts among multiple people. The perceived reduction in accountability often leads to behaviours that diverge from what an individual might display in face-to-face contexts. A well known example of this is cyberbullying which is exacerbated by anonymity (Barlett et al., 2016).

In more extreme, albeit increasingly common, cases, the interlocutor may not even be human – but rather a generative language model, bot, or other synthetic agent – which raises additional questions about the nature of interaction and the assumptions participants make about their conversational partners. The question of ‘Is my interlocutor human?’ has become increasingly salient in online environments, whereas it was rarely a consideration a few years ago and is never one in traditional face-to-face settings.

This decoupling of identity from utterance presents methodological challenges for research. When studying online conversations, we cannot assume that a given pseudonym corresponds to a single human agent. The presence of multiple participants behind one identity or synthetic interlocutors can distort measures of individual behaviour, social alignment, and interactional norms. Similarly, while the amount of synthetic content online is complicated to evaluate, it is undoubtedly growing and language resources such as X (formerly Twitter) or Reddit will become less and less reliable.

For studies that rely on human-like production, as well as for the development of dialogue systems, failing to account for such uncertainties may lead to misleading conclusions or inappropriate model be-

Parameter	Minimal	Extended description
Anonymity	Binary	Anonymous / Pseudonymous / Real name Digital / Physical

Table 2.3: Operational characterisation of dialogue resources and tasks – Enunciative status: Anonymity

haviours. Anonymity should therefore not be treated solely as a social or ethical concern; it is a fundamental parameter that shapes both the structure and interpretation of interaction, particularly in mediated environments. When analysing dialogue data, it is crucial to determine whether participants can establish a stable interactional history and, more fundamentally, whether they are human. While this dimension can be represented in a minimal way as a binary distinction, a more fine-grained characterisation is often necessary. For instance, one may distinguish between anonymous, pseudonymous, and real-name interactions, as well as between digital and face-to-face contexts, each of which entails different constraints on identifiability and continuity. These sub-parameters are summarised in Table 2.3.

2.1.2.2 Role and Persona Instantiation

In most conversations, participants act as themselves, although they may adapt their behaviour to the context. For example, a doctor will speak differently with patients than in everyday conversation, yet remains fundamentally the same person. However, many dialogue resources involve participants who are required to adopt a different role. The clearest examples are TV shows or films, where actors portray characters, but experimental paradigms such as the *Wizard of Oz* also require participants to play a role such as that of a conversational agent. While the precise influence of role adoption on conversational dynamics is difficult to quantify, it can meaningfully shape interactions. Similarly, many recent dialogue datasets instruct participants to adopt a persona (S. Zhang et al., 2018; Yamashita et al., 2023), which can lead to unnatural conversational patterns (see example (0.1) in the Prologue). Persona is subtle and deeply embedded in social interactions, making it difficult to capture in short exchanges or to manipulate experimentally.

This parameter is particularly relevant in the context of prompting LLMs or conversational AIs. Explicitly assigning a role can strongly influence model output. For example, prompts that encourage role-playing have been shown to improve performance, with instructions to ‘act human’ making a model’s responses appear more human-like. Indeed, Jones and Bergen (2026) showed that, in a three-party Turing test (Turing, 1950) setting – where one participant interacted separately but simultaneously with two interlocutors, one human and one LLM—human participants identified GPT-4.5 as the human interlocutor in 73% of the interactions when the model received a role-playing instruction. Without the persona instruction, however, the same model was mistaken for the human in only 38% of the interactions. While this substantial difference confirms that persona-infused prompts can significantly influence LLM outputs, it also suggests that our expectations of what human-produced conversation should sound like are not necessarily accurate. If participants were unable to distinguish between conversations produced by the human and the model, one would expect performance to be close to chance (50%). The significant deviation from chance level therefore indicates that participants consistently perceived differences between the two types of interlocutors, but that these perceived differences were not reliable indicators of whether the interlocutor was actually human.

Explicitly describing the role assumed by participants – human or artificial – is therefore an important parameter for characterising dialogue and studying the influence of role on behaviour. It is also critical for understanding linguistic phenomena shaped by social identity, such as *dog whistles* (Saul, 2018; Breitholtz & Cooper, 2021; Quaranto, 2022), which require knowledge of both the roles participants adopt and the

Parameter	Minimal	Extended description
Role instantiation	Binary	Self / Self specific / Assigned
Persona instantiation	Binary	Self / Self specific / Assigned

Table 2.4: Operational characterisation of dialogue resources and tasks – Enunciative status: Role Instantiation

groups with which they identify.

Role instantiation can be represented minimally as a binary parameter indicating whether the participants are playing a role or not but it can also be described more finely. For instance, participants may (i) act as themselves without specific instructions, (ii) emphasise a particular aspect of their identity relevant to the situation, or (iii) adopt an assigned role. This distinction is particularly informative when comparing human and LLM behaviour, such as how humans alter their responses when ‘putting on a doctor’s hat’ versus behaving freely, and how LLMs respond to role-specific prompts. These sub-parameters are summarised in Table 2.4.

2.1.2.3 Commitment Level

The commitment level of a participant can be defined as the degree to which they are invested in a conversation and its truth models. It reflects the extent to which participants consider themselves accountable for the content of their utterances and are willing to be held responsible for them. In typical face-to-face interaction, participants are generally fully committed: they are directly engaged with one another, receive immediate feedback, and are expected to be accountable for what they say. Although the topic of conversation may modulate the potential consequences of a statement, participants are generally assumed to endorse the truth of their utterances and to participate sincerely.

However, commitment is not always absolute. Participants may adopt a more playful, experimental, or strategic stance, in which they are less concerned with literal truth and more focused on exploring possibilities, entertaining perspectives, or sustaining the interaction. For example, in games of make-believe, participants temporarily commit to a fictional reality for the sake of the interaction, even though they do not endorse it outside that context. Clark (1996) illustrates this with the story of two children ‘playing a game of make-believe’ in which they impersonate Wild Bill and Calamity Jane, two characters from a book they have read. The characters live in ‘Deadwood, Dakota Territory, during the gold rush of 1876’. During the game, they use real life elements from the yard they are playing in to mimic gold digging and panning³. In this scenario, the children commit to the truth of their statements, their characters, and their interpretation of the physical environment for the length of the game but these commitments will not transfer outside of it. This highlights how commitment varies with context and intention, shaping both the interpretation of utterances and the coordination of conversational actions.

The notion of commitment is particularly relevant when contrasting human conversation with language produced by LLMs. While LLMs can generate coherent and contextually relevant text, they do not possess beliefs, intentions, or a sense of accountability (Bender et al., 2021; Schlangen, 2025). Their outputs are the product of statistical patterns learned from data rather than a commitment to truth or meaning. This fundamental difference has important implications for how we interpret LLM-generated text and what expectations we can reasonably have for interactions with such systems. Searle’s (1969) work on illocutionary acts offers conceptual tools to distinguish between the act of asserting in human conversation and the pattern-based generation of text by LLMs.

³Gold *panning* is a traditional method for extracting gold from river sediment with a pan.

Parameter	Minimal	Extended description
Commitment level	Binary	Status: role-playing, joke, etc. Scope: conversation, task, ..., general

Table 2.5: Operational characterisation of dialogue resources and tasks – Enunciative status: Commitment Level

Commitment level is less straightforward to operationalise than other parameters as it is not directly observable. However, the aspects directly defined by the conversational task can be a first axis to characterise it. This can be defined in a binary way by presenting each participant as either committed or not to their utterances, which could be enough for example to distinguish human and LLM participants. However, a more fine-grained characterisation can be useful to distinguish between different forms of commitments such as the one of a participant who is committed to the truth of their utterances but as a character in a game of make-believe. The scope of the commitment can also be defined in a more fine-grained way by defining a truth model that is limited to a specific context or conversation as opposed to a more general one. These aspects are summarised in Table 2.5.

2.1.3 Social Dynamics

While participation structure defines the ‘shell’ of a conversation, social dynamics determine how each participant is embodied and specialised within it. Just as a player in a game customises their character and develops relationships with allies and opponents, participants bring histories, social positions, and degrees of identifiability to the interaction. Hierarchies and past interactions shape what can be said, how it is expressed, and how it is interpreted. Examining these social factors allows us to understand not just the structural possibilities of a dialogue, but the nuanced ways participants navigate, influence, and are constrained by their social environment.

2.1.3.1 Hierarchy

In prototypical face-to-face conversations, the participants are on the same level and their contributions are overall equally distributed. However many conversational settings contain some kind of hierarchy which influences the flow of dialogue. For instance, in formal settings such as meetings, one person may take the lead in speaking, while others are expected to listen, interrupting only when permitted.

The hierarchy may be determined by explicit protocols, such as those found in workplaces or political debates, or by implicit norms and biases based on social status, age, gender, or expertise. This can affect both the content and the style of communication: one might speak more formally, with careful word choice, when addressing a superior or less familiar person, while adopting a more casual tone with close friends or peers. The ability to adapt to these varying conversational hierarchies is crucial as failing to adjust one’s behaviour, for instance through inappropriate levels of formality or overstepping conversational turns, can lead to misunderstandings or social friction.

Hierarchy can also stem from the type of conversation and the roles the participants are attributed or attribute themselves. In particular in task-oriented conversations, the task can create specific positions which influence the dialogue. For instance, participants can become an explainer-explainee pair or one can follow directions given by others. In such settings participants have an asymmetric role which changes the overall conversational dynamics. Those specific dynamics are one of the axes of study regarding task-oriented conversations since they are common in daily life. However, such conversations are also commonly used to study other unrelated phenomena or to train conversational agents and deep learning

Parameter	Minimal	Extended description
Hierarchy	Binary	Weighted list of dimensions

Table 2.6: Operational characterisation of dialogue resources and tasks – Social Dynamics: Hierarchy

models. The level of asymmetry and its influence on the way dialogue can unfold should thus be taken into account in more general purposes.

Relying on task-oriented corpora to train conversational agents exposes models primarily to asymmetrical interactions, which in turn shapes the patterns of their generated utterances. While this can be appropriate for practical agents designed for specific tasks – such as booking hotels or purchasing plane tickets – it can reinforce user biases when deployed in more general-purpose settings. For instance, conversational tools like ChatGPT have a tendency to agree with users, even at the expense of logical consistency, making them unsuitable as tools for fostering critical thinking without careful guidance. Nonetheless, such systems are increasingly relied upon for expert support in decision-making (F. Liu et al., 2024).

The hierarchical dynamics of a conversation are therefore a crucial parameter in characterising dialogue data, as they shape both the flow of interaction and the way participants engage with one another. At a minimal level, hierarchy can be represented with a binary distinction. For more detailed analyses, hierarchical structure can be operationalised across multiple dimensions, such as age, genre, social status, expertise, or institutional role. These factors are extensively studied in social psychology and sociology and can be adapted to the analysis of conversational data (P. Brown, 1987). The relative importance of each dimension may vary depending on the cultural or physical context. From a computational perspective, such hierarchical characterisations can be efficiently encoded as vectors: one representing the relevant dimensions and another representing their corresponding weights. This approach allows for flexibility in modelling conversational hierarchies across diverse datasets and applications. These sub-parameters are summarised in Table 2.6.

2.1.3.2 (Conversational) history

Conversational history refers to the previous exchanges between specific participants. It can be conceptualised along two complementary dimensions. First, the short-term history arises from the ongoing conversation itself, providing topics and context that can be reintroduced during the interaction. Second, the long-term history reflects the cumulative record of prior interactions between participants, spanning all past dialogues they have shared.

Short-term history exists in all types of conversation, regardless of the number of participants or whether they are human or artificial. The salience of particular elements depends on participants' interests, attention, and memory. Long-term history, in contrast, allows participants to achieve greater alignment and efficiency in communication as they become more familiar with one another.

Individual differences in memory and recall can significantly influence comprehension and coordination. If one participant implicitly relies on a shared history to support an utterance while another does not recall the relevant past exchanges, misunderstandings or disruptions may arise.

This phenomenon is less visible in written conversation as the history exists physically. While some tools erase older parts of the history, in most cases a long part of it is retained for quite some time. Participants can thus refer to it, in some cases by explicitly indicating what utterance they are elaborating on. There is also a lasting proof of what has been said exactly which partially prevents argumentation strategies that rely on distorting the other person's words.

Parameter	Minimal	Extended description
(Conversational) History	Scale	Distance in a weighted social graph Common topics

Table 2.7: Operational characterisation of dialogue resources and tasks – Social Dynamics: (Conversational) History

Beyond the conversation itself, a broader social or experiential history shapes interactions. Participants bring with them pre-existing knowledge, social norms, and expectations derived from their prior relationship or shared affiliations. For instance, meeting a friend of a friend involves different assumptions and ‘safe’ topics than meeting a person working in the same company or a complete stranger for the first time. This shared background provides a common ground that can facilitate conversation and guide topic selection (Clark, Brennan, et al., 1991).

Overall, both conversational and broader social histories are important parameters of dialogue, influencing content, alignment, and linguistic choices. At a minimal level, history can be represented on a simple scale reflecting the amount of shared experience. More detailed operationalisations could model social distance within a weighted network and include lists of common topics or interests, which can signal likely conversational trajectories. These sub-parameters are summarised in Table 2.7.

2.2 Language

The language used in a conversation is a central feature, shaping not only the content but also the social and cultural dimensions of interaction. Language carries information about identity, social norms, and cultural background; it influences thought patterns, expression, and interpersonal coordination. The choice of language also has practical implications for accessibility, inclusivity, and the potential reach of a conversation. While this chapter does not provide an exhaustive discussion of language, it is important to emphasise its role when characterising dialogue data. The research community has a tendency to focus on English, often implicitly, yet the world’s linguistic diversity is immense, and different languages exhibit features that can profoundly influence conversation. For example, some languages encode complex systems of politeness and formality, affecting interactional behaviour (P. Brown, 1987); others encode time, space, or aspect differently, shaping how participants discuss events; and some languages encode emotions or evidentiality in ways that influence both expression and comprehension.

The rise of large language models (LLMs) has brought further attention to language. While LLMs are trained on multiple languages, they are typically dominated by English and a small set of well-resourced languages. Neglecting language-specific characteristics can reinforce the marginalisation of low-resource languages and produce models that are poorly adapted to their conversational nuances (Bender et al., 2021). Moreover, the assumption that all speakers wish to see their languages represented in such models may overlook the sociocultural and ethical dimensions of language technology. Language is foremost a tool for *human* communication, and the development of computational tools should prioritise the needs and desires of speakers rather than the mere expansion of model capabilities (Bird, 2020).

When operationalising language as a parameter in dialogue data, it is important to specify the language(s) involved and consider how their characteristics may influence interaction. At a minimum, this can be represented as a simple list of languages, although a more detailed description of the linguistic features relevant to the conversation may be useful. For instance, we can indicate the level of control over language use, that is, the extent to which participants’ language use is constrained. Social media data, for example, are typically much less controlled than data collected specifically for a research experiment. At

Parameter	Minimal	Extended description
Resource level	List	Control degree
Conversation level	List	Proficiency Monolingual / Code-switching / Multilingual

Table 2.8: Operational characterisation of dialogue resources and tasks – Language

the level of individual conversations, we can also indicate the participants’ proficiency in the language, as this may influence both how they express themselves and how they understand one another. For example, if one participant has limited proficiency, they may have greater difficulty expressing themselves or understanding the other participants, potentially leading to misunderstandings and disruptions in the conversation. Finally, we can indicate whether a conversation is monolingual or multilingual and whether it involves code-switching. These sub-parameters are summarised in Table 2.8.

2.3 Environment and Medium

Face-to-face conversation is inherently multimodal: participants share the same physical space, allowing them to interact within a common environment, and communicate through voice, gestures, and facial expressions. These modalities are tightly intertwined, each enriching the conversation and supporting mutual understanding. The physical situation also plays an important role. It can make a conversation more or less prone to interruptions, influence the emergence of sub-threads, and provide referential resources when events or objects in the environment become part of the dialogue. Collectively, these aspects are fundamental to characterising the dynamics and structure of face-to-face interaction.

Sharing physical space enables visual co-presence, which shapes conversation in two major ways. First, it enriches interaction by providing access to non-verbal cues such as facial expressions, gaze, and gestures. These cues allow participants to identify speaker intent, gauge affective tone, and monitor the attention and comprehension of others. Second, the shared environment itself becomes a referential resource: participants can point to, gesture toward, or otherwise use elements of the surroundings to structure and support the dialogue.

In addition to visual sharing, vocal co-presence provides complementary information. Prosody, volume, pitch, and timing convey emphasis, emotional nuance, and communicative intent that would otherwise be absent in written or mediated forms of communication. Together, visual and vocal co-presence allow for fine-grained coordination, rapid feedback, and grounding of meaning.

However, these features can deteriorate depending on the communication medium. While face-to-face interaction provides simultaneous audio and visual channels, other media may reduce or remove one modality entirely, affecting how participants produce and interpret the conversation. Subsequent subsections explore visual co-presence, gestures, and vocal features in detail, highlighting both their individual contributions and their interplay in shaping effective communication. We will begin with the physical context of conversation (Section 2.3.1), which provides the shared environment in which these multimodal signals are produced and interpreted. We will then discuss the specifics of participants being able to see each other (Section 2.3.2) and finally hear each other (Section 2.3.3).

This discussion assumes typical face-to-face interaction between fully sighted and hearing participants. Situations involving sensory disabilities (e.g. blindness or deafness) alter the availability and perception of multimodal signals; we do not address these cases in detail here.

2.3.1 Physical context

Physical context refers to the physical space, i.e. the material and spatial conditions, in which interaction takes place. It is typically a shared part of the real world – though the attention of the participants can be focused on different aspects at a given time, and plays an active role on how conversation unfolds. Because participants share a setting, they co-experience its sights, sounds, and affordances, which can subtly or dramatically shape what is said, how it is said, and who is involved. This influence operates along several dimensions, ranging from the triggering of topics to the regulation of participation and privacy.

First, the environment provides natural conversational triggers. Everyday talk is often prompted by something occurring in the shared space: a loud sound, a passing event, an unexpected sight. Such phenomena can interrupt ongoing talk or lead to spontaneous topic shifts, demonstrating that conversation is dynamically responsive to physical context. Similarly, visible objects can become resources for situated joint reference, enabling interlocutors to point, gesture, or comment on what they both perceive (Clark, 1996; Goodwin, 2000). This capacity to anchor talk in the here-and-now creates a rich field for shared meaning-making that is often absent in mediated communication (Clark, Brennan, et al., 1991), though virtual shared spaces can partially restore this functionality in some cases (Asher et al., 2020).

Second, the environment affects who can participate and thus the dynamics of the interaction. Spatial layout determines how permeable a conversation is to new participants. Conversations held in open areas, near circulation spaces, or around standing tables are easier for others to overhear and join, creating opportunities for multi-party talk and dynamic shifts in alignment and topic (Oldenburg, 1989). Conversely, conversations in secluded corners or enclosed rooms tend to remain private and are less prone to interruptions and intrusion of additional participants, supporting more intimate or sensitive exchanges. Hall's (1966) theory of proxemics highlights that interpersonal distance and seating arrangements signal levels of intimacy, formality, and accessibility, regulating who feels entitled to approach or contribute.

Along the same line, privacy and publicness are key environmental dimensions that influence topic selection and conversational style. In acoustically transparent settings such as open-plan offices, people often self-censor, avoid disclosing sensitive information, or switch to less identifiable vocabulary because of the risk of being overheard (Cavanaugh et al., 1962). By contrast, spaces that provide good speech privacy – through partitioning, sound masking, or physical isolation – encourage more candid exchanges. Lighting and seating configuration can also modulate the involvement in conversation of communication vulnerable people (Stans et al., 2017).

Environmental factors also shape conversational effort. Acoustics play a major role in intelligibility and cognitive load: high reverberation or competing speech can force speakers to raise their voice, repeat themselves, or simplify messages, affecting both efficiency and emotional tone. Similarly, visual conditions such as obscured sightlines can make it harder to monitor the listener's reactions, leading to more verbal checks for understanding or less precise timing of turn-taking.

Crucially, these environmental affordances are absent or altered in mediated conversation. Participants may be located in different environments, connected only through a technical channel that transmits language but not the material world in which it is produced. In written online exchanges, for instance, the shared physical environment effectively disappears, replaced by a virtual interactional arena defined by the interface itself. In a phone call, participants no longer share the same perceptual environment, preventing them from jointly noticing events in their surroundings. As a result, when a disruption occurs (such as a knock at the door or a sudden noise), the affected participant must explicitly explain it ('Sorry, someone just came in') for the other to understand the break in attention (Clark, Brennan, et al., 1991). Similarly, references to objects or features of the surrounding environment must be described verbally rather than simply indicated by pointing, increasing the effort required to establish common ground. Even in video-mediated conversation, spatial relations remain fragmented: participants appear as framed projections rather than embodied agents capable of acting within a shared environment. De-

Parameter	Minimal	Extended description
Physical context	Binary	Co-located / Shared virtual / Separate Shared perceptual field

Table 2.9: Operational characterisation of dialogue resources and tasks – Environment & Medium: Physical context

ictic acts such as pointing or manipulating objects lose their immediate interpretability unless they refer to objects visible within the camera frame or are reformulated linguistically. The result, in both phone calls and video-mediated interaction, is a form of contextual abstraction in which spatial and material relations are reconstructed discursively rather than jointly enacted. This shift underscores the extent to which face-to-face conversation depends on the tacit sharing of physical context.

For researchers and practitioners working with dialogue corpora and language models, the absence of physical context shapes what kinds of conversational phenomena can be observed and modelled. Many corpora used in dialogue research are collected from phone calls or text chats, media that lack references to shared surroundings and omit environmental triggers that would normally shape topic flow or participation. This means such datasets may under-represent phenomena like topic shifts prompted by co-experienced events, repair sequences caused by shared distractions, or subtle changes in formality induced by spatial privacy. For analytic purposes, scholars must therefore be cautious when using corpora where the physical environment is absent: conversational structure may appear more topic-stable, more verbalised, and less sensitive to interruptions than in natural co-present conversation. Incorporating multimodal or situated datasets – such as meeting recordings with video, or corpora annotated for spatial layout – can reveal how much the environment contributes to turn-taking, participation, and content selection, and can help develop models that better approximate human conversational adaptability.

Across all forms of dialogue, the physical context determines how participants orient to space, objects, and the organisation of joint action. As interaction becomes increasingly mediated or artificial, physical co-presence gives way to representational or simulated forms of context. Understanding how dialogue adapts to these material and spatial conditions provides a foundation for examining other dimensions of co-presence – most notably, the visual forms of mutual accessibility that underpin gaze, gesture, and embodied alignment, discussed in the following section.

This aspect should therefore be recorded when characterising dialogue resources and tasks, as it shapes the way participants produce and interpret the conversation. As with many of the other parameters, a binary distinction may be sufficient in simple cases. However, a more detailed description may specify the degree to which the physical context is shared, distinguishing between participants who are physically co-located, who share only a virtual interactional space, or who are entirely separated. Their perceptual fields may also be characterised to distinguish situations in which participants have access to the same aspects of the environment from those in which they do not. For example, in a car, the driver and passengers occupy the same physical space but do not share the same perceptual field: the driver must focus on the road, whereas the passengers are not subject to the same attentional constraints. These parameters are summarised in Table 2.9.

2.3.2 Visibility

Many signals in face-to-face conversation are a combination of linguistic and non-linguistic modalities (Clark, 1996, p.156). The possibility to see one another during interaction enables participants to exploit facial expressions, gaze direction, and gestures as complementary communication channels, thereby facilitating coordination and grounding. Indeed, according to Oviatt (1999), ‘multimodal pen/voice lan-

guage is briefer, syntactically simpler, and less disfluent than users' unimodal speech' (based on Oviatt, 1997). Visual co-presence is thus a core feature of natural conversation, fundamentally shaping both its structure and efficiency.

As mentioned earlier, the discussion in this chapter assumes fully sighted and hearing participants. However, it is important to stress that sign languages rely entirely on the visual modality. In sign languages, visual cues are not merely complementary but essential for conveying meaning and even encode information that would otherwise be expressed vocally, such as prosody (Sandler, 2010). Although we do not discuss sign languages further in this chapter nor thesis, they illustrate that the visual modality does not always play the same role or carry the same communicative importance across different forms of language use.

2.3.2.1 Gaze

Gaze is a central organisational resource in conversation, shaping how participants coordinate turns, monitor one another's attention and cognitive state, and display social or affective stances. Even in simple dyadic exchanges, interlocutors are simultaneously speakers and listeners – speakers monitor their interlocutor to assess understanding while listeners transmit signals of attention, alignment, and readiness. In this reciprocal ecology, gaze functions both as a perceptual act and as a communicative signal. Early work by Kendon (1967) showed that speakers tend to avert their gaze at the beginning of an utterance and re-establish it at its end, a pattern interpreted as reflecting the cognitive load of planning and the need to recruit the interlocutor's uptake at a transition-relevant point. Subsequent studies have repeatedly replicated this dynamic: Duncan et al. (1979), Ho et al. (2015), and Rossano (2012), among others, find that gaze helps shape the timing and attribution of turns, facilitating the smooth transfer of the floor. Listeners generally display more sustained gaze towards the speaker, offering something akin to visual backchannels that signal attention and availability. Yet these patterns are not mechanistic: a significant proportion of turn transitions occur without the expected gaze cues, and gaze becomes more influential in situations of social uncertainty, with unfamiliar interlocutors or sensitive conversational topics. Research also reveals that interlocutors may strategically manipulate gaze to influence the organisation of the floor. Kendrick et al. (2023), for example, show that speakers who wish to retain the turn at a point where transfer is possible may avert their gaze or reorient their body in ways that discourage interruption. Conversely, sustained gaze near a potential completion point can encourage a response, subtly nominating a next speaker.

Beyond regulating turns, gaze provides crucial monitoring capacities. Goffman (1966) emphasised that sustained visual access allows interlocutors to monitor one another's behaviour, adjust their explanations, and pre-empt failures of understanding. This monitoring is cognitively demanding, which helps explain why gaze aversion reliably accompanies thought-intensive activity. Studies by Glenberg et al. (1998), Doherty-Sneddon et al. (2002), and Doherty-Sneddon and Phelps (2005) show that when individuals must remember information, plan complex responses or perform reasoning tasks, they withdraw gaze, seemingly to reduce cognitive load. Allen and Guy (1977) further demonstrate that confident speakers tend to look at their interlocutor more, whereas hesitations and uncertainties correlate with gaze aversion. Together, these findings position gaze as an embodied index of cognitive and attentional states, one that interlocutors draw on to adjust their contributions, calibrate timing and modulate the level of explicitness or elaboration required.

Gaze is equally implicated in the expression of social and affective meanings. It conveys intimacy, dominance, embarrassment, confidence, and stance, often in subtle and culturally mediated ways. Doherty-Sneddon and Phelps (2005) identify gaze aversion as a marker of self-consciousness and emotional strain, particularly when discussing personal topics. Argyle et al. (1974) show that frequent – but not continuous eye contact increases perceptions of friendliness and engagement, whereas sustained eye contact

can feel intrusive and unnatural. Cook (1977) observes that eye contact rarely consists of simple staring; instead, participants perform rapid sweeps across one another's faces, with these patterns adjusting dynamically in response to changing social relations. The expressive dimension of gaze thus interacts continuously with its regulatory and monitoring functions, contributing to the layered and affectively charged nature of conversation.

Although most of this literature concerns dyadic talk, gaze becomes even more pivotal in multi-party interaction, where participants must coordinate not only turns but also recipient and next speaker selection as well as varying levels of engagement. The larger amount of addressees entails a shorter time spent by the speaker gazing at them. To partially compensate for this, speakers tend to gaze more, in proportion, to their addressees than in dyadic exchanges (Vertegaal et al., 2001). Work by Goodwin (1981), Lerner (2003), and Stivers and Rossano (2010) shows that speakers often direct their gaze toward the intended next speaker around a possible completion point, and that recipients so addressed are statistically more likely to respond. Through such practices, gaze helps differentiate those treated as expected recipients from those merely overhearing, but also enables the speaker to indicate the next speaker they have selected in a non vocal – and hence less conversationally disruptive – way. However, this strategy only works if both the selected next speaker and the non-selected attendees have seen the speaker's gaze. Jokinen et al. (2013) stress the significance of gaze to identify when the current speaker wishes to hold their turn or leave it to the next speaker. Regarding schisming, experimental work by Vertegaal et al. (2001) suggests that people gaze more at their direct addressees than more peripheral participants for a given topic. In larger groups, gaze also helps segment interaction into simultaneous conversational threads, as participants establish local mutual gaze with temporary side-partners while withdrawing from the main floor, a phenomenon described by Egbert (1997).

The importance of gaze becomes particularly salient when this channel is removed or degraded. Mediated communication, such as audio-only calls, eliminates visual access entirely, and even video calls distort mutual gaze because the camera and screen are misaligned, making it difficult to determine precisely where someone is looking. They can also complicate coordination because of delays between audio and video. Participants can infer some attentional cues from head orientation and posture, but these cues are considerably less reliable. Although conversational modalities are generally redundant – Cook and Lalljee (1972) show that audio-only communication preserves many conversational behaviours, with differences arising primarily in utterance length – it would be difficult to argue that the loss of gaze information has no effect on conversation. Findings from virtual reality environments reinforce this point: even when avatars simulate eye movements, the fine-grained timing and mutual contingencies characteristic of face-to-face gaze are all maintained, disrupting familiar turn-taking patterns (Lücking et al., 2025). Dialogue systems face an even greater challenge, as they lack genuine perceptual monitoring and cannot ground their gaze behaviour in a real-time assessment of attentional states or mutual orientation. Consequently, the gaze behaviours of avatars and embodied agents resemble the impoverished cues available in video-mediated communication, but without the human inferential capacities that compensate for missing information.

Overall, gaze constitutes a densely multifunctional resource that structures the temporal, cognitive, and social organisation of conversation. Its absence or distortion reconfigures interaction in ways that remain only partially understood, underscoring the need to characterise dialogue data not merely in terms of modality but also in terms of the extent to which participants can access and interpret one another's gaze. While a binary description of this parameter can already provide some information about the dialogue setting of a resource, a more detailed characterisation can specify the degree of attention that participants can devote both to directing their own gaze and to monitoring that of their interlocutors. Although participants may be physically co-present, task demands can constrain their ability to establish or maintain mutual gaze. For example, a driver and passengers occupy the same physical space, but the driver cannot devote sustained visual attention to the passengers. In mediated communication, the fidelity of gaze cues

Parameter	Minimal	Extended description
Gaze	Binary	Attentional allocation Fidelity: accurate / approximated / distorted

Table 2.10: Operational characterisation of dialogue resources and tasks – Environment & Medium: Gaze

constitutes another important dimension, as some media, such as video calls, systematically distort gaze direction because of parallax and camera placement.

2.3.2.2 Gestures and Facial Expression

Gestures are often defined as visible bodily movements that accompany speech and contribute to the communicative act. They include movements of the hands and arms, but also head movements, gaze behaviour, and facial expressions, though the latter category is sometimes distinguished for its closer association with affect and stance. Kendon’s (2004) theory of the ‘gesture-speech ensemble’ emphasises that gesture and speech form a unified system: gestures are temporally and semantically integrated with the spoken utterance, and listeners interpret them in conjunction with linguistic content rather than as parallel or extraneous information. This integration is reflected in semiotic classifications dating back to Peirce (1934), who distinguished icons, indexes, and symbols as fundamental sign types. Contemporary gesture typologies build on these foundations. Deictic gestures function indexically, pointing to objects, people, or locations in the physical or conversational space. Iconic gestures depict aspects of referents through visuospatial resemblance, while metaphoric gestures extend this depictive capacity to abstract concepts, often via metaphorical mappings. Beat gestures, by contrast, are rhythmic, non-representational movements aligned with prosodic peaks; they serve to highlight or structure discourse rather than to introduce referential content.

The functions of gestures in conversation are diverse and extend beyond the merely expressive. A growing body of research has demonstrated that gestures contribute semantic information that may not be recoverable from speech alone. Recent experimental work shows that people interpret the same spoken utterance differently depending on the visual signal accompanying it, and that the interpretation of complex multimodal signals cannot be predicted from a simple combination of their components (Trujillo & Holler, 2024). Temporal coordination also plays a critical role: gesture strokes typically align with specific lexical items or prosodic prominence (McNeill, 1992; Mehler & Lücking, 2012). This synchrony contributes to the perception of a cohesive communicative unit and facilitates incremental interpretation by the listener.

Beyond semantic enrichment, gestures can bear illocutionary force and directly participate in conversational dynamics. Speakers use gestures to request clarification, to signal uncertainty, to soften or reinforce the force of a verbal act, or to cue turn-taking intentions. They also play an important role in discourse management. Gestures can introduce new discourse referents, disambiguate pronouns or spatial descriptions, and maintain coherence in referentially dense segments of talk (Debreslioska & Gullberg, 2020; Parrill et al., 2022). In these respects, gestures function as conversation facilitators, scaffolding the interaction in ways that are not easily replaced in purely verbal communication.

The absence or reduction of gesture and facial expression in mediated communication often leads participants to adopt surrogate semiotic resources. In text-based communication, emojis, stickers, GIFs, and images serve as partial substitutes for facial expressions, affective displays, or iconic gestures (Gandolfi & Pickering, 2023). Some video-conferencing platforms provide ‘reaction’ buttons intended to mimic backchannels or feedback cues. These resources reveal a persistent communicative need to compensate for the loss of visual signalling in a setting where the speaker may not be able to monitor the gestures and

Parameter	Minimal	Extended description
Gestures & Facial expressions	Binary	Attentional allocation Interpretability

Table 2.11: Operational characterisation of dialogue resources and tasks – Environment & Medium: Gestures and Facial expressions

expressions of all the receivers. This suggests that such signalling is not merely embellishment but an essential component of meaning-making.

At the same time, these substitutes bring a new layer of complexity to the interpretation of dialogue. Emojis, for instance, are inherently polysemous and context-sensitive: their interpretation depends on cultural norms, interpersonal familiarity, and conversational context, much like the interpretation of backchannels or facial expressions. Their form is unstable across platforms, as the same Unicode code point may appear with divergent visual designs on different operating systems. Furthermore, the descriptive labels provided by Unicode do not constrain actual usage and probably rarely reflect it, and user-generated ASCII art – like ;-) or T_T – has no conventionalised description at all. These properties complicate interaction and can lead to misunderstandings – both for human interlocutors and, increasingly, for computational dialogue systems that must interpret these signs.

For researchers constructing or analysing dialogue corpora, these observations raise important methodological considerations. A common pre-processing step in large-scale text corpora is to remove emojis and other ‘non-lexical’ tokens, in a manner reminiscent of the practice of removing backchannels as noise. Yet the communicative weight of these elements suggests that their removal may obscure important aspects of interactional meaning. Retaining, annotating, and analysing them is therefore essential for a complete representation of the conversational event.

The significance of gestures is also increasingly recognised in computational modelling and human-computer interaction. Traditional dialogue systems lack physical embodiment and therefore have no access to the multimodal channels that humans rely on. Research in robotics and embodied conversational agents attempts to model the generation and interpretation of gestures, synchronising them with speech to produce more naturalistic behaviour. Formal semantic frameworks, such as the Type Theory with Records (TTR) approach developed by Ginzburg and Lücking (2021), provide ways of integrating gestures into the semantic and pragmatic representations underlying dialogue. These efforts reflect a growing understanding that gestures are not an optional add-on to communication but a fundamental resource shaping how dialogue unfolds.

Altogether, gestures and facial expressions constitute a core parameter in characterising conversation, influencing meaning, interactional structure, and the interpretability of utterances. Their presence, absence, or substitution materially affects how dialogue works and how it should be analysed. A comprehensive account of conversational phenomena must therefore incorporate these multimodal resources rather than treating them as peripheral or removable elements. For this reason, researchers should include the following questions in their corpus description or selection process: Did participants have access to gestures, facial expressions, and other visual cues? If not, did they rely on substitute semiotic systems – such as emojis, GIFs, or reaction icons – and were these systems conventionalised prior to the dialogue or their meanings were locally negotiated over the course of the conversation? Are these systems interpretable from outside the interaction by the analysts? Similarly to gaze this parameter could be basically described in a binary way or in more details with the degree of attention put into the gestures and facial expressions, and the interpretability of the visual cues in case they rely on a semiotic system. These parameters are summarised in Table 2.11.

2.3.3 Auditory co-presence

Auditory co-presence – the possibility to hear each other – is another central parameter for characterising conversation. In face-to-face interaction, speech is never simply a sequence of words: it is shaped by intonation, rhythm, loudness, voice quality, and other auditory cues that convey affect, stance, epistemic alignment, and interactional intentions. When dialogue shifts to modalities without access to these auditory signals, the interpretive landscape changes correspondingly. As with the visual modality, the attenuation or loss of prosody affects how utterances are understood, how actions are recognised, and how interlocutors coordinate their interaction.

In what follows, we first discuss the role of prosody and tone in shaping interaction (Section 2.3.3.1), before turning to the influence of noise and signal degradation (Section 2.3.3.2).

2.3.3.1 Prosody and Tone

Although linguistics has long focused on words and grammatical structure, everyday interaction demonstrates that meaning is strongly modulated by tone of voice. Utterances such as ‘hmm’, ‘oh’, or ‘really’ can exhibit different meanings and functions depending on their realisation. Work on discourse markers and interjections has shown that intonation is essential for distinguishing the action performed by these items (Heritage, 1984; Ward, 2006; Norrick, 2009). Tone can therefore convey more than words alone, and prosody plays a key role in shaping the illocutionary force of an utterance.

The importance of auditory information has long been recognised in conversation analysis and phonetic research. CA-informed transcription systems such as the Jeffersonian conventions incorporate pitch movement, stress, elongation, loudness, and voice quality because these features are necessary for understanding how turns are designed and interpreted (Jefferson, 2008). Prosody is central to how interlocutors classify utterances as questions, assertions, continuations, or completions for example. While grammar can mark interrogatives, in many cases questions in conversation are realised through prosody rather than syntax, such as a rising pitch contour on an otherwise declarative sentence (Gunlogson, 2004). Languages differ in the extent to which syntax or prosody performs this work, but the general point holds: auditory cues shape action recognition (Hirschberg, 2006).

The absence of auditory co-presence in text-based communication has important consequences for how speakers interpret each other. Without prosodic cues, it becomes more difficult to detect sarcasm, irony, joking, or playful teasing, to assess degrees of certainty or commitment, or to distinguish genuine queries from rhetorical ones. These aspects of stance and action are usually signalled through pitch, timing, and voice quality (Attardo et al., 2003; Bryant, 2011). When such cues are missing, interlocutors may attribute unintended intentions to one another. Misunderstandings can persist for longer than they would in spoken dialogue because the absence of real-time feedback reduces opportunities for immediate repair. In face-to-face interaction, interlocutors constantly monitor each other’s responses through subtle changes in prosody – hesitations, micro-pauses, and shifts in tone – that prompt clarification or modification. In text-based media, these signals are not available, and speakers must compensate by being more explicit about their intended meaning. Yet it is not always clear when one’s message may be misinterpreted, and communicators often lack sufficient mutual monitoring to detect emerging problems. This can lead to unintended perceptions of coldness, abruptness, or irritation, even when none was intended.

Auditory co-presence also interacts with broader questions of synchronicity. Asynchronous voice messages – an increasingly common medium, especially in mobile communication – constitute an intermediate case between synchronous speech and text. Users often choose voice messages when they want to convey longer or more affectively charged content, reflecting that speaking can be easier and more expressive than typing. However, receiving voice messages is often less convenient: listeners cannot skim or search the content as easily as text, and partial re-listening is cumbersome. Research on mobile messag-

Parameter	Minimal	Extended description
Prosody / Tone	Binary	Attentional allocation Fidelity

Table 2.12: Operational characterisation of dialogue resources and tasks – Environment & Medium: Auditory co-presence

ing shows that vocal messages are perceived as bridging some limitations of text-messaging though their use is polarising (Haas et al., 2020). This again highlights that communicative modalities offer different trade-offs between expressivity and convenience.

Despite the centrality of prosody in human communication, most dialogue systems do not yet use the auditory modality in a human-like way. Work on expressive speech synthesis and empathetic dialogue agents recognises that emotional meaning is carried extensively through tone of voice (Mozziconacci, 2002), but computational modelling of prosody remains limited. Systems often rely solely on text, losing access to the rich paralinguistic cues that humans use to interpret intent. This reduces their robustness in detecting sarcasm, uncertainty, or subtle conversational actions, and it limits the naturalness of machine-produced speech.

Taken together, these considerations demonstrate that auditory co-presence is a vital dimension for characterising conversational settings. The presence or absence of prosody shapes the interpretation of utterances, the recognition of social actions, the management of stance and affect, and the likelihood of misunderstanding. As with visual modalities, the differences across media call for careful attention when collecting and analysing conversational data. Corpora that lack auditory information should be annotated accordingly, and the limitations of such data must be acknowledged when analysing interactional phenomena that depend heavily on tone of voice. Conversely, the presence of auditory co-presence – even in asynchronous form – provides communicative affordances that materially change how dialogue unfolds. This parameter should thus be included in any comprehensive characterisation of dialogue resources and tasks, with attention to the degree of prosodic information available and the extent to which participants can actually rely on it for interpretation. As for most parameters a basic binary distinction can be made but a more detailed characterisation would specify the attention participants put into prosodic cues and the fidelity of these cues, especially in mediated communication where technical factors can distort or degrade the auditory signal. For example, in video calls, latency and compression can affect the timing and quality of prosodic cues, making them less reliable for turn-taking or stance recognition. In asynchronous voice messages, the lack of real-time feedback may reduce the communicative value of prosody, as listeners cannot adjust their interpretation based on immediate responses. These nuances are important for understanding how auditory co-presence shapes conversational dynamics across different settings. These parameters are summarised in Table 2.12.

2.3.3.2 Noise

Beyond structured vocal signals, auditory co-presence also involves the quality and reliability of the acoustic channel through which speech is transmitted. In real-world interaction, speech is rarely perceived under ideal conditions: background noise, signal degradation, transmission artefacts, and environmental interference can all affect how utterances are heard and interpreted.

Noise influences dialogue by introducing uncertainty at the perceptual level, which in turn shapes interactional dynamics. Participants may fail to hear parts of an utterance, misperceive words, or require repetition and clarification. This leads to an increased frequency of repair sequences, slower turn progression, and more explicit confirmation strategies. In this sense, noise does not merely degrade the signal

Parameter	Minimal	Extended description
Noise	Binary	Degree Sources

Table 2.13: Operational characterisation of dialogue resources and tasks – Environment & Medium: Auditory co-presence

but actively reshapes the structure of the interaction.

The impact of noise varies significantly across conversational settings. Face-to-face interaction in a quiet environment allows for fine-grained prosodic distinctions to be perceived, whereas crowded or noisy settings reduce the availability of these cues and increase reliance on redundancy and multimodal signals. In mediated communication, technical factors such as compression, latency, or packet loss can similarly affect auditory quality. By contrast, text-based communication eliminates acoustic noise entirely, but at the cost of losing prosodic information altogether.

Many tools that deal with speech such as ASR systems tend to be designed to operate in ideal conditions, and their performance degrades significantly in noisy environments. This can lead to increased error rates, misrecognitions, and failures to capture the intended meaning. For dialogue systems that rely on speech input, this presents a significant challenge: they must be robust to noise and capable of handling the resulting ambiguities in user input. Research in robust speech recognition and noise reduction techniques is therefore crucial for improving the performance of conversational agents in real-world settings.

From the perspective of dialogue characterisation, noise is therefore a crucial parameter because it conditions how reliably auditory information is transmitted and how much effort participants must devote to maintaining mutual understanding. High-noise environments tend to favour shorter utterances, increased repetition, and more robust grounding strategies, whereas low-noise environments support more efficient and nuanced interaction. This parameter can be operationalised in a binary way – presence or absence of noise – but a more detailed characterisation could specify the degree of noise and its sources. For example, one could distinguish between environments with intermittent background noise (e.g. a café) versus constant noise (e.g. a factory floor), or between technical noise (e.g. poor audio quality) and environmental noise (e.g. traffic). Such distinctions would provide a richer understanding of how auditory conditions shape conversational dynamics. These parameters are summarised in Table 2.13.

2.4 Interaction

Dialogue is more than the mere production of utterances in proximity; it is the joint activity of talking *to* and *with* one another. It requires participants not only to speak but to attend, listen, understand, and coordinate – whether this occurs in face-to-face interaction or through mediated text. Conversation is thus fundamentally co-constructed: speakers continuously shape one another’s contributions by signalling attention and comprehension, adapting to emerging meanings, and managing turns in ways that allow the interaction to flow. Crucially, much of this coordination unfolds simultaneously, with production, monitoring, and feedback interwoven in real time. At the same time, dialogue is inherently ephemeral. Its unfolding is tied to the participants’ momentary cognitive and affective states, and while we may record or transcribe an interaction, what we preserve is only an imprint of a process that vanishes as it is produced. Understanding dialogue therefore requires attending both to the mechanisms that enable its co-construction and to the temporal constraints that make it a fleeting, situated form of human activity.

Thus in this section we first discuss the co-construction aspect of dialogue and the different mecha-

nisms that enable it such as coordination signals, turn-taking and mutual completion (Section 2.4.1). We then consider the time-bound nature of conversation and the nature of synchrony and ephemerality on the way it is produced and interpreted (Section 2.4.2).

2.4.1 Co-construction

Conversation is a social matter involving several participants who need to cooperate with each other in order to reach their goal (see Section 2.5.2). According to Grice (1975), speakers rely on the assumption that others will act cooperatively in order to interpret utterances and infer intended meanings – the basis of his Cooperative Principle and conversational maxims. Beyond this, Clark (1996) emphasizes that dialogue is organised as a series of *joint projects*, where meaning is not simply transmitted but actively co-constructed through mutual contributions. This process presupposes constant attention to the partner’s perspective: participants provide feedback to display understanding and alignment (E. Schegloff, 1982) and manage the flow of conversation through systematic turn-taking mechanisms (Sacks et al., 1974). Co-construction in dialogue therefore rests on a dynamic interplay of cooperation, feedback, and turn-taking, through which participants jointly build and negotiate meaning.

In this section we focus on the mechanisms of co-construction in dialogue. We first discuss the coordination signals that enable participants to monitor and adapt to each other in real time (Section 2.4.1.1), then we examine the turn-taking system that structures the flow of conversation (Section 2.4.1.2), and finally we consider mutual completion, a phenomenon where participants collaboratively complete each other’s utterances or thoughts (Section 2.4.1.3).

2.4.1.1 Coordination signals

face-to-face conversations enable speakers to monitor their addressees in real-time and receive feedback from them in various forms. Whether it is through gestures, facial expressions or backchannels, feedback lets the speaker adapt to their audiences and make sure that they are adequately coordinated (Clark & Schaefer, 1992). Indeed, ignoring the other participants is a failure if one wants to have a conversation. Conversely, providing irrelevant feedback as a listener can completely disrupt the flow of conversation. In fact cooperating to understand each other reduces the mental cost compared to understanding what someone says without repairs, without having the possibility to request more information on specific points, without giving feedback as it goes, etc. (Jefferson, 1972; E. A. Schegloff, 1979; Dingemanse et al., 2015; Van Arkel et al., 2020). So this symbiotic relationship is a crucial point in conversation in the sense that alignment makes conversation smoother and less cognitively demanding than other forms of communication with solo-understanding. For this reason, being able to both adapt to the addressees based on their feedback and provide the right amount of it at the right time are important skills when it comes to conversation.

This core feature of conversation deteriorates the more asynchronous a communication form is (synchronicity is further discussed in Section 2.4.2.1). Indeed, one requires real-time feedback to be able to adapt to their addressees. It also becomes more and more complicated for the recipient to ask for clarifications as time passes since it requires re-raising points that can be far in the past from the speaker’s point of view. While it remains possible, it is heavier in terms of conversational practices and participants could give up on some forms of repairs in order to avoid disrupting the conversation too much.

A way of mitigating the loss of this feature in asynchronous forms of communication is to provide more explicit information from the outset. However, this also increases the amount of information that recipients must process. In instant messaging tools, an increasing number of features have been developed to enable users to provide explicit feedback to interlocutors, such as reacting to messages with emojis. Similar features are also available in some video-conferencing tools, even when the camera is turned off.

Parameter	Minimal	Extended description
Coordination signals	Binary	Availability of channel (vocal, textual, visual, haptic) per participant Technical ability per channel and participant Signal density per participant Feedback latency tolerance

Table 2.14: Operational characterisation of dialogue resources and tasks – (Verbal) Interaction: Coordination signals

Although these features do not fully address the challenges of highly asynchronous communication, they illustrate how tools originally designed around asynchronous interaction are increasingly attempting to reproduce aspects of face-to-face conversation.

The influence of coordination signals on cognitive resources and on the way language is produced makes them a crucial aspect of dialogue to consider when characterising and selecting resources. While a simple binary description of the presence or absence of such signals may be sufficient in some cases, several dimensions can be elaborated to provide a more detailed characterisation. For example, we can specify the availability of different feedback channels (vocal, textual, visual, haptic) for each participant, as well as the participants' technical ability to use these channels. For instance, backchannels require an oral channel, but not all oral productions function as backchannels. Conversational AI systems are often characterised by their lack of backchannels, and the resulting absence of feedback makes interaction with them different from human-human interaction. The density of signals can also be specified, with the advantage that it provides a numerical value that can be readily incorporated into computational models. Finally, the latency tolerated for receiving feedback can be important when comparing different forms of mediated conversation. In face-to-face conversation, feedback can be provided with very low latency and is crucial for maintaining conversational flow. In asynchronous forms of communication, latency can be much higher without necessarily preventing the conversation from unfolding. However, when latency becomes excessive, it can become difficult to track different conversational threads and maintain a coherent dialogue. These different dimensions for describing coordination signals are summarised in Table 2.14.

2.4.1.2 Turn-taking

In conversation, people take turns speaking, typically with very short gaps and relatively little overlap. This phenomenon, known as turn-taking, has been extensively studied in Conversation Analysis. Sacks et al. (1974) established a set of loose organisational principles that describe how turns are allocated. Although these principles have been criticised for their lack of strictness, the general idea they capture remains valid: in the vast majority of cases, the floor changes at what have been called transition-relevance places (Sacks et al., 1974, p.703), and the next participant is either selected by the prior speaker or the floor becomes available for whoever takes it first. The idea that there are few to no overlaps, however, is not entirely accurate. While overlaps between substantive utterances – those that contribute information relevant to the ongoing topic – are relatively rare, overlaps between the current speaker and feedback from other participants are frequent. As discussed in the previous section (Section 2.4.1.1), such feedback is a crucial feature of conversation.

Historically, these types of utterances were often considered errors and were not studied as part of the intended content of conversation. However, current theories acknowledge their importance. Nevertheless, overlaps, whether resulting from feedback, false starts, or mismanagement of turn-taking, create

challenges for the study of conversational data. In many cases, for practical reasons such as privacy constraints and ease of storage, linguists rely on transcribed conversations, which provide a reduced representation of the original interaction. Since transcriptions are often the primary form in which conversations are analysed, the transcription schemes employed are crucial for preserving as much information from the original dialogue as possible. Jefferson (2008) proposed a transcription scheme that enables the representation of, among other conversational features, overlapping speech. While this scheme is now widely used, it is not universally adopted because it requires more effort than simpler transcription approaches.

With advances in technology, new methods for storing and transcribing conversational data have emerged. For instance, tools now exist that allow transcripts to be aligned with audio recordings, making it possible to preserve overlaps and represent turn-taking mechanisms in greater detail. However, such representations are less straightforward to process than linear sequences of utterances, which are the format assumed by many tools developed in NLP. As a consequence, speaker diarisation systems still struggle to handle overlapping speech, limiting their ability to process multi-party conversations (Addlessee et al., 2020). This is particularly problematic because researchers increasingly rely on automatic tools to prepare and structure data, as these approaches are substantially cheaper and less time-consuming. As a result, future datasets may increasingly omit interactional features that are considered central to the organisation of human conversation.

Turn-taking is mostly reserved for oral communication modes with the exception of what Earnshaw, 2017 calls super-synchronous chats where the messages appear keystroke by keystroke and participants can technically type and read at the same time. Regarding the number of participants, turn-allocation is simpler and more organised in duologues as participants speak one after the other but it can become very chaotic in multilogues unless the genre of conversation comes with some structure. For example in meetings or at court, the turn-taking system follows established rules which makes the exchange more organised despite the potentially high number of participants.

In written conversation, the quick succession of turns is lost in favour of potentially longer and more numerous turns before getting an answer from the addressees. The written aspect lets participants hold more threads of conversation at the same time since they do not have to remember everything, and this is probably all the more common when switching to asynchronous mode (Section 2.4.2.1). Participants can start several topics without closing the previous ones, and expect responses to some or all of them. Indeed, one can send an informational message without necessarily expecting a response; depending on the mode of communication, one may not even expect acknowledgment feedback.

Multi-party written conversation probably works similarly in the sense that sub-conversations appear to have a more classical turn-taking system when separated (Hunter et al., 2015). However both for written and oral multi-party conversation, the subdivision phenomenon becomes more complicated to study when the subsets of participants have a non-empty intersection, i.e. when one or more participants are taking part in more than one sub-conversation at the same time. This is probably complicated to maintain, especially orally due to the high informational load but this type of situation is not impossible, in particular with the development of text message applications as (2.2) illustrates. Studying such conversations by separating the sub-dialogues and not considering the global exchange would represent a big loss of information in terms of what participants perceive of the conversation at a given time.

Launching several conversations in parallel in written communication may be a form of response to the loss of efficiency of this communication mode compared to face-to-face. Since we cannot carry a conversation through to the end in one go, we carry several of them out at once to settle as many things as possible in the shortest possible time, the idea being to make the most of the presence of the interlocutor(s) when they are there. This is made possible by the fact that the entire content of the conversation is recorded in the communication medium, which lowers the memory barrier. It would be interesting to investigate if this kind of behaviour remains when using messaging tools where the messages disappear once they have been read or where one cannot highlight which message they are answering to.

Parameter	Minimal	Extended description
Turn-taking	Binary	Overlap possible Turn-allocation method Average overlap time

Table 2.15: Operational characterisation of dialogue resources and tasks – (Verbal) Interaction: Turn-taking

In summary characterising the turn-taking system of a conversation is important because it is one of the core structural features of dialogue and it shapes the way conversation unfolds by allowing synchronous feedback or not, by enabling or not the quick succession of turns, etc. A simple binary parameter indicating the presence or absence of turn-taking can be sufficient in some cases especially to indicate very clear types of systems such as face-to-face conversations where turn-taking takes its most classical form, or written conversations where turn-taking is mostly absent. However, in many forms of mediated conversations turn-taking can be present but slightly different from its face-to-face form such as video calls where the delay can make it more difficult to coordinate turns. In such cases a more detailed description could indicate whether overlaps are at all possible and the turn-allocation method. A resource that focuses on turn-taking could also indicate the average overlap time, a positive one indicating that gaps are more common than overlaps and a negative one indicating the opposite. These parameters are summarised in Table 2.15.

2.4.1.3 Completion

Completion occurs when one participant completes another participant's utterance. It is a common feature of face-to-face conversation and can be understood as a form of co-construction, in which participants collaboratively build an utterance or formulate a thought. Completion may arise when a speaker struggles to find the appropriate word or when an interlocutor anticipates and supports what the speaker intends to say. However, it can also be interpreted negatively: when the participant completing the utterance prevents the original speaker from continuing, it may signal dominance or a lack of respect. The interpretation of completion therefore depends on the context, the nature of the interaction, and the relationship between participants (Sacks et al., 1974; Jefferson, 1986).

Completion is primarily observed in oral communication, as it relies on real-time feedback and the ability to respond to an unfolding utterance. It is generally absent from written communication, where contributions are typically produced in full before being presented to other participants.

From a modelling perspective, completion is important because it demonstrates that utterances may be co-constructed by multiple participants rather than produced by a single speaker alone. This has implications for dialogue systems, automatic speech recognition (ASR), and computational analyses of conversational data, as such systems must be capable of representing and transcribing partial and overlapping contributions.

In terms of operationalisation, completion can be represented minimally as a binary parameter indicating whether it occurs, while a more detailed representation could measure its density, i.e. the frequency of completed utterances per conversation or per turn, and indicate the function of the completion, e.g. support or competition. This allows for numerical analysis and integration into computational models of interaction. This parameter is summarised in Table 2.16.

Parameter	Minimal	Extended description
Completion	Binary	Density Function (support, competition)

Table 2.16: Operational characterisation of dialogue resources and tasks – (Verbal) Interaction: Completion

2.4.2 Time

All conversations unfold under temporal constraints that shape how they are produced, interpreted, and remembered. Face-to-face dialogue is typically synchronous: participants must coordinate speaking, listening, and understanding in real time, with each turn shaped by the immediacy of what has just been said and by anticipation of what may come next. These temporal pressures structure both the pace of interaction and the kinds of actions that are possible within it. At the same time, conversation is transient. Its full richness – including fleeting cues, momentary cognitive states, and the evolving context in which turns are produced – cannot be preserved in any stable form. What we record or transcribe captures only traces of a process that is constantly disappearing as it unfolds. Understanding the role of time in dialogue therefore requires examining both its real-time dynamics and its inherent ephemerality.

We thus discuss the synchrony of production and interpretation in dialogue and how it is affected by different communication forms in Section 2.4.2.1, then how the degree of planning participants can put into their contributions affects the way they produce and interpret each other in Section 2.4.2.2, before proceeding to Section 2.4.2.3 where we examine the ephemerality of conversation and how it affects the way we can study it.

2.4.2.1 Production and interpretation

In face-to-face conversation production and interpretation happen simultaneously. The speaker interprets themselves as they talk to self-correct and adapt to their audience if needed and the addressees interpret the utterance in real time which enables them to provide appropriate feedback. This symbiosis contributes to the fluidity of conversation as we already discussed in the previous section.

However, this synchrony deteriorates or disappears in many mediated communication forms. In video calls for example, a bad internet connexion can create a delay between the users which makes the production and interpretation slightly asynchronous. This can make it more complicated for a speaker to properly adapt to their addressees because the feedback they would receive would not be as well aligned with their production as they would expect. Indeed, even though we know that a delay is possible, it is quite complicated to estimate exactly how desynchronised the different participants are. However the most striking examples are written forms of communication which are in most cases asynchronous. Even though they can be used in more synchronous ways if the participants coordinate well, there will always remain a delay between production and interpretation since a message, e-mail or letter is first entirely produced and then sent. This deteriorates the usual feedback and adaptation mechanisms since one has to explicitly state which part was unclear if they require clarifications. Providing quick feedback is also more complicated in written communication since nodding or backchanneling are face-to-face mechanisms. Nevertheless, people tend to develop ways to remedy this and narrow the gap between written and face-to-face communication. For example one can break down an e-mail in small pieces and write their answer to each point separately, which in some ways resembles real-time feedback. Similarly, more and more messaging apps provide ways to react to messages very quickly through emojis. Again this supports the idea that face-to-face conversation is the simplest and most efficient communication form

Parameter	Minimal	Extended description
Synchrony	Binary	Time between production and interpretation

Table 2.17: Operational characterisation of dialogue resources and tasks – Time: Synchrony

and that mimicking its features in other forms makes them more efficient as well.

Interpreting utterances at a time different from that of their production means that the speaker and the addressee may no longer share the same context. While this does not necessarily prevent mutual understanding, contextual information can be crucial in certain situations. If the speaker has not made the relevant aspects of the context explicit, the receiver must reconstruct the speaker's context in order to fully interpret their contribution. Otherwise, misunderstandings may arise and disrupt the conversation.

The loss of synchrony also makes clarification requests more complicated. In a synchronous setting, participants can interrupt the ongoing interaction to clarify a specific point. However, when there is a delay between production and interpretation, it may become more difficult to identify precisely which aspect of an utterance requires clarification. The initial speaker may also more easily misunderstand the purpose of a clarification request when the original utterance is no longer recent, particularly because they may have partially forgotten the intended meaning or context behind what they said. Moreover, if the clarification concerns information that is necessary for interpreting subsequent contributions, the inability to resolve it immediately can complicate the interpretation of the rest of the conversation. Dialogue can therefore become more difficult to manage when speakers do not provide sufficient contextual information from the outset.

Studying language use in asynchronous settings thus changes the cooperation modalities since participants have to adapt to the fact that they will not be able to align with each other in real time. Alignment is made more complex by the increased cost of clarification requests, which can lead participants to try harder to understand on their own and possibly come up with hypotheses that enable them to interpret the conversation in a coherent way. Nevertheless those hypotheses are in most cases not verified. This position has similarities to the one we are in as language researchers since we often study language use retrospectively rather than in real time situations.

Asynchronous conversations also have a more or less important delay between interventions which means that the participants can be in different mindsets and physical contexts at different states of the conversation. This can make past contributions less relevant for the speaker when the addressee actually interprets them. Asynchronous conversations can last for extended periods of time, to the point that it makes it complicated to determine their exact beginning and end. This has implications regarding coherence since contradictions can appear in very long conversations without necessarily being disruptive if the contradictory statements are far enough from each other. However this distance is not straightforward to evaluate since it depends on the density of the conversation, its time rate and the flexibility of participants towards contradictions.

This parameter can be summarised in binary terms or with a more fine-grained value indicating the average time between production and interpretation. This value can be used to compare different communication forms and to determine the level of synchrony of a conversation. For example, face-to-face conversations would have a very low value while e-mails would have a much higher one. This parameter is summarised in Table 2.17.

2.4.2.2 Planning

When using language, we always plan our utterances to some extent. In face-to-face conversation, we plan them while the current speaker is talking and continue planning on the fly as we speak. In many cases, we

Parameter	Minimal	Extended description
Planning	Binary	Opportunities for content planning Opportunities for utterance formulation

Table 2.18: Operational characterisation of dialogue resources and tasks – Time: Planning

cannot afford to fully plan our next utterance before beginning to speak, as doing so may cause us to lose the floor. The amount of planning time available, however, depends on the type or medium of interaction. Different forms of written communication, for example, allow for varying degrees of advance planning. In an email exchange, we often have ample time to formulate a response, although some email threads can become sufficiently rapid to resemble instant messaging. Likewise, when participants are simultaneously engaged in an instant messaging conversation, the demands on planning can closely resemble those of face-to-face interaction. By contrast, when engagement is lower or the exchange is more asynchronous, participants have considerably more time to plan their responses.

The level of planning can influence the way we produce language. For example, when we have more time to plan, we can afford to be more careful and to check our contributions before sending them. We can also take the time to think about the best way to express ourselves and to choose our words more carefully. On the other hand, with less time, we may be more likely to make mistakes or to use less precise language, thus leading to more repairs.

In LLMs, although we can prompt the model to produce an output that appears more or less spontaneous, such prompts do not fundamentally alter the way the model generates its response. Rather, they primarily affect the statistical distribution of the generated output. Newer models, however, often include a ‘thinking’ mode, in which the model performs an intermediate reasoning phase before producing the response shown to the user. This intermediate stage can be interpreted as a form of planning, insofar as it allows the model to refine its response before outputting it. Thinking mode has been shown to improve results on commonsense reasoning benchmarks (Wei et al., 2022); however, its underlying mechanisms should not be too readily equated with the planning processes involved in human conversation.

Planning is an important parameter for distinguishing between different forms of conversation, whether they occur between humans or involve conversational AIs. In its simplest form, it can be represented as a binary distinction indicating whether planning is possible before producing an utterance. A more informative characterisation, however, distinguishes between two complementary dimensions: the opportunity to plan the content of an utterance and the opportunity to plan its formulation. While these dimensions often correlate, they need not coincide. For example, a politician participating in a live interview may have had ample time beforehand to anticipate questions and prepare the content of their answers, yet have only limited opportunity to refine the wording of each response during the interaction. By contrast, if the interview is conducted in writing or recorded with the possibility of multiple takes, the speaker can also revise and optimise the formulation of their utterances before they are delivered.

This parameter is summarised in Table 2.18. It is operationalised in terms of *opportunities* rather than *time* because elapsed time is only a proxy for the opportunities available to plan content and formulate an utterance. Although additional time generally affords greater scope for planning, these opportunities also depend on the constraints imposed by the interaction itself. This formulation accommodates both human and AI dialogue, where planning may be enabled by mechanisms that are not naturally expressed in temporal terms, such as the ability to revise a written response or, in the case of an LLM, to perform an intermediate reasoning phase before generating the final output.

2.4.2.3 Ephemerality

Conversational interaction is by nature ephemeral. Each exchange takes place at a particular moment and under unique conditions –material, cognitive, and affective–that cannot be identically reproduced. The state of the physical environment (see Section 2.3.1) and the interlocutors’ mental or emotional states jointly shape what can be said and how it is interpreted. As conversation is a situated and embodied activity (Goodwin, 2000; Mondada, 2018), its occurrence is inseparable from the circumstances of its production. Once the moment passes, the precise configuration of contextual, perceptual, and psychological factors that made the interaction possible also disappears. This irreversibility constitutes a fundamental property of dialogue: it unfolds in real time and cannot be re-enacted under identical conditions (Sacks, 1995).

Add a paragraph on the distinction between media that preserve access to the past interaction and media that do not. For example, in a face-to-face conversation, once a turn is produced, it is gone and cannot be retrieved. In contrast, in a chat or email exchange, the previous turns are preserved and can be accessed by participants at any time. This distinction has implications for how participants manage their contributions and interpret the conversation. In ephemeral media, participants must rely on memory and immediate feedback to maintain coherence, while in persistent media, they can refer back to previous messages to clarify meaning or resolve ambiguities. [Amandine]

Ephemerality must therefore be taken into account when characterising dialogue resources and tasks for two main reasons. First, it shapes the interaction itself by determining the resources available to participants when producing and interpreting conversational contributions. Second, as discussed in Sections 2.3.3.1 and 2.4.1.2, it conditions how conversations can be observed, recorded, and analysed, since researchers never have direct access to the original interaction but only to one of its representations.

The first aspect can be characterised by a binary parameter indicating whether the conversation is ephemeral or not. A more detailed description, however, can distinguish different dimensions of ephemerality, such as message persistence, editability, and whether participants have access to the edit history. These properties influence language production in mediated communication. For example, the inability to edit a previously sent message may encourage greater caution during composition or lead participants to adopt repair strategies, such as preceding a message with an asterisk (‘*’) to indicate a correction of the previous message.

The second aspect concerns the representation of the interaction. It can be characterised by the type of recording available (e.g., audio, video, or transcript) and, in the case of transcripts, by their level of detail. A richer characterisation may also specify whether participants are aware that their contributions will persist and for how long. This information is important both from an ethical perspective, since conversations that are publicly accessible are not necessarily intended to be reused for research or language model development, and from an interactional perspective, as speakers may adapt their behaviour according to their expectations about the persistence of their contributions. These different dimensions of ephemerality are summarised in Table 2.19.

2.5 Function and Structure

Traditionally, conversation has often been characterised by its goal – the set of information that must be exchanged for the interaction to be considered successful. This task-oriented perspective provides a clear framework for analysing dialogue in terms of steps, sub-goals, and the logical progression of questions and answers. However, other approaches, particularly Conversation Analysis, emphasised the social dimension of dialogue: how participants manage turn-taking, alignment, repair, and the subtle social work of maintaining relationships. To some critics focused on argumentative or goal-directed

Parameter	Minimal	Extended description
Ephemerality	Binary	Persistence time
		Editability time
		Edition history
	Recording form	Third-party persistence time Third-party persistence awareness

Table 2.19: Operational characterisation of dialogue resources and tasks – Time: Ephemerality

aspects, these social phenomena were deemed secondary or ‘uninteresting’ (Wooffitt, 2005), yet they are central to understanding how conversation actually unfolds (Goffman, 1967). Most real-world dialogue blends these dimensions: practical and social purposes coexist, shaping both what is said and how it is organised.

In this section, we characterise conversation in terms of the function and structure that guide its unfolding. We begin in Section 2.5.1 with epistemic grounding, which concerns the set of assumptions, truths, or shared facts on which the dialogue is based, including both the world setting and its stability. We then turn to the purpose of the conversation in Section 2.5.2, considering the practical and social goals that shape what participants aim to accomplish and how they organise their contributions. Finally, we examine activity in Section 2.5.3, which captures the broader context in which the interaction occurs, including the constraints imposed by the medium, the task, or the social norms of the situation. By analysing these three dimensions together, we can better understand how content, interactional organisation, and context interact to structure real-world dialogue.

2.5.1 Epistemic Grounding

Epistemic grounding refers to the set of assumptions, facts, or truths upon which a conversation is based. It is the framework within which participants judge the plausibility, relevance, and appropriateness of utterances. Conversations do not always occur in the ‘real world’: participants may discuss fictional worlds, hypothetical scenarios, or modified versions of reality. Two parameters are particularly relevant for operationalising epistemic grounding: world setting, discussed in Section 2.5.1.1, and world stability in Section 2.5.1.2.

2.5.1.1 World Setting

The world setting of a conversation defines the framework of assumptions, facts, and realities within which participants interpret and produce utterances. Conversations may take place in the real world, where interlocutors share common knowledge, cultural norms, and factual assumptions. Characters in fiction can have conversations pertaining to their real world which is fictional for the readers. Alternatively, participants may talk about a fictional or hypothetical environment, such as in role-playing games or thought experiments. In such cases, certain facts and rules are treated as given within the conversational frame, even though they do not correspond to objective reality. Participants typically rely on these assumed truths to coordinate meaning, allowing the conversation to progress smoothly despite the constructed nature of the world.

Even when discussing fictional or hypothetical worlds, interlocutors do not define every aspect and rule of the environment explicitly, and no work of fiction can be exhaustive in its coverage. Instead, participants rely on partial assumptions and conventions that can be negotiated implicitly or explicitly during the interaction. This phenomenon has been extensively studied in the semantics and pragmatics of

Parameter	Minimal	Extended description
World setting	Real (contemporary or not) / Fictional	Time period Location Conventions (cultural, social, scientific, moral, etc.)

Table 2.20: Operational characterisation of dialogue resources and tasks – Epistemic grounding: World setting

fiction fields, which examine how meaning is established and maintained when participants engage with non-real worlds (D. K. Lewis, 1978). For instance, moral laws are the kind of rules we typically import implicitly into fictional worlds, and they are more complicated to deconstruct than other assumptions such as physical laws. Alignment in world setting is crucial for comprehension: misalignment may result in misunderstandings, breakdowns in coordination, or divergent interpretations of the same utterance.

The world setting also plays a pivotal role in interactions with conversational AI systems. In these contexts, the assumptions and reasoning patterns of the system may differ from those of human participants. If the AI does not share or recognise the intended world frame, its contributions may appear incoherent or irrelevant, highlighting the need to model world assumptions explicitly when analysing or designing dialogue systems.

In terms of descriptive parameters, the most basic characterisation would need to indicate whether the conversation takes place in the contemporary real world or not. A more detailed description could specify the time period and location of the world setting, as well as any relevant cultural, social, or fictional conventions that participants are expected to adhere to. This description matters not only for fictional or role-playing contexts but also for real-world interactions, where correctly anchoring the conversation in its temporal and cultural context is essential for accurate interpretation. The reusability of dialogue resources over time is therefore dependent on the clarity and specificity of the world setting. This parameter is summarised in Table 2.20.

2.5.1.2 World Stability

World stability refers to the extent to which the assumptions and truths underpinning a conversation remain fixed throughout the interaction. While some conversations rely on a stable shared reality, in many interactions the world is dynamic, evolving as participants introduce new information, reinterpret prior statements, or negotiate the meaning of terms. For example, the course of a joint storytelling task may shift as new narrative details are introduced, and in problem-solving conversations, answers to questions may modify the shared understanding of the scenario.

Not all changes to the conversational world are immediately accepted. Participants may need to negotiate, clarify, or explicitly acknowledge new information for it to be incorporated into the shared framework. This reflects the inherently collaborative nature of conversation, where interlocutors continuously co-construct the epistemic environment that enables coordinated action (Clark, 1996; Clark, Brennan, et al., 1991).

The stability of the conversational world therefore shapes multiple aspects of dialogue. It influences how topics develop, how turns are allocated, and how participants achieve alignment. In interactions with artificial agents, a lack of shared world stability can lead to misaligned dialogue strategies or incoherent responses. Consequently, both the setting and stability of the world must be considered as central parameters when characterising dialogue, particularly in research on human–machine interaction or computational modelling of conversation.

Parameter	Minimal	Extended description
World stability	Binary	Planned changes Frequency of changes Source of changes (participants, external agent, environment)

Table 2.21: Operational characterisation of dialogue resources and tasks – Epistemic grounding: World stability

Some experiments for building dialogue resources explicitly manipulate world stability by providing participants with a persona at the beginning of the conversation and changing it later (S. Zhang et al., 2018). Precisely describing such manipulations is crucial for understanding how participants adapt to these changes and how they affect the flow of conversation. For example, if the persona is changed without any warning, it may lead to a breakdown in coordination and a significant shift in the conversation’s trajectory. If the change is introduced gradually or with some form of explanation, participants may be able to adjust more smoothly, maintaining coherence despite the evolving world. Operationalising world stability can be done with a binary parameter indicating whether the world is stable or not, but a more detailed description could indicate the degree of stability, the frequency of changes, and the manner in which changes are introduced and negotiated among participants. This would provide a richer understanding of how world dynamics influence conversational structure and content.

Similarly to the COMMITMENT parameter discussed in Section 2.1.2.3, the operationalisation of world stability heavily depends on the specific task and context of the conversation. In many cases, a binary distinction may suffice to indicate whether the world is expected to remain stable or is subject to change. However, in more complex scenarios the rely on the manipulation of the epistemic grounding of the conversation, the types of changes that are intended to occur and the mechanisms by which they are introduced may need to be specified in greater detail. In particular, the source of changes – whether they are introduced by the participants themselves, by an external agent, or by the environment – is often a descriptive parameter of the task in corpora collected for research purposes. Table 2.21 summarises these parameters.

2.5.2 Purpose

We engage in conversation for multiple, overlapping reasons. A long tradition in linguistics, pragmatics, and philosophy of language has treated dialogue primarily as a vehicle for accomplishing tasks or exchanging information: theories of speech acts, reasoning about intentions, and collaborative discourse treat conversation as a coordinated effort to reach shared goals, track commitments, or solve problems together (Grice, 1975; Austin, 1975; Clark, Brennan, et al., 1991; Clark, 1996). Cognitive psychology has similarly shown that speaking with others helps organise one’s own thoughts, offload cognitive work, and refine emerging ideas through social scaffolding (Vygotsky, 1978; Kuhn, 1991). Yet conversation is not only a tool for information management or practical cooperation: it is also a deeply social activity. Work in anthropology, sociology, and CA has emphasised the role of talk in maintaining relationships, affiliating with others, and enacting social identities (Goffman, 1967; Sacks, 1995; R. Dunbar, 1996). Practices sometimes labelled ‘gossip’ (R. Dunbar, 1996) highlight that talk can serve bonding functions even when no task needs to be achieved, though the term itself still presupposes an informational exchange. In practice, task-oriented and socio-emotional dimensions are rarely separable; most conversations blend both, with social alignment supporting joint activity and joint activity reinforcing social cohesion. The following section examines how conversational purpose – practical, social, or both – shapes the organisation of dialogue and how different theoretical traditions have selectively foregrounded one dimension over

the other (Section 2.5.2.1). It also dives into the intentions participants can have in conversations (Section 2.5.2.2).

2.5.2.1 **Practical Goal**

Conversation is central to carrying out tasks. We talk to order bread in a bakery, to coordinate a meeting, to decide what to cook for dinner, or to jointly solve a problem. Even when alone, people often talk to themselves when reasoning or organising their thoughts, highlighting the cognitive scaffolding role of dialogue-like behaviour.

Task-oriented conversations have been foundational in linguistics, pragmatics, cognitive science, and human-computer interaction. Tasks serve two methodological functions. First, they elicit talk while constraining its possible content, enabling systematic analysis and corpus sharing. Classic paradigms – including tangram description tasks (Clark & Wilkes-Gibbs, 1986) and maze-navigation tasks (Thompson et al., 1993) – have provided controlled environments for studying reference resolution, grounding, and coordination. Second, they offer a natural setting in which collaborative reasoning and information management can be observed, forming the empirical basis for information-state theories and logic-based dialogue models.

But task constraints also shape conversation in ways that diverge from everyday interaction. Highly specific tasks can narrow the scope of permissible actions and reduce the social flexibility of the encounter. Participants may allocate more cognitive resources to the task than they ordinarily would, or focus disproportionately on task-relevant cues at the expense of interactional nuance. When learning a task for the first time, speakers often struggle to balance understanding the task structure with attending to turn-taking, gaze, or politeness norms. With increasing familiarity, however, participants can take greater liberties: once a routine is well-practised, less verbal coordination is required, allowing social talk to emerge alongside task execution.

A simple everyday example illustrates this. In the restaurant of my laboratory, the cashier must register whether each customer's dish is vegetarian and record the sides they selected while the customer pays – often with a contactless badge. Although this procedure requires coordinated action, experienced customers and the cashier can execute it with minimal or no spoken instruction. Conversation thus shifts toward the social domain: greetings, small talk, or remarks about the day. The task is still performed, but its cognitive and interactional demands are low enough that social purposes dominate. When something unexpected happens, however – insufficient credit on the customer's account or confusion about the order – the social layer retracts and participants temporarily reorient toward the task. This flexible prioritisation is characteristic of natural conversation but is poorly captured by logic-based formal models concerned primarily with propositional content and goal-directed structure.

For dialogue systems, the lesson is significant. A system that rigidly assumes task orientation may misinterpret a user who does not seek problem-solving but simply wants to vent, chat, or affiliate. Conversely, social behaviours can meaningfully influence task outcomes – for example, in persuasion, negotiation, or customer service – yet remain under-modelled in information-centred frameworks. Understanding the task-social balance is therefore essential for designing emotion-aware, context-sensitive agents.

Many linguistic theories historically sidelined the social dimension of conversation, treating it as noise that obscures the 'real' work of information transfer or argumentation. From Gricean pragmatics to formal semantics and much of computational dialogue modelling, the focus has been on truth conditions, rational inference, and the efficient transmission of content.

However, the social dimension is not peripheral – it may well be part of the evolutionary foundation of human language. Anthropological accounts (R. Dunbar, 1996) propose that conversation emerged as a scalable mechanism for social bonding in increasingly large groups, replacing earlier forms of one-to-one bonding such as physical grooming. Regardless of the original function, language today clearly plays a

central role in maintaining social cohesion, managing impressions, and navigating interpersonal relations.

Social talk comes with its own norms and expectations, distinct from those underlying task-oriented dialogue. Consider the following conversation I had with friends on the way to go bowling. One of us had picked up a parcel and was carrying it, which led to an exchange as in (2.3)⁴:

(2.3) **G** What is in your parcel?

V A folding bowling bowl.

The ensuing dialogue involved collectively elaborating on the absurd properties of this fictional object – hydrogen boosters to make up for its lack of weight and strength, collapsible mechanisms to fit it in the small parcel – until the original factual question became irrelevant. While this conversation started with a genuine question meant to satisfy **G**'s curiosity, **V**'s answer and the collective agreement of the participants to follow **V**'s lead completely changed the purpose of this multilogue, making it purely social. From the perspective of rational information exchange, this conversation violates most of Grice's maxims (Grice, 1975). Yet every participant recognised and embraced its social function: play, affiliation, and shared humour. By adding to the conversation, everyone contributes to grounding this topic and making the true answer to **G**'s question increasingly irrelevant. Formal theories that evaluate moves in terms of truth, relevance, or goal advancement – QUD frameworks, belief-update models, argumentation-based approaches – struggle to characterise such interactions without deeming them defective, despite their clear social success.

While mainly social conversations are less studied in linguistics, they seem to be quite common in daily life and also follow some kind of rules. Small talk, or gossip, which has been more studied is a form of social conversation. While it is often considered as a way to exchange information, I would argue that the truth value of the shared information is not crucial and that the actual purpose of small talk is social bonding. However it does not exist in all cultures and can prove to be complicated to participate in when one is not used to it (missing reference). This shows that small talk follows some rules even though they seem different from the ones we usually assign to conversation. Similarly, carrying out conversations such as the bowling bowl one where everything is clearly false can prove to be complicated for some people, namely those struggling with non-literal interpretations. Those conversations are thus worth analysing to understand how they differ from more task oriented exchanges and how they are processed by people.

Humans have continuously developed new ways to communicate, especially when face-to-face interaction is not possible. From letters to telephones, smartphones, and videoconferencing, each advancement has sought to bridge the gap between individuals separated by distance. Interestingly, some of these communication methods, initially designed for convenience or practicality, have evolved to increasingly replicate the dynamics of in-person conversation. Features like typing indicators, read receipts, and on-line presence flags simulate aspects of face-to-face feedback, catering directly to social expectations for responsiveness and mutual attention. This phenomenon underscores the fundamental social purpose of communication: to bond with others. Despite being mediated through technology, these interactions still fulfil a deep-seated need for connection, highlighting that, at its core, conversation is about establishing and maintaining social ties.

Recent trends – such as fine-tuned language models imitating loved ones or celebrities – exploit the affiliative pull of conversation, raising ethical questions about deception, attachment, and the erosion of boundaries between human and machine interaction. Proposals to use LLMs to reduce social isolation, particularly among the elderly, similarly require scrutiny: replacing human interaction with artificial dialogue may have profound effects on well-being, identity, and influenceability. Understanding the social dimension of conversation is therefore essential for evaluating the risks and benefits of these technologies.

⁴The conversation originally happened in French and I do not have a precise recollection of the exact words that were used but its meaning is here.

Parameter	Minimal	Extended description
Practical goal	Task-oriented / Social / Information request	Proportions

Table 2.22: Operational characterisation of dialogue resources and tasks – Practical goal

As the laboratory restaurant example illustrated, natural conversations typically integrate task and social purposes rather than belonging exclusively to one or the other. When a task becomes routine, social talk often expands to fill the interactional space; when an unexpected problem arises, participants swiftly retract into task-oriented coordination. Even within a single activity type, the social affordances vary. A bakery interaction allows for light social exchange, but the amount tolerated depends on contextual factors such as queue length. A furniture salesperson may use socially skilled rapport-building to influence a customer's choices – a phenomenon that standard task-based models of 'slot filling' overlook entirely. Thus, although the underlying task (ordering bread, buying furniture) is consistent, the flow and structure of the conversation differ dramatically depending on social positioning, interpersonal goals, and emotional stance.

Some theories, such as Segmented Discourse Representation Theory (Asher & Lascarides, 2003), begin to capture aspects of these complexities by modelling relations like Elaboration, Commentary, or Acknowledgement, which often encode social actions rather than propositional advancement. Yet even these frameworks have limited capacity to represent shifting participation, playful departures from truth-conditional content, or the negotiable boundary between task and social purposes.

Designing dialogue systems therefore requires more than modelling the informational structure of tasks: it demands recognising when social talk is expected, when it supports the task, when it conflicts with it, and how participants flexibly transition between the two. Understanding this meshing is crucial for analysing natural dialogue and for creating conversational agents that behave in ways users perceive as appropriate, respectful, and contextually attuned.

The operationalisation of this parameter should thus explicit the type of purpose of the conversation, whether it is mainly task-oriented, social or with a focus on information gathering. A more detailed description could also indicate the proportion of each type of purpose in the conversation. This parameter is summarised in Table 2.22.

2.5.2.2 Intention

Warning – Ongoing work

This section requires more work and references. It is a draft for now.

Communicative purpose in conversation is fundamentally intentional. Every contribution reflects not only *what* a speaker chooses to communicate but also *how* they choose to communicate it. As mentioned repeatedly, at the most general level, conversation is typically analysed under the assumption that participants are cooperative, following the principles described by Grice (1975). This cooperative stance constitutes a useful default for explaining how interlocutors produce and interpret utterances. However, many conversational situations intentionally depart from these expectations. Speakers may seek to persuade rather than merely inform, entertain rather than accurately describe, or even mislead. Such departures are not exceptions to communication but integral parts of conversational behaviour. They are closely related to the notion of illocutionary force, namely the communicative act that an utterance is intended to perform (Searle, 1969; Austin, 1975), while extending beyond it to encompass broader strategic goals.

One important dimension of communicative intention is *argumentation*. From the speaker's perspec-

tive, argumentation provides principles for structuring conversational contributions in order to justify a claim, convince an interlocutor, or support the transmission of information. Formal approaches to argumentation and discourse organisation, such as Rhetorical Structure Theory (Mann & Thompson, 1988), describe the relations that connect discourse units and explain how larger argumentative structures emerge from local exchanges. Beyond these structural aspects, argumentation also involves adopting a particular stance towards a proposition and anticipating how an interlocutor is likely to evaluate the evidence presented.

This naturally leads to the distinction between informing and persuading. While informative argumentation primarily aims to communicate accurate information, persuasive discourse is concerned with obtaining the interlocutor's adherence to a particular viewpoint. Classical rhetoric has long recognised that effective persuasion depends not only on the logical structure of arguments (*logos*) but also on the credibility of the speaker (*ethos*) and the emotional response elicited from the audience (*pathos*) (missing reference: Aristotle). More recent theories of argumentation likewise emphasise that successful persuasion requires modelling the beliefs, preferences, and expectations of one's interlocutor. Consequently, communicative intention includes anticipating what the other participant is likely to find convincing and adapting one's discourse accordingly.

Deception can be viewed as a natural extension of these strategic intentions. Rather than simply presenting information, the speaker deliberately attempts to influence the interlocutor's beliefs while concealing their own epistemic state. This requires maintaining multiple representations of the conversational situation: the world model that the interlocutors are expected to accept and the speaker's own representation of reality. Producing deceptive dialogue therefore involves carefully managing commitments, consistency across turns, and the risk of revealing contradictions. This additional representational burden distinguishes deceptive interaction from ordinary cooperative communication.

Communicative intentions also encompass the speaker's intended stance or framing of an interaction. A conversational contribution may aim to be humorous, ironic, sarcastic, playful, or provocative without altering its propositional content. These intentions fundamentally shape how an utterance should be interpreted. For example, the earlier example (2.3) about the folding bowling bowl illustrates how the same factual response can be framed in different ways depending on the speaker's intended stance. In its original context, it was considered a playful and humorous contribution to a social exchange. The same interaction with a child or someone who was not expected to understand that the folding bowling bowl was a fictional object could have been interpreted as an attempt to deceive or exploit the listener's naivety. Such interpretations depend on the interlocutors' ability to infer the speaker's intended frame and relate closely to theories of common ground, relevance, and pragmatic inference.

These various intentional dimensions are neither equally salient nor simultaneously relevant in every conversation. Their importance varies depending on the interactional context, participants, and communicative task. Nevertheless, they substantially influence both language production and interpretation. Recognising another participant's intentions enables appropriate pragmatic interpretation, whereas failures of intention recognition frequently lead to misunderstandings, conversational breakdowns, and subsequent repair sequences. Rich interactional settings such as face-to-face conversation facilitate intention recognition through multimodal cues which complement linguistic content and provide redundant evidence for inferring speakers' goals.

The importance of modelling communicative intentions has also become increasingly apparent in human-AI interaction. While LLMs cannot have intentions in the human sense, the outputs they generate are often interpreted as if they were intentional due to their sycophantic style, leading to potentially harmful judgements from users (Cheng, Lee, et al., 2026). Highlighting the difference between displayed and actual intentions could help users better understand the capabilities and limitations of AI systems.

From an operational perspective, communicative intentions may be represented at different levels of granularity. A minimal characterisation could simply indicate whether a conversational setting pre-

Parameter	Minimal	Extended description
Intention	Cooperativeness Truthful / hidden Chosen / imposed	Weighted dimensions for each ordered participant pair

Table 2.23: Operational characterisation of dialogue resources and tasks – Purpose: Intention

supposes particular strategic intentions beyond ordinary cooperative interaction, including whether the participants are truthful or have hidden intentions, and whether those intentions were imposed to them within the conversational setting. A richer representation would instead identify the intention dimensions relevant to each participant and distinguish between a participant's actual intentions and their beliefs about the intentions of others. The specific dimensions would naturally depend on the application and an in-depth operationalisation of this parameter would require further insights from pragmatics, sociolinguistics, argumentation theory, and rhetoric. One possible formalisation is to associate each ordered participant pair with a vector of weighted intentional dimensions – such as informing, persuading, deceiving, entertaining, or provoking – capturing both communicative goals and the interlocutor's intended interpretation of those goals. This characterisation is summarised in Table 2.23.

2.5.3 Activity

Warning – Ongoing work

This section requires more work and references. It is a draft for now.

While communicative purpose captures *why* participants engage in a conversation and the intentions that guide their contributions, activity characterises *what* they are doing together through interaction. It provides a more organisational perspective on conversation by specifying the type of joint activity being carried out – such as buying bread, conducting an interview, or solving a problem – and the expectations that this activity creates for how the conversation unfolds. This perspective is closely related to work on discourse genres and speech activities, which emphasises that different activities give rise to different interactional conventions.

The conversational activity influences several structural aspects of an interaction. It constrains the vocabulary that participants are likely to use (Section 2.5.3.1), shapes the topical structure through which information is organised and progresses (Section 2.5.3.2), and determines the success conditions by which the conversation can be considered to have fulfilled its purpose (Section 2.5.3.3). Together, these dimensions describe how an activity organises conversational behaviour beyond participants' immediate communicative intentions, providing a complementary perspective on the functional characterisation of dialogue.

2.5.3.1 Vocabulary

The activity in which a conversation takes place strongly influences the vocabulary that participants are likely to use. While language always exhibits lexical variation, conversational activities constrain the set of concepts, expressions, and linguistic registers that are expected to appear. As a result, conversations associated with different activities may differ substantially in their lexical characteristics, even when they involve similar participant settings or communicative goals.

Parameter	Minimal	Extended description
Vocabulary	Genericity degree	Proficiency level Domain Community

Table 2.24: Operational characterisation of dialogue resources and tasks – Activity: Vocabulary

A first factor is the proficiency level expected by the activity. Conversations involving experts or highly educated participants often employ a richer and more specialised vocabulary than interactions aimed at novices or children. Likewise, educational or technical activities may require precise terminology, whereas everyday conversations generally favour more common lexical items and simpler formulations. The expected proficiency therefore influences both the complexity of the vocabulary and the assumptions speakers make about their interlocutors' lexical knowledge.

A second factor is the domain of the conversation. Activities centred on medicine, law, engineering, finance, or other specialised fields naturally rely on domain-specific terminology that may be uncommon outside those contexts. Even when participants are not experts, the topic itself introduces concepts and expressions that are characteristic of the domain under discussion. Domain-specific vocabulary is consequently an important source of lexical variation and often represents a challenge for both human analysts and computational dialogue systems.

Finally, vocabulary is shaped by the community to which speakers belong. Social groups, professions, online communities, or regional speech communities frequently develop their own jargon, abbreviations, slang, or shared expressions that facilitate communication among members while remaining opaque to outsiders. Such community-specific vocabulary contributes to the identity of the group and reflects shared experiences, practices, and cultural references. It also illustrates that lexical variation cannot be explained solely by topic or expertise, but is equally influenced by social affiliation.

These dimensions are particularly important when analysing conversational data. Highly specialised or community-specific vocabulary may reduce the ability of external annotators to correctly interpret the interaction, increasing the risk of annotation errors or inconsistent analyses. From a computational perspective, dialogue resources collected within a particular domain or community may also exhibit limited generalisability, as language models trained on such data can struggle when applied to conversations involving different lexical conventions. Conversely, general-purpose dialogue corpora with broad vocabulary coverage may be less informative for specific applications, as they may not capture the specialised terminology or social nuances relevant to a particular activity.

Explicitly characterising the expected vocabulary of a conversational activity therefore facilitates both the interpretation of dialogue resources and the assessment of their applicability to downstream tasks. From an operational perspective, a minimal characterisation of vocabulary may simply indicate its degree of genericity, ranging from broadly accessible language to highly specialised terminology. A richer representation can further specify the factors responsible for lexical variation, including the expected proficiency level of the participants, the domain of the interaction, and the communities whose linguistic conventions shape the conversation. These dimensions are summarised in Table 2.24.

2.5.3.2 Topical structure

Topics – understood in the intuitive sense of *what the conversation is about* – are a central organising dimension of dialogue. They shape what contributions are relevant, what inferences can be drawn, and how participants decide whether an utterance fits the ongoing flow. An off-topic remark can disrupt coordination or require repair; conversely, a well-timed topical shift can reorient the group or open a new line

of joint action. Despite their importance, topics remain notoriously difficult to define and formalise (e.g. G. Brown & Yule, 1983; Schiffrin, 1987). In this section, we adopt the pragmatic, participant-oriented notion commonly used in conversation analysis and discourse studies: topics are the locally recognised domains of talk that participants themselves treat as current, shifting, or suspended.

In open, informal, face-to-face conversation, topical structure is typically fluid. Topics may develop gradually, shift abruptly, or fragment into loosely connected strands. In multiparty settings these dynamics become especially complex: each participant can add new associations, reinterpret previous turns, or shift attention to a neighbouring sub-group. Once groups reach four or more participants, it is common for multiple simultaneous ‘floors’ to emerge (Egbert, 1997), each sustaining its own topic while remaining loosely linked to other strands through participants who move between them. Even in dyadic conversation, more than one topic can be ‘open’ at once – for instance, when interlocutors conduct a task while engaging in parallel social talk. These are not simultaneous in the acoustic sense, but they coexist as latent threads to which speakers can return.

Written and computer-mediated communication expands this even further. Asynchrony allows interlocutors to pursue several topics at once, reply to older turns out of chronological order, or maintain long-lived topical threads that re-emerge after hours or days. This environment makes explicit a capacity that also exists in spoken interaction, though typically in subtler forms.

Topic dynamics also depend strongly on the activity type. In task-oriented dialogues, the set of admissible topics is shaped by the task structure and can be more or less rigid. In institutional settings – court hearings, political debates, news interviews – topical scope is heavily regulated: participants may be prohibited from entering certain topics, required to remain on a prescribed agenda, or allowed to shift topics only through specific, institutionally sanctioned moves. Here, the topical structure is part of the interaction’s normative framework.

Computational approaches have typically operationalised topic through segmentation tasks. *Linear* topic segmentation aims to divide a transcript into contiguous segments, each associated with a distinct topic (Hearst, 1997). This is useful for meeting analysis, summarisation, or information-retrieval applications, where the goal is to locate sections of interest. However, linear segmentation presupposes that each stretch of talk belongs to exactly one topic at a time – an assumption that fails in many natural conversational settings, especially multiparty dialogue, where overlapping or interleaved topics are common.

Hierarchical topic segmentation attempts to capture nested structure, but it is less commonly used due to its complexity and still does not easily track *parallel* topic threads. Villing (2015) shows that participants can sustain several topical trajectories simultaneously and rely on context and common sense to identify the right topic for each utterance. In addition, speakers often produce meta-comments about the conversation – remarks on relevance, transitions, humour, or framing – without leaving the current topic, further complicating formal segmentation.

These observations point to the need for theories of topical structure that can accommodate concurrency, rapid shifts, embedded commentary, and the interpretive work participants perform to maintain coherence.

NLP has long acknowledged the importance of topic for natural language understanding, yet computational treatments remain disproportionately focused on task-oriented settings. Topic modelling and segmentation algorithms were developed largely on expository text, technical documents, or structured meetings. Attempts to adapt these methods to casual, open-domain conversation have been limited by the highly dynamic, interaction-driven nature of topical organisation: shifts are subtle, boundaries are fuzzy, and contributions often rely on shared context rather than lexical cohesion.

While linear segmentation has clear usefulness in meetings or customer-support interactions, it transfers poorly to conversations with fluid purposes and multiple concurrent strands. As conversational agents grow more common – and as applications increasingly require systems to sustain long-term, open-domain interaction – modelling topical structure becomes indispensable. Systems must track what topics are cur-

Parameter	Minimal	Extended description
(Topical) structure	Constrained?	Agenda Scenario

Table 2.25: Operational characterisation of dialogue resources and tasks – Activity: Structure

rently active, when a user reopens an older thread, when an utterance is meant as a meta-comment, and which topical expectations are at play.

The challenge is substantial as the topical relevance of an utterance depends on many parameters like the conversational setting, the number and roles of participants, the medium, and the broader activity in progress. Humans are remarkably adept at inferring topical connections even in unpredictable or elliptical contexts; capturing this interpretive flexibility in computational models remains an open problem.

A first attempt at characterising the topical structure of a conversation is proposed in Table 2.25. A minimal characterisation could indicate whether the topical structure is constrained by the activity or not. A more detailed description could include the skeleton induced by the task like an agenda in meetings, or a formal structure like the one used in interviews. However those scenarios are very task-dependant and a more generic parameter like the degree of structure and predictability of a conversation would allow us to compare different types of dialogues, which is one of the primary goals of the descriptive framework proposed in this chapter. This direction is thus the focus of Part II.

2.5.3.3 Success Condition

Conversational activities are generally associated with conditions under which they can be considered successful. These success conditions follow naturally from the purpose of the interaction and provide criteria for determining whether the activity has achieved its intended outcome. In formal dialogue frameworks, success is often defined explicitly, allowing systems or interactions to be evaluated against predefined objectives. However, the nature of these objectives varies considerably across conversational activities.

For task-oriented dialogues, success conditions are often relatively straightforward. A customer obtains the requested information, a travel reservation is completed, or a collaborative problem is solved. In these settings, success can frequently be measured by whether the expected information slots were correctly filled or whether the task was successfully completed. In contrast, social conversations pursue interpersonal rather than instrumental goals. Their success therefore depends on less tangible outcomes such as maintaining rapport, strengthening social bonds, entertaining participants, or creating a pleasant interaction. These objectives are inherently more difficult to formalise. For example, in the bowling bowl conversation from (2.3), success would not simply consist in exchanging information, but rather in sustaining the playful make-believe frame so that all participants enjoyed the interaction and experienced a sense of social connection. A contribution that prematurely disrupted this shared frame could reasonably be perceived as a conversational failure, even if the factual content of the exchange remained perfectly understandable.

The notion of conversational success is particularly important for dialogue systems because it directly determines how their performance should be evaluated. Some success conditions, such as maintaining fluency or producing contextually appropriate responses, are relevant across many conversational settings. Others are activity-specific and must be defined with respect to the intended purpose of the interaction. Coherence is often cited as a universal criterion, yet everyday conversations routinely contain inconsistencies, brutal topic shifts, unfinished utterances, and locally incoherent sequences without being regarded as unsuccessful. Judging a conversation solely according to formal linguistic criteria therefore overlooks important aspects of human interaction. Likewise, evaluating conversational agents exclusively through

Parameter	Minimal	Extended description
Success condition	Description	?

Table 2.26: Operational characterisation of dialogue resources and tasks – Activity: Success condition

task completion ignores the dialogical abilities that make interactions effective and enjoyable. If conversational systems are intended to replace or complement existing interfaces, they should ultimately be assessed not only on whether they complete a task, but also on whether they provide an interaction that users perceive as successful according to the goals of the activity.

Operationalising conversational success remains an open research challenge. The rapid development of conversational AI has led to the emergence of numerous benchmarks and evaluation frameworks, including both automatic metrics and human or LLM-as-a-Judge evaluation protocols. However, the dimensions they measure are often only loosely defined and differ substantially across studies, making results difficult to compare. As we will discuss in Chapter 3, the design of evaluation frameworks would benefit from stronger connections with psychometrics and measurement theory, which provide well-established principles for constructing reliable and valid assessment instruments. While automatic evaluation remains attractive because of its scalability, metrics that fail to capture meaningful aspects of conversational success offer limited practical value. Due to the open nature of the issue, we deliberately adopt a minimal operationalisation of this parameter. Rather than proposing a detailed taxonomy of success criteria, we simply advocate for an explicit description of the success condition(s) associated with the conversational activity, as summarised in Table 2.26. This first step is intended to encourage more systematic consideration of success conditions in the design of dialogue systems, while recognising that further research is needed to develop robust and generalisable evaluation metrics, as we further discuss in Chapter 3.

2.6 Summary and Conclusion

This chapter has proposed a set of dimensions along which conversations – and, by extension, conversational datasets – can be characterised. The framework is organised around five main parameters: PARTICIPANTS, LANGUAGE, ENVIRONMENT AND MEDIUM, VERBAL INTERACTION, and CONVERSATIONAL ACTIVITY AND PURPOSE. Together, these dimensions capture complementary aspects of dialogue, ranging from who is involved and how they relate to one another, to the linguistic resources they mobilise, the constraints imposed by the interactional setting, the mechanisms through which turns are coordinated, and the goals that structure the exchange. As summarised in Table 2.27, each parameter can be described at different levels of granularity, from minimal to more detailed operationalisations, allowing the framework to be adapted to a wide range of research needs.

These parameters are relevant because they make explicit many of the factors that shape conversational behaviour but are often left implicit in the design and use of dialogue resources. Face-to-face interaction, which serves as the prototypical conversational setting, combines a rich set of features across all five dimensions. However, due to the practical challenges of collecting and annotating such data, much of dialogue research relies on mediated or task-specific corpora. As a result, important aspects of interaction may be absent, attenuated, or transformed. Being able to characterise these differences systematically is therefore essential for assessing the scope of empirical findings, comparing datasets, and identifying potential limitations in generalisability. Moreover, as illustrated throughout the chapter, these parameters do not operate independently: their interactions can produce complex effects on conversational dynamics, sometimes in ways that are not immediately apparent.

Although the LANGUAGE parameter has not been discussed as extensively as others, it remains a core

feature of dialogue. Language encodes cultural, social, and cognitive patterns that fundamentally shape how interaction unfolds. The predominance of English and a small number of well-resourced languages in current research risks obscuring this diversity and its implications. Explicitly accounting for the language(s) involved in a dataset is therefore crucial, both for interpretability and for the development of more inclusive and robust dialogue systems and formal models.

Beyond the analysis of human dialogue, the proposed dimensions also provide a foundation for evaluating and designing dialogue systems. They can inform the definition of system capabilities, guide the selection of appropriate evaluation metrics, and help clarify what kinds of interaction a system is expected to support. For instance, a system designed for task completion in a constrained environment will require a different configuration of parameters than one intended for open-domain social interaction. Making these assumptions explicit is particularly important as conversational agents are increasingly deployed in complex, real-world settings.

At the same time, this framework is not intended to be exhaustive. The parameters presented here should be understood as a starting point rather than a definitive taxonomy. Each dimension connects to extensive bodies of work in fields such as interactional linguistics, pragmatics, psycholinguistics, sociology, and communication studies, where more fine-grained analyses and alternative conceptualisations have been developed. Specialists in these areas will undoubtedly be able to refine, extend, or challenge the operationalisations proposed here. The aim is therefore not to replace existing approaches, but to provide a common ground that facilitates dialogue across disciplines and encourages more explicit descriptions of conversational data.

In practical terms, we envision this framework as a tool for constructing ‘identity cards’ for dialogue resources. By systematically describing datasets along these dimensions, researchers can more easily compare resources, assess their suitability for specific tasks, and make informed decisions when selecting or designing data. This is particularly relevant in NLP and computational linguistics, where the choice of training and evaluation data has a direct impact on system behaviour. Adopting a broader perspective on the properties of dialogue may help mitigate mismatches between datasets and intended applications, and ultimately lead to more robust and context-sensitive systems.

In the next chapter, we examine how dialogue is currently studied within the NLP and computational linguistics community. We show how the framework proposed here can contribute to improving the description and selection of resources, and how it can support a more reflective approach to the design and evaluation of conversational models.

Parameters		Minimal	Extended description
Participants			
Participation structures	Number of participants	Number	Number of simulated participants
			Number of workers
			Workers - Simulated participants mapping
	Functions	List	Time-aligned list
			Entropy over function transitions
Enunciative status	Anonymity	Binary	Anonymous / pseudonymous / real name
			Digital / physical
	Role instantiation	Binary	Self / self specific / assigned
	Persona instantiation	Binary	Self / self specific / assigned
	Commitment level	Binary	Status: role-playing, joke, etc. Scope: conversation, task, ..., general
Social dynamics	Hierarchy	Binary	Weighted list of dimensions
	(Conversational) History	Scale	Distance in a weighted social graph
			Common topics
Language			
	Resource level	List	Control degree
	Conversation level	List	Proficiency
			Monolingual / code-switching / multilingual
Environment & Medium			
	Physical context	Binary	Co-located / shared virtual / separate
			Shared perceptual field
Visibility	Gaze	Binary	Allocation degree
			Fidelity: accurate / approximated / distorted
	Gesture / Facial expression	Binary	Allocation degree
			Interpretability
Audibility	Prosody / Tone	Binary	Allocation degree
			Fidelity
	Noise	Binary	Degree
			Sources

Table 2.27: Parameters for an operational characterisation of dialogue resources and tasks

Chapter 2. Parameters for Characterising Dialogue

Parameters		Minimal	Extended description
(Verbal) Interaction			
Co-construction	Coordination signals	Binary	Availability of channel (vocal, textual, visual, haptic) per participant
			Technical ability per channel per participant
			Density per participant
			Latency tolerance
	Turn-taking	Binary	Overlap possible
			Turn-allocation method
			Average overlap time
	Completion	Binary	Density
	Time	Synchrony	Binary
Planning		Binary	Time allowed for production
Ephemerality		Binary	Persistence time
			Editability time
			Edition history
Recording form			Third-party persistence time
			Third-party persistence awareness
Function and Structure			
Epistemic grounding	World setting	Real (contemporary or not) / fictional	Time period
			Location
			Conventions (cultural, social, scientific, moral, etc.)
	World stability	Binary	Planned changes
			Frequency of changes
Source of changes (participants, external agent, environment)			
Purpose	Practical goal	TO / social / information request	Proportions
	Intention	Cooperativeness	Weighted dimensions for each ordered participant pair
		Truthful / hidden	
		Chosen / imposed	
Activity	Vocabulary	Genericity degree	Proficiency level
			Domain
			Community
	(Topical) structure	Constrained?	Agenda
			Scenario
	Success condition	Description	?

Table 2.27: Parameters for an operational characterisation of dialogue resources and tasks (continued)

Dialogue Research in Practice: A Critical Analysis of Contemporary NLP Approaches

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Update all mentions of 'recent' papers?

[Amandine]

In the previous chapter, we approached dialogue from a theoretical standpoint. Drawing on linguistics, philosophy, psychology, sociology, conversation analysis (CA), and computational modelling, we examined how dialogue has been conceptualised across disciplines. Although we do not claim to be exhaustive in our account, especially since dialogue studies are so intrinsically interdisciplinary, we sought to highlight a central insight shared across traditions: dialogue is not reducible to a linear exchange of sentences. It is a complex coordination of cognitive, linguistic, social, and environmental processes, whose mechanisms interact in dynamic and sometimes unpredictable ways.

We also emphasised that dialogue admits a vast range of instantiations: more or less multimodal settings, synchronous or not, with different degrees of task structure, etc. This diversity allows researchers to focus on specific aspects of interaction, but it also requires careful delimitation. Without explicit characterisation of what kind of dialogue is being studied, under what constraints, and for what purpose, comparisons can become fragile and theoretical claims risk overgeneralisation.

To address this issue, we proposed a descriptive framework organised around five main axes: PARTICIPANTS, LANGUAGE, ENVIRONMENT AND MEDIUM, VERBAL INTERACTION, and CONVERSATIONAL ACTIVITY AND PURPOSE. Each axis is further articulated into more specific parameters that can be described with varying degrees of granularity depending on the research question at hand. The framework is not intended as a theory of dialogue in itself, but as a meta-descriptive tool: a way of making explicit what can sometimes be left implicit in resource design, experimental setup, and system evaluation.

However, a framework remains abstract until it is confronted with concrete research practices. The present chapter therefore moves from theoretical articulation to application. We examine how the proposed framework can be mobilised to analyse current dialogue research, clarify methodological choices, and illuminate blind spots. In doing so, we also begin to explore the interdependence of dialogue parameters: how constraints imposed along one axis (for example, task structure or medium) reshape other dimensions such as participation roles, interactional dynamics, or evaluative criteria.

To ground this discussion empirically, we first take stock of contemporary dialogue research through a meta-review of recent publications (Section 3.1). We then situate this analysis within the broader NLP research pipeline as it currently operates, particularly in the era of LLMs, where data availability, evaluation practices, and system architectures are undergoing rapid transformation (Section 3.2). In Section 3.3, we demonstrate how the framework can guide both analytical reflection and practical interventions – ranging from resource selection to delimitation of the scope of a study or application. Finally Section 3.4 shows how it can be used for targeted adaptation of language models along specific dialogue dimensions before we conclude in Section 3.5 with a discussion of future directions and open challenges in dialogue research.

3.1 Contemporary Landscape of Dialogue Research

This section is based on the work Decker et al. (2026) that was presented at the Fifteenth Language Resources and Evaluation Conference (LREC 2026).

Many new types of dialogues have emerged in the past few years. Covid-19 (Joskowicz, 2023) made video-calls and hybrid meetings between face-to-face and remote participants more typical, while the rise of conversational agents and LLMs increased the usualness of human-machine interactions, in particular since the release of ChatGPT (Qi et al., 2024), which made this type of tool well known to the general public. These changes are bound to have an impact on the NLP/CL community. Several surveys have been conducted in the past few years regarding multiple tasks. Kusal et al. (2022) surveyed the different approaches to AI-based conversational agents, identifying several gaps in this field at the time such as the lack of harmonised evaluation processes which is still an issue nowadays as we will discuss in Section 3.1.4. Y.-P. Chen et al. (2024) focused on personalised dialogue generation which consists in inferring and/or modelling the user’s persona on the one side, and in having the agent display a coherent and relevant persona throughout the dialogue on the other side. They highlight the complexity to collect great quality datasets for this task due to the artificiality of actively displaying a persona, when it is typically a more subtle phenomenon. Approaches to personalised dialogue generation also tend to work on modelling the agent’s and the user’s persona separately which somewhat disregards the interactional nature of dialogue. Finally they raised concerns regarding the evaluation approaches for this task which are overall too basic compared to the complexity of dialogue generation. On a similar note, Yang et al.

(2026) provided a critical analysis of the trends in NLG evaluation from 2020 to 2025, assessed through the automatic retrieval of information regarding the task and evaluation approaches – automatic metrics, LaaJ, and human evaluation – used in papers from four major CL venues. They focus in particular on four tasks, including dialogue generation and question answering. This mostly automatic approach allows the authors to study trends and shifts in evaluation practices over several years, thus highlighting turning points but also challenges in the field.

While these surveys provide valuable insights into specific tasks, they do not offer a comprehensive overview of the broader landscape of dialogue research in NLP/CL. Hence in this section, we take a meta-review approach to investigate how conversation is currently studied in the field by analysing papers from the ACL ANTHOLOGY 2024. We focus on papers written in English from major venues indexed in the anthology (excluding workshops) using metadata accessed via the ACL API⁵. Our goal is not to provide an exhaustive survey, but to provide a snapshot of where dialogue research stands today, what tasks are being studied, what resources they rely on, and how these tasks are evaluated. This enables us to identify the current gaps in the field and define future research directions.

We begin by outlining our approach to examining the current state of dialogue research (Section 3.1.1). We then detail the main dialogue-related tasks addressed in the NLP/CL community (Section 3.1.2), discuss the dialogue resources used and created (Section 3.1.3), and review widely used evaluation methods for DSs (Section 3.1.4). This leads us to the next section where we situate these practices within the broader research pipeline and discuss their implications for the future of dialogue research in the LLM era.

3.1.1 Methodology of the Review

The ACL ANTHOLOGY⁶ is an open-access digital repository of research in CL and NLP. Maintained by the Association for Computational Linguistics (ACL), it hosts conference and workshop proceedings, journal articles, and other scholarly materials dating back to the 1960s and is updated continuously. Its open-access nature and structured bibliographic metadata have made it a widely used corpus for bibliometric studies, meta-analyses, and research on scientific trends (Vogel & Jurafsky, 2012; Omodei et al., 2014; Gábor et al., 2016; Pramanick et al., 2023).

In this section, we present an analysis of a targeted subset of the ACL ANTHOLOGY to understand how conversation has recently been studied in the NLP/CL community. The ACL ANTHOLOGY was chosen both for practical purposes, as its API allows us to retrieve the papers and their metadata easily, and because it gathers conferences that are considered important in the field. We restrict our survey to papers published in 2024 and indexed under the anthology’s major venues. Our selection is presented in Table 3.1. While it does not cover journals, which typically allow for more in-depth studies thanks to the longer page limits, the restriction to conference papers allows us to capture the trends that are showcased in the faster-paced conference cycle, which is particularly relevant in the current era of rapid developments in dialogue research.

Stats on this

[Amandine]

Conferences have indeed shorter review cycles which makes them more attractive in institutional contexts that induce a pressure to publish, whether for hiring, promotion, or funding decisions (Franceschet, 2010; D. Van Dijk et al., 2014).

We used the ACL ANTHOLOGY API to retrieve metadata for all papers published in 2024, excluding venues labelled as workshops in the metadata. Our goal was to identify papers dealing with dialogue. Due to the absence of keywords in the metadata we had to rely on another filtering method. We selected

⁵<https://aclanthology.org/info/development/>

⁶<https://aclanthology.org/>

Venue	#	%
TACL	3	3.2
EMNLP	52	3.6
ACL	36	3.8
NAACL	29	4.4
EACL	14	5.0
JEP/TALN/RECITAL	8	6.0
LREC/COLING	105	6.8
Findings	112	7.8
INLG	5	9.3
CODI	3	17.6
SIGDIAL	40	60.6
Total	408	6.1

Table 3.1: Number (#) of retrieved paper for the selected venues and portion (%) out of the entire venue in 2024

papers whose titles contained the stems of ‘conversation’, ‘dialogue’, and ‘discourse’. While this method may miss some relevant papers and include some marginal ones, it provided a practical and transparent approach for assembling a representative sample. We then excluded the ones that were not actually focusing on dialogue (e.g. focus on monological discourse, or papers on *conversion*). In the end we retrieved 407 papers, which represents about 6.1% of the papers published in the selected venues. Once retrieved, one person manually reviewed the papers to determine the conversational task addressed, the corpora used and the evaluation methods employed.

3.1.2 Tasks

The NLP/CL field covers a wide range of tasks, so as a first step we annotated each paper with the primary task it addressed and then grouped these tasks into broader categories. The categories we defined are described in the paragraphs below, and their distribution – both in absolute number of papers and as a proportion of the total – is shown in Fig. 3.1. Some papers naturally span multiple categories: for instance, a benchmark proposal or a paper on generative systems may also introduce a new corpus. In such cases, we selected the most salient aspect when assigning the task category, but the absolute counts should be interpreted with caution. Nevertheless, the overall distribution reflects the dominant research trends in the surveyed set.

Generative Systems. Across the ACL ANTHOLOGY 2024, the majority of papers dealing with conversation focus on generative systems (44%). This category encompasses dialogue systems, QA, and Dialogue State Tracking (DST), as well as some papers presenting new language models (LMs) or ways to improve existing ones.

Dialogue systems are programs designed to interact with humans through natural language. Within this category, task-oriented dialogue (TOD) systems used for e.g. booking restaurants, are especially dominant. These systems typically aim to assist users in completing specific goals, and their performance is usually measured through metrics like joint goal accuracy, which tracks whether the system correctly identifies all necessary slots in a dialogue (i.e. place, time, etc.). Less constrained tasks are also tackled such as emotional support, negotiation and persuasion, or tutoring. Both for domain-specific and more

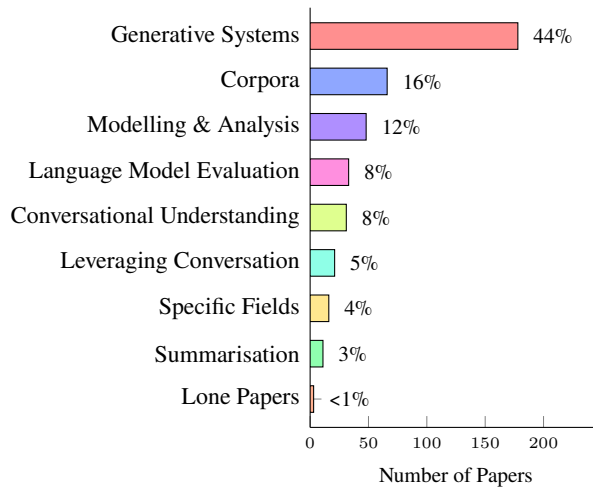


Figure 3.1: Distribution of papers by task category.

open tasks, the conversational dimension is a crucial part of the system. It is however harder to evaluate (see Section 3.1.4) than the task-specific outcomes.

Question-answering (QA) is a sub-category of dialogue systems where users can ask questions and get answers in natural language. The system can also encourage the user to refine their question or ask for more information if the answer is not precise enough, providing more flexibility than querying a web interface (Biancofiore et al., 2024). This branch of research focuses notably on query rewriting and knowledge retrieval.

DST involves monitoring and recording the user’s set of queries and goals throughout a conversation (Jacqmin et al., 2022). It plays a pivotal role in developing TOD systems. Recent studies have focused on enhancing DST models through techniques like zero-shot learning, the aim being to reduce the cost of dialogue collection and annotation.

Language Model Evaluation. The growth of generative systems in research, but also in society, requires solid evaluation to understand their strengths and limitations. Slightly more than 8% of the papers we surveyed focused on this aspect, whether through the production of benchmarks or empirical evaluations. Most of them probe LMs capacities with regard to different aspects, from very specific ones such as their negotiation capacity or ability to generate slot-values, to more general conversational dynamics like multi-turn and/or multi-party dialogue capacity. Security is also a concern where the risk that LMs leak private information is investigated as well as their ability to resist misinformation and tendency to hallucinate or generate harmful content. Another angle is the limits of current LMs when it comes to languages other than English.

In some cases, LLMs were themselves used as evaluators for certain tasks, including tasks that were based on LLMs. While this can provide scalable and context-aware assessment, it is important to recognise the potential biases and limitations inherent in using a model to evaluate another model. We discuss this aspect in more details in Section 3.1.4.

Socio-Pragmatic Understanding. Similarly the growth of LMs has increased the need to identify users’ emotional states, communicative intentions, and stances in conversation, but also subtler aspects of interaction such as humour and sarcasm identification, personality profiling, harmful behaviour detection, and social meaning recognition, to enable more contextually aware and socially appropriate dialogue systems. These tasks fall under the broader umbrella of conversational understanding and were the focus of

about 8% of the surveyed papers. The venue SemEval 2024, which we surveyed but did not include in this work to remain more general, also hosted 2 shared tasks (tasks 3 and 10) revolving around conversational emotion recognition.

Summarisation. Summarisation is a classical NLP task that is more challenging for conversations than for non-interaction-based discourse (X. Feng et al., 2022). Less than 3% of the papers tackled this particular task. Most approaches leverage LLMs, some evaluate hallucinations of LLMs specifically in summaries. Summarisation is indeed particularly prone to hallucination in LLMs (Ramprasad et al., 2024): since the model must compress and rephrase content, it often generates details that are plausible-sounding but not supported by the source.

Leveraging Conversation. The recent hype around LLMs has renewed interest in conversational AI, inspiring new directions for traditional tasks, exploiting interactive dialogue to provide more context-aware, personalised, and adaptive experiences that leverage conversation as a primary mode of interaction. This represents slightly more than 5% of the surveyed papers. Most of them reflect the emergence of paradigms such as conversational search and conversational recommender systems, where dialogue is used to iteratively refine user queries or provide personalised suggestions, improving task efficiency and user experience.

A key issue with most of those papers, independently of their contribution to the scientific community, is that they often fail to assess dialogue quality in a meaningful way, or rely on metrics that do not align with established principles from conversation theory. For instance, the Dist-n metric (J. Li et al., 2016), which measures the number of distinct tokens in generated utterances, captures lexical diversity but provides little insight into the coherence, relevance, or appropriateness of a response within the conversational context.

Beyond these more common applications, a smaller number of studies explore alternative uses of conversation, such as solving complex tasks through multi-turn dialogue (X. Wang et al., 2024) or generating explanations of news claims (Hsu et al., 2024). While these papers are a minority in our classification, they overlap with the dialogue system category which may undervalue the actual amount of such papers.

Corpus Creation and Resources. About 16% of the surveyed papers focus on describing resources or proposing methods for resource generation. The presented datasets range from human-human interactions to human-system dialogues, as well as fully synthetic datasets generated by LLMs and some handcrafted by humans without actual interaction. Several papers present corpora created by translating existing datasets or reproduce well-known ones in other languages – a reflection of the fact that NLP/CL research remains heavily focused on English, making new resources essential for studying other languages. Some contributions extend or merge existing resources by adding new annotations, either human-produced or automatically generated by LLMs. We further discuss the role of resources in the ACL ANTHOLOGY 2024 and the typology of datasets in Section 3.1.3.

Modelling and Analysis. Approximately 12% of the surveyed papers focus on modelling and analysing conversations, a category that seeks to uncover the mechanisms of human communication rather than merely build systems. This research is essential for advancing our general understanding of how people interact – from turn-taking patterns, feedback and grounding procedures, to topic-shift and redirection strategies or the ways intentions and information are conveyed. A well established task of this category is dialogue discourse parsing, which consists in segmenting a conversation into discourse units and identifying the relations between them (e.g. question-answer, elaboration, clarification request, etc.), thereby

modelling the hierarchical and functional structure of dialogue (J. Li et al., 2022). Such insights are critical for identifying conversational practices used to manipulate, mislead, or deceive, which is particularly important in the current era of widespread misinformation (Lazer et al., 2018; Vijayan et al., 2025) and populism (Algan et al., 2017; Norris & Inglehart, 2019; Kaltwasser & Taggart, 2025), where identifying manipulative or deceptive strategies is critical. They should also form the conceptual backbone of conversational research, providing the empirical foundations on which robust and socially responsible systems can be built.

Despite its importance, this line of work appears under-represented at top-tier venues, representing only 6% and 4% of papers at ACL and EMNLP (ranked A* according to the ICORE Conference Portal⁷), and 3% and 7% at NAACL and EACL (A). On the other hand, SigDial, LREC (ranked B) and the Findings⁸ from ACL, EACL, NAACL and EMNLP contain between 10 and 18% of modelling and analysis papers. This disparity likely reflects the time-consuming nature of such research – requiring detailed annotation, fine-grained analysis, and often interdisciplinary collaboration – which may make it less immediately appealing than system-building papers in the current high publication pressure environment.

Specific Fields. All of the categories of papers discussed thus far also appear in domain-specific contexts representing almost 4% of the surveyed papers, where the different systems are adapted to the needs of particular fields such as medicine and education. In these settings, generative systems and summarisation models are developed with a more targeted focus based on the needs of the fields. This specialisation often introduces additional challenges, including the need for domain-specific data. Some papers indeed focus on domain specific corpus generation methods (J. Wang et al., 2024; Arana et al., 2024; Bychkova et al., 2024). While medicine and education are usual application domains of NLP, both having their own dedicated workshops indexed in the ACL ANTHOLOGY, one paper was also dedicated to dialogues in motor-sports (Isaka et al., 2024) which highlights the broad applicability of NLP across diverse fields. Such diversity underscores the need for multidisciplinary approaches and collaboration, as different domains bring unique knowledge requirements, evaluation criteria, and contextual challenges that must be addressed to develop effective systems.

Lone Papers. The rest of the papers (less than 1%) present a tool meant to help with dialogue collection and annotations (A. Li et al., 2024) and papers on automatic speech recognition (W. Lee et al., 2024; Htun et al., 2024).

Limitation of the selection. Due to the selection of venues we operated, papers presenting results in the context of shared task do not appear in our survey. They are however a crucial motor in our community and cover varied tasks including ones that are under-represented in the venues we covered.

3.1.3 Datasets and Resources

Two aspects were taken into consideration regarding corpora in our review: the datasets that were *created* (Section 3.1.3.1) and the ones that were *used* (Section 3.1.3.2).

3.1.3.1 Created Datasets

In terms of creation, about 1/5 of the surveyed papers presented a new dataset. More than half of the new ones are in English but other languages are also included. However, in more than a third of the papers

⁷The last ranking for the mentioned venues was made in 2026.

⁸Findings are not indexed in the ICORE Conference Portal but are by definition considered of lesser importance than the main conference.

describing new resources, the language has to be inferred from the examples – when there are some – or from the collection or evaluation method. Some other papers mention the language in the appendix or limitations only, even though this information is crucial. English seems to be the language in most of these cases but, as it has been noted several times in the past decade, English is not equivalent to all the other languages and the language of the study should always be clearly stated (Bender, 2011; Duce et al., 2022).

Some papers tackle multiple languages at once, reaching up to 150 languages in a paper presenting XSGD (Tu et al., 2024), the translation of the English Schema-Guided Dialogue dataset (Rastogi et al., 2020). These efforts show that the community is seriously trying to go beyond English. Translating existing corpora can be a pragmatic path for extending coverage to low-resource languages, but several caveats apply. Conversation reflects culture: idiomatic expressions, norms of politeness, turn-taking cues, notions of relevance or appropriate responses are culturally embedded, and may not transfer cleanly via translation (M. Zhang & Toral, 2019; Majewska et al., 2023). Automatic translators can help with scale, but should not be used without the input of native speakers, not only as post-hoc reviewers but as active collaborators in the translation process.

Most of the created datasets (~61%) feature pairs of participants engaged in short- or long-term conversations. A few datasets consist of pairs of adjacent utterances extracted from longer dialogues which potentially involve more speakers. Three datasets contain conversations with imbalanced roles with one main speakers and several sub-speakers or listeners, including one study where a single individual played the role of multiple speakers in a virtual reality setting (Lai et al., 2024). About 13% of the created datasets involve more than two speakers with some papers clearly stating the number and intentionally designing their studies around multilogues, while in others multi-party dynamics appear incidentally without being a primary focus.

Studying written data remains the most favoured modality (~57%) but multimodality also has a large place in the created datasets (~28%). This includes datasets that combine written conversations with related video or speech (e.g. YouTube comments), as well as conversations transcribed from videos or speech-only recordings, sometimes including additional data such as eye-tracking or physiological signals.

Almost half of the created datasets contain data produced by humans with different degrees of interaction and freedom. This includes 24% of human-human dialogues and 10% of conversations taken from the internet such as social media exchanges or YouTube comment threads for the most interactive and least constrained ones. Data gathered from the internet is increasingly hard to confirm as human produced as we have no reliable way to distinguish between human and machine-generated content. The created datasets also include 7% of dialogues from TV shows, which can resemble spontaneous conversation to varying degrees. In particular, scripts tend to be less naturalistic than transcripts of what was actually said by the actors. Moreover, 8% of the datasets rely on hand-crafted dialogues, i.e. dialogues produced by humans in isolation given a set of instructions, like screen-writers write dialogues for their characters. However real-life conversations contain backchannels, disfluencies, overlaps, and other interactional phenomena, as opposed to most fictional dialogues (Chepinchikj & Thompson, 2016). Typical conversational dynamics are crucial for the participants to understand each other and build their contributions, but they are inconvenient from a reader's point of view. Thus, we are typically not exposed to naturalistic looking conversation transcripts, which makes it complicated to produce human-like scripted or handcrafted conversations (Pilan et al., 2024). This is not an issue when conversational dynamics are not in the scope of the study, but making sure that degraded dialogue does not influence other characteristics the study is interested in requires a careful assessment.

Similarly, about one third of the created datasets were generated by LLMs. These are much easier to produce than datasets involving real human interactions, which are costly and time-consuming, especially given current data demands. However, LLM-generated conversations should not be considered equivalent

to human–human ones. Their creation settings impose structural constraints: LLMs typically produce clean, alternating turns, unlike the irregular patterns of real dialogue. Moreover, they are mostly written, despite aiming to mimic spontaneous conversation. Crucially, LLMs ‘learn’ and generate language in ways fundamentally different from humans (Bender & Koller, 2020); lacking grounding, interaction, and shared attention, their outputs cannot reliably represent human communicative behaviour, and thus corpora generated in this way cannot serve as reliable evidence for studying how people communicate. Nonetheless, such resources can support analytical work comparing human and machine dialogue. Three created datasets included both types to study their differences (Y. Li et al., 2024; Occhipinti et al., 2024) or improve LLMs (Sato et al., 2024). Human–machine corpora also account for about 9% of all created datasets, consistent with the prevalence of DSs. However, the strong research focus on these technologies warrants reflection. Access to DSs is not universal, and they should not be seen as substitutes for human contact, particularly for isolated populations such as the elderly. Assessing actual user needs remains essential when developing such tools (Kann et al., 2022).

The rest of the created datasets contained no conversation except for one which consists of non-player characters dialogues from a video game (Weir et al., 2024).

Moreover, 7% of the surveyed papers extended existing datasets with human-made and/or synthetic annotations. A few papers (about 4%) also present generation methods for producing synthetic data.

Five papers present benchmarks for evaluating LLMs, each targeting different aspects ranging from specific elements like social norm incorporation to broader traits such as human-likeness or multi-turn ability. They cover various languages: two address English and Chinese (Duan et al., 2024; Ou et al., 2024), one focuses solely on Chinese (Zhan et al., 2024), one on Korean (Jang et al., 2024), and one omits language specification (X. Li et al., 2024) but is based on the English Wizards of Wikipedia (Dinan et al., 2019). While evaluating LLMs is essential given their growing prominence, releasing fixed benchmarks risks rapid obsolescence: they suit pre-existing models but may later enter training data, compromising validity. Similarly, benchmarks derived from existing corpora can introduce bias. Nonetheless, producing evaluation datasets in languages other than English may yield a positive side effect – enriching resources for under-represented languages and promoting greater linguistic diversity in future LLMs.

3.1.3.2 Used Datasets

Regarding usage, the most widely used corpora in the papers we reviewed were variants of the MultiWOZ (Budzianowski et al., 2018; Eric et al., 2020; Zang et al., 2020) dataset, which is unsurprising given the high proportion of papers on TOD systems. Indeed, these corpora contain task-oriented human–human chat-based dialogues collected in an adaptation of the Wizard-of-Oz (WOZ) setting, where one participant acts as a customer performing a task (e.g. booking a restaurant), and the other participant has access to a database to respond, which mimics what the system would have access to. In the original WOZ setting, the first participant was told that they were talking to a machine and the second was expected to play the system, which is not the case in MultiWOZ. Originally in English, MultiWOZ has since been translated into several languages. While reusing such established corpora is methodologically convenient, recent LLM-based approaches make this practice increasingly problematic. Because these datasets are used for both training and evaluation (see Section 3.1.4), building a TOD system often reduces to adapting an LM to reproduce the sentences expected in the WOZ corpora, rather than generalising beyond them, i.e. adapting an LM to a dataset rather than generalising from a corpus. Moreover, it seems reasonably likely that those corpora were included in the training set of most currently used LLMs which reinforces this circularity of data use and makes the evaluations based on ground truths less reliable.

A broad range of other corpora are also employed across different studies. Among the most frequently cited are DailyDialog (Y. Li et al., 2017) and SGD (Rastogi et al., 2020; H. Lee et al., 2022) – another TOD corpus. Zahir and Choi (2018) and Poria et al. (2019), based on the TV-show *Friends* and IEMO-

CAP (Busso et al., 2008) – an acted multimodal multi-speaker corpus – are widely used for conversation emotion recognition. Various datasets are used to enhance different aspects of dialogue systems: PersonaChat (S. Zhang et al., 2018) – persona-based conversations in pairs – for more personalised chit-chat systems, ESConv – emotional support conversations (S. Liu et al., 2021) between two crowdworkers – for building emotional support dialogue systems, ReDial (R. Li et al., 2018) – movie recommendation dialogues – for recommender systems, QReCC (Anantha et al., 2021) – hand-crafted dialogues with QA pairs grounded in web pages – for QA, and DSTC (Anantha et al., 2021) – phone calls between bus passengers and dialogue systems – for dialogue state tracking. All of these most used datasets are in English.

Both in terms of used and created resources, a recurring issue is the lack of information w.r.t. the datasets used. Many papers omit essentials such as the language of the data, whether the dialogue is handcrafted, scripted or naturally occurring, and whether it is human-human or human-machine. This lack of transparency has implications not only for reproducibility but also for the validity of the resulting models. For instance, DailyDialog is often described as a dataset for everyday chit-chat, but the dialogues were handcrafted by annotators rather than collected from real interactions. While using handcrafted data does not necessarily entail poor quality, it means that models are being trained and evaluated on data which reflects the *representation* we have of what a natural conversation is, rather than what it actually is, and people tend to underestimate the amount of e.g. pauses, repairs and other disfluencies (Bard & Lickley, 1997), which are distinctive markers of conversation. This observation points to a broader need: we must better define what conversational features are desirable in a model, and these criteria are likely to be highly task-dependent. For example, while hesitations and pauses may increase human-likeness, artificially inserting them into TOD system might not be necessary to satisfy users. On the other hand including them may help users to process information more easily. In any case, those trade-offs require investigations through a joint effort of the community.

3.1.4 Evaluation

Our findings (unsurprisingly) show that conversational AI is the main focus of current research. We therefore focus on the categories Generative systems and LM Evaluation defined in Section 3.1.2 for our assessment of the current evaluation practices in the community.

Two aspects are relevant when it comes to conversational generative systems: their ability to tackle the specific task they were designed for, and their conversational competency. The practical evaluation of task performance depends on the application and lies outside our scope in this section.

Regarding the conversation evaluation, we noted that many studies (45%) rely on reference-based automatic metrics such as BLEU (Papineni et al., 2002), ROUGE (C.-Y. Lin, 2004), and METEOR (Banerjee & Lavie, 2005). These metrics compare the system’s responses to predefined gold-standard utterances based on lexical overlap. While they are convenient and widely used, they are limited in their ability to capture the interactive and context-sensitive nature of conversation. For instance, these metrics may penalise valid but diverse or creative responses that differ from the reference, even though such variation is a core feature of natural dialogue. Nevertheless, about one third of the studies do not use additional metrics to evaluate conversation which is problematic because the conversation quality assessment is multifaceted and should go deeper than the semantic comparison with a ground truth. These basic evaluations prevent us from accurately determining when a dialogue system is genuinely useful and more effective than a simpler alternative.

The increasing complexity of conversational tasks has led to the use of LLM-based evaluators, a practice usually called LLM-as-a-judge (LaaJ; Y.-T. Lin and Chen (2023) and Bavaresco et al. (2025)), which can flexibly score multiple conversational dimensions. Among papers on dialogue systems, about 12% rely on such evaluation⁹. Yet, more than one fourth do not validate the LLM judgements with

⁹We do not include BERTScore (T. Zhang et al., 2019) in the LLM-based evaluators but rather prompt based approaches

human input, even though LLMs might have divergent ‘expectations’ of conversational quality in relation to humans. Moreover, as stated earlier, it is reasonable to think that many of the corpora we use to train and evaluate our models were included in the training data of the LLMs we use, which biases the evaluation. Similarly, many models are evaluated using benchmarks that may already be included in the training data of the models themselves, raising concerns about data leakage and the validity of results.

Regarding human evaluation, the most assessed dimensions are coherence (18% of the papers on dialogue systems including human evaluation) and fluency (16%). Other dimensions such as relevance, consistency and naturalness are also assessed (each 8-9%). This shows the need to break down the evaluation in more objective features which human could assess as external evaluators. Overall conversational quality is also assessed (9%), but this dimension remains difficult to capture in a controlled and reproducible way due to its inherently subjective nature. Crucially, very few systems are evaluated by end users during interaction ($\leq 6\%$), even though extrinsic evaluations in controlled settings may not reflect real-world user experience.

Our findings align with those of Yang et al. (2026) who conducted a large survey on NLG evaluation practices in general. They collected all the papers from ACL, NAACL, EMNLP and INLG between 2020 and 2025 (14171 papers). Information about the task and the evaluation approaches (automatic metrics, LaaJ, human evaluation) are retrieved automatically by several LLMs and aggregated through a majority vote. 61% of all the retrieved papers were labelled as NLG tasks by the models, 110 of those papers (about 1%) were further annotated by hand to evaluate the agreement with the automatic retrieval. One of the findings of this manual annotation was that most disagreements between the different models occurred for tasks that are not strictly NLG, such as code generation or mathematical reasoning.

After keeping only the top-30 ‘real’ NLG tasks they are left with 3,334 papers, 78.1% of which belong to the following four most studied categories: summarisation, translation, dialogue generation and question answering. The two last ones are of particular interest for us. This paper confirms our finding that BLEU is the most widely used metric for dialogue generation, and QA is mostly evaluated through string matching. There is an increase in LaaJ especially in dialogue generation and despite common assumptions there seems to be differences between human evaluation and LaaJ in particular for higher level features like coherence and fluency. They note, like us, that when both LaaJ and human evaluation are used, there is not always an explicit comparison between them. Indeed, 58.7% of the papers that use both approaches explicitly compare them, indicating that the rest either does not report a comparison or measures different aspects.

All of this suggests that without careful reconsideration of evaluation practices, the field risks measuring progress in ways that may be convenient but ultimately misleading, reinforcing the need for more rigorous and context-aware assessment methods.

3.2 Pipeline for NLP/Dialogue Research

The epistemic pipeline followed when doing research on dialogue can be seen as a process to go from conversational phenomena to their analyses through formal tools and implementation in computational systems. One view of this pipeline is represented in simplified form in Fig. 3.2. The process often begins with an intuition about a particular dialogue behaviour, e.g. a recurring interactional pattern, a coordination mechanism, a structural regularity, or a communicative difficulty. This intuition either motivates the creation of new dialogue resources or emerges from the close analysis of existing ones.

From these resources, researchers attempt to construct a model of the observed phenomenon. Nowadays, this modelling phase typically involves both a theoretical and a computational component. A formal

with models such as LLaMa, Mistral, ChatGPT, etc.

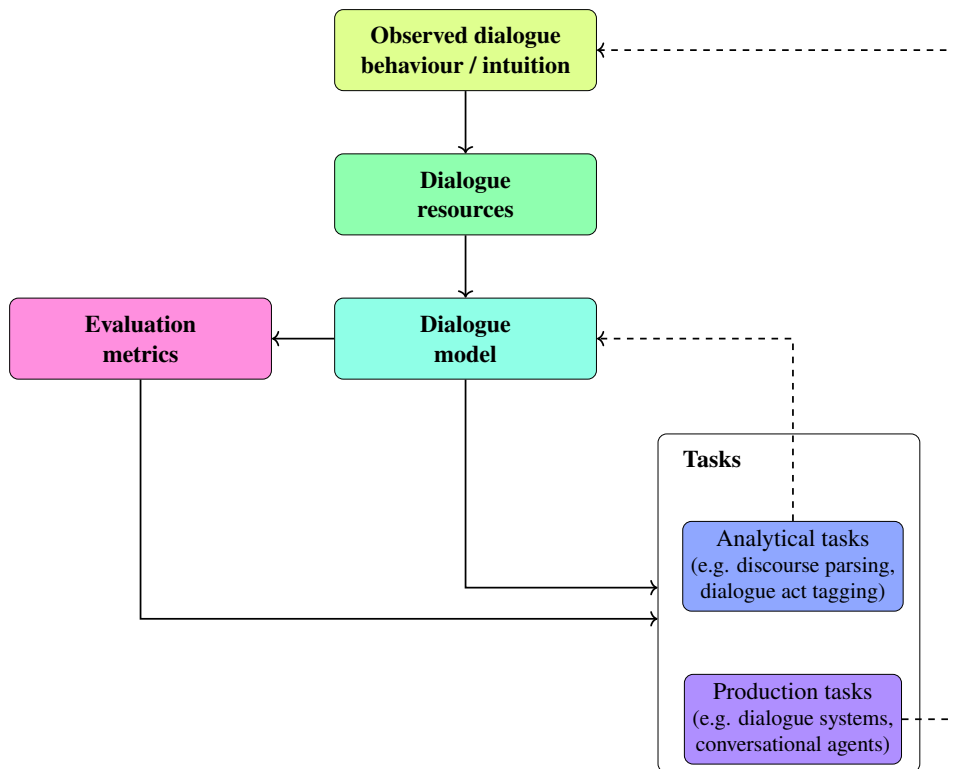


Figure 3.2: Description of (dialogue) research pipeline in NLP/CL

account would first specify the structure or mechanism under investigation; computational implementations would then allow systematic testing and confrontation with empirical data.

On the basis of these models, research tasks can be defined. We can divide them into two broad families: analytical tasks and production tasks. The distinction does not lie in the technical tools employed – both may rely on rule-based systems, statistical models, or LLMs – but in their epistemic goals and evaluation logic.

Analytical tasks, such as discourse parsing, dialogue act tagging, or turn-taking prediction, aim to provide insight into dialogue structure. Their purpose is descriptive or explanatory. They can also participate in refining theoretical models by testing their adequacy against annotated data.

Production tasks, by contrast, aim to reproduce or simulate aspects of dialogue for practical purposes. These include the development of dialogue systems, conversational agents, or synthetic resource generation. Their goal is generative and application-oriented. At the same time, they may also challenge the underlying model: if system outputs fail to match real-world expectations, some modelling assumptions may need to be revised.

Crucially, these two task families require different evaluation strategies. Analytical tasks are typically evaluated through alignment with annotated ground truth and category agreement. Production tasks require broader criteria, including task success, usability, user satisfaction, and interactional quality. Conflating these evaluation logics risks obscuring both scientific insight and practical performance.

3.2.1 From theoretical models to black-box ones

Historically, theoretical dialogue modelling included explicit formal frameworks such as dialogue games (D. Lewis, 1979) or speech acts and plan-based theories (Austin, 1975; P. R. Cohen & Perrault, 1979). These

models participated to producing structured representations of grounding mechanisms and meaning negotiation among others. For instance, Type Theory with Records (TTR; Cooper (2005)) offered a formalism to represent the dynamic and context-sensitive nature of dialogue, including the management of shared knowledge and the resolution of ambiguities. Some of these theoretical models have been implemented in computational systems; the dialogue manager based on information state proposed by Larsson and Traum (2000) is one such example. Yet they were not merely engineering tools but theoretical commitments about how dialogue works.

In the current LLM-dominated landscape shaped by large-scale transformer architectures (Vaswani et al., 2017) and their scaling paradigm (T. Brown et al., 2020), however, there is a growing tendency to assume that such theoretical structure will be implicitly captured by neural models trained on massive corpora. As a result, the explicit modelling layer risks becoming invisible. Instead of articulating formal assumptions about dialogue structure, participation roles, or grounding mechanisms, researchers increasingly rely on pre-trained models as black boxes, later adapted through fine-tuning or instruction tuning (Ouyang et al., 2022) for specific tasks.

This shift brings undeniable practical advantages: scalability, adaptability, and impressive performance across benchmarks. Yet it also entails epistemic costs. When modelling choices are replaced by reliance on opaque pre-trained systems, it becomes difficult to determine what is actually being learned (Rogers et al., 2020). Apparent improvements may reflect memorisation or training data overlap (Carlini et al., 2021), in particular when the same benchmarks keep being used over the years, or the exploitation of surface-level distributional regularities rather than a principled modelling of dialogue phenomena (Bender & Koller, 2020; Bender et al., 2021). More broadly, the assumption that scale can substitute for explicit theoretical structure remains insufficient when the objective is not only performance, but understanding the mechanisms underlying dialogue.

3.2.2 Evaluation and the risk of circularity

Since modelling has become less explicit, evaluation has become both more central and more fragile. The current trajectory of dialogue research reveals a tendency to prioritise convenience over suitability, particularly in the selection of data and evaluation methods.

Systems without rigorous evaluation are meaningless, yet the field often relies on metrics chosen for convenience or historical precedent, even when these metrics are known to be imperfect (Sai et al., 2022). Automatic evaluation cannot replace careful, task-specific assessment, and the cost of human evaluation should not be used to justify weak or inappropriate metrics. Researchers must articulate system expectations clearly and break them down into measurable aspects. Similarly, using LLMs as proxies for human evaluation must be done carefully and cannot be generalised across tasks; validation in one context does not ensure validity in another. Without human validation, reliance on LLM-based evaluation risks creating a self-reinforcing evaluation loop, particularly when the evaluation corpora themselves may have been included in the models' training data. Such circularity can lead to an overestimation of system performance and obscure meaningful limitations.

Generative systems, while increasingly sophisticated, display conversational abilities that diverge sharply from human interaction. Conversations produced by these systems are typically structured in strict alternation and constrained by intents defined by their designers, who optimise for engagement and utility rather than natural interaction. Unlike humans, these systems do not rely on scaffolding through backchannels, overlapping speech, or incremental feedback. Users often adjust their behaviour to accommodate the model, subtly shaping their conversational patterns to align with machine expectations rather than engaging in natural dialogue. Even if models could perfectly mimic surface behaviours, they would not capture the grounded, embodied, and socially situated nature of human language. LLMs, due to their design, represent an 'average' over their training data, which cannot reflect the nuances of any individual

or even population. Moreover while their training data does not capture the full variety of language use in the world (Bender et al., 2021), imagining a way to build a representative dataset would not be enough to make such models produce a rich and varied representation. Their design is such that they often rely on the same structure (Namuduri et al., 2025) and use vocabulary patterns that are different from the ones used by humans (Durward & Thomson, 2024). While this behaviour can be slightly adjusted in LLMs through prompting, their underlying structure will remain fundamentally different from humans' language acquisition and production mechanisms.

The question of human-likeness in dialogue further illustrates these limitations. LLMs can sometimes pass restricted forms of the Turing test (Jones & Bergen, 2026), yet evidence shows that evaluators may prefer machine-generated outputs over human dialogue in controlled settings. This does not imply that models are indistinguishable from humans. Rather, it highlights the unreliability of human judgements of conversational human-likeness, particularly under novel or brief evaluation conditions. The emergence of LLMs has also shifted public perception: whereas face-to-face interactions rarely prompt doubt about human interlocutors, non-face-to-face interactions with unknown partners are now more often approached with scepticism. Real human-human conversations are noisy, filled with overlaps, hesitations, repairs, and subtle cues that often go unnoticed when we participate in a conversation. We are indeed not often exposed to the actual look of a conversation. While repairs and backchannels are crucial for us to be efficient in conversation, they do not need to be *remembered* to follow the conversation. Thus, showing a transcript *a posteriori* to people and asking them to rate it for human-likeness will probably not reflect what they would have said *during* the conversation. Even placing someone in the role of a judge while they are having a conversation could change the way they interact and make them notice behaviours they would not have consciously analysed normally, making them doubt the quality of the conversation. Paradoxically, interactional patterns may possibly penalise authentic interactions when they are evaluated externally, while LLMs would produce a smoother dialogue.

3.2.3 Re-centring theory, data, and purpose

These considerations have important implications for the design and purpose of generative conversational systems. Researchers must critically assess whether the ultimate goal should be to replicate human-like interaction, given the associated risks, including manipulation, misinformation, or undue emotional attachment. While the conversational paradigm can presumably make many applications, such as coding assistance or information retrieval, more approachable, such systems do not need to emulate human behaviour, they must instead perform complex tasks effectively. The conversational features reproduced in such systems should be ones that facilitate conversation without triggering emotional engagement. Differentiating between those features requires rigorous theoretical analysis of the notion of conversation. This would also provide solid grounds to create more dimension-specific metrics, capable of capturing distinct aspects of dialogue, and thus enable the development of evaluation frameworks better adapted to specific tasks. A formal characterisation of conversation would also help our community to clearly distinguish between prototypical human-human dialogue and other forms of communication, including scripted, synthetic, or human-machine interactions.

While dialogue generation was one of the most studied tasks among papers focusing on dialogue in the ACL ANTHOLOGY in 2024, the need for more careful resource description is more general. Our field relies heavily on corpora and being able to clearly describe their properties and limitations is essential to ensure that they are used appropriately. Data should be at the heart of conversational research – and NLP/CL research in general, yet datasets are too often treated merely as tools to train models rather than objects of careful study in their own right. While most dialogical corpora are not produced in natural, face-to-face interactions – due to practical constraints – this does not mean that all proxies are equivalent. Hand-crafted dialogues, movie scripts, human-machine exchanges, and LLM-generated conversations

each reflect different interactional dynamics and limitations. Recognising and explicitly defining these limits is crucial. A clear understanding of the scope of a corpus should precede any model development or evaluation, ensuring that the data aligns with the intended task.

By returning data and evaluation to the centre of our research, developing theoretically grounded evaluation metrics, critically assessing human-likeness, clarifying the goals of generative systems, and grounding our decisions in the theory and practice of human conversation, the community can avoid pursuing progress that is only apparent. Without these reflections, the field risks running ever faster on a ‘hamster wheel’, producing models and metrics that give the illusion of advancement while overlooking the subtleties of human dialogue. Moving beyond the allure of human-like replication toward well-defined, task-oriented objectives will allow the field to achieve meaningful progress while mitigating the risks associated with deploying these technologies.

The limitations outlined above are not merely methodological concerns; they reveal the need for clearer descriptive tools. If resources differ fundamentally in their participation structures, environments, purposes, and interactional constraints, then treating dialogical behaviours as universal is unreasonable.

This is precisely where our theoretical framework becomes operational. By decomposing dialogue along explicit axes – PARTICIPANTS, LANGUAGE, ENVIRONMENT AND MEDIUM, VERBAL INTERACTION, and CONVERSATIONAL ACTIVITY AND PURPOSE – we can systematically describe both datasets and tasks. Such a structured description makes it possible to articulate what kind of dialogue a resource represents, what constraints a task imposes, and which dimensions are being modelled, evaluated, or ignored.

In the following section, we apply this framework to three contrasting dialogue scenarios. These case studies illustrate how the framework can not only clarify the nature of different dialogue data but also reveal the specific constraints they impose on resource selection.

3.3 Applying the Descriptive Framework

The three scenarios examined in this section operate at different descriptive levels. Turn-taking (Section 3.3.1) concerns the interactional organisation of dialogue and constitutes a domain-general mechanism that cuts across genres, activities, and institutions. Task-oriented dialogue (Section 3.3.2), by contrast, designates a broad activity type, defined primarily by the presence of an explicit practical goal, but encompassing a wide range of interactional configurations and degrees of social engagement. Clinical interviews (Section 3.3.3) represent a specific form of institutional interaction: they are goal-directed, like task-oriented dialogue, but are further constrained by professional norms, asymmetrical roles, and institutional definitions of success, while remaining open-ended in their interactional unfolding. The goal they are directed towards is not as clearly defined as in task-oriented dialogues and relates much more to the social function of dialogue.

3.3.1 Scenario 1: Turn-taking

Turn-taking refers to the organisation of participation in spoken interaction whereby interlocutors take turns speaking (Sacks et al., 1974). In ordinary conversation, transitions between speakers are typically seamless, with little to no overlap or extended silence. The ability to determine when it is one’s turn to take the floor is crucial for the planning of dialogue: participants must interpret an ongoing utterance incrementally and prepare their own contribution in advance in order to respond at an appropriate moment. Failing to do so entails interactional risks, such as losing the opportunity to speak if another participant takes the floor more quickly, or having a silence interpreted by the current speaker as signalling misunderstanding, hesitation, or a request for elaboration.

Turn-taking is therefore one of the most fundamental properties of dialogue. It is not merely a surface

feature describing the alternation of speakers, but a structuring mechanism that enables participants to coordinate action, negotiate meaning, and maintain the coherence of an interaction over time. In this sense, turn-taking contributes directly to the form of dialogue: it makes participation visible, organises sequential dependencies between contributions, and allows interlocutors to project upcoming actions.

Importantly, turn-taking emerges from spontaneous language use in real time. It relies on fine-grained temporal coordination, anticipation of others' actions, and the management of overlaps, pauses, and repair phenomena. These properties make turn-taking both central to dialogue and difficult to model, observe, or reproduce outside of naturally occurring interaction. As such, it provides a particularly revealing case for examining how dialogue research conceptualises its objects of study, selects its resources, and evaluates its tools.

In what follows, turn-taking is used as a wide scenario to question practices in dialogue research, examined successively from an analytical, tool-building, and productive perspective.

3.3.1.1 Analytical perspective: constraints for studying turn-taking

Studying turn-taking requires making explicit a number of constraints on both the data and the interactional setting. Turn-taking is inherently relational and temporal: it only exists through the coordination of multiple participants over time. As a consequence, resources suitable for its study must preserve interactional dynamics with minimal post hoc normalisation. In particular, temporal properties of the interaction – such as precise alignment of speech segments, access to pauses and overlaps, and the preservation of simultaneity – are essential. Resources derived from scripted dialogue, heavily edited recordings, or post hoc reconstructions do not satisfy these requirements, even if they involve multiple speakers and conversational content.

These constraints become especially salient when considering mediated forms of spoken dialogue. While the organisation of turn-taking in face-to-face interaction has been extensively described in CA, the emergence of new interactional settings calls for renewed investigation. For instance, video-mediated conversations differ from face-to-face interaction in several aspects that are directly relevant to turn-taking. In particular, such settings introduce varying degrees of latency between the production of an utterance and its reception, as well as restricted or distorted access to visual cues such as gaze direction or body posture. Phenomena such as parallax effects or partial visual representations can further complicate the coordination of turns, potentially making turn-taking less efficient or more error-prone. In the following, we will thus focus on the differences between face-to-face and video-mediated interaction with respect to turn-taking.

Understanding these divergences matters not only from an epistemological perspective but also for practical reasons. Video-mediated interaction is generally easier to record than face-to-face conversation and thus constitutes an attractive alternative for building large dialogue resources. The Candor corpus (Reece et al., 2023), for instance, provides a large collection of casual video-mediated dyadic interactions. However, if video mediation significantly alters turn-taking behaviour, then resources of this type cannot be straightforwardly treated as proxies for face-to-face interaction. Since turn-taking supports many other aspects of dialogue co-construction, its modification may have cascading effects on the interaction as a whole.

Using the descriptive framework introduced in Chapter 2, three parameters emerge as particularly critical in this scenario: PARTICIPANTS and VERBAL INTERACTION due to the inherent characteristics of turn-taking, and ENVIRONMENT AND MEDIUM because of the specific video-mediated setting. Table 3.2 describes the relevant sections from the framework to characterise the constraints on dialogue resources for studying turn-taking in a video-mediated setting. At a minimum, the interaction must involve more than one participant talking spontaneously to each other. Beyond these requirements, many participant-related sub-parameters – such as role stability or participation symmetry – are not strictly fixed by the

task itself. *VERBAL INTERACTION* is the studied parameter, in particular turn-taking, which requires a synchronous setting that allows turn-taking to be freely decided. While other co-construction parameters are not constrained they are very likely to strongly interact with turn-taking mechanisms and should thus be left as free as possible. In contrast, the medium is directly constrained: the resource must consist of video-mediated interaction. Nevertheless, even within this category, the exact instantiation of the medium can vary substantially with different degrees and qualities of visibility and audibility.

Parameter	Sub-	Subsub-	Constraint
Participants			Humans
	Participation structures	Number	≥ 2
	Enunciative status	Spontaneity	✓
Environment & Medium			Video-mediated
(Verbal) Interaction	Co-construction	Coordination signals	<i>free</i>
		Turn-taking	✓
		Completion	<i>free</i>
	Time	Synchrony	✓

Table 3.2: Constraints on the dialogue resource for studying turn-taking in a video-mediated setting

Crucially, the specific instantiation of participant-related and medium-related sub-parameters is likely to affect the resulting interactional dynamics. A comprehensive study of turn-taking in video-mediated dialogue would therefore need to explore a range of settings, from highly organised to more open ones. More controlled settings could involve task-oriented dyadic interactions, allowing for partial control over participation patterns and role switching. With respect to the medium, not all video-mediated configurations are equivalent. Turn-taking practices depend on the availability and quality of communicative channels, and video-mediated interaction introduces latency, restricted visual fields, and altered feedback mechanisms to varying degrees depending on the tool. A highly controlled setting might involve high-quality video streams showing the interlocutor directly, as opposed to avatars or choppy images, combined with quiet physical environments and minimal external distractions. Starting from such a controlled configuration, the setting could then be progressively complexified, for instance by introducing noisier environments, more open-domain dialogue, or additional participants.

Studying turn-taking in video-mediated interaction also calls for systematic comparison with face-to-face dialogue. Such a comparison is necessary to determine what is lost, transformed, or compensated for when interaction moves from a co-present to a mediated setting. The descriptive framework helps clarify the conditions under which such a comparison is meaningful. Both resources must satisfy the basic constraints required to study turn-taking, and differences in modality must be treated as part of the phenomenon rather than as noise. Latency, reduced access to visual cues, and altered feedback loops must therefore be explicitly characterised, as they directly shape turn-taking behaviour.

Ideally, such a comparison would rely on two corpora or sub-corpora that are closely matched with respect to participant configuration, activity type, and interactional purpose, differing primarily in the medium of interaction. While it is difficult to isolate turn-taking entirely from other co-construction mechanisms – such as coordination signals or completion phenomena, which are tightly interrelated and also influenced by the medium – other parameters should be controlled as strictly as possible. This requirement further highlights the need for a solid characterisation of turn-taking in face-to-face interaction, which serves as a reference point for studying adaptations to mediated forms.

In summary, this analytical scenario imposes strong constraints on the medium of interaction while

leaving a range of possibilities open for other parameters, provided that these are aligned across resources when comparisons are made. Making these constraints explicit is essential to delimit the scope of empirical findings and to avoid conflating genuine interactional differences with artefacts of data collection or processing.

3.3.1.2 Tool-building perspective: ASR, transcription, and the degradation of interactional structure

While the analytical study of turn-taking depends on rich interactional data, much dialogue research relies on tools that substantially transform, and sometimes degrade, the phenomena of interest. Automatic Speech Recognition (ASR) systems are a particularly salient example. ASR has become a ubiquitous component of dialogue processing pipelines, yet it is typically optimised with respect to lexical accuracy, most commonly measured through word error rate. In this optimisation process, interactional features central to turn-taking – such as overlaps, interruptions, and fine-grained timing – are frequently flattened or suppressed.

In practice, ASR outputs often miss overlapping turns, in particular short utterances and vocal coordination signals (Liesenfeld et al., 2023), or assign them to the wrong speaker, thus producing a flat and misleading representation of interaction. From the perspective of turn-taking, such representations are not neutral transformations: they erase precisely the features that make the phenomenon observable.

Transcription practices raise similar issues. Transcriptions remain the primary working support for much dialogue research, yet they are inherently degraded representations of spoken interaction. Different transcription schemes preserve different aspects of dialogue, and the choice of scheme has direct consequences for what can be analysed. Minimal orthographic transcriptions may suffice for lexical or semantic analysis, but they offer little access to turn-taking phenomena. Richer interactional transcription schemes, which encode overlaps, pauses, and prosodic cues, preserve far more information but are costly to produce and require more specialised expertise.

A face-to-face interaction, characterised by multimodality and fine-grained temporal coordination, undergoes a substantial change when converted into a transcription. Depending on the transcription scheme, parameters such as temporal resolution, simultaneity, and participant coordination may be partially or entirely lost. Table 3.3 uses our framework to make these transformations explicit from a face-to-face interaction to a richly transcribed version following a scheme such as the one from CA (Jefferson, 2008), and a basic lexical transcript that includes only sanitised linguistic content.

While the original interaction allows the participants to perceive all the parameters like the participation structure and the environment, the transcription process necessarily degrades this information. A rich transcript can preserve some of these aspects by describing the context for example, like stage directions in a theatrical script, but it is not the case for all the parameters and it is not as good as the original interaction. The lexical transcript, on the other hand, completely flattens the interactional dynamics, leaving only the observable linguistic content. Regarding verbal interaction, Jefferson (2008) proposes conventions to represent overlaps and interruptions in transcriptions which can be used to preserve the turn-taking dynamics to a certain extent. Nevertheless the transcription process will always require to make choices about what to preserve and what to discard, but also about how to represent the preserved features which is not always an objective process. For instance, the decision to use a certain punctuation to represent at least partially the structure carried by prosody in spoken interaction can change the perceived meaning of an utterance and thus the way it is interpreted in the context of the interaction (Hammersley, 2010). The transcription process is thus not a neutral transformation but an active interpretation of the interaction, which can lead to a significant loss of information and a distortion of the original phenomena (Johnson et al., 2025). While not all parameters of the original interaction are necessarily relevant for a given study, knowing what is preserved and what is lost in the transcription process is essential to ensure the validity

of the research process.

Parameter	Sub-	Subsub-	Original	Rich transcript	Lexical transcript
Participants	Participation structures	Function	Observable	∅	∅
Environment & Medium	Physical context		Observable	Partially transcribed	∅
	Visibility	Gaze	Observable	∅	∅
		Gesture / Facial expression	Observable	∅	∅
	Audibility	Prosody / Tone	Observable	Partially transcribed	∅
(Verbal) Interaction	Co-construction	Coordination signals	Observable	Transcribed	∅
		Turn-taking	Observable	Transcribed	∅
		Completion	Observable	Transcribed	∅

Table 3.3: Degradation of dialogical features from original occurrence to transcription

These issues also raise questions about the use of LLMs in dialogue research. While they can process structured text and may be capable of handling enriched transcription schemes, they are typically trained on dialogue data that has already been flattened and normalised. This raises doubts about their sensitivity to interactional features such as turn-taking, and about the extent to which they can meaningfully model phenomena that are poorly represented in their training data.

3.3.1.3 Productive perspective: generating dialogue with turn-taking

Recent years have seen an increasing emphasis on the automatic production of dialogue resources, both written and spoken. From the perspective of turn-taking, this trend raises important questions about what it means to generate dialogue that preserves interactional structure.

In written dialogue generation, when producing data that is intended to simulate spoken interaction, incorporating turn-taking in any meaningful sense requires adherence to a transcription or annotation scheme that encodes interactional features. Simply alternating speaker labels is insufficient to capture phenomena such as overlap, interruption, or repair. Achieving richer representations would require training data transcribed with appropriate schemes, as well as models capable of learning and reproducing these conventions.

In spoken dialogue generation, the challenges are even greater. Generating turn-taking behaviour involves not only deciding when speakers alternate, but also modelling timing, anticipation, and occasional simultaneity. Current spoken dialogue systems largely rely on turn-based architectures that assume clear boundaries between speakers' contributions. While this assumption simplifies system design, it departs significantly from the dynamics of human interaction.

These limitations suggest that, despite recent progress in generative models, we are not yet in a position to produce dialogue data that exhibits human-like turn-taking behaviour in a robust way (Mayor et al., 2025). This observation does not imply that synthetic dialogue resources are without value. Rather, it underscores the importance of treating generated data as a distinct type of resource, with its own constraints and limitations, rather than as a cheaper replacement for human-produced interaction.

3.3.1.4 Key takeaways and implications

Turn-taking functions as a revealing stress test for dialogue research practices. As a structuring property of dialogue, it is central to interaction yet fragile with respect to data processing and tool design. When turn-taking is ignored or flattened, resources may still support a wide range of tasks, but the scope of the phenomena that can be legitimately studied is reduced.

The descriptive framework proposed in Chapter 2 provides a means to make these limitations explicit. By characterising how resources, tools, and representations constrain access to turn-taking, it becomes possible to delimit the generalisability of analytical findings and the claims made on the basis of generated dialogue. In the following scenarios, a similar approach will be applied to other aspects of dialogue research, where constraints do not necessarily bear on the form of dialogue but are no less consequential.

3.3.2 Scenario 2: Task-oriented Dialogue

Task-oriented dialogue constitutes one of the most widely studied categories of dialogue in computational linguistics and NLP. In the literature, the term typically refers to interactions that are structured around the completion of a practical goal, such as booking a train or restaurant, solving a collaborative task, or achieving a shared objective within a constrained environment. As discussed in Chapter 2, the goal of a dialogue can be characterised along several dimensions, including practical purpose, information exchange, and social bonding. Task-oriented dialogues are primarily defined by the predominance of a practical purpose, though in practice these dimensions coexist to varying degrees.

The category of task-oriented dialogue itself covers a broad range of interaction types. Some tasks are highly constrained and admit a clear success condition, often formalised as a set of slot values to be filled, as in classical booking scenarios. Others are more open, such as map tasks (Thompson et al., 1993) or tangram dialogues (Clark & Wilkes-Gibbs, 1986), where success depends on mutual understanding and coordination rather than the completion of a predefined form. Still others, such as service encounters in everyday settings (e.g. bakery dialogues), combine a practical goal with elements of social bonding and politeness.

Task-oriented dialogues are commonly used to study specific linguistic or interactional phenomena, for instance the production and interpretation of referring expressions in tangram tasks. More broadly, they have become a dominant type of resource in dialogue research, in part because they are comparatively easy to collect. The presence of an explicit task provides participants with a clear interactional frame, reducing the need to sustain conversation for its own sake with potentially unknown dialogue partners. Moreover, task-oriented settings tend to limit the range of personal information exchanged, which facilitates anonymisation (Amblard et al., 2014) and makes data collection more straightforward than in casual, open-ended conversations.

However, the very constraints that make task-oriented dialogue attractive as a research resource can also influence a wide range of other dialogue parameters in less obvious ways. For this reason, task-oriented dialogue provides a particularly instructive scenario for examining how dialogue purpose interacts with resource selection, task formulation, and evaluation practices, especially in the context of contemporary, LLM-based systems.

3.3.2.1 Analytical perspective: task-oriented dialogue as a basis for discourse modelling

Task-oriented dialogues have long served as a privileged empirical basis for studying discourse structure and rhetorical relations. A rich body of work has investigated how contributions in such interactions relate to one another in terms of explanation, clarification, elaboration, or argumentation, often with the explicit goal of supporting discourse parsing or rhetorical relations tagging. Corpora such as STAC (Asher et al., 2016), which consists of chat-based, task-oriented multi-party dialogues, and Molweni (J. Li et al.,

2020), derived from Ubuntu forum discussions, have played a central role in this line of research. These resources, whose features are described in Table 3.4, are frequently used to operationalise and evaluate theoretical frameworks such as SDRT (Lascarides & Asher, 2007) or information state update models in dialogue settings.

The STAC corpus provides a particularly rich illustration of how task constraints shape interactional organisation. As a reminder, it consists in multi-party dialogues in which participants play a board game that requires them to negotiate and trade resources. They play online through a virtual interface and they can communicate with each other through a written chat system. The participation structure is thus strongly influenced by the game setting: players repeatedly engage in sequences where one participant requests a resource (e.g. clay or sheep) and attempts to acquire it through negotiation or trade. This recurring pattern yields highly regular discourse sequences, typically involving a main speaker addressing all other players as potential interlocutors. At the same time, STAC dialogues also contain parallel threads that are not directly oriented toward game progress, in which some participants temporarily adopt a bystander role. As a result, the corpus exhibits a complex organisational structure combining focused task-oriented exchanges with more peripheral conversational activity. (3.1) illustrates such a situation, where a side conversation about the trials for a league starts in parallel with the main thread (normal font) of interaction that revolves around resource trading. This parallel thread further splits into two sub-threads (marked in bold and italics).

- (3.1) **LJ** anyone want sheep for clay?
GWFS got none, sorry :(
GWFS **so how do people know about the league?**
WM no
LJ **i did the trials**
TK **i know about it from my gf**
GWFS [yeah me too,]_a [*are you an Informatics student then, lj?*]_b
TK **did not do the trials**
WM has anyone got wood for me?
GWFS **[I did them]_a [because a friend did]_b**
GWFS lol wm, you cad
GWFS afraid not :(
LJ [*no, I'm about to start math.*]_a [*I just hang around appleton a lot*]_b
TK sry no
GWFS my single wood is precious
WM what's a cad?

[STAC corpus (Asher et al., 2016, Ex.1)]

Several additional parameters contribute to this complexity. In terms of enunciative status, participants are engaged in a game and may therefore display a persona that differs from their everyday conversational self; this distinction may itself correlate with differences between game-related and non-game-related threads. The conversational history is also non-trivial: since the game is organised as a tournament, participants can encounter each other several times, thus creating interactional histories between them. There may also be a more subtle form of shared background in terms of inter-personal familiarity and topical ease linked to recruitment practices. In (3.1) all the participants seem to be university students which gives them shared topics to talk about. Linguistically, we can imagine that participants were instructed to interact in English though full control cannot be guaranteed.

Parameter	Sub-	Subsub-	STAC	Molweni
Participants			Humans	Humans
	Participation structures	Number	4?	2-9 (avg. 3.51)
		Functions	Main speaker - Addressees depending on requested information	Main speaker - Open / Addressees
	Enunciative status	Anonymity	Pseudonymous	Pseudonymous
		Role instantiation	Self in game	Self
		Persona instantiation	Self in game	Self - Self as expert
		Spontaneity	✓	✓
		Commitment	✓	✓
Social Dynamics	Hierarchy	∅	Information seeker - holder	
	Conversational history	Game session	∅?	
Language			English	English
	Control		✓?	✗
Environment & Medium			Chat & Virtual board game	Online forum
			Physical context	Virtual boardgame
Visibility	Gaze	∅	∅	
	Gesture / Facial expression	ASCII emojis	ASCII emojis	
Audibility	Prosody / Tone	∅	∅	
	Noise	∅	∅	
(Verbal) Interaction				
Co-construction	Coordination signals		∅	∅
	Turn-taking		∅	∅
	Completion		∅	∅
Time	Synchrony	Semi-synchronous	Asynchronous	
	Ephemerality	Saved for the task	Persistent by design	
Conversational Activity and Purpose			Board game	Ubuntu information seeking
			Epistemic grounding	World setting World stability
Purpose	Practical	Task + Social	Information exchange	
	Intention	?	?	
Activity	Success condition		?	Information acquired?
	Vocabulary		Partially game specific	Technical
	Structure		Negotiation phases	QA

Table 3.4: Description of the dialogical setting in the STAC and Molweni resources

Although STAC is a text-based corpus, it nevertheless involves a shared physical – or more precisely, virtual – context: participants interact while jointly manipulating a game board. This shared environment plays an important role in grounding discourse, as demonstrated by work in SDRT that explicitly integrates references to ‘physical’ context in the analysis (Hunter et al., 2018). Epistemically, the dialogue unfolds across two worlds: a fictional but fixed game world, defined by the rules and state of the game, and the real world, which participants occasionally invoke in side conversations. The fictional world is fixed in the sense that its properties are determined by the game mechanics, though it may still be subject to local interpretation or negotiation.

Molweni represents a different but complementary type of task-oriented resource. Derived from Ubuntu technical support forums, it consists of written exchanges oriented toward problem solving and information provision. With respect to participants, the corpus is old enough that one can reasonably assume that the contributors are human users. However, this assumption is becoming increasingly fragile as AI-generated content proliferates online, raising the risk that newer forum-based datasets may unknowingly include synthetic dialogue with no reliable method for detection.

The participation structure in Molweni is also quite complex as most of the dialogues involve multiple participants. Duologues represent about 21% of the corpus, the rest involving 3 to 9 participants. Functionally, the information seeker can be analysed as a main speaker addressing a broad but implicit audience, while respondents self-select based on their expertise, temporarily becoming both main addressee and main speaker. The enunciative status of participants is pseudonymous, with the additional possibility that a single username may correspond to more than one individual. Although contributors are not explicitly role-playing, answerers may adopt a ‘teacher-like’ or expert stance, subtly shaping their language and interactional behaviour. Notably, this shift in stance occurs seamlessly in human interaction, in contrast with LLM-based systems where such behaviour must be explicitly prompted.

Several other parameters introduce analytical challenges. Due to the open nature of forum interaction control over the language is difficult to maintain, though the original paper does not mention the presence of non-English content. Co-construction, while often associated with spoken dialogue, is also relevant here: participants can compose replies simultaneously, leading to overlapping or partially incompatible contributions depending on posting order. Timestamps therefore become a crucial resource for interpreting interactional dynamics. This asynchrony complicates annotation within frameworks such as SDRT or information state update models, which typically rely on a well-defined sequential order of utterances.

STAC and Molweni exemplify a class of task-oriented interactions that are particularly amenable to analytical modelling. They involve explicit goals, relatively stable participation structures, and interactional trajectories that can be interpreted in terms of information exchange and collaborative problem solving. As such, they provide a favourable testing ground for models that conceptualise dialogue as a sequence of information state updates, where each contribution incrementally modifies a shared representation of the task, the domain, or the interlocutors’ commitments. At the same time, they introduce non-trivial complexity: STAC through multilogue interaction and a shared virtual environment, Molweni through strong topical constraints and asynchronous written exchange.

Compared to more sanitised resources such as DailyDialog (Y. Li et al., 2017) – which presents itself as modelling everyday conversation but in practice exhibits highly organised and simplified interactional patterns – STAC and Molweni offer a richer and more challenging empirical basis. They have enabled the community to develop and extend discourse theories, including SDRT, by confronting them with multilogue structure, virtual grounding, and asynchronous coordination. Nevertheless, both corpora remain fundamentally goal-oriented, and this orientation stabilises the interaction in ways that distinguish them from more casual forms of conversation. Moreover, the text-based medium allows for organisational patterns, such as parallel topical threads and non-adjacent responses, that do not always reflect the mechanisms participants rely on in synchronous spoken interaction.

A key limitation shared by these resources concerns the treatment of social purposes in dialogue.

Task-oriented interaction is well suited for identifying and annotating rhetorical relations, as well as for deriving dialogue-level structures such as QUDs. In practice, QUDs are easier to define when they relate to task progress or information needs, but much harder to formalise when the interaction serves social goals that are diffuse, implicit, or only loosely connected to a practical outcome. This limitation becomes particularly apparent when considering everyday conversational activities involving social bonding or low-stakes coordination, such as talk accompanying shared leisure activities. While some theoretical approaches – SDRT in particular – arguably leave more room for accommodating such phenomena, they still presuppose identifiable communicative intentions, leaving certain forms of social interaction difficult to capture despite their prevalence in daily language use.

In this sense, task-oriented dialogue illustrates how analytical convenience can shape theoretical coverage. By privileging interactions structured around practical goals and information exchange, dialogue research risks developing models that are robust within a narrow domain but fragile when confronted with the diversity of everyday conversational practices. Making these constraints explicit through a descriptive framework is therefore essential, not only to delimit the scope of existing analyses, but also to clarify what remains to be accounted for when moving beyond task-oriented interaction.

3.3.2.2 Tool-building perspective: data-collection platforms and induced constraints

Tools designed specifically for data collection within the research community have also played an important role in gathering task-oriented dialogue resources. A wide range of platforms have been created to elicit task-oriented interactions in a controlled yet scalable manner, including chat-based interfaces (Healey & Mills, 2009; Decker et al., 2024), collaborative games, and experimental environments designed to support corpus construction (Chalamalasetti et al., 2023). Examples include generic chat tools developed for dialogue data collection, as well as more specialised setups such as game-based interaction frameworks used in resources like STAC.

In the context of this thesis, we developed a conversational tool (Decker et al., 2024) inspired by the system proposed by Healey and Mills (2009), using the instant messaging application Cinny, which is based on Matrix, a real-time open communication protocol. Cinny is available as a web, desktop, and mobile application, making it convenient for participants to use. It also allows us to host our own server, giving us full control over the experimental data. Our tool consists of Python scripts that connect to the server, create user accounts and chat rooms for participants, and monitor conversations in order to modify some messages. Each participant has a real account as well as an associated bot account. For each conversation, a dedicated chat room is created for each participant, containing the participant and the bot associated with their interlocutor(s). In a two-person conversation, for example, two rooms are created: one for each participant and the bot representing the other participant. Messages are therefore initially received only by the corresponding bot account, allowing the scripts to process and modify them before relaying them to the other room through the bot, making them appear as if they had been sent directly by the interlocutor.

Fig. 3.3 illustrates this process. This architecture enables controlled modifications of messages before transmission, allowing us to introduce specific disturbances into conversations and investigate phenomena such as participants' resilience to misunderstandings. Since the interaction takes place through separate rooms, participants perceive their messages as being sent normally and remain unaware of the manipulations. The scripts also store all conversation data in an SQLite database.

Another advantage of this framework is the ability to integrate widgets into chat rooms, i.e. web pages that participants can interact with while chatting. These can be used to design more interactive experimental tasks, such as cooperative mini-games.

These tools are valuable precisely because they allow researchers to gather dialogue data efficiently and at scale. By embedding participants in a task, they provide an interactional frame that facilitates

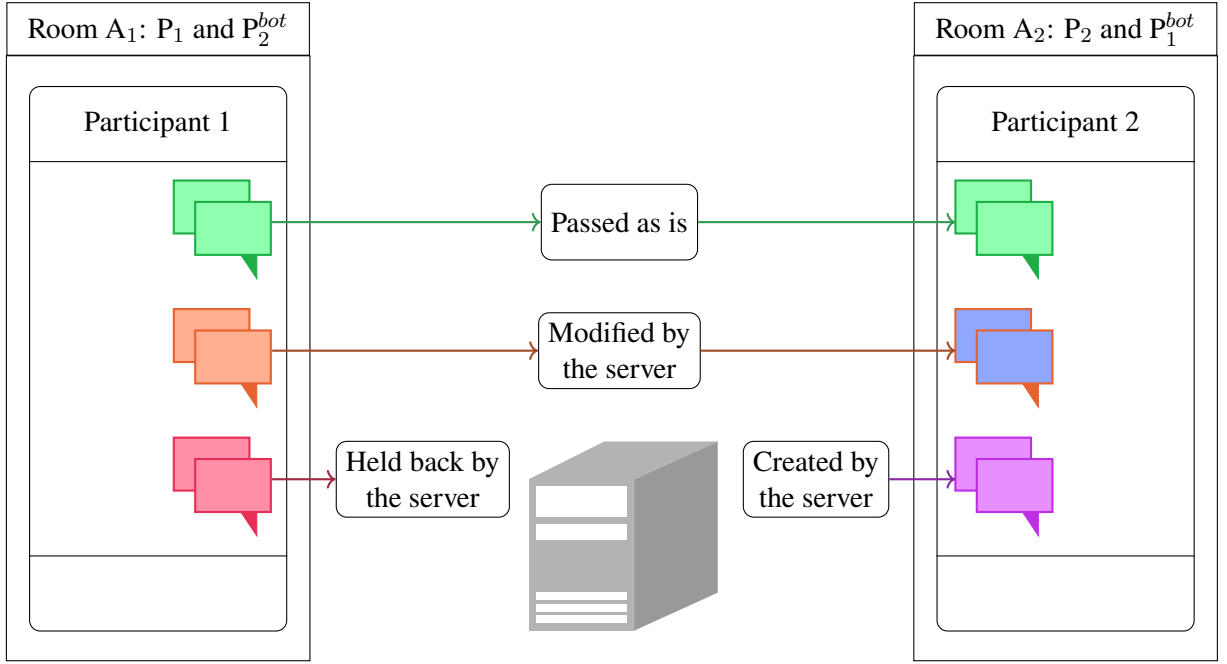


Figure 3.3: Representation of the exchange and manipulation of the messages with our ChatTool

sustained dialogue while reducing the need for extensive prompting or moderation. In the current context, where LLMs increasingly absorb vast quantities of publicly available text, the ability to generate new, well-documented dialogue resources under controlled conditions has become particularly important for the research community.

At the same time, data collection tools are not neutral instruments. By design, they impose constraints on the dialogue they elicit, shaping both the interactional structure and the range of behaviours that can emerge. The task definition, the interface, the turn-taking mechanism, and the available communicative channels all influence how participants engage with one another. From the perspective of the descriptive framework introduced in Chapter 2, such tools typically strongly constrain the environment and medium, while leaving other parameters – such as social dynamics or interactional flexibility – only partially accessible.

These induced constraints are not inherently problematic, provided they are made explicit and taken into account when resources are reused. Task-oriented data collected through specialised tools can be highly appropriate for analytical studies focusing on information exchange, coordination, or rhetorical structure, as well as for training or fine-tuning dialogue models whose intended use aligns with the properties of the resource. However, problems arise when such data is treated as a generic representation of dialogue, or when the constraints imposed by the collection tool are overlooked.

In this sense, tool-based data collection exemplifies a broader methodological trade-off. On the one hand, the community needs efficient ways to produce new dialogue resources in order to maintain empirical grounding and avoid over-reliance on pre-existing, widely circulated datasets. On the other hand, the very tools that make data collection feasible also shape the dialogue they produce. Our descriptive framework provides a means to document and reason about these effects, ensuring that the resulting resources are used in contexts where their constraints are compatible with the research goals.

3.3.2.3 Productive perspective: ultra-exploitation of benchmarks

From a productive perspective, task-oriented dialogue has become the dominant paradigm for training and evaluating conversational agents, most notably through the downstream task of next-utterance prediction. This paradigm underlies the majority of contemporary chatbots and conversational AI systems and has been widely operationalised through benchmark corpora such as MultiWOZ (Budzianowski et al., 2018) and related datasets. In this setting, a system is trained to generate an appropriate next contribution in an ongoing task-oriented interaction, typically framed as a dialogue between a human user and a machine agent.

The tasks targeted in this paradigm are generally defined by a relatively clear and finite set of parameters or slot values that must be retrieved, such as dates, locations, preferences, and availability constraints in domains like travel planning, restaurant booking, or hotel reservation. This formalisation makes task-oriented dialogue particularly attractive from a production standpoint. Success conditions can be specified with relative clarity, evaluation metrics can be derived from task completion, and dialogue progress can be modelled as incremental constraint satisfaction. Historically, such systems relied on rigid architectures such as finite-state machines or hand-crafted dialogue policies. In current practice, however, LLMs dominate this space, offering substantially greater surface-level flexibility while remaining constrained – either explicitly through prompting and system design, or implicitly through training data and evaluation metrics – by the underlying task structure.

The widespread adoption of LLM-based task-oriented systems raises questions that extend beyond technical performance. In particular, it invites reconsideration of what kind of conversation users expect, or are willing to engage in, when interacting with a machine even in narrowly task-constrained settings. Unlike graphical interfaces or web forms, conversational agents invite interactional behaviours that resemble human dialogue, despite the fact that the underlying purpose remains strictly practical. Empirical observations suggest that users adapt their behaviour accordingly: they may simplify their language, adjust politeness strategies, or attribute communicative intent and understanding to the system. As a result, the dialogues produced and consumed in this context cannot be treated as straightforward instances of human-human task-oriented interaction. Rather, they constitute a distinct interactional genre that warrants analysis in its own right.

Table 3.5 illustrates several structural tensions of the task-oriented dialogue paradigm by contrasting the properties of the MultiWOZ corpus with those of contemporary task-oriented dialogue system outputs, which are now very often trained and evaluated on this benchmark.

At the level of participants and interactional structure, MultiWOZ dialogues are presented as dialogues but may in fact be produced by several successive pairs of participants. Each new participant is required to familiarise themselves with the entire dialogue history before continuing the interaction. While this procedure ensures surface-level continuity, it may affect intention attribution, persona consistency, and longer-range discourse coherence. These properties make such data less suitable for modelling interactional phenomena that depend on stable participant identities or paralinguistic cues inferred from transcripts, such as fine-grained intent recognition. By contrast, deployed task-oriented systems are overwhelmingly designed for one-to-one interaction between a single human user and a machine agent. This asymmetry is not structurally necessary but reflects a dominant design choice aligned with efficiency and usability constraints.

Related tensions emerge with respect to roles, commitment, and enunciative status. In MultiWOZ, participants play roles within hypothetical scenarios, committing linguistically to speech acts despite the explicitly fictional nature of the interaction. This produces a form of commitment that is coherent within the corpus but does not straightforwardly transfer to deployed systems, where machine agents cannot sustain the same pragmatic or social commitments as humans. The original Wizard-of-Oz paradigm partially addressed this mismatch by telling the ‘user’ they would be interacting with a machine while

3.3. Applying the Descriptive Framework

Parameter	Sub-	Subsub-	MultiWOZ	LLM-based TODS
Participants			Humans	Human-Machine
	Participation structures	Number	2 (≥ 2 workers)	2
		Functions	One speaker - addressee pair	One speaker - addressee pair
	Enunciative status	Anonymity	Anonymous	?
		Role instantiation	Fake information seeker - holder	Self
		Persona instantiation	Self - Self as fake expert	Self
		Spontaneity	✗	✓
	Commitment	✓?	✗	
	Social Dynamics	Hierarchy	Information seeker - holder	User - Tool
		Conversational history	✗	∅?
	Language			English
	Control		✓?	✓?
Environment & Medium			Chat	Chat
	Physical context		∅	∅
	Visibility	Gaze	∅	∅
		Gesture / Facial expression	ASCII emojis	Emojis
	Audibility	Prosody / Tone	∅	∅
		Noise	∅	∅
(Verbal) Interaction				
	Co-construction	Coordination signals	∅	∅
		Turn-taking	∅	∅
		Completion	∅	∅
	Time	Synchrony Ephemerality	Semi-synchronous Saved for the task	Semi-synchronous Persistent by design?
Conversational Activity and Purpose			Booking	Booking
	Epistemic grounding	World setting	Hypothetical?	Real
		World stability	Fixed	Fixed?
	Purpose	Practical	Information exchange + Task	Information exchange + Task
		Intention	?	?
	Activity	Success condition	Valid slots, successful booking	Valid slots, successful booking
		Vocabulary	Free	Free
		Structure	Some kind of plan	Some kind of plan

Table 3.5: Description of the dialogical setting in the MultiWOZ resource and in TODS outputs

the other participant was asked to play the system. However this may also introduce artefacts whose interactional consequences remain underexplored. The pretence is abandoned in the MultiWOZ corpus, making the participants closer to their real selves but producing this mismatch in possible commitments between the training and evaluation data, and the actual model output.

Hierarchy and role asymmetry further complicate the transfer from training data to deployed systems. In human-human task-oriented interaction, one participant typically seeks information or assistance, while the other provides help. In contrast, a chatbot is primarily perceived as a tool rather than as a social peer. Training on human-human interactions may therefore induce expectations or behaviours that do not align with actual usage conditions. This mismatch affects not only surface-level phenomena such as politeness or initiative-taking, but also deeper aspects of dialogue organisation and user accommodation strategies. Similarly, although task-oriented dialogue is often conceptualised as information-seeking interaction, real usage scenarios vary considerably. Some users may already know exactly what they want and merely require execution rather than guidance. Others may expect recommendation, negotiation, or confirmation. These differences in user stance have implications for dialogue initiative, grounding strategies, and system behaviour, yet they are not always adequately represented in training data.

Epistemic grounding constitutes another source of discrepancy. In MultiWOZ, the interaction is grounded in a hypothetical yet realistic scenario: the domains are familiar, and the databases are derived from real-world information. However, the task itself is never actually performed; only its linguistic simulation is realised. By contrast, deployed systems are expected to act in the real world by executing bookings or retrieving live information. This disparity can influence dialogue structure, grounding strategies, and user expectations. It is further compounded by the fact that, while the database is fixed during training, LLM-based systems are prone to hallucination, potentially inventing entities such as hotels, restaurants, or transport options that do not exist. Although recent models have improved in this respect, hallucination cannot be fully eliminated by design and directly affects the reliability of such task-oriented dialogue systems.

Recent developments further complicate this picture. The increasing use of personalisation mechanisms such as persistent user profiles, cookies, or interaction histories, means that task-oriented systems may be treated as having some form of continuity across interactions. While such mechanisms can improve efficiency and user experience, they also raise ethical and interactional concerns. In particular, they risk fostering an illusion of familiarity or social continuity with a system that cannot sustain genuine long-term commitment. Given current tendencies for LLM-based conversational agents to be designed with anthropomorphic features (Xiao et al., 2025), these issues must be addressed explicitly when defining both training resources and system behaviour.

Finally, the environment and medium of interaction remain strongly constrained in current systems. Most task-oriented systems operate through text-based interfaces, with no access to shared physical context and no multimodal cues such as gaze, gesture, or prosody. Under these conditions, reliance on text-based dialogue data is a reasonable design choice, but it necessarily limits the interactional phenomena that can be modelled. As development will undoubtedly move toward spoken or multimodal conversational agents, additional dimensions such as turn-taking, synchrony, incremental understanding, and coordination signals will become central. Spoken dialogue systems have historically struggled with these aspects (Addlesee et al., 2020), and while recent advances have improved performance, co-construction processes and fine-grained interactional timing remain active areas of research.

Evaluation practices further reveal the limits of the current paradigm. Task success is typically assessed using slot-filling accuracy or task completion metrics. While these measures appear reasonable, they presuppose that the goal of the interaction is unambiguous and recoverable from the dialogue. In practice, however, the actual goal may not align neatly with the intent imposed by the experimental setting. In real usage, it is the user's self-determined intent that matters, making misalignment between training data, evaluation protocols, and deployment scenarios a significant issue.

More critically, dialogical quality is often evaluated using metrics such as BLEU or other semantic similarity scores. These metrics do not correspond to any dimension of the descriptive framework proposed in Chapter 2 and rest on assumptions that are fundamentally at odds with the nature of dialogue. Dialogue does not admit a single optimal next utterance against which all others can be ranked. Rather, it unfolds through negotiation, accommodation, and co-construction. While some utterances may appear incoherent in isolation, it is ultimately up to participants to accommodate, repair, or reject them. Evaluating dialogue quality through similarity to a gold sentence therefore misrepresents interactional flexibility and obscures core dialogical properties.

Although the proposed framework does not directly resolve the evaluation crisis currently faced in dialogue research, it helps make visible the gap between training resources and system outputs. As illustrated in Table 3.5, systematic differences exist between the properties of the MultiWOZ corpus and those of the dialogues generated by systems trained on it. Many of these differences stem from inherent properties of current model architectures and cannot be addressed through fine-tuning alone. A more promising direction for evaluation would therefore be to assess how such variations affect the dialogical experience of real users, rather than relying solely on abstract similarity metrics.

Overall, this scenario demonstrates that task-oriented dialogue generation is far from a solved problem, even in its most established settings. While the task goal provides a clear anchor for system design and evaluation, nearly all other parameters require explicit modelling choices. The descriptive framework proposed in Chapter 2 offers a way to make these choices visible, to align training resources with intended system behaviour, and to situate technical decisions within a broader understanding of dialogue as interaction rather than mere information exchange.

3.3.2.4 Key takeaways and implications

This scenario highlights that task-oriented dialogue research still faces significant challenges across analytical and productive dimensions. Our descriptive framework proves particularly valuable not only for specifying desired system behaviour in production tasks, but also for guiding resource selection and design choices, and for reflecting on the limitations and ethical implications of the tools and applications we develop.

Task-oriented settings are attractive because they offer greater control and reproducibility: clear goals, bounded domains, and explicit success criteria make systematic experimentation feasible. At the same time, this control imposes empirical limitations. Such settings should therefore be regarded as an initial step in understanding dialogue phenomena, rather than as fully representative of natural interaction. Yet even within constrained tasks, participants display complex behaviours – adaptive strategies, negotiation, coordination, and social signalling – which demonstrate that task-oriented dialogue can still reveal rich interactional patterns.

A critical insight emerging from this analysis is the strong interdependence between dialogue parameters. Constraints imposed by the task, the medium, or participant roles inevitably affect other aspects of interaction, including initiative, grounding, commitment, and coherence. Studying these dependencies requires careful experimental design: selectively constraining some parameters allows us to observe their ripple effects across the interaction. In this matter clinical interviews, the focus of the next section (Section 3.3.3), can be interesting to study because they allow us to study interaction features in a disfunctional setting, thus requiring adaptations from the participants. They are also a clear example of dialogue whose purpose goes way beyond a clearly defined task or information request, highlighting that conversation is rarely reducible to simple metrics or slot-based evaluations.

Overall these observations underscore the need to continuously question and adapt our practices as we work on increasingly complex tasks. Task-oriented research is not a final destination, but a structured lens that allows us to explore phenomena systematically, anticipate interdependencies, and ground technical

and ethical decisions in a broader understanding of dialogue as dynamic social interaction.

3.3.3 Scenario 3: Clinical Interview

The clinical interview designates a specific form of institutional dialogue that emerged alongside the development of clinical medicine and, later, clinical psychology and psychiatry. The term clinical derives from the Greek *klinikos* that relates to the practice of medicine at the patient's bedside, emphasising direct observation and situated interaction. Within psychology and psychiatry, the clinical interview became a central methodological and therapeutic instrument: a structured yet flexible conversational exchange aimed at understanding and alleviating psychological suffering.

A clinical interview typically involves a dyadic interaction between a therapist and a patient. Its overarching purpose is neither simple information exchange nor the completion of a practical task, but the exploration and transformation of the patient's experience. The goal is curative in a broad sense: to relieve distress, clarify difficulties, or support psychological change through attentive listening, interpretation, and guidance.

This interaction is intrinsically asymmetrical. The therapist occupies a professional role defined by expertise, responsibility, and ethical obligation, while the patient is invited – sometimes with difficulty – to articulate personal experiences, vulnerabilities, and sources of suffering. Unlike ordinary reciprocal conversation, the therapist is not expected to disclose comparable personal material. The asymmetry is functional: it structures the interaction so that one participant's inner world becomes the focal point of sustained attention.

Clinical interviews are also sensitive interactions. They require patients to verbalise experiences that may be painful, confusing, or only partially accessible to consciousness. Expression may be hesitant, fragmented, or indirect. Patients may struggle to find words, may contradict themselves, or may avoid certain topics altogether. In this sense, clinical dialogue goes beyond the 'basic' cooperation assumed in ordinary conversation. It relies on the therapist's ability to interpret implicit meanings, attend to hesitations and reformulations, and integrate verbal and non-verbal cues into a coherent understanding of the patient's situation.

Although clinical interviews can be described as goal-oriented as there is an intention to improve the patient's condition, they are orthogonal to task-oriented dialogue in an important sense. Their purpose is not practical execution but transformation of subjective experience. They also provide a particularly rich site for examining the limits of language: in many clinical contexts, communicative abilities may be affected by psychological or neurological conditions, forcing both participants to adapt their interactional strategies.

In what follows we examine clinical interviews through the three complementary perspectives we have explored in the previous scenarios. Firstly the analytical perspective enables us to shed light on what this type of data can teach us about language use and yet how complex it is to work with. Secondly a productive perspective leads us to question recent trends of building psychological counselling models, which seek to automate a complex type of interaction that goes much further than words. Finally we discuss tools that can be seen as a bridge between those two perspectives where they are meant to support therapists and clinicians in their analysis. Language is very complex as we said and NLP/CL approaches can help highlight some features of particular importance in a clinical monitoring, thus helping them focus on particular aspects they may not have otherwise noticed or they might have had an intuition about but the more systematic study NLP tools allows would further validate.

3.3.3.1 Analytical perspective: studying dialogue phenomena through the lens of dysfunctions

Clinical interviews occupy an intermediate position between ordinary conversation and institutional discourse. On the one hand, they resemble everyday dialogue in that topics are often introduced by the patient and may evolve in relatively open-ended ways. The therapist encourages narrative development, association, and reflection rather than strictly directing the exchange toward predefined informational goals. The interaction can appear relatively unstructured, allowing associations, hesitations, and reformulations. On the other hand, the interaction is clearly institutional: participant roles are asymmetrical, expectations normatively defined, and the encounter embedded in a professional, ethical, and sometimes medical framework.

From an interactional perspective, clinical interviews provide a privileged setting for studying meaning-making beyond literal content. Patients may express distress indirectly, through metaphor, silence, laughter, or narrative fragmentation. They may require several attempts before approximating what they really want to say, and they may never fully achieve a formulation they consider satisfactory. The therapist's task is to integrate these partial cues into an evolving interpretation, while remaining aware that convergence between representations can never be guaranteed. Although all dialogue involves alignment and accommodation, clinical interviews intensify this process: therapeutic progress depends on the therapist's capacity to approximate the patient's experiential world as closely as possible without presuming full access to it.

Clinical interviews are also particularly informative when communicative capacities are altered. Certain mental disorders affect coherence, turn-taking, referential clarity, or pragmatic appropriateness. In such cases, participants must develop compensatory strategies to maintain mutual understanding. Studying these adaptations can help reveal the interdependence of dialogue parameters: when one dimension of interaction is weakened, others are reconfigured. Clinical dialogue thus allows us to observe how dialogue functions under strain and which mechanisms prove most resilient.

This analytical richness, however, comes with methodological and ethical challenges. Clinical interviews are sensitive and confidential by nature. Access to such data is constrained by ethical requirements designed to protect patients' privacy and well-being. Recruitment may be difficult, attendance inconsistent, and longitudinal data collection fragile. Ethical review processes are essential but may impose constraints that complicate experimental control. Furthermore, methodological issues arise when attempting to compare clinical populations with control groups: certain structural aspects of clinical interviews, such as the therapeutic asymmetry or the emotional stakes, are difficult – if not impossible – to replicate authentically in non-clinical settings.

The SLAM (Amblard, 2018) corpus illustrates these tensions. One of its main objectives was to collect spontaneous clinical interviews between a therapist and patients with schizophrenia, while also gathering dialogues from a control group to highlight distinctive dialogic signals. The therapist's role was primarily to sustain the interaction rather than strongly guide it.

Yet aligning the two settings proved challenging in practice. Table 3.6 details some of the similarities and differences between the patient and control conditions, focusing on the parameters of our framework that are most directly related to the dialogical configuration.

At first glance, the participant structure appears symmetrical. In both groups, interactions are dyadic, involving one therapist and one interlocutor. The therapist rarely occupies the role of main speaker as their function is to support and sustain the dialogue rather than dominate it. When the therapist does take on a more active verbal role, this tends to be a response to individual reluctance or conversational breakdown rather than a group-specific characteristic. The enunciative status is also comparable: participants know who they are speaking to and are not instructed to adopt fictional roles. In that sense, both settings involve authentic self-presentation rather than role-play.

However, more subtle dimensions are harder to control. The persona displayed by participants is not

experimentally fixed, yet it is shaped by the institutional asymmetry inherent in clinical interviews. The therapist occupies a position of professional authority, and this asymmetry may induce desirability biases, particularly in the control group. Control participants, aware of being observed in a quasi-clinical setting, may orient their discourse toward socially acceptable or consensual topics. Patients with schizophrenia, by contrast, may be less influenced by such strategic self-presentation, partly due to impairments in theory of mind or altered social-cognitive processing. Although the asymmetrical structure is formally identical across groups, its interpretation and its interactional consequences may differ significantly.

In theory, some of these biases could be mitigated through double-blind procedures, for example, by ensuring that the therapist does not know whether their interlocutor belongs to the patient or control group, and that participants are unaware of the specific hypotheses under investigation. In practice, however, such designs are difficult to implement. The sensitive nature of clinical encounters, combined with the ethical requirement of fully informed consent – especially when working with vulnerable populations – limits the feasibility of masking group membership. Furthermore, practical constraints make true blinding unrealistic.

Indeed the physical environment provides a concrete illustration of this difficulty. Although interviews in both groups were conducted in dedicated rooms, the location differed. Patients were seen in the hospital, whereas control participants were either interviewed at the university or, in some cases, at the hospital if they were affiliated with it. The setting itself thus functioned as an implicit marker of group membership. Standardising location would have required either bringing all participants to a neutral space outside the hospital – which would raise ethical and logistical complications for clinical patients – or inviting all control participants into the hospital environment, which presents its own institutional constraints.

In addition, the relational histories differed subtly between groups. While the therapist and their interlocutor met for the first time in both conditions, the broader institutional background was not equivalent. For patients, the therapist could be perceived as part of an existing therapeutic system – a continuation of medical or psychiatric care – thereby creating a familiar framework in which speaking about one-self is expected and legitimised. Control participants, in contrast, did not share this therapeutic context. Instead, they often shared academic or professional affiliations with the recruitment environment, particularly when recruited from a psychology department. This difference in background may influence topic selection and discursive orientation: patients more readily engage in discussions centred on internal states and personal experiences, whereas controls tend to gravitate toward more neutral, socially shared topics.

These divergences may not all play a role in the differences found in terms of verbal interaction or gaze patterns, as described in Lecomte et al. (2022), but they should be taken into account for their analysis. They also highlight the additional complexity associated with building a resource involving patients, not only in terms of institutional constraints but also due to the intrinsic asymmetry between patients and controls.

3.3.3.2 Tool-building perspective: Supporting the analysis of complex resources

If we follow the pipeline from data collection to analysis followed to gather the SLAM corpus, tool-building quickly appears as a central component of the research process. The corpus does not consist solely of verbal exchanges: it combines audio and video recordings with eye-tracking data. Analysing such material therefore requires a layered technical infrastructure capable of processing multiple modalities and synchronising them with sufficient precision to preserve their interactional relevance.

One aspect concerns transcription and signal processing. Audio recordings must be converted into analysable linguistic material, which could be done with ASR systems. However, clinical dialogue presents challenges that differ from those encountered in standard conversational corpora. Speech may be disfluent, fragmented, or marked by atypical prosody. Overlapping speech, long pauses, low vocal intensity, and environmental noise – due to hospital settings or recording constraints – further complicate

Parameter	Sub-	Subsub-	Patient	Control
Participants			Humans	Humans
	Participation structures	Number Functions	2 Main speaker - Main addressee as ‘continuer’	
	Enunciative status	Anonymity	Physical co-presence	
		Role instantiation	Self	
		Persona instantiation	Self* - Self as therapist	
		Spontaneity	✓	
		Commitment	✓	
	Social Dynamics	Hierarchy	Patient - Therapist*	
		Conversational history	First meeting Part of the therapeutic circuit	Common study background
Environment & Medium			Dedicated room	
	Physical context		Face-to-face in hospital	Face-to-face in university or hospital
Conversational Activity and Purpose			Clinical interview	
	Purpose	Practical Intention	?	?
	Activity	Success condition		

Table 3.6: Description of the dialogical setting in the SLAM (patient and control) corpus. The asterisks indicate parameters that are formally identical across groups but whose instantiation may actually differ between the groups due to the inherent institutional asymmetry of clinical interviews.

recognition. Importantly, many of these ‘difficulties’ are not incidental noise but potentially meaningful signals, especially when working with populations affected by psychiatric disorders. ASR systems must therefore be evaluated not only for word error rate but also for their ability to preserve clinically relevant phenomena.

The second layer concerns eye-tracking data. Mapping gaze patterns onto conversational flow requires precise temporal alignment between modalities. This involves synchronising gaze coordinates with utterances, identifying fixation points within a dynamic physical environment, and distinguishing between meaningful gaze behaviour and artefacts produced by imperfect calibration or participant movement. In practice, eye-tracking setups in clinical environments are rarely optimised laboratory installations. Equipment may not always be configured by technical specialists, and data may contain substantial noise. Cleaning, aligning, and interpreting gaze data thus demands both computational tools and methodological caution.

Once multimodal data has been processed, the analytical phase introduces further challenges. Existing tools for discourse parsing, dialogue act tagging, or syntactic analysis have typically been developed and trained on more canonical corpora. Clinical dialogue, by contrast, may be less structured, more elliptical, and more context-dependent than task-oriented or scripted interaction. Topic shifts may be subtle, referential chains may remain partially implicit, coherence may unfold across extended stretches of narrative rather than through explicit markers. Off-the-shelf models may therefore fail to capture interactional dynamics that are central to therapeutic exchange.

A structural constraint further complicates tool development: clinical data is not easily shareable. Ethical regulations and privacy concerns prevent uploading recordings to external proprietary services. Privately owned LLM-based APIs cannot simply be used for transcription, annotation, or exploratory analysis. Secure, locally deployable solutions become necessary, which increases the technical burden and costs on research teams. Moreover, the size of clinical corpora is often limited. Recruitment constraints and ethical considerations restrict the amount of data that can be collected, making large-scale deep learning approaches less directly applicable. Tool-building in this domain therefore requires careful adaptation to small, sensitive datasets rather than reliance on data-hungry architectures.

Finally, the question of automation must be addressed explicitly. While LLMs and other AI tools can assist with transcription, annotation suggestions, or pattern detection, they cannot replace the interpretive work performed by trained clinicians. Clinical dialogue involves reading between the lines, integrating contextual knowledge, and recognising subtle shifts in affect. Tool-building in the clinical domain should therefore be framed as instrumentation, not substitution. Computational systems can support researchers and therapists by organising, visualising, or highlighting patterns in complex multimodal data. They cannot, and should not, be considered equivalent to the situated, responsible expertise required to interpret and respond to vulnerable interlocutors.

3.3.3.3 Productive perspective: risks and limits of automated counselling

Recent years have witnessed growing interest in developing computational systems for psychological counselling. These range from emotion-recognition tools designed to support various tasks such as opinion mining or detecting distress signals online to conversational agents explicitly intended to simulate therapeutic dialogue (Y. M. Cho et al., 2023; H. Li et al., 2023; Sarkar et al., 2023; Pereira et al., 2024).

The argument often advanced in favour of such systems is pragmatic: access to mental healthcare is limited, counselling is costly, and many individuals lack adequate support. From this perspective, automated systems appear as scalable supplements capable of partially alleviating structural shortages. Yet this framing raises concerns. If technological substitution is normalised as an acceptable response to systemic underinvestment in public healthcare, models may gradually be presented as adequate replacements rather than provisional aids. Such a shift risks weakening collective efforts to secure broader access to

qualified human professionals.

From an interactional standpoint, counselling systems attempt to automate a form of dialogue that extends far beyond lexical exchange. Therapeutic interaction involves responsibility, ethical commitment, long-term relational work, and a capacity to interpret subtle affective and contextual cues. Although confiding in a machine may feel less intimidating to some users – particularly if it reduces fear of judgement – this apparent advantage is ambiguous.

Indeed, inherent differences exist due to the structural architecture of conversational AI as we describe in Table 3.7. First the lack of actual self for the machine makes it different in terms of spontaneity and commitment. It also aims at mimicking a therapist through its linguistic patterns thanks to a specifically chosen prompt. However human therapists are way more flexible than this as they have the ability to adapt to the patient with regards to how they frame their therapist role. Some patients may require more or less of a stereotypical counsellor and it is part of the therapist's job to adapt to it. While we can improve LLMs to make them more aware of such things, it will remain that they require some kind of prompting to restrain the type of role and persona they are supposed to mimic. Iftikhar et al. (2025) show that LLM counsellors typically follow very stereotypical patterns of therapeutic dialogue which leads to 'oversimplified and irrelevant template advice'.

A further structural risk concerns the tendency of conversational AI systems to accommodate the user. Depending on the topic at hand and the user's state of mind, this could lead to the reinforcement of confirmation bias but also validate harmful beliefs or behaviours (Iftikhar et al., 2025; Cheng, Hawkins, & Jurafsky, 2026). In the opposite direction, a model that challenges the user may be dismissed precisely because it lacks human authority and accountability. In the recent years social media have seen a rise in content on mental health and psychological troubles, which is a good thing if it can help destigmatise this topic, encourage people to seek help, or provide support and acceptance to individuals. However, a part of this content also pushes towards self-diagnosis through the description of relatable so-called symptoms of various disorders – ADHD being a particularly popular one – while the described behaviours are common ones not exclusive to a disorder (Hartnett & Cummings, 2024). For individuals already inclined toward a particular diagnostic label, seeking confirmation from a language model may amplify misconceptions.

This is particularly worrying in the current context where many of these models are developed by private companies on which we have no influence and whose motivations are not necessarily aligned with the well-being of users. The pharmaceutical industry is known to have financial ties with the psychiatric world, through contributors to the Diagnostic and Statistical Manual of Mental Disorders (DSM-5-TR) for example (Davis et al., 2024), which can influence how diagnoses are made and what drugs are prescribed. It is not inconceivable to imagine that companies developing LLMs used for psychological counselling would also have financial interests in the pharmaceutical industry, which could lead to a conflict of interest in the advice given by the model.

Beyond these epistemic and institutional issues, the conversational configuration itself differs markedly between human and machine counselling. Traditional therapy typically takes place in the therapist's office, although online counselling has become increasingly common, particularly since the COVID-19 pandemic, and some therapists also conduct sessions outdoors or in other non-traditional environments. In all of these settings, modalities beyond linguistic content, i.e. the physical context as well as visual and auditory channels, remain much more present than in contemporary approaches to conversational AI counselling, which largely rely on text-based chat interfaces. This difference affects not only the expressive resources available to participants but also the dynamics of turn-taking, emotional attunement, and relational depth.

Privacy and ephemerality further distinguish the two contexts. In human therapy, confidentiality is generally institutionally regulated, although the specific requirements may vary across countries. While therapists may take notes, their storage, access, and use are typically governed by professional and legal standards. Interactions with conversational AI, by contrast, often involve data that is accessible to the

service provider and may be subject to secondary uses. In a political climate where minority rights are increasingly vulnerable in some regions, the possibility that sensitive disclosures could be stored, analysed, or repurposed raises serious concerns.

Differences also emerge at the level of purpose and intentionality. A therapist operates within an explicit ethical framework and works toward collaboratively defined, evolving therapeutic goals. A language model, by contrast, has no intrinsic intentions and its ‘success condition’ is determined by training objectives and reward functions. Its representation of the world is limited to patterns inferred from data, and users can, in principle, negotiate or reshape that world within the interaction.

This difference in intentions and purposes is also reflected in the way conversational history is handled. Therapy typically takes more than one session and the therapist takes notes and plans the counselling journey based on the patient’s unique traits and needs. In opposition to this, conversational AI works with a context window which can be more refined than a simple dialogue history with summarisation and attention but remains fundamentally different from the structured, longitudinal planning that characterises human therapy. Setting up counselling goals is not as simple as choosing a final goal when learning a language or a new skill, a treatment plan is carefully tuned and refined through interactions where the patient themselves do not have full control over the final destination of their journey.

3.3.3.4 Key takeaways and implications

Clinical dialogue highlights a fundamental limitation of purely statistical modelling: therapy is not reducible to generating plausible next sentences. It involves responsibility, accountability, and long-term relational work that cannot be easily encoded in current architectures, and should arguably not be detached of a genuine human bond.

Nevertheless NLP and CL can provide practical support in the analysis of pathological conversations. On the one hand, NLP can facilitate the preprocessing of such data through robust ASR systems or multi-modal synchronisation tools. On the other hand, CL can assist in identifying particular dialogical features that may be clinically relevant, such as recurring speech patterns, gaze behaviours, and linguistic markers associated with specific conditions, or tracking thematic trajectories across sessions. However, these systems should be understood in analogy with specialised tools in medicine: they require trained professionals for interpretation and decision-making. Just as a scalpel presupposes surgical expertise, analytical tools for clinical dialogue must remain under qualified human control. Their outputs can inform clinical reflection, but they cannot replace the interpretive judgment and ethical responsibility of practitioners.

In the current political and economic climate, where public services are increasingly subject to cost-reduction pressures, the development of tools that move toward automated counselling risks being misinterpreted as a substitute for human professionals. Research in this area must therefore remain explicit about its scope, its intended use, and its limitations.

3.4 Targeted Dialogue Generation Control

Beyond the theoretical support that the framework provides for the analysis and description of dialogue data, it can also be leveraged to guide the development of dialogue systems. This section discusses two complementary strategies that could support NLP research in this direction. The first, presented in Section 3.4.1, is to use the framework to guide the adaptation of dialogue generation toward particular interactional configurations. The second, considered in Section 3.4.2, is to use it to inform the design of evaluation metrics that are more sensitive to specific dialogical competencies.

Parameter	Sub-	Subsub-	Therapist	conversational AI
Participants			Humans	Human - Machine
	Participation structures	Number	2	
		Functions	Main speaker - Main addressee as continuer	Main speaker - Main addressee?
	Enunciative status	Anonymity	Physical co-presence	Self - No self
		Role instantiation	Self - Self as expert	Self - Self as expert's mimicry
		Persona instantiation	Self - Self as therapist	Self - 'Universal' therapist
		Spontaneity	✓	?
		Commitment	✓	✓ - ✗
	Social Dynamics	Hierarchy	Patient - Therapist	Machine
		Conversational history	Therapeutic follow-up	Synthetic
	Environment & Medium			Therapist's office*
	Physical context		Shared	∅
Visibility	Gaze	✓	∅	
	Gesture / Facial expression	✓	Emojis	
Audibility	Prosody / Tone	✓	∅	
	Noise	✗	∅	
(Verbal) Interaction			Maximal	∅
	Co-construction	Coordination signals	✓	∅
		Turn-taking	✓	∅
		Completion	✓	∅
	Time	Synchrony Ephemerality	Synchronous Notes	Semi-synchronous Persistent by design
Conversational Activity and Purpose			Psychological counselling	
	Epistemic grounding	World setting	Real	?
		World stability	Fixed	Negotiated
	Purpose	Practical	?	?
		Intention	?	?
	Activity	Success condition	?	?
		Structure	Clinical interview	?

Table 3.7: Description of the expected dialogical setting in therapist-patient conversation vs conversational system-patient. The asterisk indicate a parameter that can vary due to the multitude of possible real-life settings in which therapy can take place.

3.4.1 Generation control

The evolution of dialogue systems from rule-based architectures to neural networks and large language models (LLMs) has completely changed the paradigm of dialogue generation. While statistical approaches have enabled impressive advances in natural language generation, with more varied outputs and less rigid structures, they have also introduced new challenges. One of the most pressing is the difficulty of controlling the generation in a reliable way. LLMs, in particular, are known for their unpredictability and sensitivity to prompts (Zhuo et al., 2024; Razavi et al., 2025), which can lead to outputs that deviate from desired interactional configurations. The unreliability of prompt-based control has motivated research into model adaptation techniques, including knowledge-enhanced generation (Yu et al., 2022; Hu et al., 2024), which aims to integrate structured information into the generation process to improve control and alignment with specific dialogue scenarios. In this perspective, the descriptive framework we proposed in Chapter 2 can serve as a guide for the design of such knowledge-enhanced generation systems. In the following, we explain how it could be used to inform the application of an ontology-based approach to conversational control for LLMs, proposed by Gendron et al. (2025) (Section 3.4.1.1). We then discuss how this approach could be extended to synthetic dialogue generation, which requires to produce *all* utterances in a conversation rather than relying on a human interlocutor to provide some of them (Section 3.4.1.2).

3.4.1.1 Dialogue agents control

An ontology is a structured representation of knowledge that defines concepts, relationships, and constraints within a specific domain. In the context of dialogue systems, an ontology can be used to formalise the concept of conversation and to provide a structured framework for controlling the generation of dialogue. Gendron et al. (2025) propose an ontology-based approach to conversational control for LLMs, which, for a given parameter to control for, leverages a decision tree trained on annotated data to guide the model's output. The idea consists in describing categories of utterances in an ontology, defined with respect to the parameter of interest, and then inferring the ontology class of the next utterance based on the previous one during generation.

In practice, they suggest to use a dataset annotated with the categories that operationalise the parameter to train a decision tree that can predict the class of an utterance based on various meta-information. For example, they illustrate this approach with the parameter of language proficiency, that they represent with the Common European Framework of Reference for Languages (CEFR) levels. The classes are thus the six CEFR levels, *A1*, *A2*, *B1*, *B2*, *C1*, and *C2*, and they are modelled thanks to a wide range of linguistic features such as the 'average word length' or the 'pronoun density' (Gendron et al., 2025, p.3). By fitting a decision tree on this annotated data, they infer quantitative rules from these linguistic features, which are then used to define the ontology classes. This decision tree can then be used to predict the ontology class of an utterance, which is wrapped in the corresponding label in the form $[o.C : C_i] u_j [o.C : C_i]$, where *o* is the ontology, *o.C* is a concept from *o*, *C_i* is the *i*th sub-class of *C* in *o*, and *u_j* is the utterance from the *j*th data sample. These 'label-wrapped data samples' are used to fine-tune the LLM that is meant to be used as dialogue agent. A conversation strategy, which specifies how to predict the class of the next utterance based on the current one, is then used to control the generation of the dialogue agent.

This approach allows to control the generation of dialogue agents along a specific dimension, as long as a definition of the parameter can be formalised in an ontology and a dataset annotated with the corresponding classes is available. The example presented by Gendron et al. (2025), i.e. language proficiency, is very suited to this approach as it can be defined in a relatively straightforward way. However, conversation involves much more complex dynamics as we have discussed in Chapter 2. Even apparently simple parameters such as the `LANGUAGE` can involve complex interactional dynamics. While the method

proposed by Gendron et al. (2025) can be used very directly to control the language of the generated utterances, hence participating in reducing unwanted language switches LLMs are known for due to their not-so-universal multilingualism, accounting for code-switching within an utterance is more complex. Some languages are well-known for their tendency to involve code-switching and the code-switching versions are sometimes labelled with their own name such as Hinglish, the combination of Hindi and English. While including these languages in the ontology classes would be a possible option, it would lack flexibility when facing unplanned code-switching patterns. A more flexible approach would be to rely on weighted classes rather than unique ones, allowing to represent the probability of each language to be employed in an utterance. This would still require large-scale data collection and analysis work in order to fine-tune the model on realistic code-switching patterns (Ahn et al., 2020; Khan & Noble, 2022; Sinclair & Fernández, 2023).

This highlights the importance of carefully selecting relevant data, which is what we have been advocating for since the beginning of this work. Indeed, training the decision trees at the root of the ontology-based approach on data that looks nothing like the target scenario is unlikely to yield satisfactory results. The more theoretical and descriptive application of the framework we proposed in Section 3.3 can help formulate the objectives of dialogue generation control and to select data accordingly.

Controlling dialogue generation can also help enhance synthetic dialogue production, which we discuss in the next section.

3.4.1.2 Synthetic dialogue generation control

Generating synthetic dialogue data has become increasingly common in NLP research, as it makes it possible to produce large-scale datasets at reduced cost for the experimenters¹⁰ in a short amount of time. It also theoretically allows researchers to produce data that highlights specific phenomena of interest, which is where the ability to control the generation of dialogue can be particularly useful. However, the production of synthetic dialogues is more complex than the generation of single utterances as was the case for the dialogue agents described in Section 3.4.1.1. Indeed, the generation of a dialogue requires generating *all* utterances in a conversation, which means that each utterance is conditioned by those that precede it, which are themselves synthetic, and no human input is present to guide the process. For this reason, the dialogue strategies must be designed with a holistic view rather than a local utterance-to-utterance one.

If we focus on an apparently simple parameter from our descriptive framework, such as the number of participants, it quickly becomes apparent that that it is not sufficient to indicate how many participants the generated conversation should involve. Indeed, while current synthetic dialogues tend to follow a very rigid alternation between participants, real-life conversations are much more complex and involve various patterns of turn-taking and varying levels of participation. Therefore, controlling the number of participants is not enough to ensure that the generated dialogues reflect realistic interactional dynamics. Instead, corpora matching the intended use of the synthetic data should be analysed to characterise speaker behaviour. This includes the distribution of turn lengths before speakers yield the floor, as well as levels of participation in multilogues. Indeed, not all participants are always necessarily active, and reproducing subgroup dynamics is another important aspect in synthetic data production. Another challenge posed by multilogues is that turn allocation becomes much less straightforward. Distinguishing between utterances that explicitly select the next speaker and those that leave the floor open or only restrict the set of possible next speakers is crucial. These reflections show that the number of participants cannot be considered individually but rather as one component of a net of parameters that influence each other. Again, this highlights that dialogue is far from a monolithic concept and that it must be considered globally to capture

¹⁰The full cost of generative AI should not be reduced to the one paid by the experimenters, as we further discuss in Chapter 8.

its richness.

Of course not all parameters are of equal importance depending on the use-case. ENVIRONMENT AND MEDIUM, though rather irrelevant in terms of scope of dialogue agents in their early years, is going to become increasingly crucial with the rise of multimodal and embodied agents, bringing new challenges in terms of control, truth grounding, and hallucination mitigation.

Find the relevant reference for this (based on Claire Bonial’s intervention at the DMR 2026 panel): LLM agents display fundamentally different strategies from humans when linguistic information conflicts with perceptual evidence. Humans tend to privilege perception over language in such situations, whereas LLM outputs show a much heavier reliance on linguistic information. [Amandine]

3.4.2 Evaluation metrics

Our goal when framing our descriptive framework as a way to produce evaluation metrics is not to circumvent the need for data, which we consider foundational in NLP/CL. Nor is the aim to establish a corpus or paradigm as gold standard for evaluating dialogue. Although the dimensions of our descriptive framework have been outlined and explained in terms of how different communicative settings differ from face-to-face dialogue, we certainly do not wish to imply that all theories and applications should be evaluated by comparison with face-to-face interaction. One of the fundamental principles on which we base our approach is that the scope and objectives of the task must determine the type of data involved. Consequently, dialogue evaluation metrics must be able to reflect this flexibility while remaining sufficiently descriptive to effectively characterise the dialogues being evaluated.

We envision our framework as a tool to identify relevant aspects to investigate when comparing the produced data with an evaluation corpus that displays the intended characteristics. For example, if we consider the example of the PARTICIPANTS and multilogue turn-taking discussed earlier (Section 3.4.1.2), one aspect to quantitatively assess is the order in which participants take turns speaking. In a three participant dialogue for instance, is the same sequence of speakers repeated again and again or is there a form of randomness to it? A measure of sequential regularity can provide a quantitative evaluation of this aspect.

Formally, we can model each dialogue as a first-order Markov chain over the set of participants and compute the average entropy rate to quantify the degree of regularity in speaker turn-taking. For a dialogue of N utterances and K participants, the sequence of speakers can be represented by the sequence $[S_1, S_2, \dots, S_N]$ where S_t denotes the speaker at turn t . The transition probability from speaker i to speaker j , i.e. the probability that speaker j speaks after speaker i is estimated as:

$$P_{ij} = \mathbb{P}(S_t = j \mid S_{t-1} = i) = \frac{\mathbb{P}(S_t = j \wedge S_{t-1} = i)}{\mathbb{P}(S_{t-1} = i)},$$

yielding a $K \times K$ transition matrix that characterises the turn-taking dynamics. The regularity of the dialogue is then summarised by the entropy rate of the Markov chain,

$$H = - \sum_{i=1}^K \sum_{j=1}^K \pi_i P_{ij} \log P_{ij},$$

where π_i denotes the observed proportion of speaker i in the dialogue. This quantity measures the average uncertainty associated with predicting the next speaker given the current one. A perfectly deterministic turn-taking pattern, such as the cyclic sequence $A \rightarrow B \rightarrow C \rightarrow A \rightarrow B \rightarrow C \rightarrow A \rightarrow \dots$, has an entropy rate of zero, whereas increasingly random turn-taking leads to higher entropy values.

Original	Multi-user
C Hi, can you help me find an expensive russian restaurant?	C₂ I am in a Russke mood. You like Russian food?
	C₁ I've never tried it, but I'll willing too!
	C₂ You will love it! Please look up some expensive joints for us.
S I am sorry I looked up your criteria and I do not see any expensive russian restaurants in my data base	
C Are you looking in the centre area?	C₁ You sure? They don't have any.
	C₂ That sucks. Did you check everything in the Centre?
S Yes, I apologize but no restaurants match your search.	
C How about british food instead?	C₁ Ah well, que sera, sera. What else would you like?
	C₂ Maybe British and a spot of tea. Anything fitting?
S I have 3 British restaurants that fit your criteria. I would personally recommend Fitzbillies Restaurant.	
C Fitzbillies Restaurant sounds good,will you book it for me ?	C₂ Ohhh Fitzbillies Restaurant, sounds fancy. What do you think about it?
	C₁ I want to go there. Can you book us?
S Certainly, what day and time would you like a reservation? And for how many guests?	

Table 3.8: Comparison of the mono- and multi-user version of the MultiWOZ example MUL1379.

Because the entropy rate is independent of the particular identities of the speakers and depends only on the transition structure, it provides a principled and comparable measure of turn-taking regularity across different dialogue datasets.

Jo et al. (2023) introduces a multi-user version of the MultiWOZ corpus (Budzianowski et al., 2018), i.e. dialogues from MultiWOZ have been manually re-written so that each user turn is replaced by a small chat between two users. Table 3.8 shows an example of dialogue from this corpus together with the original version.

For each dialogue, the multi-user version was rewritten by one person after a pilot study had shown the difficulty of collecting multi-user dialogues in a pairwise setting. Jo et al. (2023, p. 6) indicate that ‘laypeople are not good at creating a collaborative scenario and holding a relevant dialogue in interactive settings’. However, the ability of individuals to write realistic dialogues outside an interactive setting can likewise be questioned. In particular, the freedom given to the crowdworkers regarding the number of utterances each original contribution should be transformed into and the order of speaking of the two participants can lead to the creation of patterns that may not reflect naturally occurring dialogues. Since we do not have access to a similar dataset with naturally occurring conversations, we cannot perform such comparisons. However, we can compute the turn-taking regularity measure we just introduced for the training and evaluation sets as the crowdworkers were completely disjoint. We also include the development set though the original paper does not indicate if the crowdworkers overlapped with the other two sets.

We compute this metric in two ways. First, by considering the three participants, i.e. the two customers **C₁** and **C₂**, and the system **S**, to take into account the freedom to replace one original customer’s utterance by any number of new ones. Second, by considering only the two customers to determine how

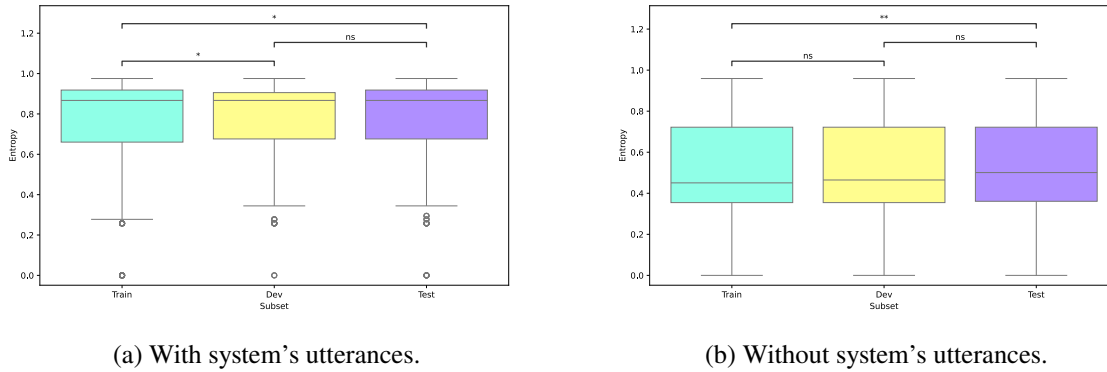


Figure 3.4: Average turn-taking entropy rate of the multi-user version of MultiWOZ.

regular the alternation between them is. The results, presented in Fig. 3.4, show that the entropy is very high when considering all three participants, around 0.8 on average for all three data subsets. It decreases to around 0.5 when considering only the two customers which is still a rather high value given the number of participants. This suggests that turn-taking is relatively irregular in this dataset and that the crowdworkers did not simply alternate between the two customers and the system. We can note that the training and test sets differ significantly in terms of entropy according to pairwise t-tests, showing that relying on disjoint sets of crowdworkers had an influence. However the differences are quite slim and the effect size appears modest which indicates that the difference is not very strong.

One advantage of using Markov chains to assess the regularity in turn-taking dynamics is that higher-order chains could be computed, i.e. the probability to have a given speaker at turn t would be computed based on the speakers in multiple preceding turns. Higher-order models could capture more fine-grained turn-taking dynamics, though a larger sample would be necessary to discover significant configurations.

This shows how our framework can be used in a more applicative way to design metrics meant to highlight differences between datasets with respect to specific parameters. Together with its potential for identifying dimensions along which dialogue generation can be controlled, this demonstrates the practical value of the framework for NLP.

3.5 Summary and Conclusion

This chapter unfolded along three complementary dimensions.

First, the survey of dialogue research published in 2024 in the ACL ANTHOLOGY offered a situated snapshot of the field. It revealed that contemporary dialogue research mostly focuses on generative systems, which is unsurprising given the rise of LLMs. Yet, their development by major technology companies in opaque ways raises fundamental questions about how we develop and evaluate conversational systems. Since reproducing LLMs at this scale is not feasible, we must seek alternative ways to regain control over our work. The solution, however, cannot be to restrict access to data or reverse the progress made toward open science. Instead, the community must focus on research methodology: clearly defining tasks, thoroughly characterising datasets, and designing meaningful, task-specific evaluation metrics.

Second, we tried to illustrate this point by applying our descriptive framework to three distinct dialogue scenarios: turn-taking, task-oriented dialogue, and clinical interviews. This comparative exercise demonstrated the analytical value of explicitly characterising dialogical configurations along multiple axes: PARTICIPANTS, LANGUAGE, ENVIRONMENT AND MEDIUM, VERBAL INTERACTION, and CONVERSATIONAL ACTIVITY AND PURPOSE. Across these scenarios, we saw that dialogue study is multifaceted and

that different settings foreground different phenomena. For example, turn-taking research highlights the temporal dynamics of interaction, task-oriented dialogue focuses on goal-directed coordination, and clinical interviews reveal the interpretive complexity of meaning-making under asymmetrical conditions. This exercise showed that methodological choices shape the phenomena we can observe and the conclusions we can draw. It also highlighted the challenges of aligning different settings, particularly when working with sensitive data or vulnerable populations. Descriptive precision is therefore not a mere formal exercise: it conditions both empirical validity and interpretability.

Third, building on these observations, we proposed a way of leveraging the framework to guide the improvement of dialogue generation and to propose evaluation metrics along specific dimensions. Rather than treating dialogue competence as a monolithic capability, we argued that targeted control could help align model behaviour with concrete pragmatic requirements. This perspective reframes model development not as a race toward generality, but as a process of structured calibration grounded in explicit dialogical parameters.

These three strands converge on a central claim: descriptive precision is foundational. A clear characterisation of dialogical settings enables us to (i) describe data more transparently, (ii) formalise task constraints more rigorously, and (iii) compare system outputs against articulated expectations rather than relying on general quality judgements. In doing so, the framework offers a basis for developing more specific and theoretically informed evaluation metrics that move beyond surface-level similarity and toward the assessment of specific dialogical competencies.

If dialogue research is to remain both scientifically robust and socially relevant, it must continue to refine its descriptive tools while critically examining its methodological practices. The framework proposed here is not intended as a definitive solution, but as a means of making explicit what too often remains implicit: the structural conditions under which dialogue emerges and the assumptions that guide its modelling. In the remainder of this thesis, we examine in greater depth one parameter of this framework, namely the `STRUCTURE` of the conversation, and in particular its topical structure, considering how it can be identified, modelled, and used as a descriptive dimension for distinguishing between different types of dialogue.

Part II

Topics and Topical Structure: Applicative analysis across various interactional contexts

Topics in Discourse and Dialogue

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The first part of this thesis was devoted to the characterisation of dialogue as an object of study. Through the examination of a wide range of theoretical perspectives, we argued that dialogue cannot be reduced to a single dimension, but should instead be understood as a multidimensional phenomenon. Different types of interaction involve different constraints on production, interpretation, coordination, and information management. To account for this diversity, we proposed a descriptive framework organised around five dimensions, allowing dialogue datasets to be situated within a broader dialogue space.

A central idea of this framework is that any given dataset instantiates only a limited region of this space. We tried to define the different dimensions of it in a sufficiently operational manner – or at least to suggest possible leads towards a computable representation – to support comparisons between datasets, to make explicit the interactional phenomena they capture, and to guide the production of new tools and evaluation metrics. This perspective has important methodological implications. Since datasets only cover specific regions of dialogue space, theories and computational tools developed on a particular dataset cannot be assumed to generalise to dialogue as a whole. Their validity and applicability depend, at least in part, on the characteristics of the data on which they are built and evaluated.

The previous chapters also highlighted the fact that the dimensions of dialogue are not independent. Rather, they interact with and constrain one another. Among the phenomena likely to be affected by these interactions is topical organisation. The way topics are introduced, maintained, and shifted throughout an interaction depends on factors such as the participants involved, the goals of the interaction, the communication medium, or the temporal organisation of the exchange. Because topical organisation is a pervasive feature of dialogue, systematic differences between dialogue types are likely to be reflected in systematic differences in their topical structure.

At the same time, topical structure presents a particular challenge for dialogue analysis and modelling. Unlike more readily observable properties such as the number of participants, turn-taking patterns, or the presence of overlaps, topics are not directly encoded in the interaction. They must instead be inferred

from the unfolding conversation and reconstructed from the linguistic and interactional cues available to the analyst. Yet, they play a central role in the organisation of dialogue, constraining the contributions that participants can make and providing a basis for interpreting new interventions. This makes topical structure both a fundamental and a difficult aspect of dialogue to study.

The second part of this thesis investigates the relationship between dialogue type and topical structure. More specifically, it examines the extent to which approaches and theories traditionally developed for relatively structured forms of discourse remain applicable when confronted with more complex types of dialogue. This chapter introduces the key notions required for this investigation. We begin by discussing the concepts of topic and topical structure in Section 4.1. We then present different perspectives on the study of topical structure in Section 4.2. Finally, we outline the questions that motivate the remainder of this part and discuss the relationship between dialogue type and topical organisation in Section 4.3.

4.1 Topics and Topical Structure

Studying topical organisation covers two related but distinct aspects. The first concerns the topics themselves, that is, what a discourse or dialogue is about. The second concerns the way these topics are organised throughout the interaction, which we refer to as *topical structure*. While these two aspects are closely related, they raise different questions and require different analytical tools. In this section, we discuss each of them in turn.

4.1.1 Topic

The notion of *topic* is widely used in linguistics, yet its precise definition remains a matter of debate. The term has been employed to refer to several related but distinct concepts, reflecting the diversity of phenomena that can be described in terms of what a piece of language is ‘about’. Broadly speaking, topics have been studied both at the sentence level (Bolinger, 1952; Firbas, 1964; M. A. Halliday, 1967; Givón, 1983) and at the discourse level (Keenan & Schieffelin, 1976; G. Brown & Yule, 1983; Asher, 2004).

At the sentence level, a *sentence topic* is typically defined as the entity or proposition about which the sentence provides information (Hockett, 1958). In many cases, this corresponds to a noun phrase occupying a privileged syntactic position. For instance in the sentence ‘Bob loves dragons.’, the noun phrase ‘Bob’ is the topic, as it is the entity about which the sentence conveys information. While this notion is useful for describing information structure within individual sentences, it does not readily extend to phenomena that emerge beyond sentence boundaries. If ‘Bob loves dragons.’ is one sentence among others in an enumeration of Bob’s interests, then we could say that Bob is indeed the topic of the piece of discourse, or more precisely that Bob’s interests are. However, if the enumeration relates to the interests of a group of people, then the sentence topic ‘Bob’ will not correspond to the global topic. In conversation, the notion of sentence topic is further complicated by the fact that participants may not always explicitly mark the topic of their contributions. For instance, the topic of an intervention may be established through gestures, gaze, or shared situational context rather than through linguistic material alone. More importantly, sentence-level approaches do not account for the interactive and incremental nature of topic development in conversation, where each participant can add to the context and thus contribute to the evolution of the topic.

For the purposes of this thesis, we are primarily concerned with *discourse topics*. These are generally understood as what a stretch of discourse or interaction is *about*, although the notion of *aboutness* itself remains difficult to formalise. Informal accounts often relate topics to participants’ intuitions about the current focus of the conversation. This perspective is compatible with Grice’s Maxim of Relation (Grice, 1975), according to which contributions should be relevant to the ongoing interaction. Remain-

ing on topic can be viewed as one way of maintaining relevance, while topic shifts are interpreted under the assumption that participants continue to cooperate and contribute meaningfully to the conversation. Similar intuitions can be found in Conversation Analysis, where topics are not treated as a fixed subject matter but as an interactionally accomplished object of discourse that participants themselves establish, maintain, transform, and close through the sequential organisation of talk (Button & Casey, 1985).

Despite their intuitive appeal, such definitions have been criticised for their limited operationalisability. The complexity of conversational interaction led G. Brown and Yule (1983, p.68) to ‘suggest that formal attempts to identify topics are doomed to failure’. Nevertheless, these authors also acknowledge the usefulness of notions such as ‘speaking topically’ or ‘the speaker’s topic’, suggesting that topics, even if difficult to define rigorously, play an important role in the organisation and planning of discourse.

More formal accounts have attempted to characterise topics based on the notion of ‘question of immediate concern’, i.e. the question or set of questions an utterance is a response to. The discourse topic is then defined as the proposition or set of propositions presupposed by this question (Keenan & Schieffelin, 1976). For instance, the question of immediate concern of this current section could be defined as ‘What is a topic?’, and the discourse topic would be the proposition ‘Topics exist.’.

These approaches provide the foundations for Question Under Discussion (QUD) theories (Van Kuppevelt, 1995; Ginzburg, 1995; Roberts, 2012), which model discourse as a structured set of questions and answers. Within this framework, each contribution either answers an existing question, introduces a sub-question, or redirects the interaction towards a different issue. Topics can therefore be viewed as the questions currently under discussion and whose resolution contributes to maintaining discourse coherence.

However, QUD-based approaches do not fully resolve the difficulties associated with topic identification. Questions of immediate concern are often implicit and may not be directly recoverable from the linguistic material alone. Like the topics themselves, they must frequently be inferred by analysts. Empirical studies support this observation: when annotators are asked to reconstruct QUDs from interview data, they achieve only moderate agreement (Mulligan & Rawlins, 2025). This suggests that multiple formulations may adequately capture the same conversational concern and that identifying a unique underlying question is not always straightforward. As already noted by G. Brown and Yule (1983, p. 72), evaluating alternative descriptions of a topic would itself require explicit criteria whose formulation may depend on the type of interaction under consideration.

These difficulties become particularly apparent in less information-seeking oriented forms of interaction. In casual conversation or small talk, topics may shift rapidly and in seemingly unexpected directions. While identifying a single question of immediate concern in such interactions can be challenging, this does not imply that they are incoherent. On the contrary, Schneider (1988) argues that topics remain an important organising principle even in phatic communication. Thus, despite numerous attempts at formalisation, discourse topics remain to some extent intuitive constructs (T. A. Van Dijk, 1977).

For the present work, we adopt a broad understanding of topics as organising units that contribute to discourse and dialogue coherence. The topic under discussion constrains the range of contributions that participants can reasonably make and provides a basis for interpreting new interventions. In written discourse, where opportunities for clarification are limited, topical organisation is often expected to be explicitly reflected in the structure of the text. Dialogue is more flexible in this respect, as participants can negotiate meaning and repair misunderstandings as they arise. Nevertheless, speakers generally orient towards what the interaction is currently about and attempt to produce contributions that are relevant to that evolving context (E. A. Schegloff & Sacks, 1973; Grice, 1975).

Unlike the topic of a written text, the topic of a dialogue is rarely fixed. Multiple topics may be discussed over the course of an interaction, their boundaries are often difficult to define precisely, and their development depends on the contributions of all participants as well as on the surrounding context. Topics in dialogue should therefore be understood as dynamic, incremental, and jointly constructed phenomena.

This dynamic character also reflects a broader difference in how coherence is conceived across language uses. In written discourse, coherence is often understood as the property of a text forming a meaningful whole (Williams & Nadel, 1989). As W. S. Cho et al. (2019) observe, the different parts of a coherent text fit together ‘the way all the pieces in a puzzle add up to the picture on the box’. In dialogue, by contrast, coherence emerges through interaction and refers to the participants’ ability to sustain a mutually interpretable course of action (Schiffrin, 1985; Anderson, 1995). Lautamatti (1982) describes this duality as a distinction between *propositional* and *interactional* coherence. The former is concerned with information organisation and the development of semantic content, while the latter is concerned with the illocutionary force of the utterances and how they contribute to the ongoing interaction. She argues that both types of coherence are present in any type of discourse but that they are more or less prominent and easily observable depending on the level of formality. She describes the features of language use that influence the prominence of each type of coherence, which can be mapped with the descriptive framework developed in Chapter 2. In particular she includes the shared context, which we capture in ENVIRONMENT AND MEDIUM, the possibility for immediate feedback, whether repairs are a part of the discourse itself or can be edited out of it, and the time allowed for advanced planning, which together form our (VERBAL) INTERACTION category. She also mentions the shared knowledge that can be assumed which corresponds to the SOCIAL DYNAMICS entry of our PARTICIPANTS category. The most extreme types of language use along this cline would be casual conversations for the most interactional and least formal side, and scientific texts or novels for the more propositional and formal side. Our interests in this thesis lie closer to the interactional end of the spectrum, where the dynamic and incremental nature of topical organisation is more apparent, and the propositional coherence is much harder to observe from an external perspective. Lautamatti (1982, p.124) emphasises that, in such cases, participants rely heavily on a shared context and background to formulate and interpret their contributions, which ‘makes it difficult for anybody outside the speech event to analyse the propositional coherence or to follow the development of the discourse topics’. This will be a central concern of this second part, in which we shall attempt to abstract away from the internal construction of topics and focus instead on more observable characteristics of the interaction in order to study topical organisation in dialogue.

4.1.2 Topical structure

Because topics themselves are difficult to observe directly, topic shifts often constitute a more accessible object of analysis (Howe, 1991). Several types of markers and heuristics have been proposed to identify topic shifts, including discourse markers such as ‘by the way’ in English, long pauses, or changes in the vocabulary. We discuss the technical aspects in more detail in Section 4.2.2. Identifying the different topics present in a text or interaction makes it possible to divide the discourse into relatively coherent stretches devoted to separate concerns. This is particularly useful for tasks such as information retrieval as it allows to focus on the most relevant parts of a discourse. It also provides a basis for studying the relations between topics and the mechanisms through which participants move from one topic to another.

The most common approach to topic segmentation is linear, where the discourse is divided into a sequence of contiguous segments corresponding to successive topics. Figure 4.1 illustrates the topic segmentation of a text that summarises the rise and fall of an hypothetical application called ‘OintMind’ that promises to detect mental health issues based on the analysis of the user’s speech (Choi et al., 2024). The text is formatted as a short news article¹¹, making it similar to the types of text often used in topic segmentation studies. The text is represented as a sequence of paragraphs in Fig. 4.1a, and segmented linearly in Fig. 4.1b. Each segment is represented by a different colour spanning one or more paragraphs. This approach is quite basic and does not provide information about the content of the topics but it can be

¹¹The full text is available in Appendix (REF: To be added in the final version).

completed by labelling the segments as in Fig. 4.1c. In this segmentation, all the topics are represented at the same level and no relations between them are made explicit.

Yet the tension between the notions of sentence topic and discourse topic discussed in Section 4.1.1 makes apparent the idea of granularity in topical organisation, where a piece of text or conversation can be divided in very high-level topics that can be further divided in more specific topics and subtopics. Within this perspective, Van Kuppevelt (1995) defines discourse topics as the highest level of topical organisation, with topics and subtopics forming lower levels of the *hierarchy*. Figure 4.1d proposes a hierarchical topic segmentation of the same example we used for linear segmentation. We represent the structure as a tree where the root node corresponds to the main topic and each branch leads to a more specific subtopic. This representation overlooks the actual relations between the topics and subtopics which can be more complex than a simple hierarchy. For example, the two subtopics ‘Testimony’ are illustrations of the ‘data collection’ and ‘privacy issue’ respectively, while there is a logical relation of implication between the subtopic ‘Data collection issue’ and its subtopic ‘Consequences’. Formalisms such as Rhetorical Structure Theory (Mann & Thompson, 1988) or Segmented Discourse Representation Theory (Asher & Lascarides, 2003), typically used at the sentence level, can be extended to model relations between topics and subtopics at the discourse level.

Focusing on dialogue, Van Kuppevelt (1995) states that task-oriented or *bound conversations* are typically organised around a single discourse topic, whereas less constrained interactions may involve several discourse topics. In this framework, shifts between discourse topics are often treated as structurally incoherent because they move the interaction outside the current organisational structure. This view, however, raises questions when applied to more open-domain dialogues. While abrupt topic shifts may indeed be perceived as disruptive in highly structured interactions, they are commonplace in many forms of everyday conversation. Participants frequently introduce new topics, return to previous ones, or connect apparently unrelated subjects through interactional work. Some topic shifts can also be more gradual, where two parts of the dialogue concern different topics but the transition between them is not clearly marked. In such cases, the notion of a topical boundary can be intrinsically fuzzy, and attempting to impose a strict hierarchical structure may not accurately reflect the dynamics of the interaction. Overall in conversation, what matters is often not the existence of a topic shift itself, but whether participants can successfully interpret and accommodate it. The ability to identify relevant places for topic shifts and to manage the transition between topics is therefore a crucial aspect of conversational competence. From this perspective, topical coherence emerges not only from structural relations between topics, but also from the participants’ ability to negotiate and maintain mutual understanding throughout the interaction. One of the aims of this second part is to examine the limitations of existing approaches to topic segmentation when confronted with complex dialogical data.

4.2 Studying Topical Structure

As discussed in Chapter 3, CL and NLP produce an epistemic duality between analytical and predictive works. The former aims to describe and explain phenomena, producing theories and models that account for the observed data. The latter aims to anticipate future developments, producing models that can predict what will happen next. These two perspectives are complementary: analytical accounts provide hypotheses about how topics are introduced, maintained, and shifted, whereas predictive models support applications such as dialogue systems but also offer a way of testing whether these hypotheses capture regularities that are sufficiently systematic to support prediction. In this section we first discuss the different perspectives (Section 4.2.1) and computational approaches (Section 4.2.2) to studying topical structure, with a particular focus on the challenges posed by dialogue data compared to written monologues.

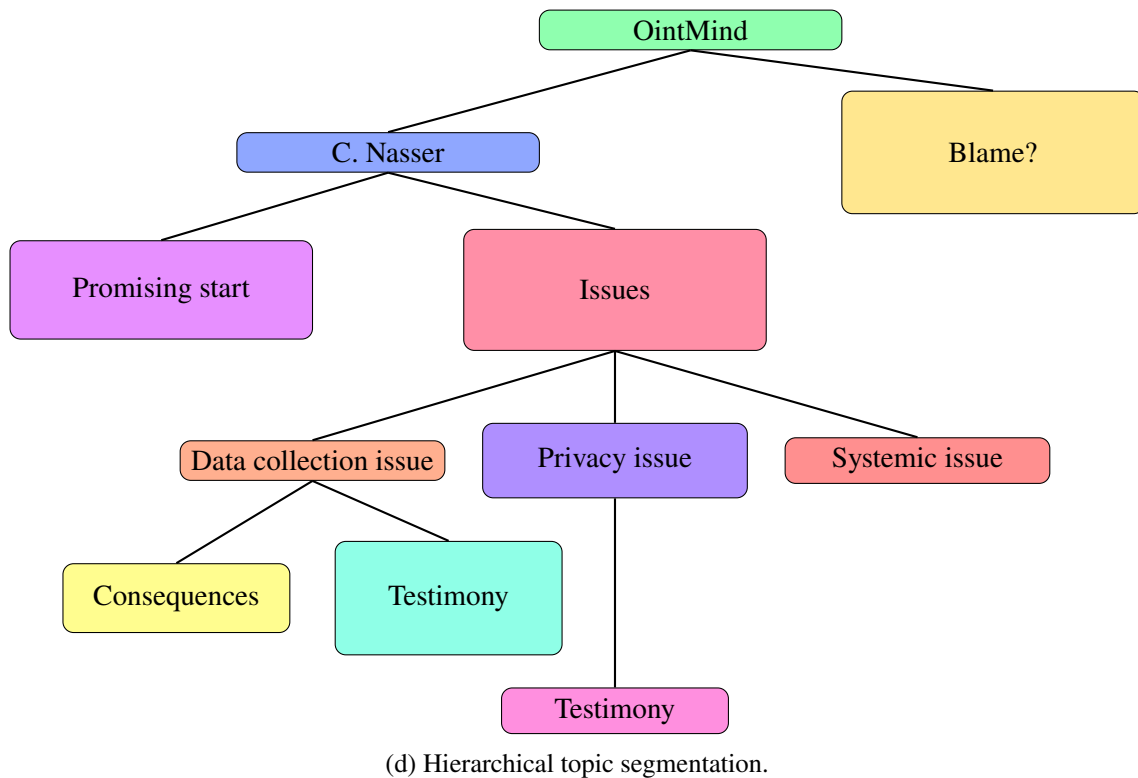
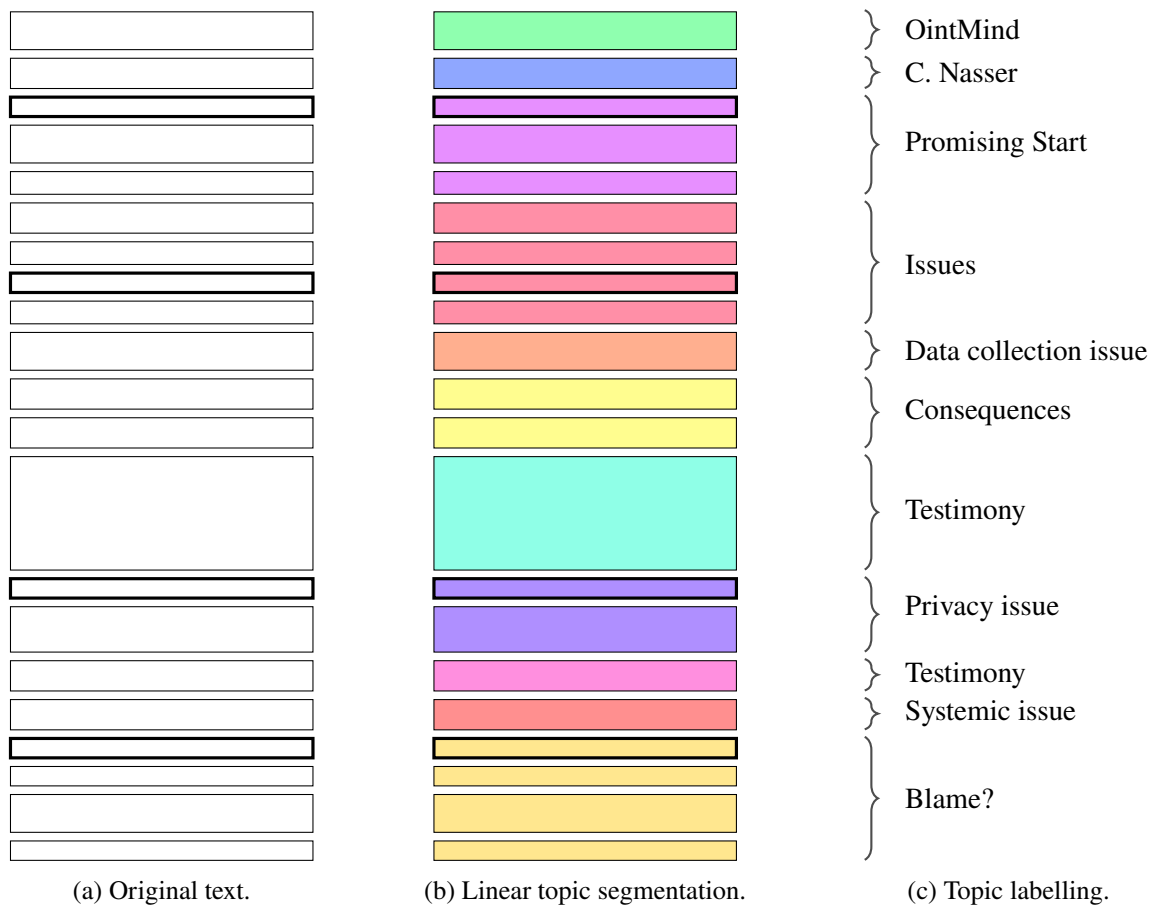


Figure 4.1: Representation of the text ‘OintMind, the Mental Health Scandal’ (Choi et al., 2024) segmented in topics. The text is represented as a sequence of paragraphs, with titles marked in bold.

4.2.1 Analytical and Predictive Perspectives

The study of topical organisation can be approached from two complementary perspectives. The first is *analytical*: it seeks to describe and explain how topical structures emerge in naturally occurring discourse and dialogue. The second is *predictive*: it seeks to model topical organisation in a way that makes it possible to anticipate future developments in the interaction. While these perspectives differ in their objectives, they are closely related. Analytical accounts provide hypotheses about how topics are introduced, maintained, and shifted, whereas predictive models offer a way of testing whether these hypotheses capture regularities that are sufficiently systematic to support prediction.

From an analytical perspective, topical organisation is studied *a posteriori*, on already produced discourse. The goal is to identify the mechanisms that structure a conversation and to characterise the relations between its different topics. Several computational tasks contribute to this objective.

One of the most prominent is *topic segmentation*, which consists in identifying boundaries between topics within a document or dialogue. By dividing a discourse into coherent topical units, topic segmentation makes it possible to study how information is organised and how topics evolve over time. As discussed in Section 4.1, segmentation can be approached either linearly, by identifying successive topical segments, or hierarchically, by modelling relations between topics and subtopics. Linear segmentation remains the most common approach because it is comparatively easy to annotate and evaluate, and because it is often sufficient for highly structured data. However, more complex forms of dialogue, particularly open-ended conversations, frequently exhibit gradual topic shifts and intertwined topical trajectories that challenge strictly linear representations.

A related task is *topic labelling*, which aims to assign descriptive labels to identified segments. Topic labels provide a compact representation of a segment's content and facilitate navigation, retrieval, and analysis. Determining an appropriate level of granularity remains challenging, as different applications require different degrees of specificity. Existing works suggest that annotators tend to agree more readily on coarse-grained descriptions than on fine-grained ones (Banerjee & Rudnicky, 2007), highlighting the inherently subjective nature of topical interpretation.

The outputs of segmentation and labelling can themselves be used to support downstream applications such as information retrieval (Shtekh et al., 2018) but also more epistemic tasks. Indeed, the resulting segmentations and labels can be analysed and annotated with a rhetorical or discourse structure to identify the underlying principles of topical organisation. Such analyses provide a better understanding of typical patterns of topical organisation, as well as the mechanisms used by participants to manage changes of topic and maintain coherence. They can also be used to identify disfluencies or breakdowns in topical organisation, which may indicate cognitive or communicative difficulties.

The predictive perspective approaches topical organisation from a different angle. Rather than reconstructing the structure after the interaction has taken place, it attempts to anticipate how that structure will develop. In computational applications, this perspective is particularly relevant for conversational agents and dialogue systems, which would benefit from the ability to predict upcoming topics, anticipate topic shifts, and maintain coherent interactions over time. Predictive models can therefore be used to generate more appropriate responses and improve dialogue management in these systems.

Beyond their practical applications, predictive models also play an important methodological role. If a theory of topical organisation claims that certain conversational moves constrain the range of possible continuations, then these constraints should improve our ability to predict what happens next. Predictive performance can therefore serve as an indirect way of evaluating analytical accounts. Conversely, prediction failures may reveal limitations in existing theories and motivate their refinement.

Questions Under Discussion (QUD) frameworks illustrate particularly well the complementarity of these two perspectives. They are often used analytically to reconstruct the coherence structure of a dialogue by identifying the questions that organise the interaction. At the same time, because active QUDs

constrain the range of relevant future contributions, they can also be used predictively in the form of potential questions (Onea, 2016) to anticipate upcoming moves. If such predictions can be consistently validated by empirical data, QUDs can be considered as a cognitively relevant construct used when producing and interpreting discourse.

Beyond the distinction between analytical and predictive perspective, the approach used to tackle them is also important. In particular, the choice between symbolic and statistical methods, as well as supervised and unsupervised techniques, has implications for the explainability of the resulting model. In this thesis, our goal is to develop a better understanding of topical organisation in dialogue, which requires both to investigate topics in conversations a posteriori, and to determine whether they can be predicted throughout the interaction in naturalistic settings. Identifying topically coherent spans of discourse or dialogue is therefore a first logical step towards this goal, as it provides a basis for analysing topical structure and thus also for imagining predictive models of topical organisation. In the following subsection, we therefore review the main computational approaches that have been proposed to segment discourse into topically coherent units.

4.2.2 Approaches to Modelling Topical Organisation

The computational study of topical organisation has produced a wide range of methods for identifying topically coherent segments of discourse. Although these approaches differ in their technical implementation, they generally rely on the assumption that topical organisation leaves observable traces in discourse. Topic shifts are therefore treated as detectable events, whether through changes in lexical content, latent semantic representations, discourse structure, or interactional dynamics.

Existing approaches can broadly be divided into unsupervised methods, which infer topical organisation from regularities in the data, and supervised methods, which learn topical boundaries from annotated examples. Unsupervised techniques typically rely on lexical cohesion (Hearst, 1997; Galley et al., 2003) or automatically learned latent topical distributions (Eisenstein & Barzilay, 2008; Riedl & Biemann, 2012; Du et al., 2013), whereas supervised approaches formulate topic segmentation as a classification task (Arguello & Rosé, 2006; Takanobu et al., 2018). However, the effectiveness of these methods depends heavily on the nature of the language data under consideration, as we shall see in Chapters 5 and 7. The assumptions that work well for written discourse do not necessarily hold for spontaneous conversation. Identifying and understanding these assumptions is therefore essential for assessing the applicability of existing approaches to different dialogue types.

4.2.2.1 Written Text and Discourse

Topical organisation is comparatively easier to identify in written discourse because texts often provide explicit structural cues. Chapters, section headings, paragraphs, and typographical conventions guide the reader through the progression of ideas. Discourse markers may also signal transitions between topics. Since readers cannot negotiate meaning with the author, written discourse generally relies on clear organisational principles to ensure coherence.

Many computational approaches exploit these regularities. Content-based methods identify topic boundaries by measuring lexical cohesion between adjacent segments. Their underlying assumption is that vocabulary remains relatively stable within a topic and changes when a new topic is introduced (M. A. K. Halliday & Hasan, 1976). TextTiling (Hearst, 1997) is perhaps the best-known example of this family of methods. It compares lexical distributions across sliding windows and identifies topic boundaries at points where similarity sharply decreases. Topic segmentation has also been formulated as a clustering problem, where semantically similar text segments are grouped together. Dotplot approaches (Church & Helfman, 1993; Reynar, 1994), for instance, construct self-similarity matrices that reveal coherent blocks

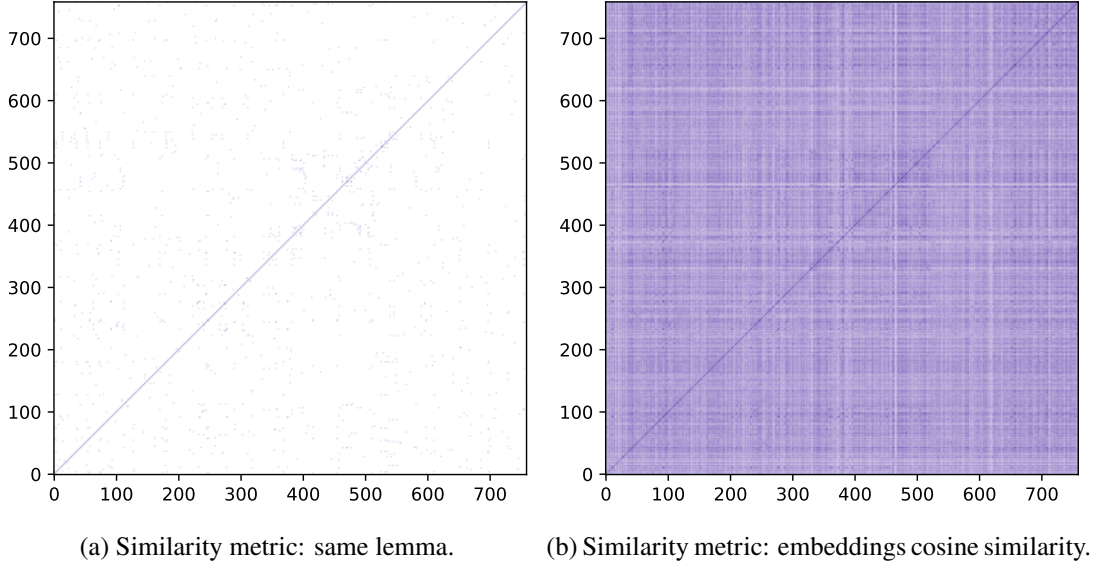


Figure 4.2: Dotplot representation of the text ‘OintMind, the Mental Health Scandal’ from Choi et al. (2024). Each point represents a similarity between two tokens.

of discourse through characteristic visual patterns. Figure 4.2 illustrates two dotplot representations of the text ‘OintMind, the Mental Health Scandal’ from Choi et al. (2024) whose topic segmentation was previously depicted in Fig. 4.1. After a pre-processing step where the text is tokenised, only interjections, nouns, verbs, numerals, and adverbs are kept in their lemmatised form. The resulting sequence of tokens is then compared to itself to produce a square matrix where each axis corresponds to the position of a token in the text. Each point in the matrix represents a similarity value between two tokens, with darker points indicating higher similarity. In Fig. 4.2a, similarity is a binary value, depending on whether two tokens share the same lemma, while in Fig. 4.2b, it is computed using cosine similarity between their embeddings (based on the SpaCy model `en_core_web_lg`). On the first representation (Fig. 4.2a), a few faint squares can be seen along the diagonal between tokens 400 and 600 but no striking patterns appear. On the second representation (Fig. 4.2b), the almost uniform distribution of points outside the diagonal means that the vocabulary is not sufficiently specific to each topic to clearly isolate them.

The first implementation of this approach obtained clear separations when applied to the concatenation of four *Wall Street Journal* articles, showing that a basic self-comparison can reveal distinctions at a very coarse level. Nevertheless, (Malioutov & Barzilay, 2006) obtained better results when segmenting individual spoken lectures by using a more sophisticated similarity metrics at the sentence level.

Other approaches model topical organisation more indirectly. Generative models assume that texts are produced from sequences of latent topics and attempt to infer these hidden structures from the observed words. Early implementations relied on Hidden Markov Models and topic-specific language models to estimate the most probable segmentation of a document.

Topic segmentation has also been treated as a classification task. Rather than modelling topics directly, these approaches predict whether a given position constitutes a topic boundary. Features used to do so may include lexical similarity, discourse markers, paragraph structure, or sentence position. Machine-learning models range from traditional classifiers such as Support Vector Machines (Cortes & Vapnik, 1995; Georgescu et al., 2006) to transformer-based architectures fine-tuned on datasets that include topical information (Xing & Carenini, 2021). The most recent approaches mostly rely on LLMs and treat topic segmentation as a generation task, where the model is prompted to produce a segmentation

of the input text (Fan, Jiang, et al., 2024; Maraj & Martin, 2026). While these methods can integrate a wider range of signals, they require annotated training data and often sacrifice interpretability.

Spoken monologue introduces additional complexity. As discussed in Chapter 2, speech contains information that is absent from written text, including prosody, rhythm, pauses, and, in multimodal settings, gestures and facial expressions. These cues often contribute to the signalling of topic transitions and may compensate for linguistic markers that would otherwise be required in written discourse. Despite these additional challenges, spoken monologue generally remains more structured than conversation, and current methods achieve relatively strong performance on high-resource languages (Purver, 2011; Ghinassi et al., 2024).

4.2.2.2 Dialogue

Dialogue introduces a qualitatively different challenge because topics are no longer organised by a single producer. Instead, they emerge through the interaction between participants. Topic management becomes a collaborative activity in which speakers jointly negotiate what the conversation is about, whether a shift is acceptable, and how new contributions relate to previous ones.

As a result, topical organisation in dialogue depends on factors that extend beyond linguistic content. Participants draw on common ground, social roles, shared histories, and situational goals when evaluating the relevance of a contribution. A seemingly abrupt topic shift may be entirely coherent when interpreted against shared knowledge or an implicit objective. Conversely, contributions that appear topically related at the lexical level may be interactionally inappropriate. The relationship between topical coherence and conversational success is therefore considerably more complex than in monologic discourse.

Furthermore, dialogue lacks many of the structural markers available in written texts. There are no paragraphs to indicate boundaries, utterance segmentation is itself often ambiguous, and discourse markers may fulfil multiple interactional functions. Topic shifts can also occur gradually, making boundaries difficult to localise precisely.

These characteristics pose significant challenges for computational models. Early dialogue segmentation systems, such as LCSEg (Galley et al., 2003), adapted discourse-based methods by relying on lexical chains and cohesion measures. While effective in relatively structured settings, such approaches struggle to capture the pragmatic and interactional dimensions of topical organisation.

The development of distributional semantics and machine-learning techniques has enabled more sophisticated representations of topical relatedness. For example, Xing and Carenini (2021) exploit adjacency pairs (E. A. Schegloff & Sacks, 1973) to learn an unsupervised similarity metric before applying a TextTiling-style segmentation procedure. Their approach performs well on task-oriented datasets such as Stanford Dialog Dataset (Eric et al., 2017) or MultiWOZ (Budzianowski et al., 2018), and highly structured ones such as Doc2Dial (S. Feng et al., 2020). However, similar methods perform substantially worse on open-ended multiparty conversations, where topical organisation is less constrained and lexical cohesion provides a weaker signal (Decker & Amblard, 2024a).

As with written text, recent work increasingly relies on LLMs, framing topic segmentation as a generation or reasoning task rather than a segmentation problem (Fan, Jiang, et al., 2024). Topic tracking and topical representations are also frequently incorporated as intermediate components in dialogue generation (Fan, Li, & Zhu, 2024; Rebedea et al., 2024) and summarisation systems (Hua et al., 2024; Tang et al., 2024). While these approaches often achieve strong empirical performance, they remain largely opaque. Their decisions are difficult to interpret, and their training data are often inaccessible, making it challenging to assess the extent to which benchmark datasets have already influenced the models' internal representations. The rising amount of LLM-generated data in the training corpora will most likely decrease the diversity of patterns models are exposed to and reinforce their biases. Namuduri et al. (2025) show that LLM-generated obituaries, creative writing texts, and long-form texts display more similarities

at the level of discourse structure when compared with other model-generated texts than with human-written ones.

Shaib et al. (2024) for semantic templates, apparently learned quickly so less diversity would have a quick impact during training [Amandine]

In addition to the resource costs and ethical implications associated with LLMs – opacity, undisclosed training corpora, intrinsic biases (Ducel et al., 2026) –, their use for research purposes raises increasingly pressing concerns regarding generalisation, evaluation and scientific validity.

From the perspective adopted in this thesis, these limitations are particularly important. Most contemporary approaches are evaluated primarily in terms of performance, whereas our objective is also to understand how topical organisation varies across dialogue types. If topical structure is shaped by the constraints imposed by different forms of interaction, then methods developed on highly structured datasets may not generalise to more open-ended conversational settings, as we discuss in the following section.

4.3 Dialogue Type and Topical Structure

The discussion in this chapter has shown that most computational approaches to topical organisation were originally developed for written discourse or highly structured forms of interaction. Their underlying assumptions are often reasonable in these settings as lexical cohesion, explicit discourse markers, and clearly signalled transitions provide strong cues for identifying topic boundaries. However, as interaction becomes less constrained, these cues become weaker and topical organisation increasingly depends on pragmatic and interactional factors. Topic shifts may be negotiated rather than announced, developed gradually rather than abruptly, and justified through shared knowledge and cues in the physical context rather than argumentative continuity. The success of a topical model therefore depends not only on its technical sophistication but also on the compatibility between its assumptions and the type of data on which it is applied.

This tendency to treat topical organisation as largely independent from the communicative context is rooted in classical perceptions of discourse as a rather homogeneous or at least continuous phenomenon. Chomsky’s generative framework, which has dominated large parts of linguistic theory since the 1960s, proposes a universal grammar underlying all human language (Chomsky, 1969). In this view, observable differences between dialogue and monologic discourse are attributed to performance factors – memory limitations, distractions, or other extralinguistic influences – rather than to fundamental structural differences. Disfluencies, repairs, and interactional shifts, central to conversational organisation, are thus interpreted as imperfect realisations of internally well-formed sentences, rather than as inherent properties of communicative behaviour. This theoretical stance, by prioritising an abstract, sentence-based competence over situated interaction, may have indirectly reinforced the assumption in computational work that dialogue can be processed with the same tools developed for monologic text. In contrast, Bakhtin’s (2010) theory of dialogism views all language as fundamentally interactive, shaped by the presence – real or imagined – of an addressee. Even the most monologic utterance is conceived in anticipation of response and in relation to prior discourse. From this perspective, the similarities between dialogue and monologic discourse arise not from the absence of structural distinctions, but from their shared roots in inherently social and responsive language use. Both perspectives support, in different ways, the idea that discourse and dialogue can be treated in the same way: Chomsky’s (1969) model by downplaying interactional structure, and Bakhtin’s (2010) by emphasising their shared dialogical foundation.

The recent emergence of LLMs has further reinforced the perception that a single computational architecture can handle all types of language use equally well. Their apparent ability to produce fluent, plausible dialogue – alongside summaries, essays, and technical explanations – gives the impression of a unified capacity for language processing. Yet the language they produce is different from human-produced

one in various ways. LLM-generated text is easily identifiable after few expositions to it because of the frequent use of specific phrases and the repetitive discourse patterns they rely on (Namuduri et al., 2025). Moreover, written training data limits the exposure to actual conversation that includes gestures, turn-taking and grounding among others. This entails that LLMs struggle with more interactive data. Nevertheless, their success across a wide range of tasks and benchmarks (missing reference: Mention most common ones.), many of which do not isolate dialogue-specific challenges, reduces the incentive to design specialised methods for conversational data, especially when such methods would require additional annotation or data collection, and thus additional time and resources. The current climate in academia with high pressure to publish quickly and produce results that are easily identifiable as competitive further reinforces this tendency.

Nevertheless, extensive work in linguistics has shown that the interactional nature of dialogue produces structural differences that set it apart from monologic discourse and other communicative modes. Turn-taking, grounding, repair, and topic management among other features are characteristic of dialogue and contribute to mutual understanding. Different forms of communication operate under different norms and constraints. A casual conversation between friends, for example, has different turn-taking dynamics, topical fluidity, and tolerance for ambiguity than a formal pitch to a potential investor. While both may stem from a shared cognitive and linguistic capacity, their actual realisations diverge.

This perspective is particularly limiting when studying topical organisation. As Purver (2011) observes, identifying topics may appear straightforward in highly structured settings such as news broadcasts, business meetings, or court transcripts, where topical units are often tied to externally defined tasks or agenda items. Yet even in such cases, finer-grained segmentation quickly becomes difficult. Annotators frequently disagree on boundary placement, and agreement generally decreases as the desired level of granularity increases. Rather than reflecting annotation noise alone, these disagreements suggest that topical structure is partly dependent on the analytical perspective adopted and on the intended use of the segmentation.

The challenge becomes even greater in more open-ended dialogue where all the participants can continuously reshape the trajectory of the interaction. Topic boundaries may be gradual, negotiated, or retrospectively interpretable rather than explicitly signalled. Consequently, approaches developed for written discourse or highly structured interactions cannot be assumed to transfer straightforwardly to more open-ended conversational settings.

Interestingly, Questions Under Discussion (QUD) frameworks emerge from the opposite direction. Whereas most computational approaches to topical organisation originated in discourse processing and were subsequently adapted to dialogue, QUD-based theories were developed to account for conversational coherence and only later applied to monologic discourse and computational tasks. They therefore place interaction, relevance, and the management of conversational goals at the centre of their account of topical organisation. In principle, this makes them particularly well suited to dialogue. Yet despite their theoretical influence, empirical investigations of QUDs in naturalistic conversations remain comparatively limited, especially for open-ended forms of interaction where topical trajectories may be less constrained and less information-oriented.

Overall, we have discussed two facets of the works on topics in NLP/CL. On the one hand, we have computational approaches to topic segmentation and modelling that were originally developed for written discourse and rely either on surface information such as lexical cohesion or on latent representations learned by topic models and LLMs. While some of these methods have been adapted to structured forms of interaction, they are not designed for open-ended conversational data. On the other hand, we have theoretically grounded approaches to topical organisation, such as QUD-driven frameworks, which emphasise the interactive nature of dialogue but have not been extensively tested on naturalistic conversations. In both cases, there is a lack of empirical validation on spontaneous dialogue, leaving open questions about the generalisability of existing methods and theories across different dialogue types.

Taken together, these observations raise two complementary questions. First, to what extent do existing approaches to topical organisation generalise across dialogue types? If topical structure is shaped by the constraints of interaction, methods developed on highly structured datasets may not remain effective when applied to more open-ended conversational settings. Second, can topical organisation itself serve as a discriminative signal for characterising dialogue types? If different regions of the dialogue space exhibit systematically different topical dynamics, then topical structure may provide a useful perspective for comparing, describing, and ultimately classifying conversational data.

The remainder of this part investigates these questions through three complementary studies. They all dive into the `STRUCTURE` parameter of dialogue according to the typology we proposed in Chapter 2 through the analysis of topical organisation in various kinds of dialogues. The first study, presented in Chapter 5, examines the robustness of classical topic segmentation approaches by extending TextTiling-inspired methods to open-ended multiparty conversations. This very headfront study aims at questioning whether methods and tools designed mostly with monologic or organised dialogic forms of language use in mind can resist the challenges posed by more spontaneous and less structured conversational settings. Topic segmentation is a first step towards understanding topical organisation, and this study provides a first element to indicate how important the structure of the dialogue is to consider for producing robust methods. The second study, presented in Chapter 6 is rooted in the assumption that a binary approach to topic shifts, i.e. considering that a topic shift is either present or absent at a given boundary, is too reductive to capture the complexity of topical organisation in dialogue and thus to support structure analysis. Instead, it looks at topics changes in a holistic way, considering the whole topical trajectory of a conversation. It investigates whether a local topical similarity metric can reveal the underlying conversational structure of large-scale online discussions as well as differences between communities. Finally, the third study, presented in Chapter 7, takes a more abstract perspective and investigates the predictive power of QUD-based representations across a range of dialogue types. It aims to assess whether the salience of questions at a given point in a conversation can be used to anticipate the trajectory of the interaction, and whether this predictive power varies across dialogues with different number of participants and interactional settings. One of the goals of this study is to explicit the relationship between observable features, such as the participants and the task, and the `STRUCTURE` parameter, which is more latent, showing that they depend on each other.

Together, these studies aim to clarify the relationship between dialogue type and topical organisation while assessing the extent to which existing theories and computational methods generalise beyond the settings in which they were originally developed. By investigating their limitations, we hope to identify the scope in which topics can be studied depending on the type of dialogue.

5

Topic Segmentation in Open-ended Multi-party Conversation

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Todo Writing

- Intro: chapter 5 covers at least a part of it, get to the point more quickly?
- Comparison Friends / DailyDialog?
- Results: colour / bold / etc. in tables (uniform)
- Figures:
 - TextTiling: schematic representation of the algorithm, better graph with scores
 - our models (feature based and ML metric based)

Understanding how topics interact in dialogue is essential when it comes to modelling dialogue structure beyond the sequence of utterances. However, analysing this structure requires insight on the topics themselves. Being able to segment a dialogue into topically coherent segments seems to be a first step towards modelling high level dialogue structure. The segments could later be linked inside a structure that describes the interactions between them. This task, called dialogue topic segmentation (DTS), finds utility in dialogue generation (J. Xu et al., 2021) and summarising (J. Chen & Yang, 2020), among other applications.

DTS has received less attention compared to monologue or written text topic segmentation, primarily due to the scarcity of annotated data but some DTS approaches get good results on task-oriented dialogues (Takanobu et al., 2018) or conversations with a restricted set of possible topics such as meeting minutes (Hsueh et al., 2006; Georgescu et al., 2008). Xing and Carenini (2021) suggest another method to tackle more varied dialogues. They use the *TextTiling* algorithm (Hearst, 1997), that relies on a similarity metric between subsequent blocks of text to identify topic boundaries, and enhance it with a learned utterance-pair coherence scoring model based on BERT (Devlin et al., 2019) as similarity metric. They obtain good results in English and Chinese when evaluating their model on three datasets: DialSeg_711 (Y. Xu et al., 2021), Doc2Dial (S. Feng et al., 2020), and ZYS (Y. Xu et al., 2021). Even though these datasets cover different domains, they all contain task-oriented conversations. Evaluating this model on more open-domain dialogues would provide insight on the limits of its generalisation capability.

In this chapter we adapt several linear topic segmentation approaches to open-domain multi-party conversations¹². To address the lack of such annotated data, we leverage transcripts from the *Friends* TV show, which include scene annotations that we use as a proxy for topic boundaries. The approaches we adapt include two versions of the *TextTiling* algorithm: one enhanced with linguistic and dialogical features, and the other using a BERT-based utterance pair coherence scoring model inspired by Xing and Carenini (2021). We also evaluate the performance of large language models (LLMs) in a basic zero-shot setting on this task.

The motivation behind this work is twofold. First, we want to evaluate the performance of linear topic segmentation approaches on multi-party casual conversations, which are more chaotic than the task-oriented dialogues typically used for this task. Second, we want to determine what performance we can reach with a formal approach based on linguistic and dialogical features. Since we are interested in the structure of topical interactions, an explainable model would help us better understand what features play a role in perceiving topic shifts. A rule-based approach also has the advantage of minimising the amount of computation required by our model, which is more sustainable.

We begin in Section 5.1 by describing the two main studies our work relies on, including the original *TextTiling* algorithm and Xing and Carenini’s (2021) approach. We then present our experimental setting in Section 5.2, including a brief description of the models we used, the baselines we compared them to, and the evaluation metrics we used. In Section 5.3 we describe our dataset, the *Friends* TV show transcripts. Next we detail our baselines (Section 5.4), our feature-based improvements of the original

¹²Our code is available at <https://gitlab.inria.fr/adecker/topicsegmentationtexttiling.git>.

TextTiling algorithm (Section 5.5), and the application of Xing and Carenini’s (2021) approach to our data (Section 5.6). Finally in Section 5.7, we summarise and compare all the results, and discuss their implications for topic segmentation in multi-party casual conversations.

Most of the work described in this chapter is taken from Decker and Amblard (2024a) and was presented at the 5th Workshop on Computational Approaches to Discourse (CODI) in 2024. We refined the evaluation process and added an LLM-based baseline for this chapter.

5.1 Related works

This work relies heavily on two main approaches to topic segmentation in dialogue: the TextTiling algorithm (Hearst, 1997) and Xing and Carenini (2021)’s enhancement of this algorithm. We describe these two approaches in the following sections. For a more general overview of topic segmentation, in discourse and in dialogue, we refer the reader to the previous chapter (Chapter 4).

5.1.1 TextTiling Approach

TextTiling (Hearst, 1997) is a topic segmentation algorithm that predicts topic boundaries for a given *text*. This section is dedicated to describing the original approach, and we will discuss our various adaptations to make it suitable for *dialogue* in Section 5.4.2. In this section we will thus focus on the application of the TextTiling algorithm to organised monologic text which we will consider to be a sequence of paragraphs. Since the original algorithm was meant for ‘expository text’, i.e. text ‘not heavily stylized or structured’, we will explain the algorithm with an example that does not include headings or similar organisational elements. A schematic representation of such a document is given in Fig. 5.1. The TextTiling algorithm relies on lexical distribution information and its execution follows three main steps that we will describe in the upcoming sections: (1) pre-processing, (2) lexical score determination, (3) boundary identification.

5.1.1.1 Pre-processing: Distillating the lexical content

The goal of the pre-processing step is to extract the most salient lexical information from the text. In this first step, if the document contains markup information such as headings or similar organisational elements, they are removed and only the body of the text is kept due to the algorithm being designed for expository text. The text is then tokenised, i.e. split into individual lexical units, and lowercased, which corresponds to the [Amandine: Replace when figure is here] on the schematic representation Fig. 5.1.

Stopwords are then removed from the tokenised text. Stopwords are words that are very frequent in a language but do not carry much meaning, such as articles, prepositions, or conjunctions. Removing them allows the algorithm to focus on the more meaningful words that are more likely to be indicative of topic shifts.

Then the remaining tokens are stemmed or lemmatised¹³. Stemming and lemmatisation are two techniques used to reduce words to their base or root form, thus allowing the algorithm to identify different forms of the same words as being the same. For instance, the words ‘run’, ‘runs’, and ‘running’ would all be reduced to the same root form, which allows the algorithm to better capture the lexical similarity between different parts of the text.

Finally the text is subdivided into sequences of tokens of a predefined length w – stopwords are included in the computation of the length – which we will call *spans* in the following. This step is necessary

¹³The original paper describes this step as reducing the tokens to their ‘root by a morphological analysis function based on that of Karttunen et al. (1987)’ which corresponds to stemming but both stemming and lemmatisation have been attempted in the different implementations of this algorithm.

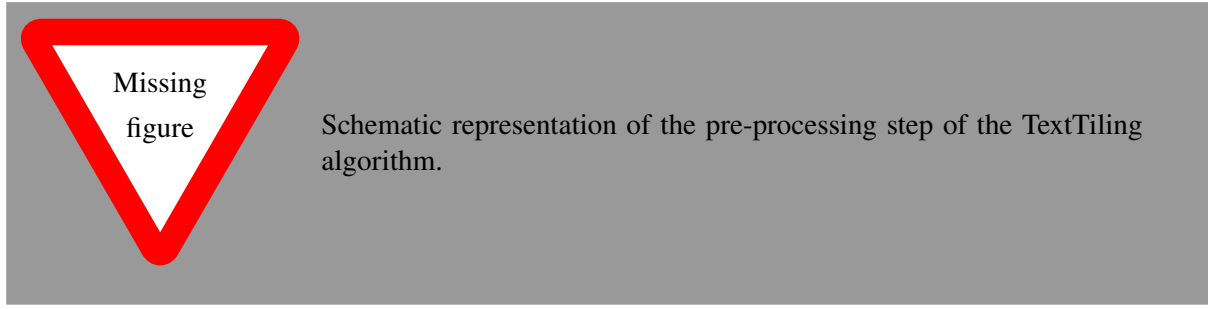


Figure 5.1: Schematic representation of the pre-processing step of the TextTiling algorithm.

to compare sequences of similar lengths when computing the similarity scores between subsequent parts of the text.

We thus end up with a sequence of spans $s_1, s_2, \dots, s_n$, each constituted of w tokens, but only the non-stopwords among these will be considered in the next step where we compute lexical scores.

5.1.1.2 Lexical score determination: Three approaches based on lexical co-occurrence and distribution

After the text has been split in spans of w tokens, a lexical score is computed at the gap between each span. For instance as represented on Fig. 5.2, for a text of n spans $\{s_1, s_2, \dots, s_n\}$, there are $n - 1$ gaps and thus $n - 1$ lexical scores to compute.

Three approaches are suggested in the original paper to measure this score: one based on the lexical similarity between subsequent blocks of text, one based on the proportion of never-yet-seen words in the two spans surrounding a gap, an one based on lexical chains, i.e. chains of related words. We briefly describe them in the following.

Block comparison The idea behind this method is that a topic is characterised by a specific vocabulary which entails that when a topic shift occurs, the vocabulary also changes. Thus, we can identify topic boundaries by looking at the lexical similarity between subsequent blocks of text. The more different the vocabulary is between two blocks, the more likely it is that there is a topic shift between them.

In the original TextTiling algorithm, the scores were computed by comparing the two blocks of k spans on each side of the gap. As Fig. 5.2 shows, the lexical score $score_{BC}(i)$ (corresponding to the gap i) would correspond to the normalised sum of the inner product between the term frequency vectors of the two blocks, where the term frequency vector of a block is a vector that counts the number of times each token appears in the block. Formally we would have:

$$score_{BC}(i) = \frac{\sum_{t \in V} tf(t, \{s_{i-k+1}, \dots, s_i\}) \times tf(t, \{s_{i+1}, \dots, s_{i+k}\})}{\sqrt{\sum_{t \in V} tf(t, \{s_{i-k+1}, \dots, s_i\})^2} \times \sqrt{\sum_{t \in V} tf(t, \{s_{i+1}, \dots, s_{i+k}\})^2}}$$

where V is the vocabulary of the text and $tf(t, B)$ is the term frequency of the token t in the block B .

This score corresponds to a cosine similarity between the term frequency vectors of the two blocks. With this measure, a low similarity score encased between higher ones would indicate a potential topic shift.

Vocabulary introduction The second approach relies on an idea similar to the first one, but instead of looking at the similarity between two blocks of text, it focuses on the introduction of new words in a segment of text. This approach is inspired from the observation of Youmans (1991) that a segment of text

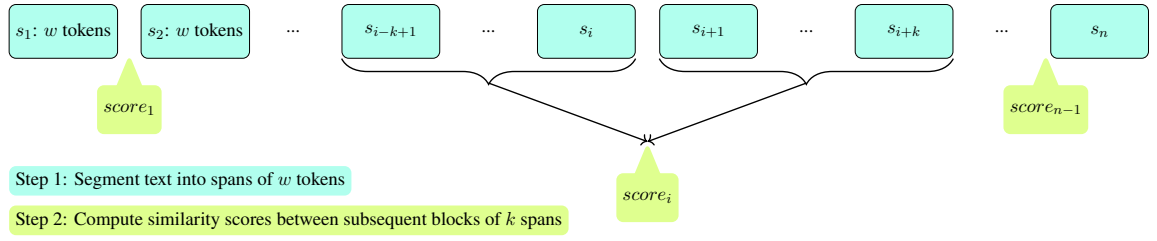


Figure 5.2: Segmentation in spans of w tokens and computation of the lexical scores in the TextTiling algorithm.

with low portion of never-yet-seen words in a segment of text followed by one with a much higher portion correlates ‘closely to constituent boundaries and information flow’ (p. 788). This idea is thus adapted to the TextTiling algorithm by computing the ratio of never-yet-seen words in an interval of 2 spans centred around the gap. For a given gap i we would thus have:

$$score_{VI}(i) = \frac{\text{number of never-yet-seen words in } \{s_i, s_{i+1}\}}{\text{total number of tokens in } \{s_i, s_{i+1}\}} = \frac{\text{number of never-yet-seen words in } \{s_i, s_{i+1}\}}{2 \times w}$$

With this measure, a sharp increase in the score would indicate a potential topic shift.

Lexical chains The last approach relies on the concept of lexical chains, i.e. chains of tokens that are considered ‘to be in a coherence relation with one another’. This idea comes from Morris (1988) who developed a dictionary-based method to identify these chains, and considered that a coherent segment of text would correspond to the extent of a chain. This approach is not directly adapted to the TextTiling algorithm for several reasons, including the practical limitations of the dictionary-based method, and the possibility to have multiple simultaneous chains sometimes overlapping each other. Hearst (1997) thus suggest a few modifications to this approach to make it more suitable for their algorithm. First they suggest to replace the dictionary with a co-occurrence based method to identify chains of morphological variants of the same word. Second, they compute all the chains and then compute the score of a gap by looking at the number of *active* chains at this gap, where a ‘chain for term t is considered active across a sentence gap if instances of t occur within some distance threshold of one another’ (p. 47).

This approach is not formalised in the original paper nor in Hearst (1994) which the authors point to for further details. This paper also shows that the block comparison approach performs slightly better than the lexical chains one, though the difference ‘may not prove significant over the long run’ (p. 14). In our own experiment, we used the Nltk implementation of the TextTiling algorithm which includes the block comparison similarity measure but not the vocabulary introduction nor the lexical chains one. We thus implemented our own version of these two metrics.

5.1.1.3 Boundaries identification: Locating the highest variabilities in the lexical scores

Once the lexical scores have been computed for each gap, the next step is to identify the topic boundaries by looking at the variability of these scores. The idea is that a topic shift would correspond to a sharp change in the lexical scores, which would be reflected in a local minimum in the scores. However, not all local minima correspond to topic shifts, and some topic shifts might not correspond to local minima. To address this issue, the original TextTiling algorithm suggests looking at the depth of the *valleys* in which the lexical scores fall. A deeper valley would indicate a more significant change in the lexical scores, which would be more likely to correspond to a topic shift. This approach is independent of the metrics

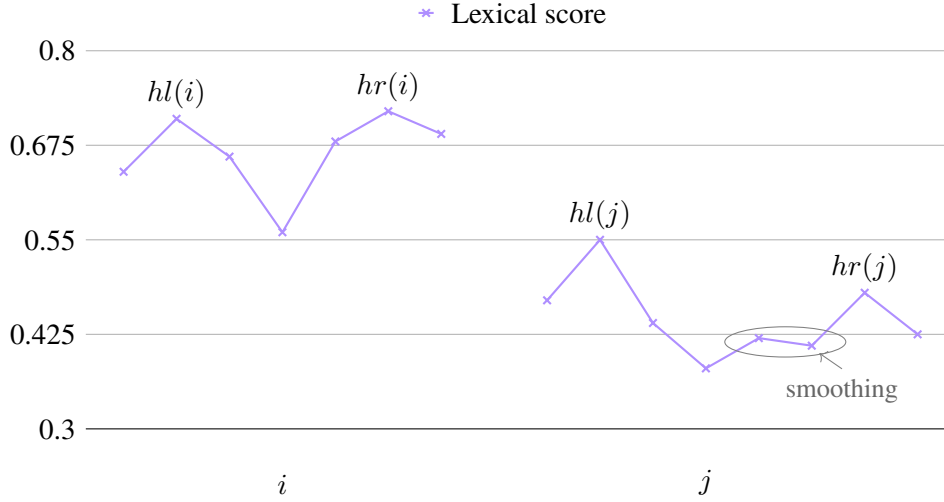


Figure 5.3: Examples of lexical scores used in the depth scores computation.

used to compute the lexical scores, and thus can be applied to any of the three methods described in the previous section as well as a combination of them or a different process as we will see in Section 5.1.2.

A prior step to identifying boundaries thus consists in computing the *depth scores* for our document. Formally, given a gap i , we measure the depth of this valley by retrieving the first lexical score on the left that forms a pic, *i.e.* $hl(i)$ such that it is greater than the score directly on its left. We retrieve $hr(i)$ similarly on the right. The depth score of the gap i is then computed by adding the depths on both sides: $dp(i) = \frac{(hl(i) - score(i)) + (hr(i) - score(i))}{2}$.

This method can be sensitive to small perturbations in the lexical scores, which can lead to identifying false boundaries. To address this issue, the original TextTiling algorithm suggests applying a smoothing function to the lexical scores before computing the depth scores. This smoothing function can be a simple moving average or a more sophisticated method such as a Gaussian filter. The goal of this smoothing is to reduce the impact of small fluctuations in the lexical scores, which can help to identify more accurate topic boundaries.

Fig. 5.3 shows two cases of depth score computation. The first one (i) is classical, where $hl(i)$ and $hr(i)$ are the first pics on the left and right of the considered score. The second case (j) illustrates the role of smoothing, as there is a very small pic between $score(j)$ and the $hr(j)$ we actually consider. Without smoothing, $score(j+1)$ would have been used to compute the depth score while it would not be representative of the real lexical similarity at this point.

The topic boundaries can then be identified based on the local maxima of the depth scores. The recommendation of the original paper is to consider that there is a boundary at gap i if the depth score at this gap is above a certain threshold that is determined based on characteristics of the text. Formally, they suggest to consider that a gap i is a topic boundary if $dp(i) > \bar{dp} - \sigma_{dp}$ where $\bar{dp}$ is the mean of the depth scores and σ_{dp} is the standard deviation of the depth scores. In practice these boundaries are shifted to the closest gap between two paragraphs because the split into spans of length w at the very beginning of the algorithm erases this structure.

5.1.2 Carenini's coherence-layer based approach

As explained in Section 5.1.1, the TextTiling algorithm relies on a similarity metric to compute the lexical scores that are used to identify topic boundaries. The original paper suggests three approaches to compute

this similarity metric, but other approaches can be used as well. For instance, Xing and Carenini (2021) suggest to use a trained utterance-pair coherence scoring model based on *Next Sentence Prediction* (NSP) BERT model (Devlin et al., 2019). In this section we describe this approach (Section 5.1.2.1) and the different datasets they used to train and evaluate their model (Section 5.1.2.2). Their work covers English and Chinese and we will discuss the small differences between the approaches which concern the datasets and the way they are used. The approach itself is theoretically language-agnostic but its reliance on NSP BERT makes it likely to be unsuitable as is for some languages, in particular less-resourced ones.

5.1.2.1 The metric: leveraging a text-pair coherence scoring task in an unsupervised way

The main idea behind Xing and Carenini’s (2021) approach is that the DTS task can be seen as the extension of a text-pair coherence scoring task (L. Wang et al., 2017; P. Xu et al., 2019; W. Wang et al., 2020). Indeed, based on (Hearst, 1997; Purver et al., 2006), they consider that ‘utterances associated with the same topic should be more coherent together than the utterances about different topics’ (Xing & Carenini, 2021, p. 167). Thus, they suggest to leverage NSP BERT to train an utterance-pair coherence scoring model that can be used as a similarity metric in the TextTiling algorithm. However, given the complexity of annotating utterance pairs with coherence scores, they suggest to train this model in an unsupervised way by using a relative ranking objective (H. Li, 2011). The idea is to feed the model with triples of utterance pairs that they indicate as more or less coherent one from the other, which allows the model to learn a distribution of coherence scores and incidentally train it to produce utterance-pair coherence scores. We detail the text-pair coherence scoring task and the relative ranking approach in the following.

Text-pair coherence scoring task The text-pair coherence scoring task consists in assigning a coherence score to a pair of texts, which in our case are two utterances. However assigning an absolute coherence score to a pair of utterances is a complex and very subjective task which would make it complicated to produce annotated data in sufficient amount and quality to train a model. Instead, Xing and Carenini (2021) suggest to input triples of utterance pairs to the model ordered according to their coherence, which allows the model to learn a distribution of coherence scores and incidentally train it to produce utterance-pair coherence scores. This requires to define a way to build these ordered triples. Xing and Carenini (2021) extend L. Wang et al.’s (2017) idea that ‘a pair of texts from the same segment should be ranked more coherent than a pair of texts randomly picked from different paragraphs’ (Xing & Carenini, 2021, p. 169) by suggesting an adaptation based on dialogue structure (E. A. Schegloff & Sacks, 1973).

They thus suggest to consider three levels of coherence: (1) adjacent pairs of utterances, (2) pairs of utterances from the same dialogue but not adjacent, and (3) pairs of utterances from different dialogues. Adjacency is a notion from Conversation Analysis (E. A. Schegloff & Sacks, 1973) that is stronger than proximity, as it also requires that the pair of utterances is produced by a pair of speakers and fits a certain dialogical structure such as question-answer or offer-acceptance/refusal.

In practice Xing and Carenini (2021) build a training triple by randomly sampling an adjacent pair of utterances from their dataset, then keeping the first utterance of this pair and randomly sampling a second utterance from the same dialogue but not adjacent to the first one, and finally keeping the first utterance and randomly sampling a third utterance from a different dialogue. This method is illustrated in Fig. 5.4 where we can see two schematic dialogues where the first one begins with an adjacent pair of utterances (A and B) which constitutes the most coherent pair, then a random utterance from the same dialogue (C) which is less coherent with A, and finally a random utterance from a different dialogue (D) which is the least coherent with A.

This method allows Xing and Carenini (2021) to produce a set of ordered triples of utterance-pair without the need for data annotated with coherence scores. Their three levels of coherence only require

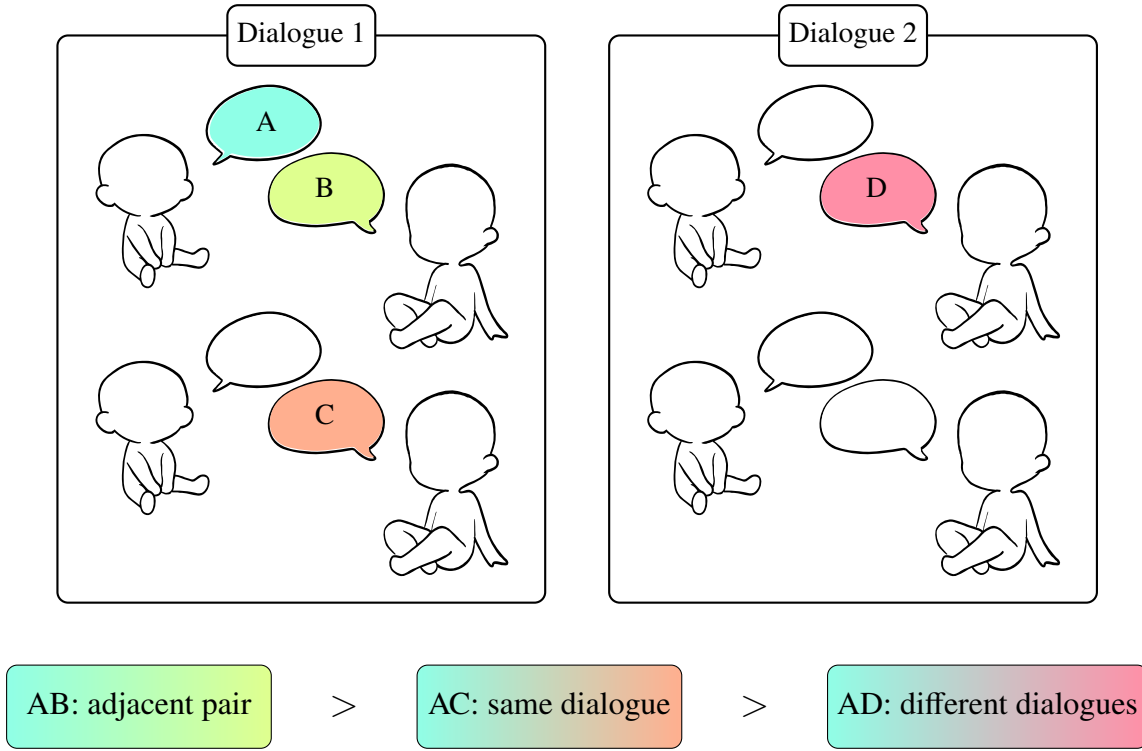


Figure 5.4: Levels of coherence considered by Xing and Carenini (2021)

annotations for the adjacency pairs.

Relative ranking objective The goal of the utterance-pair scoring model is to produce coherence scores for pairs of utterances that reflect the relative coherence of these pairs. However, given that there is no gold score for a given utterance pair, it is not possible to train the model with a classical regression loss. Indeed, as represented on Fig. 5.5, a regression loss would require to have a gold score for each pair of utterances, in order to train the model to minimise the difference between the hypothesised score and the gold score.

Xing and Carenini (2021) thus suggest to train the model with a relative ranking objective, which means that instead of considering one pair per input and rewarding the model for assigning a score that is close to a gold score, *two* ordered pairs of utterances per input are considered and the model is rewarded for assigning a higher score to the more coherent pair. This method allows the model to learn to produce coherence scores that reflect the relative coherence of the pairs of utterances, which is what is needed for the task.

The ranking loss associated with this method is described in Fig. 5.5. Since there are three levels of coherence, three inputs per triple of utterances can be built: AB vs AC, AC vs AD, and AB vs AD. The loss is then computed by averaging the individual ranking losses. In practice, Xing and Carenini (2021) use a marginal ranking loss with a margin of 1 and a reLu activation function which sets the individual scores to 0 when they were negative.

By showing the model all the possible pairs of utterance pairs, it can learn to produce a distribution of coherence scores that reflects the relative coherence of the pairs. Indeed, it learns that $coherence(AB) > coherence(AC)$, $coherence(AC) > coherence(AD)$, and $coherence(AB) > coherence(AD)$, which taken together allows the model to learn that $coherence(AB) > coherence(AC) > coherence(AD)$. At the end, the model is trained to produce coherence scores between 0 and 1 that reflect the relative

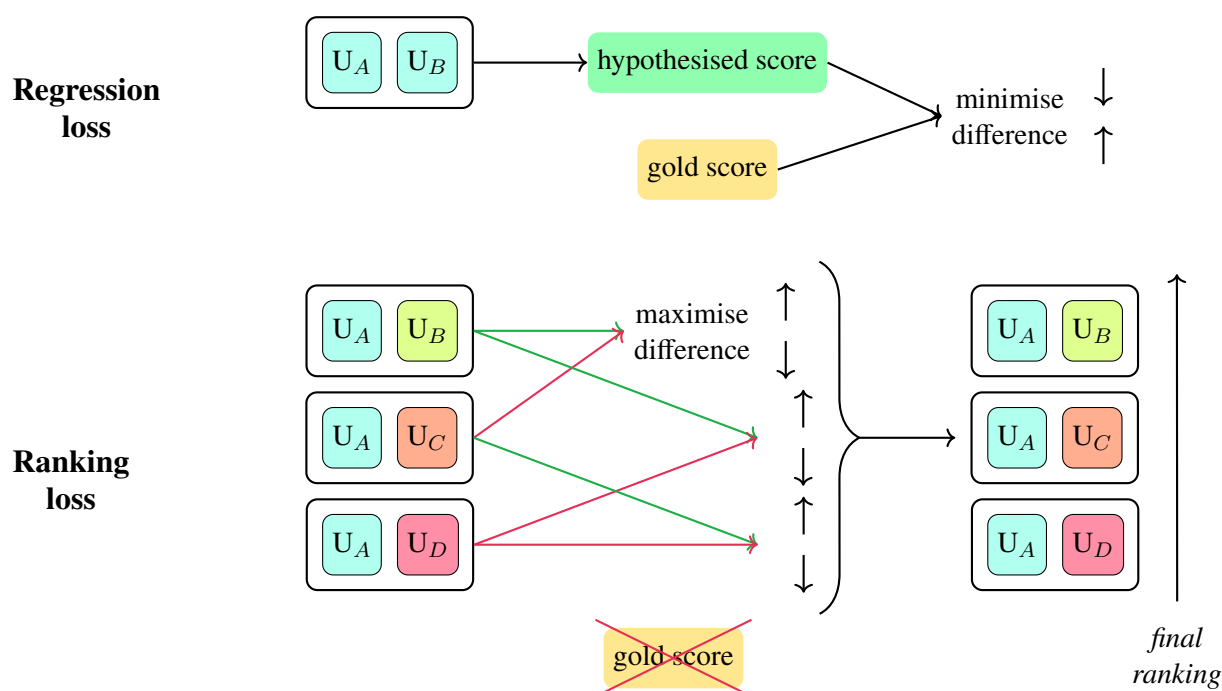


Figure 5.5: Illustration of the use of a regression loss compared to a ranking loss in similarity metric training

coherence of the pairs of utterances, which can then be used as a similarity metric in the TextTiling algorithm.

5.1.2.2 The datasets: two settings for two tasks

The utterance-pair coherence scoring task described above is only an auxiliary task to achieve DTS. Thus the datasets used to train and evaluate the model for this task are not necessarily the same as the ones used to train and evaluate the DTS model. Xing and Carenini (2021) indeed use separate datasets for these two tasks, which we describe in the following.

Training dataset: DailyDialog & NaturalConv Xing and Carenini (2021) define the ideal source corpus for the utterance-pair scoring model as ‘(1) having a fairly large size; (2) covering as many topics as possible; (3) containing both formal and informal expressions’ (p. 170). Based on these requirements they select two datasets: DailyDialog (Y. Li et al., 2017) for English and NaturalConv (X. Wang et al., 2021) for Chinese. They describe both datasets as containing ‘open-domain conversations about [multiple] daily topics’. A more detailed description of DailyDialog can be found in Section 5.3 where we compare it to the dataset we used. *[Amandine: I didn’t actually do it, wondering if it is useful]*

The main difference between the two datasets is that DailyDialog is annotated with dialogue acts, which can be used to identify adjacency pairs in the Conversation Analysis sense, while NaturalConv is not.

DailyDialog is annotated with the four dialogue acts *Inform*, *Question*, *Directive*, and *Commissive*, and the adjacency pairs are taken from the two dialogue act pairs *Question-Inform* and *Directive-Commissive*. Xing and Carenini (2021) add two additional constraints to the sampling of the utterances for the two other pairs. Among the four utterances A, B, C, and D, C must be of a different dialogue act from B, and D must be from a dialogue with a different topic from the one of A and B. These constraints make the

Dataset	DialSeg_711	Doc2Dial	ZYS	Friends
documents	711	4,130	505	236
language	English	English	Chinese	English
# sent/seg	5.6	3.5	6.4	13.2
# seg/doc	4.9	3.7	4.0	23.6
real-world	✓	✗	✓	✗

Table 5.1: Statistics of the datasets used for evaluation in Xing and Carenini (2021) and our dataset. *The statistics for DialSeg, Doc2Dial, and ZYS are reported directly from Xing and Carenini (2021).*

three pairs more different from each other, which increases the chances that the model learns to produce coherence scores that reflect these differences.

None of these strategies can be applied to NaturalConv since it is not annotated with dialogue acts, which means that the most coherent pair is simply made of subsequent utterances. Similarly the constraint for sampling C cannot be applied and an utterance is simply sampled from the same dialogue but not adjacent to A. The constraint for sampling D can be applied since NaturalConv is annotated with topics. This sampling method is thus less sophisticated than the one used for DailyDialog, but can be applied to any dataset that is not annotated with dialogue acts, which is closer to the reality of many existing datasets, including the one we used in our experiments.

Evaluation dataset: DialSeg, Doc2Dial & ZYS The final goal of Xing and Carenini (2021) is to improve the performance of the TextTiling algorithm on the DTS task. Thus, they evaluate their approach on datasets annotated with topic shifts. They use three datasets for this evaluation: DialSeg (Y. Xu et al., 2021) and Doc2Dial (S. Feng et al., 2020) for English, and ZYS (Y. Xu et al., 2021) for Chinese.

DialSeg is a real-world dataset constituted of task-oriented multi-turn dialogues from Stanford Dialog Dataset (Eric et al., 2017) and MultiWOZ (Budzianowski et al., 2018), in which topics are annotated manually. Doc2Dial is a synthetic dataset made from automatically built dialogue flows based on the structure of documents. Crowd workers then filled in these dialogue flows with natural language utterances. The topics are thus automatically defined by the structure of the documents. ZYS is a real-world dataset constituted of multi-turn dialogues from customer service phone calls and banking consultations. The topics are also annotated manually.

Overall these datasets are very task-oriented and contain rather short dialogues as reported by Table 5.1, which makes them quite different from the kind of dialogues we are interested in, which are longer and more open-ended.

5.2 Experimental setting

As explained before, DTS is typically performed on rather organised dialogues while we are interested in more open-ended and less organised dialogues. Thus, we want to see how the Xing and Carenini’s (2021) approach performs on a multi-party dialogue dataset, and whether a feature-based approach can compete with it.

We first summarise the different models and baselines that we compare in our experiments (Section 5.2.1). Then we present the evaluation metrics we use to evaluate the performance of the different models on the DTS task (Section 5.2.2).

Model	Description	Training
XingCarenini _{OG}	Xing & Carenini (2021)’s original approach trained on their original dataset	DailyDialog & NaturalConv
XingCarenini _{Friends}	Xing & Carenini (2021)’s original approach adapted to our dataset	Friends TV show
Feature-based	Our approach based on linguistic features	X
Random	Random baseline	X
TextTiling _{OG}	Original TextTiling algorithm	X
TextTiling _{Dial}	Original TextTiling algorithm with dialogue-specific adaptations	X
LLM-based	Basic LLM-based approach	Zero-shot*

Table 5.2: Summary of the models compared in our experiments. The first three models are the ones we compare for the main results, while the last four are baselines.

*While we do not perform any fine-tuning on the models for this baseline, we want to stress the training cost, both economic and environmental, of these models, which can be easily obscured by their relative ease of use and accessibility.

5.2.1 Models

Table 5.2 summarises the different models we compare in our experiments. The first three models are the ones we compare for the main results, while the last four are baselines.

Models All the models we compare in our experiments are adaptations of the original TextTiling algorithm. On the one hand we have two adaptations of Xing and Carenini (2021)’s approach, one trained on their original dataset and one adapted to our dataset, and on the other hand we have our feature-based approach which is detailed in Section 5.5.

Baselines We use three types of baselines for our experiments. First, we use a random baseline to see how much the models perform better than chance. This baseline is the same as the one used by Xing and Carenini (2021) which assigns boundaries with a probability $\frac{b}{k}$ where k is the number of utterances and $b \in \llbracket 0, k-1 \rrbracket$ is a randomly chosen number of segment boundaries. Second, we compare our results to the original TextTiling algorithm to see how much the adaptations we made to it improve its performance. We use both the original TextTiling algorithm and a version of it with dialogue-specific adaptations. Finally, we use a basic LLM-based approach to see how it can perform in a zero-shot setting on this task. All these baselines are described in more details in Section 5.4

5.2.2 Evaluation Metrics

We use two families of evaluation metrics for our experiments. On the one hand, F-measures are based on the number of correct, false positive, and false negative boundaries. On the other hand, P_k and WindowDiff perform different comparisons between the predicted and the gold segmentations within a sliding window. They are represented on Fig. 5.6 and Fig. 5.7 respectively.

F-measure The classical F-measure is calculated by comparing the predicted boundaries to the gold ones, and computing the ratio of twice the correct number of boundaries over the total number of boundaries, i.e. the number of predicted ones added to the number of expected ones. A predicted boundary is

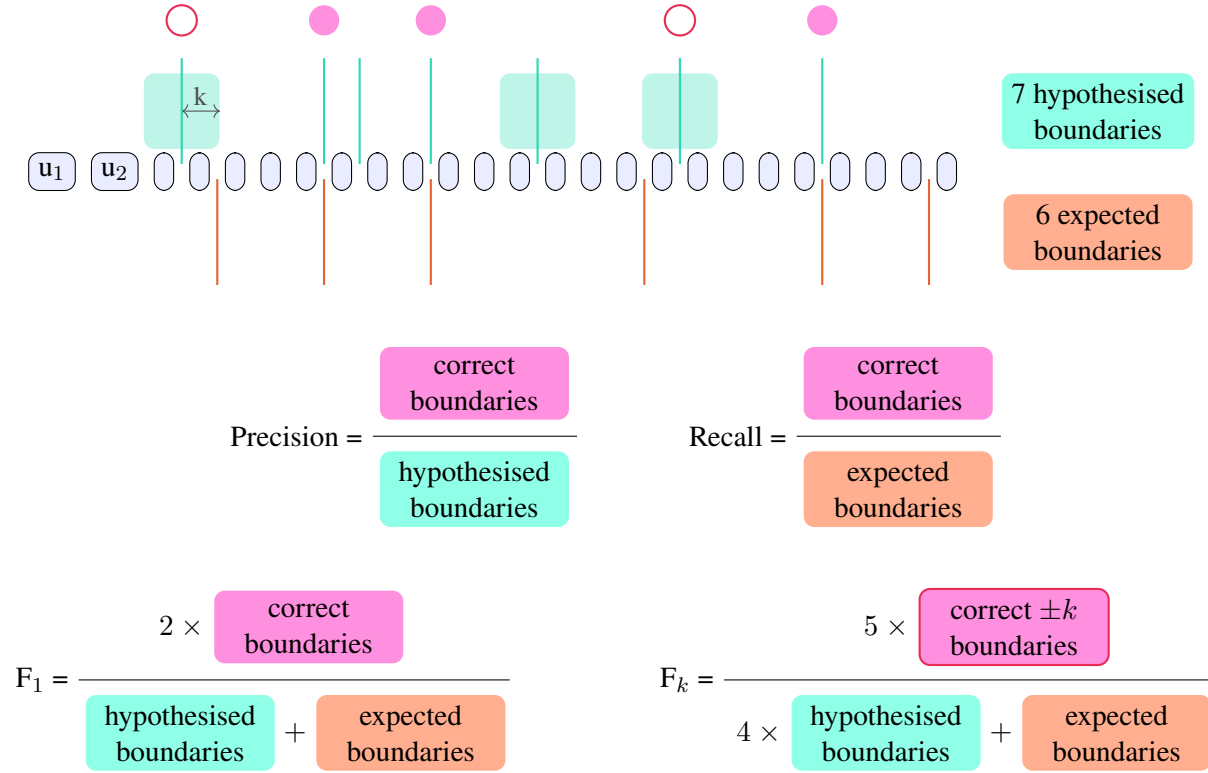


Figure 5.6: Explanation of the Precision, Recall, F₁ and F_k measures for topic segmentation

considered correct if it matches a gold boundary, which means that it is at the same position in the dialogue. This metric is quite strict since it does not give any leeway to the models for predicting boundaries that are close to the gold ones, which can be problematic given the difficulty of the task. A segmentation where each boundary falls one utterance before or after a gold boundary should be considered better than a segmentation where all the boundaries are far from the gold ones, but the classical F-measure would give the same score to both segmentations.

Thus, we also report a more relaxed version of it, which we call F_k , that considers a predicted boundary as correct if there is a gold boundary at most k utterances before or after the predicted one. This allows us to give some leeway to the models for predicting boundaries that are close to the gold ones, which is important given the difficulty of the task and the fact that there can be multiple valid segmentations of a dialogue. A gold boundary cannot count a second time once it has been matched by a predicted boundary. We also give more weight to precision compared to recall as we consider that finding right boundaries is more important than finding all of them. Formally, it means that the number of predicted boundaries is weighted by a factor of 4 in the denominator of the formula, which decreases the performance of a model that would suggest much more boundaries than expected.

Window-based metrics P_k (Beeferman et al., 1999) and WindowDiff were both proposed to address the limitations of the F-measure for topic segmentation. They are both based on a sliding window of size k that moves through the dialogue, and they compare the predicted and gold segmentations within this window. P_k considers that there is a disagreement when the predicted and gold segmentations do not agree on whether the two ends of the window belong to the same segment or not. Beeferman et al. (1999) recommend to set the size of the window k to half the average topical segment length.

While P_k solves the the strictness issue of the classical F-measure, Pevzner and Hearst (2002) point

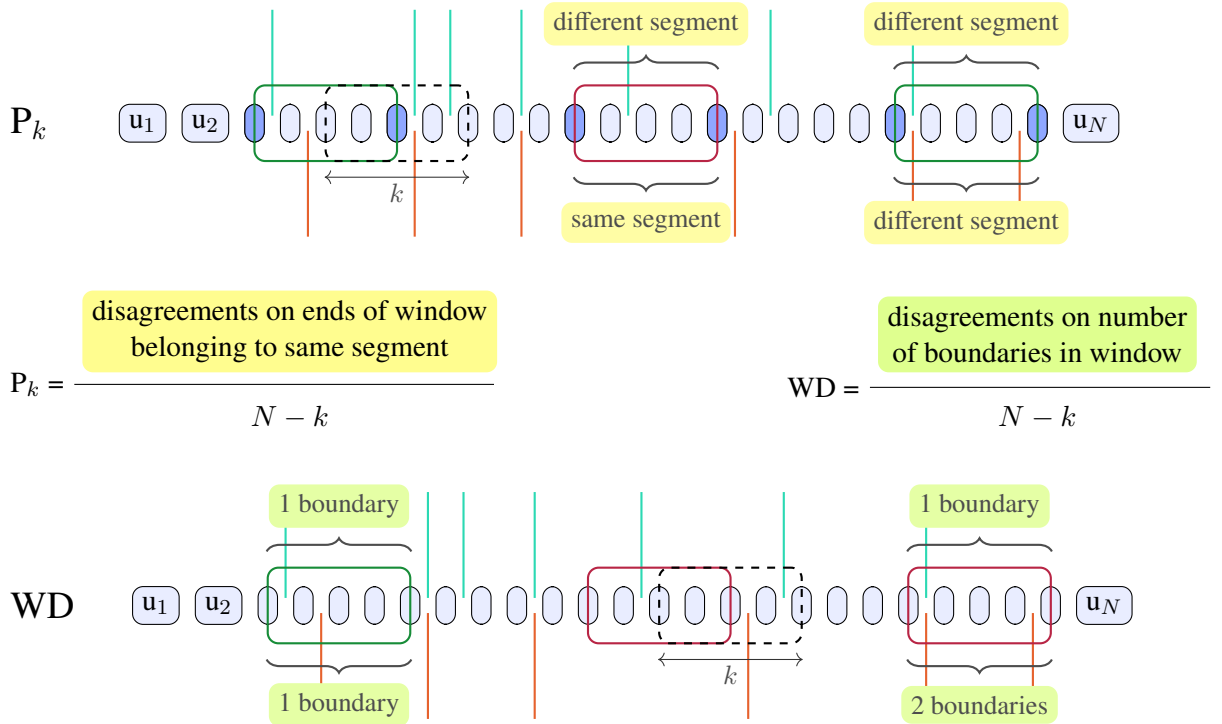


Figure 5.7: Explanation of the P_k and Window Diff (WD) metrics for topic segmentation

that it still over-penalises near-misses, and that the metric varies heavily depending on the topical segment size. They also report that false-positive, i.e. missing boundaries, are also more penalised than false-negatives, i.e. extra boundaries, though we consider this to be a desirable property for our task, as we explained when introducing F_k . They thus propose WindowDiff, which is similar to P_k but considers that there is a disagreement when the predicted and gold segmentations do not agree on the *number* of boundaries within the window, which allows it to be more stable across segment size distribution, to penalise near-misses less than other errors, and to balance the penalties for false-positives and false-negatives. They keep the same recommendation as Beeferman et al. (1999) for the size of the window, which is to set it to half the average topical segment length.

Both metrics are calculated as the ratio of the number of disagreements over the total number of windows, which is $N - k$ where N is the total number of utterances in the dialogue. Thus, a lower score indicates a better performance for these metrics, which is the opposite of the F-measure where a higher score indicates a better performance.

5.3 Dataset

For these experiments we use transcriptions in English of the episodes of the TV show *Friends*. Transcripts for all ten seasons (236 episodes), annotated in scenes and with additional notes, were used for the Character Mining project (Y.-H. Chen & Choi, 2016) and their dataset is available online¹⁴ (Apache License, Version 2.0).

Casual conversations are central to human interactions but finding suitable data to analyse them, in particular at the topical level, remains complicated (Gilmartin & Campbell, 2016). For this reason, using

¹⁴<https://github.com/emorynlp/character-mining>

transcriptions from a TV show seemed like a good idea for a first approach of our problem as it enabled us to have a sufficient amount of data to work with ML tools, while remaining close enough to real-life dialogues. Studies have indeed shown that spoken language in fiction is quite similar to spontaneous speech (Forchini, 2009).

In this section we begin by giving a detailed description of the type of dialogues that constitute our dataset using the framework we proposed in Chapter 2 (Section 5.3.1) and then we explain how we segmented the dialogues into topics (Section 5.3.2) without actually performing annotations.

5.3.1 Detailed description

The Friends corpus was chosen for the large amount of data it provides with 236 episodes of daily-life conversations, often involving multiple participants, in a variety of contexts and situations. The main characters are six friends who live in New York and interact with each other and with other characters in a variety of situations. These dynamics are quite similar to the ones we can find in real-life conversations, which makes them quite interesting for our purposes. The Friends corpus is thus rather similar to naturalistic conversations when it comes to the PARTICIPANTS dimension of our framework, as highlighted in purple in Table 5.3. The show is set in the 1990s and early 2000s, which means that the vocabulary used is probably quite different from the one used today, but it also means that it provides a good representation of the language used in casual conversations at that time, in particular given the range of topics covered that mostly revolve around daily-life situations.

However, the fact that the show is scripted means that there is a high level of control over the way things are said and the way the conversations are structured, which is quite different from naturalistic conversations where there is no such control. Indeed, the production of the dialogue content was done by a team of script writers before it was performed by the actors. Overall the TV show nature influences the degree of structure and control in the dialogues, which are rather chaotic by design, while naturalistic conversations are so because of the natural dynamics of human interactions. This is particularly visible in the LANGUAGE dimensions of our framework as we just said, but also in (VERBAL) INTERACTION and FUNCTION & STRUCTURE, which are highlighted in orange in Table 5.3. The control over the language means that there is little, if any, code-switching or use of non-standard language in the show, unless it is used for comedic effect. This is due to the actual purpose of the show, which is to entertain an audience, where conversations are used as a means to create humour rather than being the goal in themselves. This factor possibly makes the conversation more easily understandable from an external point of view than naturalistic conversations. When building a topic segmentation system, it means that despite the complex nature of these multi-party open-ended conversations, they are most likely easier to work with than real daily-life ones.

In terms of verbal interaction, the order of the turns is mostly decided in advance, though the fact that we rely on transcriptions of the episodes rather than the original scripts means that we have access to the little bits of improvisation that actors added to the show. However a drawback of using the transcription rather than the multimodal data is that it is mostly linearised in terms of turn-taking and that it is complicated to know how much of the coordination signals that may be present in the original data were actually kept in the transcription.

The last dimension of our framework, ENVIRONMENT & MEDIUM, is very relevant to the DTS task and is the one where the most differences between the original video data and the transcriptions are visible, as highlighted in teal in Table 5.3. The fact that we rely on transcriptions means that we do not have access to the visual and auditory cues that are present in the original data, which are important for understanding the conversations and for segmenting them into topics. For example, we do not have access to the gaze and gesture signals that are present in the original data, which may give cues about the topic of the conversation that are not present in the linguistic content only. While some notes are provided in the

transcriptions when important changes happen in the physical context, such as a change of location or the entrance of a new character, we did not include them in our experiments in order to compare the different models on a common ground, which is the linguistic content of the dialogues. This is also more realistic for the DTS task as it is more likely that we will have access to the linguistic content of the conversations than to the visual and auditory cues, which are more difficult to capture and to process. Indeed, while multimodal datasets have received more attention in the last few years, they remain complex to gather and to work with. We thus wanted our approach to be as easily usable as possible on other datasets in order to set a baseline for future work on DTS that can be easily compared to and built upon.

5.3.2 Segmentation in topics

The *Friends* dataset is not annotated in topics but we chose to automatically infer the segmentation from its segmentation in scenes. Additionally, we consider that the notes in the transcripts indicate an important enough change to create a topic shift. As a result, our assumption is that the topic boundaries coincide with the notes and change in scene. This annotation method is far from perfect but it has the advantage of being objective.

(5.1) is an extract of our dataset. We can see that five different characters appear in this short extract, as well as what we consider as two different topics as there is a note between the second and third intervention of this extract. The note explains that a new character enters the room, which is a sufficient disruption to create a topic shift. However in practice, the first speech turn after the note remains on the previous topic and the shift happens right after. This is quite common in the dataset and not unexpected based on real life dialogues, especially when they involve many people. Moreover, the format of the transcriptions is such that concurrent events and/or speech turns are written down in a given order and overlaps are not represented as we discussed in Section 5.3.1. This further justifies the metrics we use on top of the classical F-measure as discussed in Section 5.2.2, as judging a segmentation based on the exact position of the boundaries is not an accurate reflection of the actual performance of the model on this task.

- (5.1) **Joey Tribbiani:** Strip joint! C'mon, you're single! Have some hormones!
Ross Geller: I don't want to be single, okay? I just... I just- I just wanna be married again!
(Rachel enters in a wet wedding dress and starts to search the room.)
Chandler Bing: And I just want a million dollars!
Monica Geller: Rachel?!
Rachel Green: Oh God Monica hi! Thank God! I just went to your building and you weren't there and then this guy with a big hammer said you might be here and you are, you are!
 [Friends, Season 1, Episode 1]

In total our corpus thus comprises 236 episodes, each segmented in topics based on the changes in scenes and the notes, which gives us an average of 23.6 segments per episode and 13.2 utterances per segments. In comparison with the datasets used in Xing and Carenini (2021), our dialogues are in average four to five times longer and the topical segments span twice to four for utterances as summarised in Table 5.1. Table 5.4 gives a more detailed overview of the statistics of our dataset, which shows that there is a lot of variability in the number of segments per episode, as well as in the number of utterances per segment, which makes the DTS task more challenging and justifies once more the variety of metrics we use to evaluate the different models on this task. One of the shortcomings of P_k was indeed that it is heavily influenced by the segment size distribution, which is quite variable in our dataset, and thus we wanted to use a metric that is more robust to this variability like WindowDiff. We now detail the different baselines and the different version of the TextTiling algorithm we worked with on this dataset.

Parameters hierarchy		Original	Transcription
Participants			
Participation structures	Number of participants	6 main participants + others, variable unknown number of script writers	
	Functions	???	
Social dynamics	(Conversational) History	friends with long history + growing history with viewers throughout seasons	
Language			
Resource level		highly controlled	
Conversation level		monolingual, wide audience	
Environment & Medium			
Physical context		co-located (mostly)	some notes, removed for topic segmentation
Visibility	Gaze	face-to-face (mostly)	✗
	Gesture / Facial expression	✓	✗
Audibility	Prosody / Tone	✓	✗
	Noise	controlled	✗
(Verbal) Interaction			
Co-construction	Coordination signals	✓ (scripted)	limited (no pauses, ???)
	Turn-taking	scripted	linearised
	Completion	scripted	
Time	Planning	scripted	
Function & Structure			
Epistemic grounding	World setting	fictional, possibly fiction in fiction	
Purpose	Practical	viewers' entertainment	
	Intention		
Activity	Success condition	viewer oriented	
	Vocabulary	daily life	
	Structure	more structured than equivalent setting in real life?	

Table 5.3: Description of the Friends corpus for a topic segmentation task

Season	# Episodes	# Segments / Episode	# Utterances / Segment	# Tokens / Utterance
1	24	36.7 ± 11.6	6.6 ± 6.2	13.4 ± 13.5
2	24	11.2 ± 2.5	19.6 ± 14.7	14.1 ± 14.0
3	25	27.0 ± 9.5	9.3 ± 9.1	13.8 ± 13.5
4	24	28.8 ± 11.4	8.9 ± 8.3	13.6 ± 13.5
5	24	30.5 ± 5.4	8.2 ± 8.1	13.3 ± 12.9
6	25	27.2 ± 10.1	9.2 ± 7.8	13.8 ± 13.3
7	24	25.7 ± 7.6	9.9 ± 8.6	13.3 ± 12.6
8	24	22.4 ± 8.8	11.1 ± 9.3	13.6 ± 12.9
9	24	11.8 ± 3.5	20.6 ± 19.8	14.8 ± 13.7
10	18	11.2 ± 3.0	24.0 ± 22.3	13.2 ± 12.4
All	236	23.6 ± 11.8	13.2 ± 14.9	13.0 ± 13.6

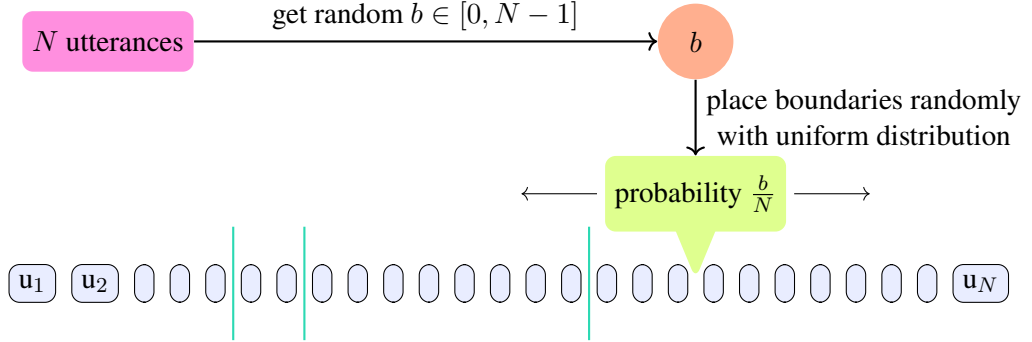
Table 5.4: Statistics of the *Friends* dataset used for topic segmentation.

Figure 5.8: Representation of the random baseline for topic segmentation.

5.4 Description of the Baselines: From random guessing to high-cost LLMs

We explore a wide range of baselines for the DTS task, that are presented in this section by increasing computational power. The most basic one is a simple random baseline (Section 5.4.1). Then we try the TextTiling algorithm in two basic versions, one for each of the approaches we evaluate in this chapter, i.e. feature-based and ML-based. Section 5.4.2 presents the results of the original TextTiling algorithm as well as a slightly modified version imagined for dialogue, and Section 5.4.3 shows the results when using a BERT model as similarity metric, as in Xing and Carenini (2021), without any fine-tuning and when fine-tuned on the dataset used by Xing and Carenini (2021). Finally we leverage open-weights LLMs in a zero-shot manner to produce topic segmentation predictions (Section 5.4.4).

5.4.1 Random baseline: a basic sanity check

Our most basic baseline is a random one taken from Xing and Carenini (2021). It consists in first randomly drawing a number of boundaries b between 0 and $N - 1$ where N is the total number of utterances in the dialogue, and then determining at each gap whether there is a boundary or not by placing a boundary with a probability of $\frac{b}{N}$. This baseline very simple and constitutes a good sanity check for the other models, as any model that does not perform better than this cannot be considered as being based on or having learned any relevant signal for the task.

Season	Pr. $\uparrow$	Rec. $\uparrow$	F ₁ $\uparrow$	F _{k1} $\uparrow$	F _{k2} $\uparrow$	P _k $\downarrow$	WD $\downarrow$
1	.15	.49	.2	.29	.34	.56	.73
2	.05	.5	.09	.12	.15	.58	.92
3	.1	.49	.15	.22	.25	.58	.81
4	.11	.47	.16	.22	.27	.58	.78
5	.12	.5	.18	.24	.29	.58	.79
6	.11	.49	.16	.23	.27	.57	.79
7	.1	.5	.15	.22	.26	.58	.81
8	.09	.49	.14	.2	.23	.58	.83
9	.05	.46	.08	.11	.14	.58	.89
10	.04	.48	.07	.1	.13	.58	.92
Total	.09	.49	.14	.2	.24	.58	.82

Table 5.5: Average results of the random baseline over 10 trials

Table 5.5 shows the average results of the random baseline over 10 trials. As expected it performs quite poorly overall but it is rather stable. The recall lies around .5 and is the metrics with the most variation which is expected since we randomly draw the number of boundaries. If the b is high we will likely have a lot of boundaries and thus increase the chances that some of them are correct, which will increase the recall, but it will also increase the number of false positives and thus decrease the precision, which is the worst performing metric for this baseline. P_k is the second best performing metric with an average of .58 (the lowest the better) which seems to support Pevzner and Hearst’s (2002) criticism as this random baseline does not rely on any relevant signal for the DTS task. This will be further discussed when comparing the results of the other models to this baseline.

To further investigate the gain brought by the approaches we want to evaluate in this chapter, we also ran a *smart* version of the random baseline where b is not drawn randomly but is set to the actual number of boundaries in the gold segmentation. This allows us to have a better idea of the differences in performance when the model must determine the right number of boundaries or not. The results of this smart random baseline are shown in Table 5.6. All the results except for the precision are significantly different between the two random baselines, though the differences are quite small for most measures. In particular the F_1 is slightly better for the first random baseline, which is due to a higher recall thanks to the higher chances of overgenerating boundaries. On the contrary, the F_{k1} and F_{k2} are better for the smart random baseline, rising from .2 to .23 and from .24 to .32 respectively, which is likely due to the higher weight of precision in these metrics. The P_k and WD are also better for the smart random baseline, falling from .58 to .47 and from .82 to .51 respectively, showing that restricting the number of boundaries allows for a sharp increase in performance even for a random model.

5.4.2 Dialogue Adapted TextTiling: a low-level feature-based model

The TextTiling algorithm has a Python implementation¹⁵ available in the NLTK library which we used as a base for our experiments. We experimented with two main aspects of the original algorithm. The first one is the way the text is split in spans prior to the computation of similarity scores, where we investigated the influence of their length on the performance of the model, and whether splitting the dialogues based on the utterances boundaries rather than ignoring the original structure was more relevant for dialogue. The second one was related to the Vocabulary Introduction similarity method, which we implemented

¹⁵https://www.nltk.org/_modules/nltk/tokenize/texttiling.html

Season	Pr. $\uparrow$	Rec. $\uparrow$	F ₁ $\uparrow$	F _{k1} $\uparrow$	F _{k2} $\uparrow$	P _k $\downarrow$	WD $\downarrow$
1	.15	.15	.15	.33	.45	.45	.49
2	.05	.05	.05	.14	.22	.48	.52
3	.11	.11	.11	.26	.36	.46	.5
4	.11	.11	.11	.27	.37	.46	.5
5	.13	.13	.12	.28	.39	.47	.51
6	.11	.11	.11	.26	.36	.46	.5
7	.1	.1	.1	.25	.35	.47	.51
8	.09	.09	.09	.22	.32	.47	.51
9	.05	.05	.05	.14	.21	.48	.52
10	.04	.04	.04	.11	.18	.47	.52
Total	.1	.1	.1	.23	.32	.47	.51

Table 5.6: Average results of the *smart* random baseline over 10 trials

completely as it did not exist in the NLTK implementation, and we tested an additional memory parameter m that would allow us to consider a token as never before seen if it did not appear in the last m utterances rather than in the whole dialogue, which we thought could be a very basic way to account for the fact that in a long dialogue, a topic can be resumed after talking about something else.

Influence of the spans length. As we explained in Section 5.1.1, one of the pre-processing steps of the algorithm consists in splitting the text in spans of w tokens. The default value of this parameter is set to 20 in the NLTK implementation but we wanted to test several values for this parameter to see if it had an influence on the performance of the model. We tested values of w between 5 and 50 with a step of 5 which seems reasonable as very short spans are likely to be little informative and very long ones are more likely to bridge over actual topic shifts. We also tested the values 13 and 26 as they correspond to the average number of tokens per utterance in our dataset, which is around 13, and to its double, which is around 26. As a sanity check we also included $w = 1$ which corresponds to the case where the text is split into tokens which allows the model to place boundaries between any two tokens, thus providing us with a baseline in terms of recall, while actually accessing very little information for the similarity computation, which should lead to a very low precision.

The average results for the 10 seasons can be seen on Fig. 5.9a. The complement to 1 of the P_k and WindowDiff metrics is plotted instead of their actual value to make the comparison with the other metrics easier. We can see that the performance of the model varies with the value of w and that the metrics display different behaviours. P_k and WindowDiff get better as w increases, with no significant differences between $w = 40$ and $w = 50$. The F-measure metrics on the other hand get better as w increases up to a certain point and then start to decrease, with the best performance around $w = 20$ or $w = 25$. The precision and recall can explain this behaviour as the recall is quite high for low values of w and then decreases as w increases, while the precision is quite low for low values of w and then increases as w increases. This is expected as with low values of w the model can place boundaries between any two tokens and thus has more chances to place boundaries in the right places, which increases the recall, but it likely places much more boundaries and thus more wrong ones which decreases the precision. We thus decided to set w to 20 for the rest of our experiments as it seems to be a good compromise between the different metrics.

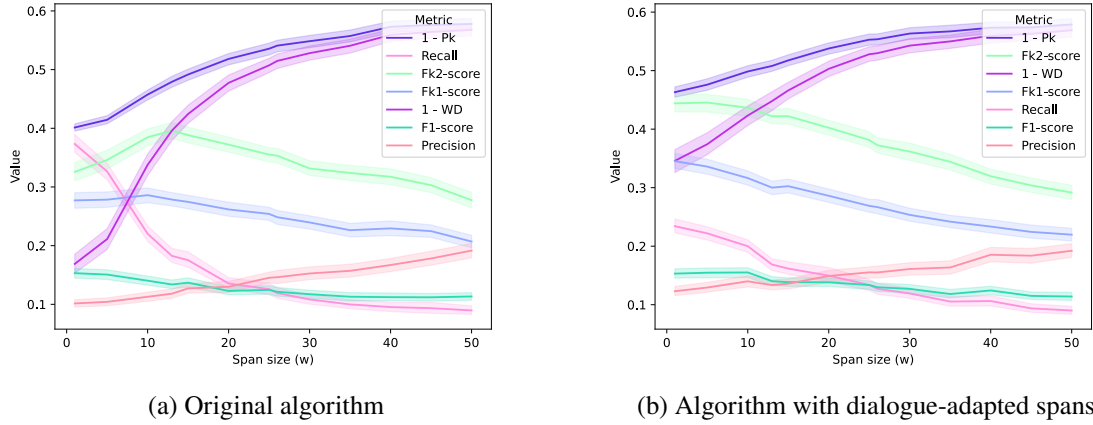


Figure 5.9: Average results of the TextTiling algorithm with different values of the spans length parameter w and the Block Comparison similarity method, evaluated on the 10 seasons of our dataset.

Splitting along utterances. The original TextTiling algorithm splits the text in spans of w tokens which can lead to spans that include partial utterances in our case. We thought that this was not the most relevant way to split our conversations because it ignored their structure. It would also be more inconvenient when including new features in the similarity computation that would rely on this structure. For this reason, we implemented a version of the algorithm that splits the text in sequences of utterances such that the number of tokens is the closest possible to w . In practice, most of the spans contain only one utterance if w is not too large, and they can contain more when the utterances are very short and are thus less likely to be informative. We performed the same tests as for the original version of the algorithm, i.e. we tried the same values of w between 1 and 50.

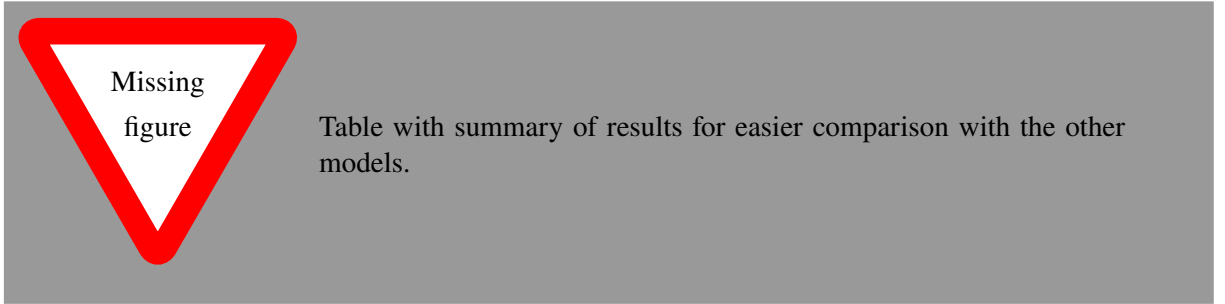
The results can be seen on Fig. 5.9b. We can see that the performance of the model is quite similar to the original version, with the precision and recall being slightly more stable with small values of w . This is completely expected as with our version of the span segmentation, the spans must at least be a full utterance long. Hence, the smallest values of w actually produce very similar span segmentations. When looking at the results in detail, the dialogue adapted method is always better than the original one when the results are significantly different – albeit often by a slight margin – except for the recall with $w = 5$ and $w = 1$ because of the higher opportunities to place boundaries with the original method. More detailed results including t-tests can be found in Appendix (missing reference: To be added in final version). We thus decided to use our version of the span segmentation for the rest of our experiments where we use $w = 20$ as decided before.

Implementing the vocabulary introduction similarity method: testing a memory parameter. Since the Vocabulary Introduction method was not implemented in the NLTK library, we implemented it ourselves and tested it on our dataset. As a reminder, this method uses the portion of never before seen words in the two spans around a gap as a similarity score, with the idea that a topic shift is likely to be accompanied by the introduction of new words. However, in a long dialogue, a topic can be resumed after talking about something else, and thus some words that were introduced at the beginning of the dialogue and are thus not new anymore can be reintroduced at the moment of the topic resumption and should be considered relevant for the detection of this topic shift.

For this reason, we added a memory parameter m that allows us to consider a word as new if it did not appear in the m spans before the two considered ones. We experiment with this parameter by testing

values of m between 5 and 50 with a step of 5. We also tested the model with an infinite memory as was suggested in the original algorithm. Surprisingly, there are no significant variations in the performance of the model with the different values of m . This suggests that most topic shifts are identified thanks to the introduction of new words that were not used in the dialogue before, and that topic resumptions are not so frequent or use a new vocabulary when they happen. This could be a consequence of using scripted dialogues and should be further investigated in other forms of dialogue, especially in naturally occurring ones, as re-introducing a topic after some time is likely less cognitively demanding when the same vocabulary is used.

We thus dropped the memory parameter for the rest of our experiments and considered only the original version of the Vocabulary Introduction method where a word is considered new if it did not appear in the dialogue before.



5.4.3 BERT for next sentence prediction: generalisability assessment

As a reminder, the model trained by Xing and Carenini (2021) is a similarity metric meant to be used in the TextTiling algorithm to perform the DTS task. Thus, the training of the model is not directly performed on the DTS task but rather on an utterance pair coherence scoring task. We use two versions of this model as baselines, one without any fine-tuning and one fine-tuned on the DailyDialog dataset as in Xing and Carenini (2021).

Without any fine-tuning, the model provides slightly better results than the random baseline but slightly worse than the smart random baseline as can be seen on the first row of Table 5.7.

When fine-tuned on DailyDialog and evaluated on the 10 seasons of our dataset, the 10 epochs of training do not change the results much. In particular, the F-measures values did not vary significantly between epochs but Recall varied between .39 and .48 with significant variations on epochs 1 to 4 compared to all the others, which led to differences in the P_k and WD scores in particular at epoch 1, 4, and also 2 specifically for WD. As with the other BERT-based models, the low precision and much higher recall indicate an over-generation of topical boundaries, which explains the results on the P_k and WD scores. The results are quite low overall and pretty similar to those obtained with the most basic random baseline, which suggests that the model trained on DailyDialog does not generalise well to our dataset. This is not very surprising given the differences between the two datasets in terms of dialogue structure and content.

A more striking result is the fact that the model without fine-tuning performs significantly better than at any of the 10 epochs of training on DailyDialog, which suggests that the training on DailyDialog actually made the model worse for our dataset, most likely diminishing its generalisation abilities.

All these results support the idea that more complex training data is needed to perform well datasets such as the *Friends* one. We will investigate this question in Section 5.6 when applying the same fine-tuning procedure on our dataset, which will allow us to have a better idea of the influence of the training data on the performance of the model.

Epoch	Pr. $\uparrow$	Rec. $\uparrow$	F ₁ $\uparrow$	F _{k1} $\uparrow$	F _{k2} $\uparrow$	P _k $\downarrow$	WD $\downarrow$
0	.11	.41	.16	.22	.28	.56	.78
1	.10	.39	.15	.20	.25	.58	.80
2	.09	.42	.15	.20	.24	.59	.83
3	.10	.48	.16	.21	.25	.59	.86
4	.09	.41	.15	.20	.25	.59	.82
5	.10	.46	.16	.20	.25	.59	.85
6	.10	.44	.15	.20	.25	.59	.84
7	.09	.45	.15	.20	.25	.60	.86
8	.09	.45	.15	.20	.25	.60	.86
9	.09	.45	.15	.20	.25	.60	.86
10	.09	.45	.15	.20	.25	.60	.86

Table 5.7: Results of 10 epochs for the model trained on DailyDialog and evaluated on the 10 seasons of our dataset.

5.4.4 LLMs: a brute-force high computational power approach

With the rise of LLMs in the recent years, an increasing amount of tasks are being solved with these models, with more or less sophisticated approaches ranging from zero-shot settings supported by prompt engineering to more elaborated ones involving fine-tuning. Thus we wanted to test the performance of LLMs on our dataset for the DTS task.

We chose a zero-shot setting to get an idea of their base performance. We thus prompted several open-source LLMs with different parameter sizes to segment the dialogues in topics. The input consisted of the **transcript** to segment where each utterance was numbered encased in instructions as can be seen on Fig. 5.10. The prompt **explicited the format of the dialogue** and indicated the **format of the expected output** with a **commented example** of such output. The very end of the prompt **repeated the requirement in terms of output format**. We include **basic politeness formulations** in the prompt as there are some indications that they can improve the performance of LLMs, though it remains hard to properly quantify the effects of prompt engineering.

We used Python and the Ollama library to test several open-weights LLMs with different parameter sizes as summarised in Table 5.8. We used only open-weights models because they can be run on a machine of our choice¹⁶ and ensure a better reproducibility of our results. We use models ranging from 1.5 billion parameters to 14 billions which is quite small compared to the largest models available but we considered this range reasonable given the computational power it requires compared to the [Amandine: *I'm looking for a word to say that our task is not so life-changing that it would justify a crazy expense of computational resources*] of our task. Furthermore, one of the models we test, Deepseek, is used in a ‘reasoning’ version, which means that during the training of the model, a phase of reinforcement learning is used where it is required to produce a ‘reasoning process’ before producing the actual answer to the input.

A comment on the improvements brought by the ‘reasoning’ version

[Amandine]

However the term ‘reasoning’ can be rather misleading due to its anthropomorphic connotations. In practice this additional phase grants the model an increased computation time which can be beneficial

¹⁶These experiments were carried out using the Nancy site of the Grid’5000 testbed, supported by a scientific interest group hosted by Inria and including CNRS, RENATER and several Universities as well as other organisations (see <https://www.grid5000.fr>).

Below is the script of one episode of the TV show Friends. Each row corresponds to a speech turn in the dialogue and is formatted as follows: "ID SPEAKER: UTTERANCE". Please provide a topic segmentation of the episode by grouping together consecutive speech turns that belong to the same topic. The output should be a list of turn IDs where a topic change occurs. For example, the output could be: [5, 7, 12], which means that the first topic includes turns 0 to 4, the second topic includes turns 5 to 6, the third topic includes turns 7 to 11, and the last topic includes turns 12 and onwards.

Here is the episode script:

<script>

Please ensure that the output is in the form of a list of integers like [5, 7, 12] where each integer represents a line number where a topic shift occurred, and that there is no additional text or explanation in the output. Thank you!

Figure 5.10: Prompt for the zero-shot segmentation with LLMs.

| Model | Version & Size | Comment |
|------------------------------|----------------------------------|--------------------|
| LLaMa (Touvron et al., 2023) | 3.1 - 8B
3.2 - 3B | |
| Deepseek (Guo et al., 2025) | r1 - 1.5B
r1 - 8B
r1 - 14B | 'reasoning' models |
| Gemma (Team et al., 2024) | 3 - 1B | |
| Mistral (Jiang et al., 2023) | 0.3 - 7B | |

Table 5.8: List of the different LLMs used as baseline for the DTS task. *Note that all the models cited here except DeepSeek were introduced in papers published solely on ArXiv which is a practice we discuss in more details in the last chapter of this thesis (Chapter 8).*

to perform the requested task but it also increases the risk of the model going off track and producing irrelevant content, which we have indeed observed in our experiments.

Forcing a model to produce an output in a specific format is not easy through the prompt alone. In our first trials, many of the outputs were not in the expected format and thus could not be evaluated directly. Some results contained a properly formatted list of integers but also additional text, but others contained a sequence of single integers encased in brackets, e.g. '[12] [28] [34]', or sequences of lists. These patterns were more or less straightforward to match with an actual output but others would have required going manually through the results to fix them. We thus chose to leverage the functionality of the ollama library that allows us to specify the expected format of the output through a json schema. This way, the model is forced to produce a list of integers as output, and if it does not, an error is raised. This allows us to easily evaluate the outputs. We set the temperature of the models to 0 for these experiments, which means that the model will always choose the most likely token at each step of the generation, which is a recommendation when forcing the structure of the outputs. It also makes the experiment fully reproducible.

The results of these experiments are shown in Table 5.9. Some of the models still produced outputs in the wrong format which in practice raised an error and there was no segmentation for the given episode. The portion of episodes with such issues is reported in the column NaN. Both gemma models were unable

| Model | Pr. $\uparrow$ | Rec. $\uparrow$ | F1 $\uparrow$ | F _{k1} $\uparrow$ | F _{k2} $\uparrow$ | P _k $\downarrow$ | WD $\downarrow$ | NaN (%) | Length (%) | Empty (%) |
|------------------|----------------|-----------------|---------------|----------------------------|----------------------------|-----------------------------|-----------------|---------|------------|-----------|
| llama3.1:8b | .11 | .24 | .14 | .30 | .44 | .58 | .71 | 0 | 66 | 0 |
| llama3.2:3b | .10 | .66 | .16 | .22 | .22 | .62 | .94 | 3 | 68 | 8 |
| deepseek-r1:1.5b | .21 | .11 | .11 | .18 | .20 | .42 | .45 | 0 | 23 | 0 |
| deepseek-r1:8b | .63 | .08 | .10 | .15 | .15 | .40 | .41 | 2 | 42 | 47 |
| deepseek-r1:14b | .13 | .52 | .15 | .18 | .20 | .54 | .75 | 0 | 66 | 0 |
| gemma3:1b | / | / | / | / | / | / | / | 100 | 0 | 0 |
| gemma3:4b | / | / | / | / | / | / | / | 100 | 0 | 0 |
| mistral:7b | .14 | .19 | .13 | .26 | .35 | .50 | .58 | 1 | 92 | 0 |

Table 5.9: Average results of the different LLMs on the 10 seasons of *Friends*, the last two columns report the portion of outputs with issues.

to produce any output in the right format for any of the episodes, which is why no values are provided in Table 5.9. The other models produced outputs in the right format for most of the episodes. However some of these outputs were actually empty, i.e. no topical boundaries were reported, which is reported in the column Empty. Deepseek 8B produced almost half of its outputs as empty, llama 3.2 3B about 8% while the other ones produced none. Nevertheless, all the models produced segmentations at impossible positions. Indeed, the expected output was a list of integers where each integer is the ID of an utterance where a topic shift occurred. All numbers must thus be contained in the interval $\llbracket 0, N - 1 \rrbracket$, where N is the total number of utterances in the episode. Yet, the models often produced numbers outside of this interval, most of the time too large but sometimes below 0 as well. The portion of outputs with such issues is reported in the column Length. All the models produced impossible numbers for a large portion of the episodes but surprisingly the smallest Deepseek model produced the least amount of such outputs. Given the large amount of impossible boundaries, we chose to simply ignore the numbers outside of the relevant interval when evaluating the models. In some cases the resulting segmentation was empty after filtering out the impossible numbers, which we did not report in this table – the portions in the column Empty do not overlap with these – but all the results can be found in Appendix(missing reference: To be added in the final version.).

The results in Table 5.9 thus only take into account the episodes where an output in the right format was produced and where we considered only the numbers in the right interval. An empty segmentation resulted in the worst value for each metric. The results are quite low overall for the F-measures with a more or less striking imbalance between the precision and recall. However this imbalance is not always in favour of the recall, as opposed to the BERT-based models, which suggests that the models do not all display the same behaviour. The P_k and WD scores are also quite bad though the two smallest deepseek models have results comparable with the smart random baseline. Nevertheless, given the large amount of outputs with issues, and the computational cost required for this method, these results should be taken with caution.

5.5 Improving the original TextTiling algorithm with Linguistic Features

As explained before we work with two versions of the original TextTiling algorithm, one of them being feature-based, and the other one being ML-based with BERT as a backbone. In this section we discuss the enhancements of the original TextTiling algorithm we worked on. Such a model has the advantage of

being explainable and of not requiring any training, as opposed the BERT-based one. We chose to explore two dialogue-specific features – its structure itself with the speakers, and the presence of questions – as well as coreference chains, which is similar to the similarity approach based on lexical chains mentioned in Hearst (1997).

All the approaches we designed are meant to be added on top of a base similarity metric. We can thus apply them on top of the block comparison method or the vocabulary introduction method. We also used the average of the two methods as a base similarity metric to see if it could provide better results.

Section 5.5.1 discusses how we include the speakers’ organisation in the computation of the depth scores. Section 5.5.2 describes how we take questions into account in the algorithm, and Section 5.5.3 explains how coreference chains were used.

5.5.1 Speakers: taking advantage of the dialogical structure

Participants’ organisation can greatly influence the topical structure of a conversation (Nguyen et al., 2012). While we chose not to include the names of the speakers in the text used as input for our task to focus on the lexical content, we wanted to find a way to take their organisation into account in the algorithm. We thus designed two approaches to include these information in the depth scores computation, one where we only consider the exact utterances where new speakers are introduced, and one where we also adjust the scores around such utterances.

Focus on speaker introduction. Given that the complete set of participants is almost always the same in an entire episode, considering only the introduction of new speakers at the scale of the episode would not be very informative. We thus use a more local approach where we consider the introduction of new speakers at the scale of a block, which is the unit used in the Block Comparison method. For each gap i , we compute the number of speakers in the span after the gap that did not speak in the block of spans before the gap. We add this number to n times the existing depth score of the gap and normalise by dividing by $2 \times n$ where n is the total number of speakers in the episode. This way, the depth score of a gap will be increased if it is followed by the introduction of new speakers, and the more new speakers are introduced, the more the depth score will be increased. The normalisation allows us to keep the depth scores between 0 and 1, which is important for the next steps of the algorithm.

Smoother changes. In this approach, we raise the depth score of a gap as well as its direct left and right neighbours by 30% if the gap is followed by the introduction of a new speaker. This way, we consider that the introduction of a new speaker is an important enough change to increase the depth scores of a gap but we do not consider that the number of new speakers matters. We chose to also increase the depth scores of the direct neighbours of the gap because the new speaker may not be the first one to speak after the gap. In this version we do not take into account the number of new speakers to produce less sharp changes. Our approach is quite basic and could be improved by applying a gaussian increase around the gap instead of a flat one like we did but the goal is to assess which features are more relevant for the task before further work could be done on the way to include them in the algorithm.

Table 5.10 shows the results of the two approaches. The approach where only one score is changed when new speakers are introduced does not improve the results significantly, while the approach where the direct neighbours are updated as well provides a significant improvement on all the scores. This suggests that the introduction of new speakers is indeed an important change that can indicate a topic shift, but that the utterance where the speaker is introduced is not necessarily the exact place of the topic shift. This makes sense in the context of the *Friends* dialogues as a new participant could barge in the conversation and change the topic, but a new scene can also start after a transition that would be obvious for the viewers and where the topic shift would not necessarily be due to the arrival of the new participant(s). The former

| Approach | Pr. $\uparrow$ | Rec. $\uparrow$ | F ₁ $\uparrow$ | F _{k1} $\uparrow$ | F _{k2} $\uparrow$ | P _k $\downarrow$ | WD $\downarrow$ |
|-----------------------------|----------------|-----------------|---------------------------|----------------------------|----------------------------|-----------------------------|-----------------|
| BC | .15 | .15 | .14 | .29 | .40 | .46 | .50 |
| VI | .12 | .12 | .11 | .23 | .33 | .50 | .53 |
| BCVI | .14 | .13 | .12 | .25 | .36 | .48 | .52 |
| BC + Sp _{sharp} | .15 | .15 | .14 | .29 | .41 | .46 | .49 |
| VI + Sp _{sharp} | .12 | .12 | .12 | .23 | .34 | .50 | .53 |
| BCVI + Sp _{sharp} | .14 | .14 | .13 | .25 | .37 | .48 | .51 |
| BC + Sp _{smooth} | .18 | .18 | .17 | .36 | .48 | .42 | .46 |
| VI + Sp _{smooth} | .15 | .16 | .15 | .31 | .43 | .45 | .49 |
| BCVI + Sp _{smooth} | .17 | .18 | .16 | .34 | .46 | .43 | .47 |

Table 5.10: Average results of the feature-based models enhanced with speaker information. Sp_{sharp} corresponds to the approach where only one score is changed when new speakers are introduced in a span, and Sp_{smooth} corresponds to the approach where the direct neighbours are updated as well.

topic shift would happen due to an interruption of the new participant while the latter one would be a natural transition between an exhausted topic and a new one.

5.5.2 Questions: a second interactional features

Questions are a very complex phenomenon in dialogue and can have different functions (Cruz Blandón et al., 2019). However we took a very simple approach to include them in the algorithm with the assumption that a topic shift would not directly follow a question. This hypothesis is rather naive but we decided to see what results a very basic implementation could produce. We thus simply smooth the gap scores that follow questions by setting them as the average of the gap score before the question and the gap score after the question.

Our question identification method is quite basic as well, as we simply consider that an utterance is a question if it ends with a question mark. In practice this method relies heavily on what the transcribers considered as questions, which can be subjective. Requests, which can also be considered as a type of question, can be formulated as statements and thus would not be taken into account with this method.

This basic approach seemed to slightly improve the results, though not significantly. Given that it did not complexify the algorithm, we decided to keep it in the end in case it would interact more strongly with other features. However it could be interesting to further explore the role of questions in the topical structure of dialogues and to find more sophisticated ways to include them in the algorithm.

5.5.3 Coreference chains: leveraging more sophisticated lexical information

A coreference chain is a set of mentions that refer to the same entity in a text (Schneidecker, 1997). For example, a character can be referred to by their name, a pronoun or a description, and all these mentions would be part of the same coreference chain. Our assumption was that a topic shift is less likely to exist inside a coreference chain. For this reason, we smoothed the gap scores between the beginning and the end of each coreference chain.

We used the *coreferee* Python library to compute the coreference chains in our dataset. This library is applicable to English, French, German, and Polish, and combines rule-based and machine learning approaches to identify coreference chains in a text. It uses a set of rules to exclude impossible coreference

| Approach | Pr. ↑ | Rec. ↑ | F ₁ ↑ | F _{k1} ↑ | F _{k2} ↑ | P _k ↓ | WD ↓ |
|------------------|-------|--------|------------------|-------------------|-------------------|------------------|------|
| BC | .15 | .15 | .14 | .29 | .40 | .46 | .50 |
| BC + SI | .18 | .18 | .17 | .36 | .48 | .42 | .46 |
| BC + SI + Q | .18 | .18 | .17 | .36 | .48 | .41 | .45 |
| BC + SI + Q + L | .18 | .18 | .17 | .36 | .48 | .41 | .45 |
| BC + SI + Q + S | .18 | .18 | .17 | .36 | .48 | .41 | .45 |
| BC + SI + Q + Co | .19 | .16 | .16 | .37 | .48 | .40 | .44 |

Table 5.11: Average results of the feature-based models enhanced with various approaches.

Block Comparison (BC), vocabulary introduction (VI), coreference chains (Co), smoother speaker introduction (SI), questions (Q), stemming (S), lemmatisation (L)

links based on grammar, e.g. in ‘*The house* stood tall. *They* went on walking.’¹⁷, the algorithm would not link ‘the house’ and ‘they’ because they do not agree in number. It also uses a machine learning model to identify coreference links based on pragmatic tendencies. This model relies on syntactic and morphological information provided by *spaCy* models, as well as manual annotations of coreference chains. *SpaCy* is a classical NLP library designed for processing natural language texts with features that include tokenisation, part-of-speech tagging and dependency parsing among others. The coreference chains are computed based on the dependency parsing of the text, which allows to identify the mentions of each entity and to link them together in a coreference chain. Unfortunately, this algorithm requires a long time to compute the coreference chains and its time complexity rises with the length of the input text. For this reason, we computed the coreference chains at the block level, which is the unit used in the Block Comparison method, instead of at the episode level. This way, we were able to compute the coreference chains in a more reasonable time and to include them in our algorithm.

As with the questions, this approach did not significantly improve the results so we dropped it for the rest of our experiments due to its computational cost. While it is possible that chains identified at the dialogue scale would have had a stronger influence, the time required to compute them seems prohibitive for such a task.

5.5.4 Summary of the Results

We summarise the results with our different approaches to improve the TextTiling algorithm in Table 5.11. We report only the results of the models based on the block comparison method as it is the one that provides the best results as a base similarity metric, but we observed similar trends with the models based on the vocabulary introduction method and the average of the two methods. These can be found in Appendix(missing reference: To be added in the final version).

The best results are obtained with the Block Comparison method augmented by the smoother speaker introduction feature. Including questions or coreference chains does not significantly improve the results. We can also see that stemming or lemmatising the text does not improve the results. This could be due to the data being artificial in the sense that scenarists may avoid repetitions when it is not for the sake of humour.

¹⁷Example taken from the documentation of the coreferee library.

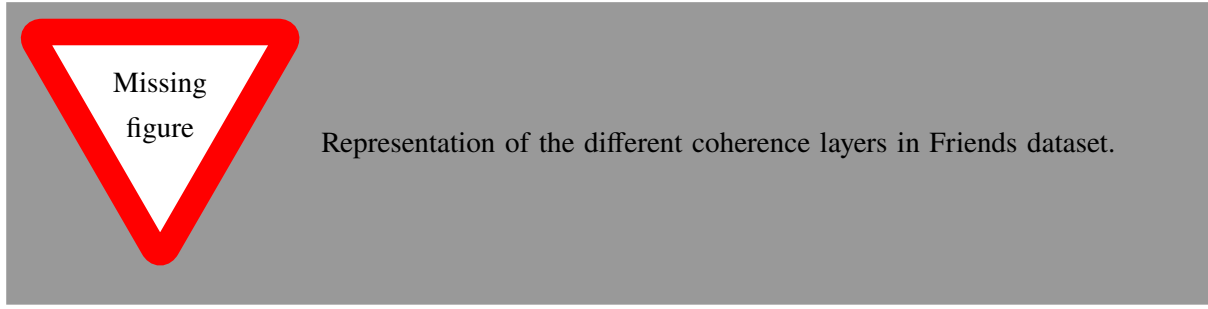


Figure 5.11: Representation of all the possible coherence layers in the *Friends* dataset.

5.6 Adapting Xing and Carenini (2021)’s BERT-based model to our Dataset

In parallel with the feature-based approaches, we also worked on adapting the BERT-based model of Xing and Carenini (2021) to our dataset. We describe this work in this section.

Training the utterance-pair scoring model on our *Friends* dataset raises an issue of data splitting. Indeed, we cannot train and evaluate the model on the same subset of the data, but the more data we use for training, the less representative our evaluation becomes compared to the other experiments. We thus proceeded in two stages, one where we trained the model on nine seasons and evaluated it on the remaining one, and another one where we trained the model on three seasons and evaluated it on the seven remaining ones. The first step gave us an idea of the performance we could get with a more diverse training set and allowed us to determine the reasonable number of epochs to train the model for as well as the organisation of the coherence layers in the model (Section 5.6.1), while the second step provided a fairer comparison with the other models (Section 5.6.2).

5.6.1 Large training set: Learning curve and coherence layers

The main difference between the training of the model on DailyDialog and on our *Friends* dataset from a technical point of view is that our dataset is not annotated with dialogue acts nor in topics, which prevents us from reproducing the exact same conditions as Xing and Carenini (2021) when building the training pairs. We thus chose an approach similar to the one they used for their Chinese dataset, NaturalConv (X. Wang et al., 2021) but without the constraint on sampling dialogues from different topics for the least coherent negative pair.

Instead, we rely on the annotation in scenes, episodes and seasons, as well as the additional notes in the transcripts, to sample what we consider to be topically different segments. As a consequence, our data has more levels of hierarchy than the one used by Xing and Carenini (2021) which allows us to consider more types of coherence layers that are represented on Fig. 5.11. In decreasing order of coherence we have:

- Layer A: subsequent utterances within the same note span;
- Layer N: randomly picked non subsequent utterances within the same note span;
- Layer C: randomly picked utterances within the same scene;
- Layer E: randomly picked utterances within the same episode;
- Layer S: randomly picked utterances within the same season;
- Layer D: randomly picked utterances within different seasons.

We thus experimented with different settings in terms of coherence layers. On the one hand, we reproduce the ‘adjacent’ and ‘same dialogue’ coherence levels from Xing and Carenini (2021) in all of our experiments by using our [A] and [N] layers. On the other hand, we try different layers for the third

coherence level – within the same scene [C] and within different seasons [D], and we also train some models with more layers: [CESD] and [CD].

Table 5.12 show the results when training one model on nine out of the ten available seasons of *Friends* for ten epochs. We can see that the results barely vary from one epoch to another. In fact, only the recall, P_k and WD display significant changes across some epochs. The recall is slightly worse at epoch 0 than the others, and even lower at epoch 3, which incidentally makes P_k and WD better at epoch 3 but not at epoch 0. This is due to the large difference between the precision and recall that show that this model over-generates topic boundaries. Hence, when the recall is lower, it indicates a version of the model that generates less boundaries than the others and thus less false positives. The models display similar variations across all layers settings with either no significant changes across epochs or slight changes on recall, P_k and WD at one or two epochs when recall gets lower. Thus in the rest of our experiments we decided to train the models for two epochs only as it seems that training for more epochs does not provide better results and can even lead to worse ones.

| Epoch | Pr. $\uparrow$ | Rec. $\uparrow$ | F_1 $\uparrow$ | F_{k1} $\uparrow$ | F_{k2} $\uparrow$ | P_k $\downarrow$ | WD $\downarrow$ |
|-------|----------------|-----------------|------------------|---------------------|---------------------|--------------------|-----------------|
| 1 | .14 | .44 | .21 | .27 | .31 | .51 | .71 |
| 2 | .14 | .47 | .21 | .27 | .31 | .53 | .73 |
| 3 | .14 | .48 | .21 | .27 | .31 | .52 | .73 |
| 4 | .13 | .38 | .19 | .27 | .32 | .50 | .66 |
| 5 | .14 | .50 | .21 | .27 | .31 | .53 | .75 |
| 6 | .14 | .51 | .22 | .27 | .31 | .52 | .74 |
| 7 | .14 | .49 | .21 | .26 | .30 | .53 | .75 |
| 8 | .14 | .51 | .22 | .27 | .31 | .53 | .75 |
| 9 | .14 | .51 | .22 | .26 | .30 | .53 | .75 |
| 10 | .14 | .52 | .22 | .27 | .30 | .53 | .75 |

Table 5.12: Results of 10 epochs for the model AND-3.

Regarding the different settings in terms of coherence layers, Table 5.13 shows that the results do not vary greatly across the different settings. The best results overall are obtained with the model trained with the [D] layer as the third coherence layer, i.e. with randomly picked utterances within different seasons. However, this model is not the best one with the P_k metric which is likely due to the large difference between its precision and recall, suggesting as we saw before that it over-generates topic boundaries. The model with the best P_k is the one trained with the [CD] layers, which is also the one with the best WD score, though most of the models have WD scores that are not significantly different from each other. The model with the most layers, i.e. CESD, performs similarly to the best one, suggesting that having more layers does not necessarily influence the performance of the model.

An interesting point is that the models [D-1] and [D-3] were trained on data based on the same coherence layers with almost the same source of data as they differed only by one season, yet they display results significantly different from each other with all the metrics. [C-1] and [C-3] have a similar dynamics except for the P_k and WD metrics where [C-3] performs better than [C-1]. [CD-1] and [CD-3] however are very similar except for the P_k metric where [CD-3] performs better than [CD-1]. The fact that the results do not vary in the same way between the models when evaluated on season 1 and season 3 suggests that the differences do not necessarily stem from the evaluation data but possibly from the models themselves and the training process, which may be quite unstable though the differences in performance are not so large. The small evaluation set is also an issue when it comes to comparing the performances of the different versions of the model between them but also with the baselines and the feature-based approach.

We further explore these questions in the next section where we train several versions of the model on a smaller training set and with different random initialisations to see how much the results can vary.

| Model | Pr. $\uparrow$ | Rec. $\uparrow$ | F ₁ $\uparrow$ | F _{k1} $\uparrow$ | F _{k2} $\uparrow$ | P _k $\downarrow$ | WD $\downarrow$ |
|----------|----------------|-----------------|---------------------------|----------------------------|----------------------------|-----------------------------|-----------------|
| ANC-1 | .18 | .56 | .27 | .35 | .38 | .53 | .71 |
| ANC-3 | .14 | .44 | .21 | .27 | .31 | .51 | .71 |
| AND-1 | .19 | .58 | .28 | .36 | .39 | .51 | .69 |
| AND-3 | .14 | .57 | .22 | .26 | .29 | .56 | .81 |
| ANCD-1 | .16 | .44 | .23 | .33 | .39 | .54 | .68 |
| ANCD-3 | .15 | .42 | .22 | .29 | .34 | .48 | .65 |
| ANCESD-1 | .18 | .54 | .27 | .35 | .38 | .52 | .70 |

Table 5.13: Results for the different layer settings. The best scores are indicated in **bold** and the results that significantly differ from this one are in *gray*. Each model was trained on 9 seasons and evaluated on the remaining one, indicated in the model name <layers>-<evaluation season>.

5.6.2 Stability and less diverse training set

For a fairer evaluation with the models that require no training, we trained the model on three seasons and evaluated it on the seven remaining ones. We trained several versions of this model with different training subsets and different random initialisations in order to get an idea of the stability of the results. We chose to work with only one layer setting for this experiment to avoid multiplying the number of models to train and evaluate, and we chose the [AND] setting as it is the one that provided the best results overall in the previous experiment.

Since the sampling of our training pairs involves a random component, two training subset based on the same input data can be different. We thus built three training subsets, α , β , and γ , based on the same three seasons 2, 3, and 4 but with different random seeds to see how much the results can vary with the sampling of the training pairs. We also trained three versions of the model on each subset with different random initialisations to see how much the results can vary. We trained these models for two epochs only, with the coherence layers [AND], following our findings from the first set of experiments.

In the end the results were not significantly different between the two epochs for the models trained on subset α , one of the versions trained on subset β had a lower recall at epoch 2 which led to a better WD but not a better P_k score, and similarly for one of the versions trained on subset γ where a lower recall at epoch 2 led to better P_k and WD. Conversely one other version of the model trained on subset γ had a lower precision and recall which led to worse performances on all metrics except F_{k2} which was not significantly different from epoch 1. Nevertheless the results do not vary greatly between epochs when the differences are statistically significant, the stronger changes appearing on WD which performs quite badly overall compared to all the other models we worked with, including our simplest baseline. This is likely due to the overgenerating tendencies of this model. For this reason we display the results of all the versions of the model only at epoch 2 in Table 5.14 to improve readability, the full results are available in the Appendix(missing reference: To be added in the final vesion.).

The results in Table 5.14 show that the behaviour of the restricted model remains quite similar to the ones trained on nine seasons. However, there is a distinct variability between the models, whether they differ in random initialisation or training subset. This variability enables us to identify the best model, which is $D_{\gamma 1}$. SoA few models display similar results on some metrics, highlighted with *italics* in Table 5.14 but $D_{\gamma 1}$ performs best overall.

| Model | Pr. $\uparrow$ | Rec. $\uparrow$ | F1 $\uparrow$ | F _{k1} $\uparrow$ | F _{k2} $\uparrow$ | P _k $\downarrow$ | WD $\downarrow$ |
|-----------------|----------------|-----------------|---------------|----------------------------|----------------------------|-----------------------------|-----------------|
| D $_{\alpha 1}$ | .12 | .56 | .19 | .24 | .26 | .56 | .84 |
| D $_{\alpha 2}$ | .09 | .39 | .14 | .19 | .24 | .59 | .82 |
| D $_{\alpha 3}$ | .11 | .37 | .16 | .24 | .30 | .55 | .74 |
| D $_{\beta 1}$ | .13 | .36 | .18 | .24 | .28 | .50 | .68 |
| D $_{\beta 2}$ | .14 | .48 | .20 | .26 | .29 | .49 | .73 |
| D $_{\beta 3}$ | .14 | .45 | .20 | .26 | .29 | .49 | .71 |
| D $_{\gamma 1}$ | .15 | .38 | .21 | .27 | .31 | .45 | .63 |
| D $_{\gamma 2}$ | .09 | .26 | .13 | .19 | .28 | .56 | .72 |
| D $_{\gamma 3}$ | .13 | .51 | .19 | .24 | .27 | .53 | .78 |

Table 5.14: Results of different models trained on three different training subsets (d $_{\alpha}$, d $_{\beta}$, d $_{\gamma}$) after the second epoch. *Evaluation carried on seasons 1, 5, 6, 7, 8, 9, and 10 only.*

Lastly, we compared these versions of the model to the ones trained on nine seasons instead of three. Only season 1 could be used for evaluation so we compared only c-1, d-1, cd-1, and cesd-1. They do not always display significant differences with the models trained on three seasons only and the ranges of values covered by the metrics generally lie in the middle of the range covered by the restricted models. This suggests that the models trained on three seasons are not necessarily worse than the ones trained on nine seasons, and that the differences in performance are more likely due to the random sampling of the training pairs and the random initialisation of the model. Detailed results are available in Appendix(missing reference: To be added in the final vesion.).

5.7 Final Comparisons and Conclusion

In this chapter we discussed several approaches to DTS on the *Friends* dataset. We mostly worked on improving the original TextTiling algorithm in two ways, one based on including more linguistic features to the lexical similarity metric and the other by replacing the similarity metric by a ML-based model for utterance-pair scoring following Xing and Carenini (2021). We also considered several baselines, including two random ones, several LLMs leveraged in a zero-shot manner, as well as the most basic versions of the two approaches we aimed to evaluate. For the feature-based approach, we considered the original TextTiling algorithm and a slightly modified version of it with adaptations to dialogical data, while for the ML-based approach we evaluated the model prior to fine-tuning and after fine-tuning on the DailyDialog dataset.

Section 5.7.1 provides a comparison of all the approaches and discusses the results before we conclude this chapter in Section 5.7.2.

5.7.1 Comparison of All the Models

Table 5.15 summarises the results of our baselines and the two best models for our feature-based approach and ML-based approach respectively. For the feature-based approach, none of the improved variants of the algorithm based on the Block comparison similarity metric displayed significant differences in results. Hence we selected the one with the highest average scores while requiring an efficient computational time, which ruled out the coreference model. We settled on the model based on Block comparison augmented with the smooth speaker introduction method and the questions feature. Regarding the ML-based approach we selected D $_{\gamma 1}$ since it was the best performing version of the restricted model overall. We

| Model | Pr. $\uparrow$ | Rec. $\uparrow$ | F ₁ $\uparrow$ | F _{k1} $\uparrow$ | F _{k2} $\uparrow$ | P _k $\downarrow$ | WD $\downarrow$ |
|-----------------------------------|----------------|-----------------|---------------------------|----------------------------|----------------------------|-----------------------------|-----------------|
| Rd | .10 | .49 | .14 | .20 | .24 | .58 | .82 |
| Rd _{smart} | .10 | .10 | .10 | .23 | .33 | .47 | .51 |
| BC | .13 | .13 | .12 | .26 | .37 | .48 | .52 |
| BC for dialogue | .15 | .15 | .14 | .28 | .40 | .46 | .50 |
| BERT _{Base} | .11 | .42 | .17 | .22 | .28 | .56 | .77 |
| BERT _{DailyDialog} | .10 | .40 | .15 | .20 | .25 | .58 | .80 |
| LLM _{mistral} | .15 | .20 | .13 | .25 | .35 | .50 | .58 |
| LLM _{deepseek} | .18 | .09 | .09 | .15 | .16 | .41 | .44 |
| BC + SI + Q | .19 | .19 | .17 | .36 | .48 | .41 | .45 |
| D _{γ1} | .15 | .38 | .21 | .27 | .31 | .45 | .63 |

Table 5.15: Comparison of the different models. *Evaluation on seasons 1, 5, 6, 7, 8, 9, 10 only.*

also include the two best LLMs, i.e. deepseek-1.5B and mistral-7B, to compare all our baselines while keeping a readable table.

Given that the ML-based model was trained on seasons two to four, we had to restrict the evaluation to seasons 1, 5, 6, 7, 8, 9, and 10 only for this comparison to ensure that the evaluation was fair and that the model was not evaluated on data it had seen during training.

When comparing the baselines, we can see that the random baseline performs quite poorly, which is expected. However, the BERT-based model fine-tuned on DailyDialog performs similarly to it, and worse than the BERT-based model prior to fine-tuning. This suggests that the fine-tuning on DailyDialog does not generalise well to our dataset, and even degrades the original generalisation capabilities of the model.

Regarding the original TextTiling algorithm, the adaptations to dialogue provide a slight improvement, and both models perform overall better than the random baselines, except for the WD metric where they are equivalent to the smart random baseline.

Our of the two best LLMs, Mistral yields results that are not significantly different from any other model, which indicates a much wider variability in its output. Deepseek performs much worse than the other models, including the simplest baselines, on the F-scores but it performs similarly to the best model, i.e. the feature-based model, on the P_k and WD metrics. It is also the only model where the precision is higher than the recall, which indicates that it generates less false positives and explains the behaviour we just described.

When it comes to the two models we developed, the feature-based model performs better than the ML-based model on all metrics except for recall, and F₁ where the ML-based model performs better, the better recall inducing the better F₁. The fact that only F₁ is better clearly indicates that the ML-based model over-generates topic boundaries, which is penalised by the F_k and WD metrics. While P_k does not penalise false positives as much as the other metrics, the ML-based model still performs worse than the feature-based one, though the difference in performance is smaller than for the other metrics. Figure 5.12 provides a boxplot representation of the results for the two models, which allows to visualise the variability of the results across the different evaluation seasons. We can see that the ML-based model displays a larger variability than the feature-based one, which shows that it is overall less consistent. Despite this variability, the differences between the two models are still statistically significant for all metrics, most likely thanks to the numerous data points we work with.

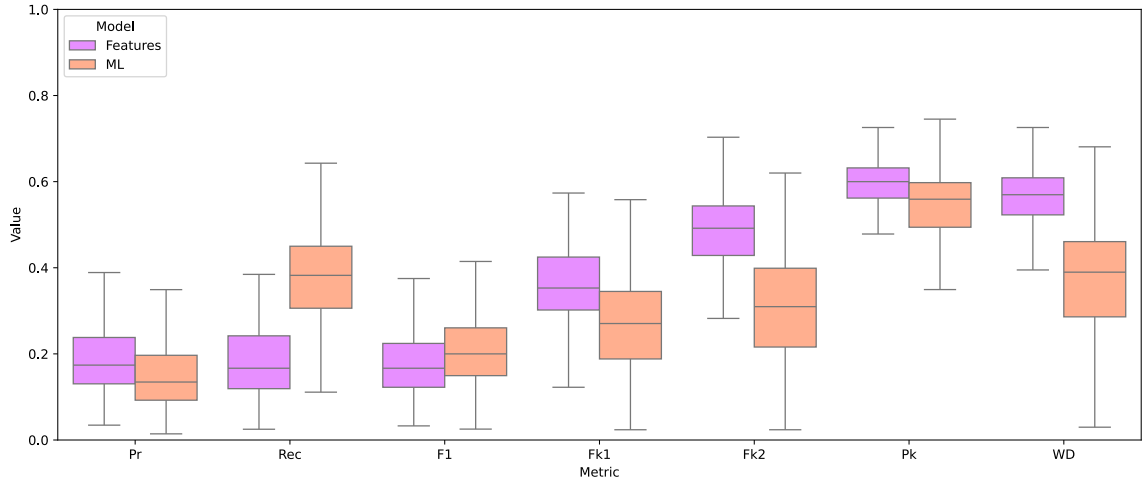


Figure 5.12: Boxplots of the results of the two improved models and the DeepSeek 1.5B model. *Evaluation on seasons 1, 5, 6, 7, 8, 9, 10 only.*

5.7.2 Final Remarks and Conclusion

In this chapter we investigated the task of linear topic segmentation on multi-party casual conversations. Since this kind of data is complicated to obtain, we chose to work on transcriptions of the TV-show *Friends* as this dataset is available online. While it is not a perfect representation of real-life conversations, the high number of speakers and the various open-domain settings of the dialogues make it interesting to work with for the DTS task. The scripted nature of these exchanges may simplify the task as the dialogues are explicitly meant for outsiders to understand, and the good quality of the transcriptions also make this dataset easier to work with than real-life conversations. However, the different approaches we tested on this dataset showed that it remains a challenging corpus for the task.

All our approaches were based on the TextTiling algorithm, which uses a similarity metric between subsequent parts of a text to identify the topic shifts. We enhanced it with more linguistic properties that could play a role in identifying topic shifts, and compared it to the same approach but enhanced with a trained utterance-pair coherence scoring model based on BERT. As BERT has been trained on the next sentence prediction task, it seemed like a relevant model for topic segmentation and in particular to improve the TextTiling approach. At the time where this work was carried out, similar but more powerful tools were models such as BART (M. Lewis et al., 2020) or T5 (Raffel et al., 2020) and they did not seem as suitable for our work as they have not been explicitly trained for the next-sentence prediction task. However, the rise of LLMs with their generalisation capabilities raised the question of whether they could be used for our task. We thus evaluated several LLMs in a zero-shot manner, but the results were not satisfactory. We could probably improve our method by using a different prompt or wrap the LLM in a larger pipeline as we did for BERT. However, the large computational power required to run these models and their high resource consumption in their training phase made us question the relevance of using them for our task.

Indeed, we tried to limit our use of heavy computational powers. Our feature-based model can run on a local machine and except for the experiments that involve co-reference chains identification, creating the topic segmentation of one episode of *Friends* takes less than a few seconds. As for the experiments using Machine Learning, we did our best to optimise the batch sizes and the number of experiments we could run in parallel to reduce the training time as much as possible. We also tried to limit the multiplication of experiments though we could have gotten a clearer picture of the stability of the results

by training more versions of the model with different random initialisations and different training subsets. We ran our experiments on the Lark servers from CLASP (Gothenburg University) where we used one Nvidia Titan RTX GPU. Each epoch took about one hour of training. The extra experiments with LLMs were carried out using the Nancy site of the Grid'5000 testbed, supported by a scientific interest group hosted by Inria and including CNRS, RENATER and several Universities as well as other organisations (see <https://www.grid5000.fr>). We chose only open-weights LLMs to avoid the use of proprietary models that would have required us to use external servers and thus to depend on the companies that have the means to develop such models, and we tried to limit the size of the models we used to reduce the computational power required to run them. However, even with these precautions, LLMs as well as ML-based experiments require a lot of resources, and the cost of initial training LLMs should not be taken out of the equation despite not being a direct part of our work.

With all these aspects in mind, the poor results of the LLMs and the ML-based model should be put into perspective. The feature-based model is more explainable, requires less resources, and does not require any training if we exclude the coreference chain feature. It also performs better than the ML-based model on all metrics except for recall and F_1 . While the final results are far from satisfactory to consider the DTS task as solved with data of the complexity level of the *Friends* dataset, the feature-based model seems to be a better working base than the ML-based model to improve the results. Including interactive features such as the participants' organisation seems promising, and a more complex utilisation of questions and possibly adjacency pair could also be relevant. These two features are somewhat language agnostic, though the notion of questions and adjacency pairs could require language-dependent adaptations depending on the sophistication of the approach. While our version of the coreference chains did not significantly improve the results, and was computationally expensive, more efficient methods could be developed in the future that would be worth testing. The shared task CRAC (Novák et al., 2025) on multilingual coreference resolution is a promising avenue for this kind of work beyond English.

The overall poor results are also a reflection of the complexity of the task and the data we worked with. The *Friends* dataset is a complex one, and the task of linear topic segmentation is not necessarily adapted to this kind of data. Indeed, the dialogues are multi-party and the topic shifts can be abrupt or subtle, which makes it difficult to identify them. The granularity of the topic shifts is also an issue, as it can vary between people and tasks. A more hierarchical approach to topic segmentation could be more relevant for this kind of data, as it would allow to capture the different levels of granularity of the topic shifts.

6

Topical Similarity at Utterance Level

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Todo Writing

- Corpus issues / limitations (find examples of very bad threads)
- Stats on the corpus
- Coherence levels hypothesis validation
- Conclusion

Todo Experiments

- Coherence levels hypothesis validation
- Exploratory structure analysis with new version of the model (margin 0.2)

Chapter 5 showed that the topical structure of a conversation is not well captured by linear topic segmentation, especially in the case of more chaotic exchanges such as multi-party casual conversations.

The approach in itself is indeed too constrained compared to the dynamics of interaction where several topics can be intertwined, a topic can drift slowly, etc. Yet the TextTiling base we used showed that there is topical information in surface level features such as lexical cohesion. Our goal in this chapter is thus to come up with a more flexible approach to extracting topical information based on more local features. The end goal is to produce a tool that would allow us to explore the topical structure of a conversation, highlighting more subtle dynamics than a binary segmentation.

To achieve this, we propose to create a topical similarity measure which would give a similarity score to two pieces of interaction in terms of topic. Using this measure could then enable us to find locally coherent pieces of interaction but also to measure the variability in terms of topic throughout the conversation. Since we cannot manually give a similarity score to pairs of dialogue interventions, we suggest to use an unsupervised method to train our measure. In particular, we will expand the idea from Xing and Carenini (2021) of training a metric, as we did in Chapter 5, but instead of using it for binary classification, where it over-generated boundaries, we will leverage it to investigate topical variability in a continuous manner rather than a binary one.

Using a dataset where the interactions are already organised in a hierarchical way provides us different levels of coherence that we can use to train our measure. The social media *Reddit* provides us with a large pool of messages, organised in tree structures, discussing various topics and presenting different formats (questions/answers, casual discussion, argumentation, etc.).

This chapter is organised as follows. We begin by describing the method we used to build a topical similarity measure for dialogue in Section 6.1, including reminders from Chapter 5 and a description of our evaluation approaches. We then present the data we used for this study in Section 6.2. Next the hypothesis that underlies our unsupervised training is discussed and evaluated in Section 6.3 before we detail the settings of our experiments and the results we obtained in Section 6.4. The focal point of this chapter is the analysis of the topical structure of conversations, which we present in Section 6.5. Finally, we discuss the implications of our findings in Section 6.6.

6.1 Method

Our goal in this chapter is to investigate the evolutions of topic in a conversation. To do so, we need a way to compute the topical similarity (1) between two pieces of conversation, and (2) in a continuous manner. That is, we are not interested in the topic boundaries themselves but rather in quantifying the degree of topical change throughout a dialogue. We thus follow a similar approach as the ML one described in Chapter 5, which is reminded in Section 6.1.1. This approach relies on an hypothesis regarding the degree of topical similarity between several pairs of pieces of conversation. We discuss the way we validated this hypothesis in Section 6.1.2.

6.1.1 Metric construction: Reminders

In Chapter 5, we tested various improvements to the linear topic segmentation algorithm TextTiling (Hearst, 1997). In particular, we adapted Xing and Carenini’s (2021) approach to our own dataset. Their idea was to replace the original similarity metrics from the TextTiling algorithm with a BERT-based similarity metric. They trained an utterance-pair coherence scoring metric in an unsupervised way, using the assumption that utterances that are close to each other in a conversation are more likely to be topically similar than utterances that are far apart. This has the advantage of not requiring any manual annotation of topical similarity, which is a very subjective task. The metric was then used to compute the similarity between two adjacent blocks of utterances in the TextTiling algorithm.

Even though this approach did not yield satisfactory results for the DTS task, the similarity metric

seemed to capture some aspects of the topical structure of the conversations as it performed better than chance. The data Xing and Carenini (2021) originally used this approach on was also very different from the Friends corpus in terms of interactional structure, length, and range of topics. DialSeg (Y. Xu et al., 2021), Doc2Dial (S. Feng et al., 2020), and ZYS (Y. Xu et al., 2021) indeed contain rather short conversations with a limited range of topic due to their task-oriented or document-grounded nature. These two aspects make them closer to the type of cues we hope to extract when comparing two Reddit comments. We thus decided to adapt Xing and Carenini’s (2021) approach to our own task and use the trained similarity metric as a measure of topical similarity between comments. The TextTiling aspect of the approach is not relevant to our case as we are not interested in linear topic segmentation but rather in a continuous measure of topical similarity between comments.

We thus needed to select a dataset where we could use the structure of the conversations to build pairs with different levels of topical similarity. Reddit is very interesting in this regard as the conversations are organised in a tree structure as we further describe in Section 6.2. We explain how we validated our coherence hypothesis and evaluated our metric in the next section (Section 6.1.2).

6.1.2 Evaluation

While the metric is only a tool aiming to support our exploratory analysis of the topical structure of conversations, we still need to validate its performance. We do this in two stages. On the one hand we try to validate our coherence ranking hypothesis, which is the basis of our unsupervised training, prior to any training. On the other hand we evaluate the ability of the metric to rank pairs of comments in the expected order of topical similarity after training.

Coherence levels hypothesis validation.

Todo Writing

Describe the approaches.

Todo Experiments

A couple ‘baselines’ ie. before training a model, we can try to see if the hypothesis is valid:

- Apply LDA to subreddit, then compute similarity of messages in the three pairs to see if they are ordered.
- BERT without training (if we are completely changing the model then probably a bad base hypothesis)
- Same as LDA with BERT topic? (cosine sim between topic repartition vectors)

Metric evaluation. In order to be as informative as possible, we report several metrics to evaluate the quality of the similarity measure. First, we compute the mean of the similarity scores for each of the three pairs (AB, AC, and AD) to determine whether the expected monotonic trend is observed. Second, we focus on the ranking of the three scores and report the pairwise accuracy to capture the quality of the ranking, i.e. the proportion of pairs that are correctly ordered. This metric represents the general quality of the ranking as it considers perfect rankings but also partial ones and includes ties. For a more in depth analysis, we also report the proportion of scores that are correctly ranked individually and compared to one another. In total the ranking analysis is covered by four types of scores summarised in Fig. 6.1. Finally we compute the mean of the margins between the scores to quantify how confidently the hierarchy is separated.

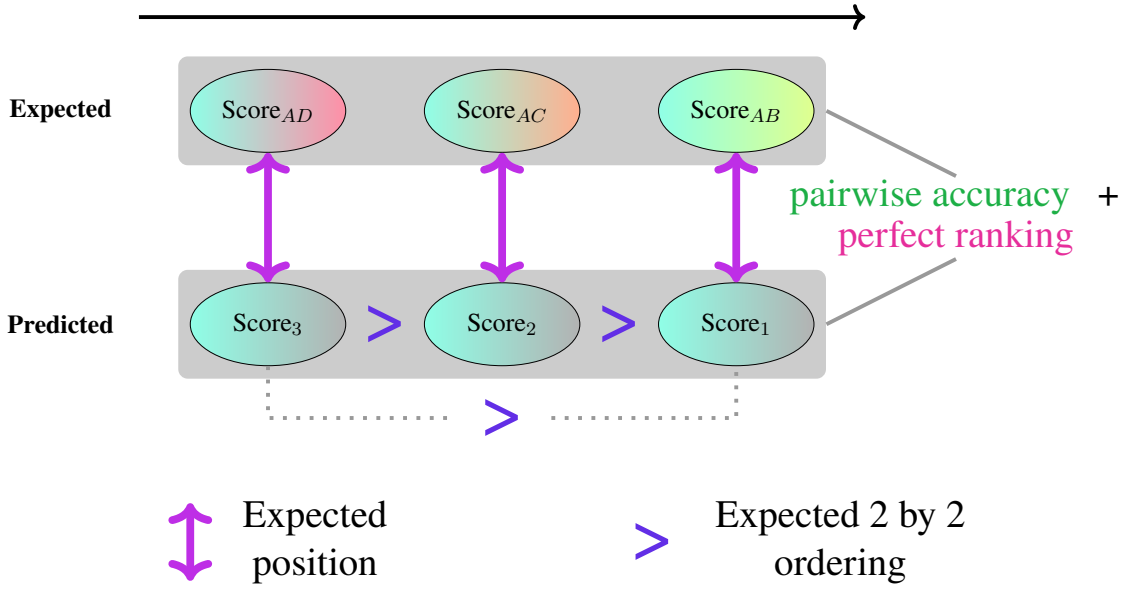


Figure 6.1: Description of the score ranking evaluation metrics.

The requirements in terms of data to determine whether our metric can allow us to observe topical variations and structure in conversations have several constraints. The first one is that the data must be organised in a hierarchical way, so that we can extract pairs of dialogue pieces with different levels of topical similarity. The second one is that the data must be sufficiently large to allow us to train a model on it. Third, since the core focus of this thesis is naturalistic open conversations, the data must be as close as possible to this type of interaction. Finally, we need to find a type of exchanges that encourages variations in topics. Given that this work is a first exploration, selecting data that is more directly suited for our analysis is preferable. The social media Reddit is particularly interesting in this regard, as we explain in the next section (Section 6.2).

6.2 Dataset: Naturalistic structured interactions

Reddit is a forum-like social media where users can post messages and other users can comment on them. This organisation allows for a tree-like structure of the interactions, where a post can have several comments and each comment can have several sub-comments. This structure is particularly interesting for our goal of investigating topical variations in conversations. Even though the interactions on Reddit are not spontaneous oral conversations, they still exhibit a dynamic of interaction that is similar to real conversations. The comments can be of various lengths and the interactions between users can involve a variable number of participants, which allows for a rich exploration of topical variations. Moreover, the written format of the exchanges removes the need for transcription, allowing us to focus on the analysis.

In this section, we first discuss the use of social media in general in NLP research (Section 6.2.1) before describing Reddit in more detail (Section 6.2.2). We then present the dataset we extracted from Reddit (Section 6.2.3) and how we constructed the different coherence levels our metric requires (Section 6.2.4).

6.2.1 Social media in NLP research

Warning – Ongoing work

This section is a direct copy-pasting from the paper (Decker & Amblard, 2024b). It will be updated and more detailed in the final version of the thesis.

Social media are widely used in research on discourse as sources of data (Hamilton et al., 2017; Baumgartner et al., 2020; Balouchzahi et al., 2023). (Jovanovic & Leeuwen, 2018) analyse the structure and genre of dialogues held on various social media platforms. (Misra & Walker, 2013) work on the identification of agreement and disagreement in social media conversations. Various studies rely on such dialogues for tasks related to mental health (Gaur et al., 2018; Turcan & McKeown, 2019; Naseem et al., 2022) and emotion recognition (Balouchzahi et al., 2023).

While Reddit is not the most studied media (Kathie Treen & Coan, 2022), its structure makes it particularly interesting for our goal. Indeed, the messages on Reddit are organised in a tree-like structure, with a main post on a given topic and answers to this post as well as answers to the different answers. We can thus investigate how the topic evolves from the first post through a sequence of answers and compare different sequences rooting from the same post. More details on the structure are given in Section 6.2.2.1.

6.2.2 Reddit detailed characterisation

Reddit is a US-based social media founded in 2005. As of March 31, 2026, it reported about 127 million daily active unique users and almost 500 million weekly ones¹⁸. In this section we describe the organisational structure of Reddit (Section 6.2.2.1), which is the reason why we chose this particular social media for our study, and detail the type of dialogue interactions we can find on it (Section 6.2.2.2).

6.2.2.1 Structure

Reddit is a forum divided into communities called *subreddits* which are defined by a topic, a hobby, a culture or even the format of the posts, and typically a set of rules. For example, the subreddit *r/boardgames* is presented as ‘The #1 source for news, information, and discussion about modern board games and board game culture.’ and describes general politeness and integrity rules such as ‘Keep Discussions Civil’, ‘Respect Intellectual Property’, or ‘Limit Usage of AI’. Some subreddits can be very specific in terms of topic and rules such as *r/poptarts* which is dedicated to the Anglo-Saxon toaster pastry and forbids mentions to ‘Toaster Strudels’. Besides the content, some subreddits’ rules also constrain the format of the posts, such as *r/Showerthoughts* which requires posters to write their entire thought in the *title* of their post.

Within a subreddit, users can post a message, called a *post*, and other users can react to it by writing a comment. A comment can be a direct answer to the post or an answer to another comment. The discussion thus created is organised in a tree structure with the initial post as the root and the comments as nodes. Figure 6.2 illustrates this structure. Posts often contain an illustration and sometimes even contain only that and a title.

What we chose to call a conversation on this platform is a *thread*, which consists of a post and all its comments. They can be of various sizes – the total number of comments – and shapes – the way the comments are organised in the tree. We further discuss this variability in Section 6.2.2.2. Before that, we present the global characteristics of Reddit interactions in Section 6.2.2.2.

¹⁸According to <https://redditinc.com>, last visited on July 16, 2026.

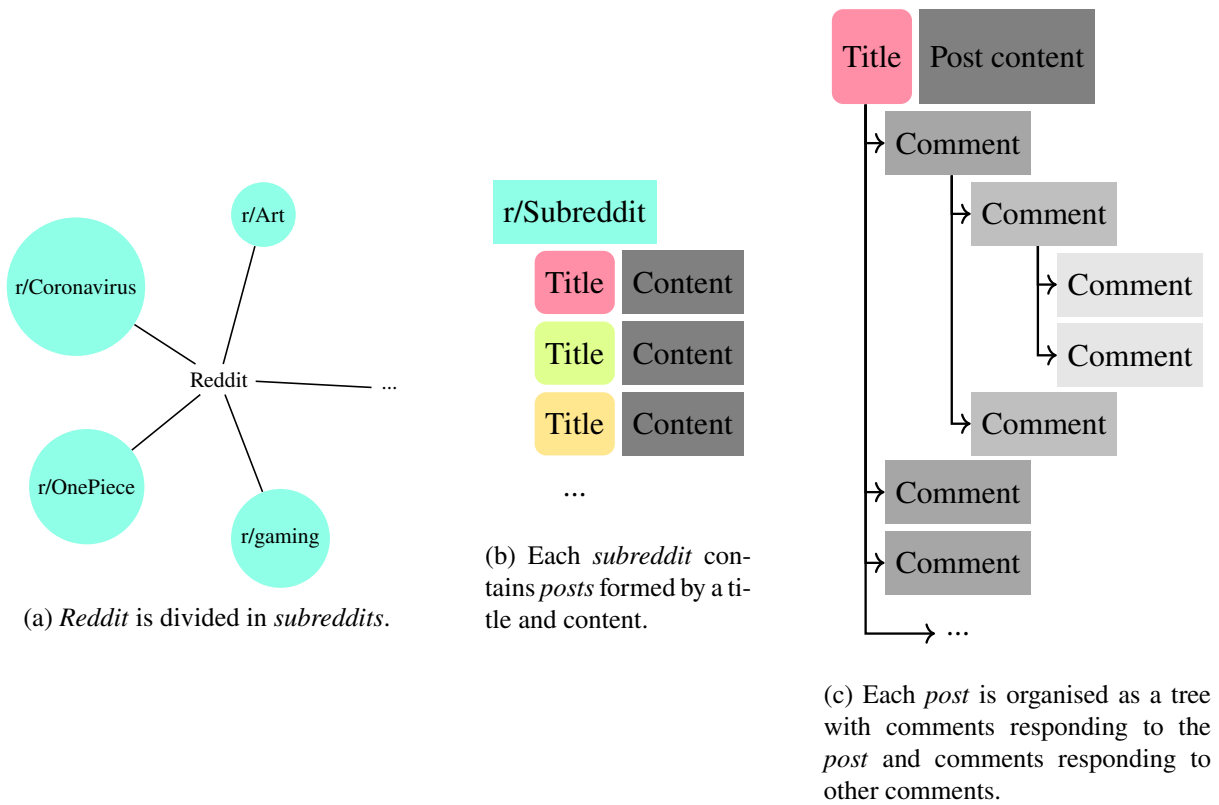


Figure 6.2: Reddit organisation.

6.2.2.2 Description

Reddit interactions are very different from face-to-face conversations for a number of reasons, the most obvious being that they are asynchronous and text-based, but they also exhibit unique social properties. Table 6.1 summarises the main characteristics of Reddit interactions according to the framework we presented in Chapter 2. The format of Reddit is the root cause of many of these characteristics. First, the number of unique participants to a *thread* is virtually unlimited as there is no upper limit to the number of users or comments that can be active under a given post nor any option to close a post other than removing it entirely. In practice, most posts stop receiving responses after a short amount of time due to the way posts are highlighted on the platform. Within a given post, the poster has a privileged position as they set the focus of the exchange. A token identifying them as the original poster is displayed next to their username when they comment on their own post within the comments tree. Other participants can also take a more important interactional function at the level of sub-conversations generated by the threads of comments.

Second, participants are identified by a pseudonym and there is no foolproof way to link a pseudonym to a real identity. In 2012, Barack Obama posted a message on r/IAmA, a subreddit where each thread is dedicated to a specific person who answers questions posted in the comments. The former president's post contained a link to a Twitter (now X) post where he announced he would be answering questions on Reddit. Contrary to Reddit, Twitter had a user identity verification system at the time which increased the credibility of the post. A picture of the former president was also linked in the post, in an attempt to provide more physical proofs, though nothing made this photograph verifiable. Anonymity goes together with the idea of commitment. Without knowing the participants' identities or motivations for joining a conversation thread, it is quite complicated to assess their level of commitment. Some participants may

be trolls, who post messages with the intention of provoking other participants. Conversely, others may be deeply involved in a topic or community.

Third, the medium has direct consequences on the dynamics of the interactions on Reddit. The only thing that participants share is the content of the post and the comments. There is no physical context, no visibility of the participants, or possibility to hear them. The exchanges are completely asynchronous, reducing the possibility of co-construction to the minimum. As a consequence, participants can technically plan their contributions for an unlimited amount of time, and they can later edit or delete them with no time limit either. Unless they choose to delete their messages, they are recorded on the platform and can be read by anyone, without necessarily owning a Reddit account, at any time. While Reddit users are supposed to be aware of this fact, they may not always be conscious of the possible way their messages could be used, whether for research or commercial purposes, which raises ethical questions about the use of such data.

Finally, the separation in communities entails subreddit specific characteristics. In terms of FUNCTION & STRUCTURE, the world setting, purpose, and activity of the conversations are typically defined by the subreddit itself. In particular, Reddit is generally steeped in internet culture, making some communities very complex to understand for outsiders. Similarly, the role and persona adopted by the participants can change depending on the subreddit, where some of them encourage participants to be truthful and authentic while others are explicitly made for role playing. The community can also have a direct impact on the social dynamics between the participants, as some people can be old members of a subreddit and have a more important position in the community than others. Some can be moderators for a given subreddit which directly grants additional power compared to other participants. Moreover, participants can develop a history of interactions with other members of a subreddit, and can even recognise each other across different subreddits, or recognise pseudonyms they have encountered somewhere else on the internet. Some participants also know each other in real life and they can import their personal history into the Reddit exchanges for instance by tagging each other in posts that relate to past conversations. Lastly, while most of the content on Reddit is in English, some subreddits have rules that require participants to post in a specific language, such as French or Spanish. In r/india, the rules explicitly state that all posts must be in English while r/france allows for both French and English. When such rules exist, they are typically enforced by the moderators of the subreddit, which increases the degree of certainty that the exchanges will be in the expected language, which is interesting from a research perspective.

| Parameter | Sub- | Subsub- | Description |
|--------------|--------------------------|------------------------|--|
| Participants | | | Humans? |
| | Participation structures | Number of participants | Virtually unlimited |
| | | Functions | Asymmetry poster – answerers, possible sub-conversations with other dynamics |
| | Enunciative status | Anonymity | Pseudonymous, digital |
| | | Role instantiation | Subreddit specific |
| | | Persona instantiation | Subreddit specific |

Table 6.1: Description of Reddit interactions.

| Parameter | Sub- | Subsub- | Description |
|------------------------|---------------------|--------------------------|---|
| | Social dynamics | Commitment level | Imbalanced? (trolls) |
| | | Hierarchy | Subreddit / Post specific |
| | | (Conversational) History | Possible within / across subreddits, possibly over the internet or even real-life interactions |
| Language | | | Mostly English |
| | Resource level | | Rules + moderation |
| | Conversation level | | No metadata provided |
| Environment & Medium | | | Online text-based forum |
| Physical context | | | Separate, share the post content |
| (Verbal) Interaction | | | |
| | Time | Synchrony | Asynchronous |
| | | Planning | No limit |
| | | Ephemerality | Recorded on the platform
Infinite edition time
Possibility to delete
Indicated as <i>edited</i> or [deleted] |
| | | | By design, as produced
Awareness of possible usage of the messages? |
| | | | |
| Function and Structure | | | |
| | Epistemic grounding | World setting | Subreddit / Post specific |
| | | World stability | Subreddit / Post specific |
| Purpose | | Practical | Subreddit / Post specific |
| | | Intention | Subreddit / Post specific |
| Activity | | Success condition | Subreddit / Post specific |
| | | Vocabulary | Subreddit / Post specific |
| | | Structure | Subreddit / Post specific |

Table 6.1: Description of Reddit interactions. (continued)

For our study, we collected conversation threads from a range of subreddits to ensure a diversity of topics and interactional dynamics. The selection is described in the next section (Section 6.2.3) together with the specificities of each subreddit.

6.2.3 Selection: a varied range of communities

Reddit’s API terms of use disclaimer. At the time the experiments described in this chapter were conducted, the Reddit’s Data API terms indicated that training machine learning models based on Reddit data was not allowed ‘without the express permission of rightsholders in the applicable User Content’. Another page indicated that using Reddit developer tools and services ‘for research purposes [was] OK provided [we] use[d] it exclusively for academic (i.e. non-commercial) purposes, and [did]n’t redistribute [the] data or any derivative products or services based on [the] data (e.g. models trained using Reddit data)’¹⁹. We thus collected our data under the assumption that the use of the API to collect data, and subsequently train LM models, was allowed in an academic context as long as neither the data nor any models trained on the data were redistributed. This restriction is also one of the reasons why we did not use the existing Pushshift Reddit Dataset (Baumgartner et al., 2020) as it was unclear whether its collection and sharing were compliant with Reddit’s policies. However, it should be noted that Reddit’s policies have evolved a lot since the time we collected our data. In particular, the use of Reddit data for research purposes is now subjected to an application to the Reddit for Researchers program. Without applying to this program, the use of the API is explicitly forbidden for academic purposes in general. Though this program did not exist at the time of our experiments, a contact form was available to request permission to share data collected from Reddit. We submitted a request in December 2023 and never received a response. We thus decided to not share our dataset or models, but we provide all our code to reproduce our experiments. However, the new policies should be taken into account for any future work on Reddit data.

The division of Reddit in communities, all with their own rules and specificities, allowed us to select a range of subreddits with different topics and interactional dynamics. We selected six subreddits, which we describe below. The selection was made based on the following criteria: (1) the range of subreddits should cover a variety of formats and expected interactional dynamics, (2) the subreddits should be active enough to provide a sufficient number of threads, and (3) the subreddits should be in English. We focused on English as in the rest of this thesis but Reddit has some communities in other languages as discussed in Section 6.2.2.2.

Art (r/Art) This subreddit is dedicated to ‘sharing finished, display ready, visual art’. The post title format is explicitly defined in the rules which makes all posts very similar in structure: a title describing an art piece and one or several pictures of it in the body of the post. We expect the comments to be more or less well-argued feedback on the art piece, thus being quite close to one another in terms of topic and differing mostly in their valence.

Coronavirus (r/Coronavirus) This subreddit was created to ‘monitor the spread of the disease COVID-19’. Due to its requirement of ‘high-quality posts and discussion’, we expect possibly long comments and rather structured discussions.

Gaming (r/gaming) This subreddit is dedicated to gaming in general and has no specific requirements except for the language which must be English, as well as some general politeness and non-toxicity rules. We expect a variety of post formats and discussion dynamics, with some posts being more argumentative and others more casual. One of the interesting aspects of this subreddit is that it is likely to contain a lot of more specialised vocabulary and references to internet and gaming culture.

OnePiece (r/OnePiece) This subreddit is dedicated to ‘Eiichiro Oda’s manga and anime series One Piece’. It contains many posts with only a title and picture but also longer ones that are meant

¹⁹Based on <https://support.reddithelp.com/hc/en-us/articles/14945211791892-Developer-Platform-Accessing-Reddit-Data> as of March 2025 in the Web Archive. We could not get back to a version of the page closer to our actual use of the API.

to trigger discussion as required by the rules. We expect a vocabulary specific to the OnePiece universe and manga culture references.

Political discussion (r/PoliticalDiscussion) This subreddit is dedicated to ‘substantive and civil discussion on political topics’. The rules require the posts to contain background information and to be formatted ‘for readability’, e.g. include paragraphs and punctuation. There are also rules regarding the comments so that the subreddit can remain ‘high quality’. We expect the exchanges to be more structured and the messages to be longer and well written compared to other subreddits.

Sports (r/sports) This subreddit is dedicated to ‘Sports News and Highlights’ from various leagues around the world. Most of the posts contain only a title and image or link but the rules require that the titles be ‘high-effort and descriptive’. Thanks to the generic nature of the topic, we expect some posts to generate comments on various subtopics.

We leverage the API to collect conversational threads from these subreddits. It does not allow us to collect all the threads from a subreddit but only the top 100 posts and their comments according to a given sorting method. We chose to collect the top 100 posts according to the number of upvotes since the creation of the subreddit as it was the sorting method that would likely provide the densest threads. Hence, in January 2024, we collected up to 100 threads from each of the six subreddits, some threads being abandoned due to the API limitations. For each thread, we collected the title and content of the main post, its ID, the number of upvotes, the pseudonym of its author, and the list of its children, that is to say all the comments in the thread. Similarly for the comments, we collected their content, their ID, the number of upvotes, the pseudonym of their author as well as the ID of their parent in the discussion tree to reconstruct the tree structure. We do not use either the number of upvotes or the pseudonyms of the authors in our experiments but we collected them in case they would be useful for further analyses. In the next section (Section 6.2.4), we describe how we leverage the tree structure of the threads to define three levels of coherence for training our topic similarity model.

Todo Writing

- Statistics on the number of threads, comments, and average depth of the trees for each subreddit.
- Pre-processing of the data: removing messages that are a url only, and deleted messages, as well as the subsequent comments

6.2.4 Coherence levels

We rely on the hierarchical structure of Reddit to build three levels of comparisons. Fig. 6.3 illustrates those levels. We consider that given a comment, one of its sons, i.e. a direct answer to it, is the most topically related. Then one of its brothers, i.e. a comment that answers the same comment is less related. Lastly a random comment from another post in the same subreddit is the least topically coherent while remaining broadly on the same topic. We initially considered a nephew comment, i.e. a comment that answers one of the brothers of the given comment, as least coherent level but we found that it did not provide enough contrast with the brother level. We never use the post itself as a base comment for the comparisons due to the variety of post formats and the fact that some posts contain only a title and a picture or link that are not described in the post, thus making most of the information in the post inaccessible to the model.

In the following section (Section 6.3), we explore several ways to validate our topical similarity ranking hypothesis.

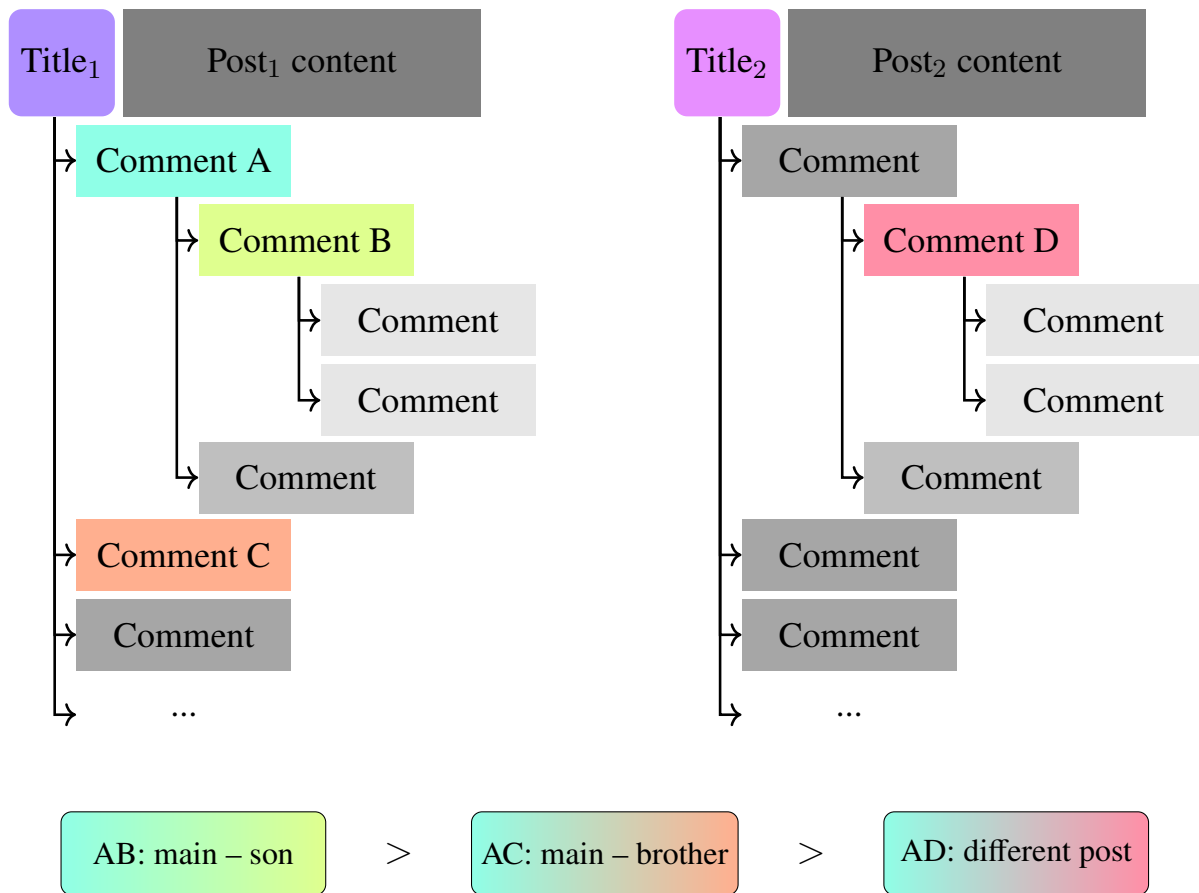


Figure 6.3: Topic similarity levels between comments.

6.3 Coherence levels hypothesis validation

Todo Experiments

- Fit LDA models (one per subreddit), compute topic distribution for each comment ABCD based on the three levels of coherence we defined. Compute the Jensen-Shannon divergence between the topic distributions of the pairs of comments and check if the divergences are in line with our hypothesis.
- Run evaluation with DSE-BERT without fine-tuning as baseline.
- BERT-Topic in a similar way as LDA?

6.4 Metrics training

Our goal is to build a topical similarity measure for dialogue that operates at a rather small scale since we are interested in the topical structure between comments. In a typical supervised learning setting, we would need to rely on a dataset of pairs of comments annotated with a topical similarity score. However, this raises the same kind of issues as the ones we discussed in Chapter 5: it is very difficult, and maybe impossible, to reliably assign such scores to pairs of comments. Instead, we leverage the hierarchical structure of Reddit threads to define three levels of topical similarity between comments and use the

same method as Xing and Carenini (2021) to train a model that learns to rank pairs of comments based on their similarity. We briefly describe the model and its training procedure in the following section (Section 6.4.1). For more details on the unsupervised training procedure, we refer the reader to Section 5.1.2. We then describe the results of our experiments in Section 6.4.2.

6.4.1 Implementation details

We adapt Xing and Carenini’s (2021) utterance-pair coherence scoring to our task and train a model that rates the similarity between two Reddit comments.

Model architecture. The structure of our model is described in Fig. 6.4. The main difference compared to the approach we described in Chapter 5 is that we replace the basic BERT model by DSE-BERT (Zhou et al., 2022), a version of BERT fine-tuned with a contrastive learning task that encourages the model to represent consecutive sentences in a similar way. Even though this fine-tuning was done on task-oriented dialogues, most of them following the Wizard of Oz paradigm – the training data is the same as for TOD-BERT and is described in C.-S. Wu et al. (2020), we hope that it provided the model with a better ability to capture small-scale similarities between sentences, which is what we are looking for in our case. This model operates with pairs of inputs instead of singular one.

Hence, our architecture first takes a pair of comments as input and passes them to DSE-BERT. The last hidden layer of the output, which is the one carrying the in-context representation, is then passed to a multi-layer perceptron (MLP) with three layers. The first layer does not change the dimensionality, the second one reduces the dimension from 768 to 6, and the last one from 6 to 1. Between the layers, we use the rectified linear activation function (ReLU) and a dropout of 10% to avoid overfitting. The final output is then passed to a sigmoid function to bring it between 0 and 1, as we want the output to represent a score of topical similarity between the two comments.

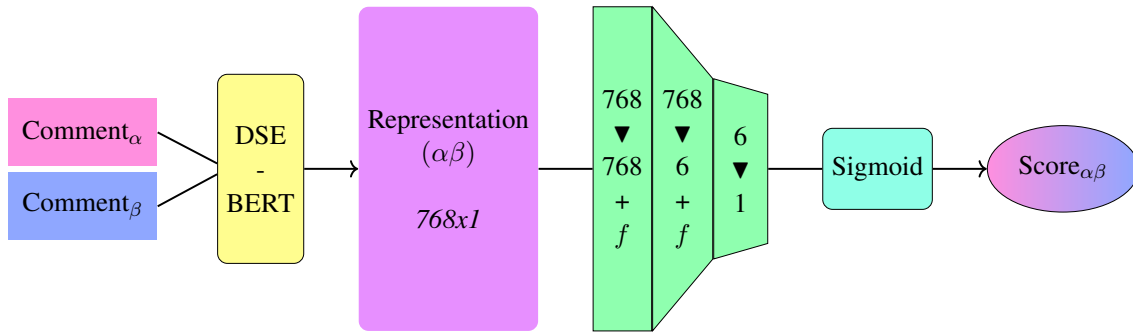


Figure 6.4: Model description. $f = \text{ReLU} + \text{dropout}$.

Loss function. As in Chapter 5, we train our model with a *marginal ranking loss* function. This means that instead of learning to predict a precise score for a given pair of comments, it learns to rank pairs of messages based on their similarity. As represented on Fig. 6.5, given the three pairs (main, son), (main, brother) and (main, random), we want the model to learn the ordering of the scores: $\text{Score}_{AB} > \text{Score}_{AC} > \text{Score}_{AD}$. Formally the function we want to minimise is the following:

$$\text{total_loss} = \frac{\text{loss}_{AB} + \text{loss}_{AC} + \text{loss}_{BC}}{3}$$

with

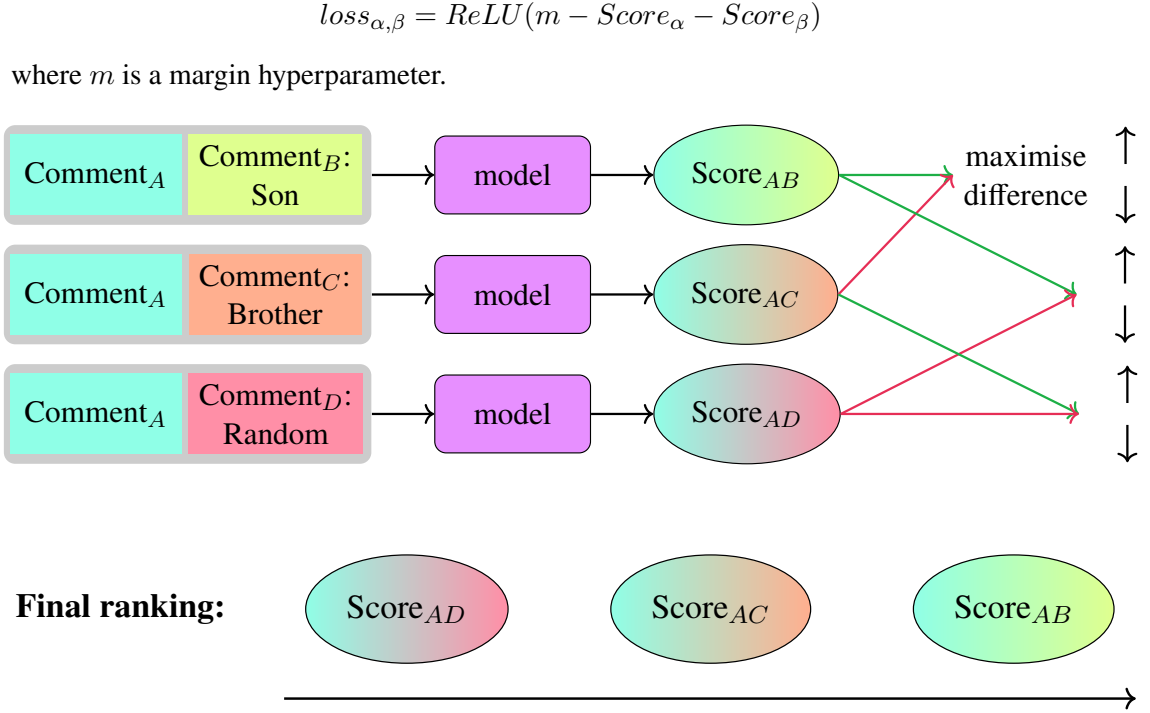


Figure 6.5: Representation of the training process.

Training setting. We trained separate models for each of the six subreddits we collected. Every time, the dataset is subdivided into a training and a test set with the proportions 80%/20%. Given the hierarchical structure of the threads, the split could not be done completely randomly as it may have separated comments from the same conversational thread into the training and test sets. This would have been especially problematic as we rely on the structure to define the three levels of topical similarity. Hence, we split the dataset at the thread level, that is to say that all the comments from a given thread, i.e. that answer to the same post, are either in the training or in the test set. In order to keep a balance regarding their size, we order them all by size, i.e. based on the number of comments they contain, and keep every fifth thread for the test set. For the training phase, we also extract 10% of the training set for validation purposes.

Regarding the generation of the triples of pairs of comments, the combinatorial nature of our approach enables us to generate a large number of triples even with a relatively small number of comments. We thus experimented with different sizes – 1,000, 7,500 and 15,000 triples – to determine a suitable amount, taking into consideration the fact that a bigger dataset requires more time and resources to train our model. The best results came with the 15,000-triple dataset and the improvement compared to the 7500-triple one was not large enough to justify testing larger datasets. We also noticed that the results were equivalent when using 15,000 triples for one epoch or 7,500 triples for two epochs. We thus decided to use the 7,500-triple dataset as the construction of the triples is time-consuming. The selection of triples is done randomly with the restriction that the same comment cannot be used twice as main comment, i.e. Comment_A in Fig. 6.5.

In terms of technical implementation, we used the optimiser AdamW (Loshchilov & Hutter, 2019) from the Python transformers library, with a learning rate of $2e - 5$ and a batch size of 16. For the loss function, we experimented with different values of the margin hyperparameter m to determine a suitable

value. We first set it to 1 as in Xing and Carenini (2021) but it made our metric take values that were very close to 0 and 1 most of the time, which was not desirable as it made it very similar to a binary classifier while we were interested in the continuous aspect. We thus experimented with different values of m and found that a value of 0.2 produced a more continuous metric while still being able to rank the pairs correctly.

6.4.2 Evaluation

Warning – Ongoing work

The results in this section were computed with the *new* models trained with a margin of 0.2.

We evaluate our metric for its ability to produce correctly ranked coherence scores for the three levels of topical similarity we defined in Section 6.2.4. We first evaluate the models trained on each subreddit independently with all the metrics we defined in Section 6.1.2. We then also evaluate their ability to generalise to other subreddits by testing each model on the test set of the other subreddits. All the evaluations were done on a test set of 100,000 triples of pairs of comments, selected randomly. We tried to select a large enough number of triples to have a good representation of the test set while keeping the evaluation time reasonable, which amounted to about an hour for each evaluation.

6.4.2.1 Independant models

The first aspect of the evaluation regards the average similarity scores for each pair of comments. As we can see in Fig. 6.6, the models trained on each subreddit are able to produce scores that are in line with our hypothesis, that is to say that the average score for the pair (main, son) is higher than the average score for the pair (main, brother), which is itself higher than the average score for the pair (main, random). The values differ between the subreddits, which is not surprising as they have different characteristics. A Student's t-test indicated that the differences between the average scores for each pair are highly significant ($p < 0.001$) for all the subreddits. Nevertheless the large standard deviations indicate that the scores are likely not always in line with our hypothesis. The high significance of the differences between the average scores is likely due to the very large number of triples in the test set, which makes it easier to detect small differences. The next step of the evaluation is thus to analyse the ranking of the scores for each triple of pairs of comments.

Table 6.2 presents the different evaluation metrics related to the ranking of the scores for each triple of pairs of comments. The pairwise accuracy is high for all the subreddits, which indicates that the models are able to rank correctly most of the pairs. When we look in more details at which of the three pairs are ranked correctly, we can see that the models are better at ranking the first pair (main, son) and the third one (main, random) than the second one (main, brother). This is confirmed by the pairwise comparisons, which show that the first and last pair are most of the time ranked correctly with respect to each other, while the second pair is less often ranked correctly with respect to the first and last pairs. The proportion of triples for which the ranking is perfect lies between .54 and .65 for all subreddits except for Coronavirus, which is higher at .76. All of these results indicate that the models are able to learn a topical similarity metric that is in line with our hypothesis, even though there is still room for improvement. The Coronavirus subreddit is the one for which the model performs best, which may be due to the fact that the comments are easier to distinguish in terms of topical similarity. But it could also mean that they are more similar to the type of data the model was pre-trained on.

Finally the last aspect of the evaluation regards the margins between the scores for each pair of comments. As we can see in Fig. 6.7, for most of the subreddits, the margin between the first and second

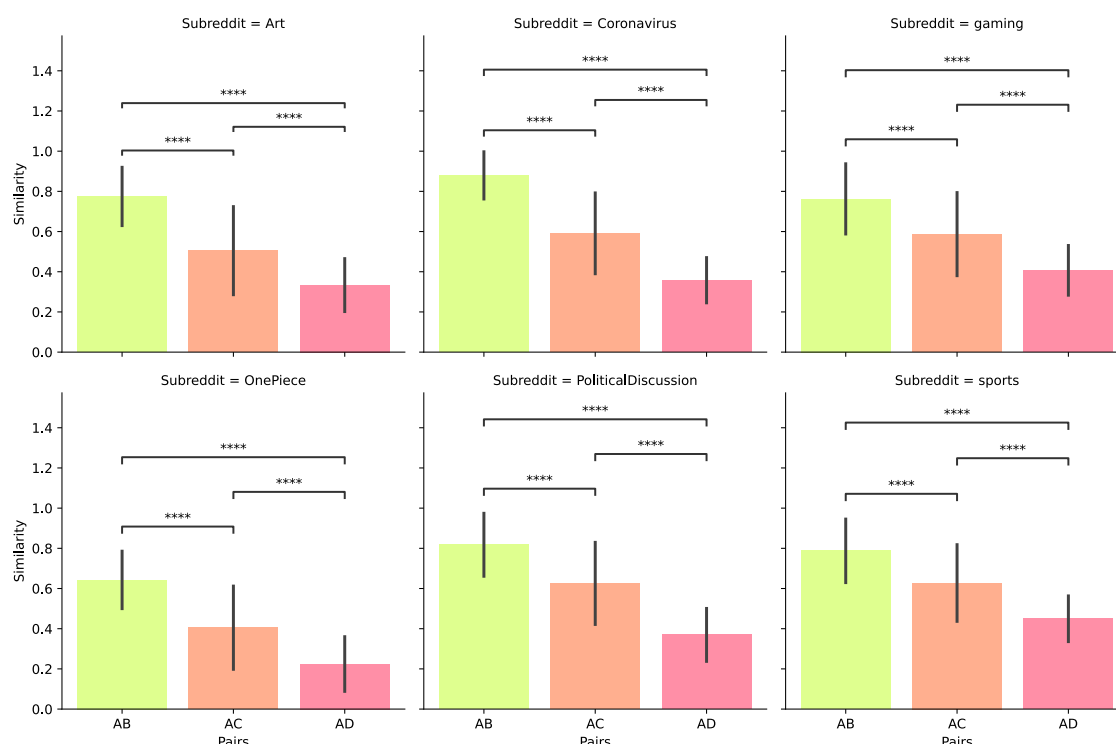


Figure 6.6: Average similarity scores for each pair computed by the models trained each on one subreddit.

| Subreddit | Acc. | 1 st ✓ | 2 nd ✓ | 3 rd ✓ | AB > AC | AC > AD | AB > AD | Perfect Ranking |
|---------------------|------------|-------------------|-------------------|-------------------|------------|------------|------------|-----------------|
| Art | .85 | .83 | .59 | .71 | .85 | .73 | .97 | .58 |
| Coronavirus | .92 | .91 | .77 | .84 | .91 | .85 | .99 | .76 |
| gaming | .82 | .73 | .55 | .73 | .76 | .76 | .93 | .54 |
| OnePiece | .85 | .82 | .60 | .74 | .84 | .76 | .96 | .60 |
| PoliticalDiscussion | .87 | .79 | .66 | .83 | .80 | .85 | .96 | .65 |
| sports | .82 | .74 | .57 | .75 | .76 | .78 | .93 | .55 |

Table 6.2: Evaluation of the ranking performed by the models trained each on one subreddit. ‘Acc.’ is the pairwise accuracy; ‘1st✓’, ‘2nd✓’ and ‘3rd✓’ are the proportions of of correctly ranked first, second and third scores; ‘AB > AC’, ‘AC > AD’ and ‘AB > AD’ are the proportions of pairs correctly ordered two by two; ‘Perfect Ranking’ is the proportion of triples for which the ranking is perfect.

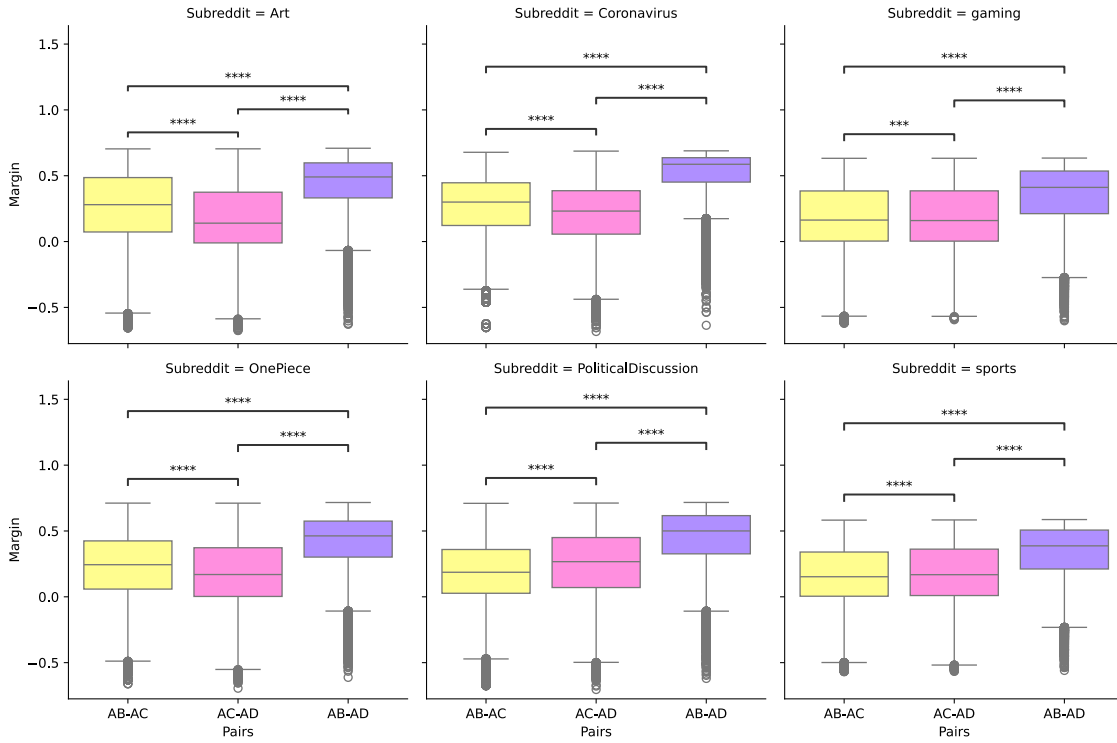


Figure 6.7: Average difference between the scores for each pair computed by the models trained each on one subreddit.

pair is larger than that between the second and third pair. This is quite surprising as the messages in the pair AB are at a distance of 1 from each other in the conversation tree, just like the messages in the pair AC though the distance is horizontal instead of vertical. The messages in the pair AD, however, are at a random distance from each other in the conversation tree, which should make them more different from each other than the messages in the pair AC. This is not what we observe in the results, and the standard deviations are quite similar for the two margins, indicating that the random pair does not vastly increase the variability of the results. This may be due to the topology of the conversation trees, which are often very large and shallow, thus making randomly selected comments quite likely to be at a short distance from the brother comment. Nevertheless, the larger margin between the first and last pair confirms that the randomly selected comment is in average quite different from the main comment, supporting our hypothesis despite the potential topological bias.

6.4.2.2 Model transfer

To assess the generalisation ability of our models, we also evaluated them on the test sets of the other subreddits. This allows us to determine if the models are able to learn a general topical similarity metric or if they are too specific to the subreddit they were trained on. The pairwise accuracies are presented in Table 6.3. Unsurprisingly, the best result for a given model or dataset is obtained with the congruent model. However, the results remain quite good when transferring a model to another subreddit. This is encouraging as it indicates that we could be able to train a single model that could be used for all the subreddits, thus reducing the training time and resources needed. Another aspect, directly relevant to our study in this chapter, is that the entirety of the threads from a given subreddit could be included in the analysis if we use a model trained on another subreddit. This is of particular interest for our exploratory

| Data \ Model | Art | Coronavirus | gaming | OnePiece | Political Discussion | sports |
|---------------------|------------|-------------|------------|------------|----------------------|------------|
| Art | .85 | .77 | .78 | .80 | .80 | .79 |
| Coronavirus | .77 | .92 | .76 | .76 | .79 | .76 |
| gaming | .81 | .77 | .82 | .81 | .80 | .80 |
| OnePiece | .81 | .78 | .79 | .85 | .80 | .80 |
| PoliticalDiscussion | .78 | .77 | .77 | .77 | .87 | .77 |
| sports | .81 | .77 | .80 | .81 | .80 | .82 |
| Average | .81 | .80 | .79 | .79 | .81 | .79 |

Table 6.3: Pairwise accuracy of the models trained on one subreddit and tested on another one.

analysis because the way we split the dataset into training and test sets, as described in Section 6.4.1, removed a large number of dense threads from the test set, which are the ones that are the most likely to contain longer sequences of comments.

When it comes to the selection of the best model overall, we can see that the model trained on the *Coronavirus* subreddit does not perform as well as we could have expected based on its performance on its own test set. It is also one of the subreddits on which the other models perform the worst. This validates our concern that the dataset has a specificity that makes it easy to *learn*, which likely led to an overfitting of the model, infringing its ability to generalise to other subreddits. The models trained on the *Art* and *PoliticalDiscussion* subreddits are the ones that perform best in average, though the differences are not large. This may be due to the fact that these subreddits contain longer comments as well as the very short ones that are present in all the subreddits.

Deepen analysis based on stats on the corpus.

[Amandine]

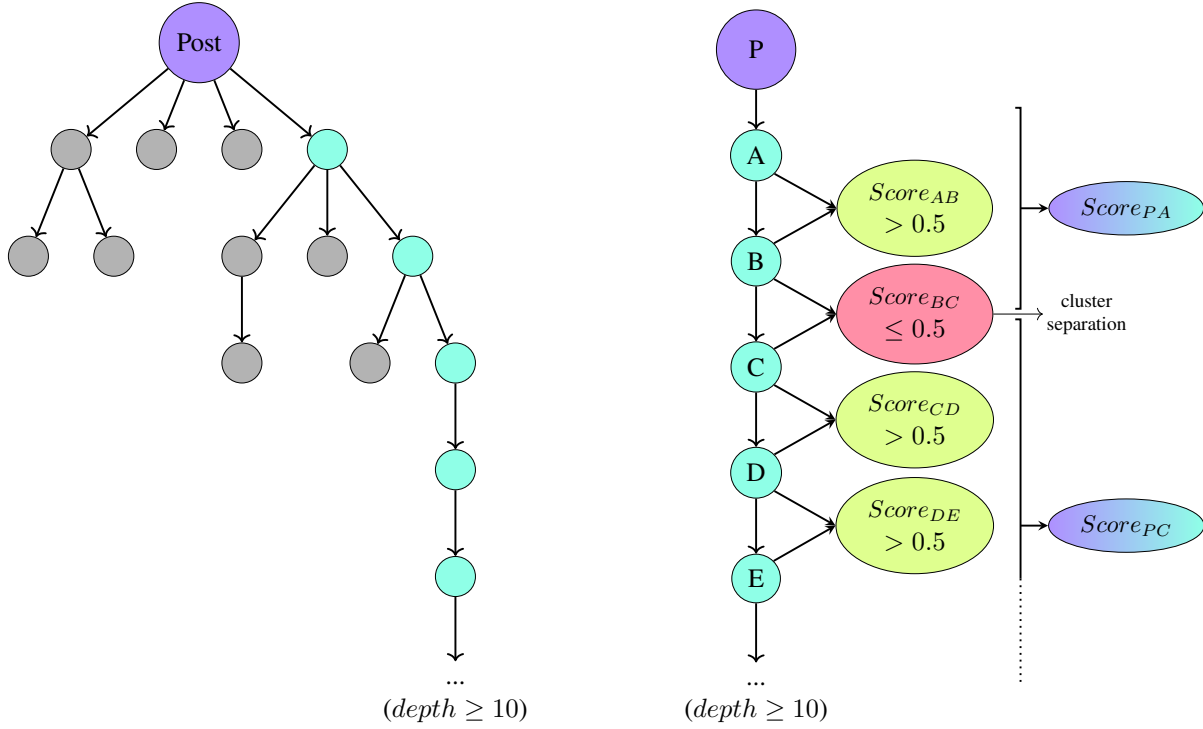
These evaluations confirmed that our models are able to produce similarity scores between comments that are, in most cases, in line with our hypothesis regarding the three levels of topical similarity. This is encouraging with respect to our goal of using the metric to analyse the topical structure of the conversations. The models are also able to generalise to other subreddits, which we will use in our exploratory analysis that are presented in the next section (Section 6.5).

6.5 Conversation modelling with a continuous topic similarity metric

Warning – Ongoing work

This section is based on the work that was done at the beginning of my PhD and presented in Decker and Amblard (2024b). At the time the model was trained with a marginal ranking loss of margin 1, which made the metric behave more like a binary classifier. The results presented here are thus not representative of what we will obtain with the new model trained with a margin of 0.2. They also do not leverage the possibility to apply one model to the *other* subreddits as discussed at the end of the previous section.

These results are included in this version of the thesis to give an idea of the kind of analysis we are interested in but they will be updated and possibly refined in the final version of the thesis. Another note at the end of the section briefly describes the ideas we have in mind for the final version of this analysis.



(a) Extraction of a comment sequence.

(b) Construction of the clusters.

Figure 6.8: Representation of the process to identify clusters of comments in a sequence.

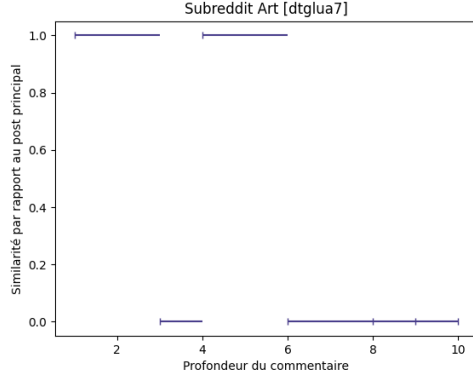
6.5.1 Clusters of comments

Our goal is to analyse the topical structure of the Reddit threads. Hence, we used our trained metric to locate coherent sequences of messages in a thread. We first extracted all the comment sequences of ten messages or more to have long enough sequences to observe thematic variations as represented on Fig. 6.8a. We then used the following method, represented on Fig. 6.8b, to identify topical variations in the sequences of comments:

- In a given sequence of comments, i.e. we do not look at the horizontal structure of the conversation trees, we compute the similarity of each message with the subsequent one;
- Every time the similarity is smaller than a threshold, that we fixed at 0.5 given the almost-binary behaviour of the metric, we consider that the subsequent message is the beginning of a new topic and carries the topic in its content;
- We compute the similarity between the main post and the first comment of each new topic to analyse how the conversation evolves.

Fig. 6.9a shows an example of coherence variation we retrieved from the subreddit *Art*. Each horizontal line represents a coherent cluster of comments and their vertical position indicates the similarity with the main post. Their length depends on the number of messages. Fig. 6.9b shows examples of comments as well as the initial post of the conversation. We see that the second cluster is reduced to the depth 3 comment and is considered thematically different from the main post ('They Don't Even Taste That Good

Anymore (I), oil on canvas, 24x30’’). This makes sense because the comment only contains the word ‘Arrow’.



(a) Length and similarity compared to the main post in the identified clusters.

Post: They Don’t Even Taste That Good Anymore (I), oil on canvas, 24x30’’

Depth 3 comment: Arrow.

Depth 4 comment: It has to be good in the first place to stop being good, though.

Depth 9 comment: Fine, as long as you have a beard, like a proper dwarf.

(b) Head comments of the identified topical clusters.

Figure 6.9: Example of a message sequence representation.

Table 6.4 presents different statistics on the sequences of messages in each subreddit. We can see that the *Coronavirus* subreddit has few sequences that we considered long enough for the clustering. All the subreddits have a similar number of messages per sequence in average but the *gaming* one has some very long sequences (at most 275 messages) and the *Art* and *sports* have rather long ones as well (about 50 messages). Regarding the evolution of the topics, we can see that about one third of the first messages after a post are topically close to it but almost 90% of the last cluster heads are far from it. We also computed the number of times where the similarity of two subsequent clusters with the main post are drastically different. In other words, we counted the number of times where a cluster is topically close to the main post and the next one is far from it or vice versa. This cannot be considered as a measure of the number of topic changes in a conversation because we compare the similarity of the clusters with the main post and not with each other. However, it gives us an idea of how many times a conversation switches from being on-topic to off-topic or vice versa.

There is in average zero or one such change depending on the subreddit, which tells us that in average the conversation is either considered off-topic by our model from the beginning or begins by being on-topic and switches to off-topic at one point without going back to the main topic. However, the very sensitive nature of our metric and its tendency to assign extreme values to the similarity of two comments makes it complicated to interpret these results.

Besides the relation of each cluster with the main post, we are also interested in the relation between the clusters themselves. We thus performed several layers of clustering to highlight a hierarchy between the topical clusters. The results of this analysis are presented in the next section (Section 6.5.2).

6.5.2 Cluster hierarchy

We went one step further in our analysis by clustering the coherent pieces of threads we retrieved with the method described above. To do this, we re-applied the same process but on the heads of the topics instead of all the comments. Hence the process is as follows:

- (i) Find clusters of coherent comments in a thread with the method described in Section 6.5.1;

| Subreddit | # sequences
(≥ 10
msgs) | #
max | # avg | Initial sim.
($<0,1 / >0,9$) | Final sim.
($<0,1 / >0,9$) | # brutal
changes |
|--------------------|-------------------------------------|----------|-------------------------|-----------------------------------|---------------------------------|------------------------|
| Art | 1556 | 51 | 11,88
($\pm 2,96$) | 63% / 34% | 87% / 11% | 1,07
($\pm 1,67$) |
| Coronavirus | 58 | 18 | 11,62
($\pm 1,95$) | 69% / 28% | 84% / 12% | 0,55
($\pm 0,90$) |
| gaming | 9409 | 275 | 11,88
($\pm 5,44$) | 64% / 34% | 88% / 10% | 0,70
($\pm 1,14$) |
| OnePiece | 908 | 28 | 11,84
($\pm 2,48$) | 61% / 37% | 91% / 07% | 0,54
($\pm 1,01$) |
| sports | 5930 | 52 | 11,71
($\pm 2,58$) | 46% / 52% | 85% / 14% | 0,75
($\pm 1,14$) |

Table 6.4: Statistics on the comment sequences and the formed topical clusters.

- (ii) Compute the similarity of the first message of each cluster with the first message of the subsequent cluster;
- (iii) If the similarity is smaller than a threshold then we consider that the second cluster is the beginning of a new topic;
- (iv) Repeat the second and third step on the new clustering until it becomes stable.

Fig. 6.10 is an illustration of this process. This enables us to highlight a hierarchy between the topical clusters where we first locate topics with a fine granularity and each additional layer of clustering is related to topics with a coarser granularity.

Following the method described in the previous section, we worked with the sequences of at least 9 messages and computed the different levels of clusters. Table 6.5 presents different statistics on the results. Regarding the number of levels of clustering, all subreddits require three levels in average. There are about four clusters in average on the highest-level layer but the high standard deviation for all subreddits shows that there is more variability on this aspect.

| Subreddit | Layers of clustering
(avg $\pm$ std) | Number of coarsest clusters
(avg $\pm$ std) |
|--------------------|---|--|
| Art | 3,03 ($\pm 1,34$) | 4,59 ($\pm 2,97$) |
| Coronavirus | 2,97 ($\pm 1,04$) | 3,86 ($\pm 2,54$) |
| gaming | 3,09 ($\pm 1,90$) | 4,41 ($\pm 3,23$) |
| OnePiece | 3,15 ($\pm 1,30$) | 4,72 ($\pm 2,90$) |
| sports | 3,14 ($\pm 1,33$) | 4,30 ($\pm 2,91$) |

Table 6.5: Statistics on the hierarchies of topical clusters.

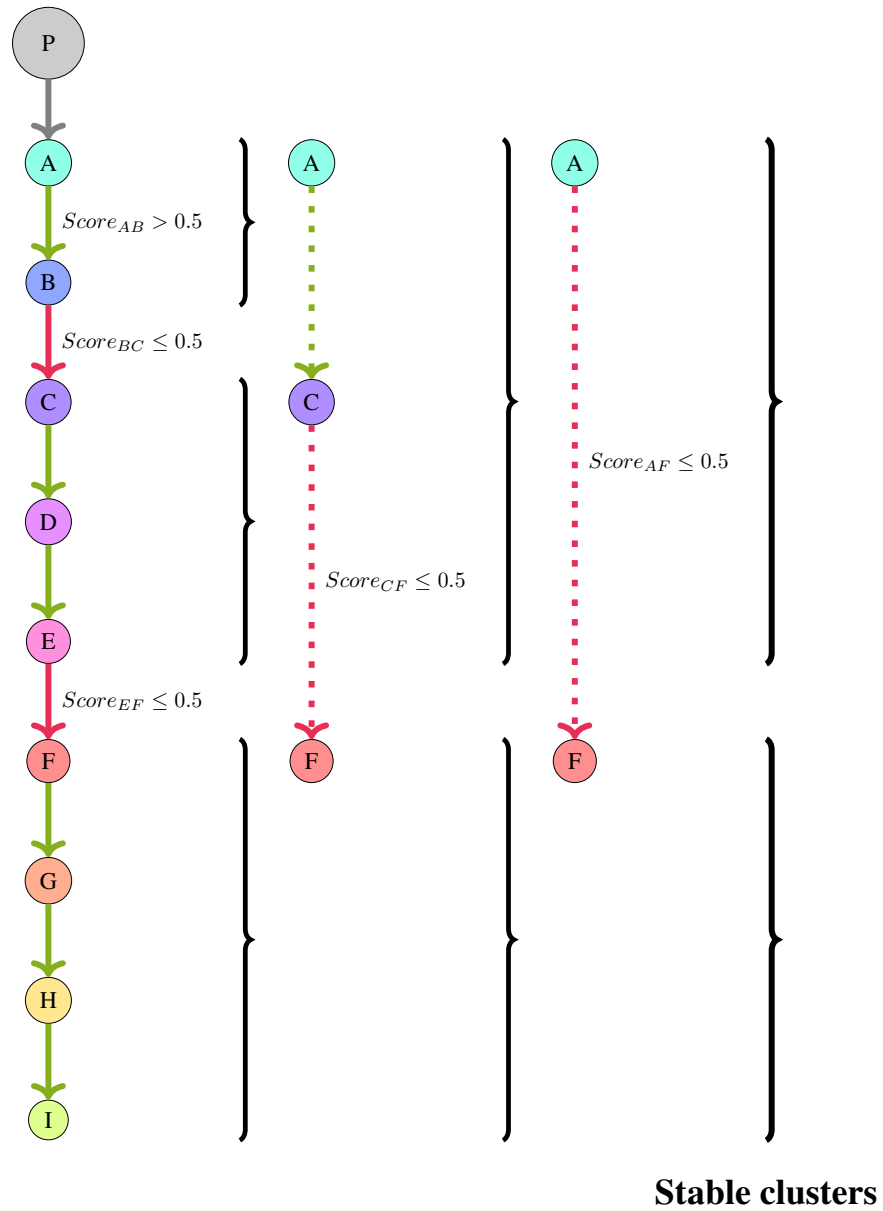


Figure 6.10: Representation of the hierarchical clustering process.

Warning – Ongoing work

Prospective plans for the final version of the analysis:

- Re-run analysis with new version of the model? The current version of the analysis is too close to binary segmentation so it is not what we are looking for but the new model may be more interesting;
- Compute distance to the root of all comments to visualise the drift from the main topic; compare across sequences from direct answers to the main post and sequences from deeper comments;
- Compute 'branching factor' of the posts when format allows it: compute distance of all depth 1 comments to the post and compared to each other

6.5.3 Challenges of the data

Todo Writing

Working with text messages gathered from the internet comes with several challenges that will be described and illustrated in the final version of the thesis. Here are some of the issues we will discuss:

- The presence of a lot of noise in the data, such as typos, abbreviations, and informal language, which can affect the performance of our model.
- Some behaviours, typically due to internet culture, can complicate the topical analysis such as long threads of messages where people repeat the same message, or count one after the other, etc.
- The format of the posts, that are often 'multimodal' with images, videos, and links, make them quite complicated to exploit in the analysis.

6.6 Discussion and Conclusion

Warning – Ongoing work

The conclusion is mostly taken from the paper Decker and Amblard (2024b) and will be updated in the final version of the thesis.

In this chapter we presented a non-supervised topical similarity measure. We showed that it enables us to identify clusters of topically coherent messages in a conversation, which corresponds to the task of linear topic segmentation. But this measure also allows us to hierarchise these clusters into a several-layer structure. Our measure can be used for a precise topic segmentation since the pieces of conversation would be logically organised on top of being separated based on their topic.

The clustering we performed was focused on vertical sequences of messages in the sense that we extracted sequences of messages instead of complete sub-trees of the conversation. The horizontal dimension could be explored to compare the evolution of topics in parallel threads and, for example, see how much the conversations tend to converge or diverge and if this behaviour varies depending on the subreddit. This analysis would allow us to highlight the relative importance of a given topic in the whole thread of conversation even if it appears little in each conversation.

Ideally we would need to validate our metric with manually annotated data, where we could ask people to rank pairs of comments based on their topical similarity and compare their ranking with the one produced by our metric. This would be a more precise evaluation of the metric than the one we performed in this chapter, that was solely based on the structure of the Reddit threads. Unfortunately we did not have the time to do this as we chose instead to focus on the analysis of face-to-face conversations as we will see in the next chapter. This remains interesting future work.

Overall this exploratory work highlighted the complexity and variability of topical structure in conversation. This led us to consider a more theoretically and empirically grounded approach to the analysis of dialogue. Moreover, even though Reddit exchanges can for the most part be considered as natural human-human conversations, they are still online exchanges and thus differ from face-to-face conversations in many ways. The next chapter (Chapter 7) brings together these data-realism (find a real word for this concept) and epistemological concerns.

Modelling Topics with QUDs

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Todo Writing

- Better inclusion of the chapter within the thesis (especially intro)
- (- Comparison of IAA and correlation between dialogue groups: figure out proper statistical way)

Discourse structure is a fundamental aspect of language use, yet it remains challenging to characterise in a way that is both theoretically precise and empirically tractable. One influential approach to modelling it is the theory of *Questions Under Discussion* (QUDs; **roberts1996information**; Ginzburg, 1995; Van Kuppevelt, 1995), which represents discourse as organised around a dynamically evolving set of questions

that conversational participants would potentially require answers for throughout the dialogue. On this view, discourse proceeds by addressing questions whose possible answers define the space of relevant contributions at each point in the interaction.

QUD-driven frameworks have the advantage of providing a unified perspective on discourse organisation: they can be used retrospectively to reconstruct the structure and content of an interaction, or predictively to model expectations about how discourse will develop. In the predictive setting, so-called potential questions (PQs; Onea, 2016) are used to represent candidate issues that may become active QUDs as the discourse unfolds.

Despite their theoretical appeal, operationalising QUDs in natural data remains difficult. The space of relevant questions is not fixed, and questions can arise at multiple levels of granularity, from local clarification requests to broader discourse-level concerns. In addition, discourse structure is shaped not only by informational content but also by interactional phenomena such as turn-taking, speaker intentions, and shared situational context, all of which complicate the identification of a well-defined set of QUDs or PQs in spontaneous data.

To address this challenge, Y. Wu et al. (2024) propose an empirical approach based on large language models (LLMs), which are used to generate PQs anchored in textual context. They show that, in monologic data such as news articles and Ted Talks, the salience of a generated question correlates with whether it is subsequently answered in the text, suggesting that discourse is organised in anticipation of the recipient’s expectations.

However, it remains unclear to what extent these findings extend to dialogue. Unlike monologic discourse, dialogue is jointly constructed in real time by multiple participants, and its structure emerges from interactional negotiation rather than a single authorial plan. To investigate whether QUDs/PQs-driven approaches generalise to naturalistic conversation, we apply the methodology of Y. Wu et al. (2024) to English dialogues exhibiting varying degrees of interactional organisation, and examine whether the relationship between question salience and later resolution persists in these more complex settings. By testing this particular method on naturalistic dialogue (ND), we identify several challenges for a QUD-driven approach to discourse structure in conversation. We also release a corpus of 7,124 generated questions, each annotated for salience and answerability by two annotators.

The remainder of the chapter is organised as follows. Section 7.1 reviews the theoretical and empirical work underpinning our study. Section 7.2 describes the conversational corpus used in our experiments. Section 7.3 presents our methodology, and Section 7.4 introduces the released dataset. We then analyse the results in Section 7.5 and conclude with a discussion of their implications in Section 7.6.

The work described in this chapter has been accepted for publication as a long paper at SemDial 2026, the 30th workshop on the semantics and pragmatics of dialogue. It will be presented on the 3-4 September 2026.

Warning – Ongoing work

Some updates to the analysis and discussion sections will be made for the final version of this thesis based on the reviewers’ comments. Descriptive notes are included in the relevant sections.

7.1 Background: Questions and Discourse structure

Questions have long played a central role in linguistic theories of discourse and dialogue structure. In particular, the QUD framework models discourse as evolving around explicit or implicit questions that guide interpretation and production. Originally developed to account for conversational coherence (Roberts 1996; Van Kuppevelt, 1995; Ginzburg, 1995), it has more recently attracted interest in NLP as a way of opera-

tionalising and modelling recipients’ expectations. This section first reviews QUDs as a theoretical and predictive framework before discussing recent computational work on question salience and the challenges posed by dialogue.

7.1.1 QUDs as predictive tools

QUD-driven approaches represent discourse as a hierarchy of questions, where each contribution either (partially) answers a question, introduces a sub-question, or shifts the focus of the interaction (Van Kuppevelt, 1995; Roberts, 2012; Ginzburg, 2012). Traditionally, QUDs were used retrospectively: analysts inferred which question must have been under discussion to make an utterance coherent. More recent work instead argues that discourse participants actively anticipate forthcoming QUDs and use them to guide interpretation and production.

A growing body of evidence supports that QUDs are a cognitively relevant construct that we used when producing and interpreting discourse. Eye-tracking experiments by Clifton and Frazier (2012) showed that comprehension is facilitated when readers’ expectations about the QUD align with subsequent text. ERP (event-related brain potential) studies by Delogu et al. (2020) further demonstrated that discourse segments failing to address an expected QUD incur higher processing costs. Extending this work computationally, Y. Wu et al. (2024) leveraged LLMs to generate PQs in news articles and TED talks and found that more salient questions are more likely to be answered later in the discourse, suggesting that authors structure discourse around anticipated recipients’ expectations.

Empirical evidence from dialogue remains more limited. Kehler and Rohde (2017) showed that grammatical context influences the QUDs evoked during dialogue interpretation, although their experiments relied on controlled completion tasks. Westera and Rohde (2019) proposed a methodology for collecting actual and potential QUDs in ND, while Mulligan and Rawlins (2025) reported only moderate agreement between annotators when inferring QUDs from interviews. Together, these studies highlight both the relevance of QUDs and the large space of plausible questions associated with a given context. However, because existing work has focused on relatively structured interactions, it remains unclear how well the framework generalises to less organised forms of conversation. We address this question by applying the methodology of Y. Wu et al. (2024) to a diverse set of ND.

7.1.2 Operationalising QUDs through question *salience* and *answerability*

Y. Wu et al. (2024) introduced the first computational framework designed to test whether salient questions are more likely to be answered in subsequent discourse. Working with English news articles (Ko et al., 2022; Huang et al., 2024) and TED talks (Westera et al., 2020), they considered questions evoked at a sentence S_k given the preceding context $S_1, \dots, S_{k-1}$ and evaluated whether these questions were answered in the remaining text.

To operationalise this idea, they leveraged LLMs to generate questions anchored in S_k and annotated each question for *salience* – its relevance at that point in the discourse – and *answerability* – the extent to which it is addressed later in the text. Figure 7.1 provides an example from their dataset. Spearman correlation between these two variables revealed strong positive relationships ($\rho = .59$ for news and $\rho = .74$ for TED talks). They also introduced QSalience, a Mistral-7B-Instruct model fine-tuned to predict question salience from context.

These findings provide a first large-scale operationalisation of QUDs as a predictive model of discourse structure. However, because the study focuses on monologic genres, it remains unclear whether the same relationship holds in dialogue. Indeed, as discussed throughout this thesis, dialogue is a more complex and multifaceted environment than monologue. Multiple participants, dependence on contextual information, and need to manage the interaction itself all increase the range of relevant PQs and may

[1] Amid skepticism that Russia’s war in Chechnya can be ended across a negotiating table, peace talks were set to resume Wednesday in neighboring Ingushetia. [2] **The scheduled resumption of talks in the town of Sleptsovsk came two days after agreement on a limited cease-fire, calling for both sides to stop using heavy artillery Tuesday.** [3] They also agreed in principle to work out a mechanism for exchanging prisoners of war and the dead.

[Q1] What other progress has been made towards peace recently?

- Is [Q1] salient given the context of [1] and the **anchor** [2]?
 - Yes, it elaborates on what is being done to establish peace in the region.
- Is it answered in the subsequent text?
 - Yes, in [3].

Figure 7.1: Example from Y. Wu et al. (2024) of a generated question based on a piece of news text.

reduce the predictability of subsequent discourse. We thus investigate whether the relationship between question salience and answerability persists in ND, and whether it varies across different interactional settings.

7.2 Source Corpus: Naturalistic Dialogues

The British National Corpus (BNC) is a 100-million-word collection of written and spoken British English designed to provide a broad representation of language use in the late twentieth century. An XML version of the corpus was released in 2014²⁰ and is freely available for research under the terms of the BNC User Licence.

The corpus comprises both written and spoken language. Since our objective is to investigate the applicability of the QUD/PQ framework to dialogue, we focus exclusively on the spoken component. We first describe this component in Section 7.2.1, then present the selection procedure used to construct our experimental dataset in Section 7.2.2, and finally provide descriptive statistics in Section 7.2.3.

7.2.1 The Spoken BNC

The spoken component of the BNC consists of an approximately 10-million-word orthographic transcriptions of a wide variety of spoken interactions, ranging from informal conversations to institutional and professional settings. The corpus was designed to achieve both demographic representativeness and contextual diversity. Because satisfying these objectives simultaneously is challenging, the spoken BNC is divided into a *demographically sampled* component and a *contextually sampled* component.

The demographically sampled component was collected through naturalistic recordings. Participants were asked to record their conversations over periods ranging from two days to one week using portable tape recorders. Whenever possible, interlocutors were informed of the recording only after the interaction had taken place. Recordings were included in the corpus only after all participants had provided informed consent. The primary objective of this sampling strategy was to obtain a balanced representation of speakers across gender, age, and social class.

The contextually sampled component was constructed to cover a broad range of communicative situations. The main categories include *education and informative* contexts (e.g. lectures, classroom in-

²⁰The XML version of the BNC can be downloaded here.

interactions, and news commentaries), *business* contexts (e.g. meetings, interviews, and trade union discussions), *public and institutional* contexts (e.g. political speeches, council meetings, and governmental discussions), and *leisure* contexts (e.g. sports commentaries, chat shows, and club meetings). A detailed description of these categories is available in the BNC User Reference Guide. Different sampling procedures were employed across contexts to ensure broad coverage of interaction types.

For the present study, we selected 60 dialogues from the spoken BNC, covering different interactional settings and varying degrees of expected conversational organisation. The selection procedure is described in Section 7.2.2.

7.2.2 Sub-selection: Varied interactional settings

One objective of this study is to examine how the relationship between question salience and answerability varies across different forms of interaction. We therefore selected dialogues that differ both in the number of participants and in their expected degree of interactional organisation. We operationalised these dimensions by defining six dialogue categories, described in Table 7.1 with the explicit constraints we applied to the BNC metadata.

The open-domain conversations were drawn from the demographically sampled component and consist of spontaneous conversations. The remaining categories were selected from the contextually sampled component using metadata-based constraints designed to capture progressively more structured interaction types. The organised conversations include interviews done in the context of oral history projects for the duologues, and meetings that were *not* classified as *leisure* for the multilogues. Finally for the public conversations we selected broadcast discussions. Both the organised and public categories contain dialogues guided by an agenda and most of the time moderated by an host or a chairperson. The most crucial difference between the two is that the public discussions are intended for an external audience and are therefore overall easier to understand as external observers.

We also specified the number of participants, exactly 2 for the duologues and 5 or more for the multilogues to ensure a clear distinction between the two types of conversations. Finally, we specified the length of the dialogues to ensure that they were long enough to contain a sufficient number of questions and answers, but not too long to be too difficult to annotate. Some of the categories did not contain dialogues that were short enough so we looked for longer ones that we manually cut at the end of a topic. For the multilogues we also confirmed that the number of participants was already at least 5 in the part of the dialogue we kept.

For each category we randomly sampled ten dialogues fulfilling all the conditions, except for public

| Group | Metadata constraints | # Ppts | Length | # |
|--------------|---|--------|-----------|----|
| Open duo | classification: spontaneous conversation | 2 | 40 – 70 | 10 |
| Open multi | classification: spontaneous conversation | > 4 | 40 – 90 | 10 |
| Org. duo | classification: interview oral history | 2 | ≥ 40 | 10 |
| Org. multi | classification: meeting, subcategory $\neq$ Leisure,
activity: does not contain ‘informal’ | > 4 | ≥ 40 | 10 |
| Public duo | classification: broadcast discussion | 2 | 40 – 90 | 10 |
| Public multi | classification: broadcast discussion, title: contains ‘radio’ | > 4 | ≥ 40 | 5 |
| | classification: broadcast discussion, title: contains
‘television’ | > 4 | ≥ 40 | 5 |

Table 7.1: Constraints for the random sampling of our corpus.

multilogues, which were split equally between radio and television discussions. When selected dialogues were too long for manual annotation, we extracted shorter excerpts while preserving the interactional characteristics of the original conversation. Excerpts were truncated manually at the end of a topic and, for multilogues, were checked to ensure that the intended number of participants remained active.

Table 7.2 summarises the characteristics of the three degrees of interaction – open, organised and public – in our corpus with the duologues and multilogues combined. The participants are human beings interacting in real life. In the duologues, the two participants take turn speaking with a more or less balanced repartition depending on the setting, where the interviews and radio broadcasts are less balanced since the host or interviewer typically asks questions and the other participant answers sometimes with long responses. Similar discrepancies exist in the multilogues with the open ones being more balanced and involving most participants at all times while in the organised ones – council meetings – a sub-group of participants is typically more active for a given point of the agenda. In the public multilogues, the floor is typically taken by one person at a time for longer interventions with the host selecting the speaker, which characterises the turn-taking behaviour. In terms of hierarchy, the most striking difference is the presence of a host or chairperson in the organised and public settings, which is absent in the open ones. This influences the structure of the interaction and the way topics are introduced and discussed. Other hierarchical aspects exist in some of the conversations independently of the setting, for example in the open conversations where participants can be family members, in some interviews where a participant can be much older than the host, or in the council meetings where a participant can be an expert of a topic on the agenda. The conversational history is a very relevant factor for our task since it can make the dialogues more complicated to understand from an external point of view. In the open conversations, the participants are typically familiar with each other and do not always clearly explain the events or people they refer to. The organised multilogues are council meetings where we do not have access to the entire history of meetings before the one we selected, which creates a similar situation. The remaining categories are more self-contained; public dialogues because they are intended for an external audience and organised duologues because they are interviews where the two people seem to have never met before. Nevertheless in all the non-open settings, a topic of discussion was defined prior to the interaction, which constrains the range of possible contributions. Conversely, the open conversations are completely free and thus much more chaotic. This aspect is very likely to play a role in the predictability of each type of dialogues.

All the conversations happen in a face-to-face setting though some specificities exist for each setting. In the open conversations one or several of the participants are sometimes realising a task in parallel to the conversation, which can reduce the attention they give to monitoring the coordination signals. In the television multilogues, the physical setting of the show possibly makes it complicated for some participants to directly look at each other. The critical aspect regarding the ENVIRONMENT & MEDIUM parameter is that while the participants interacted in a format that is overall quite close to a prototypical face-to-face conversation, the transcriptions we have access to take away a lot of the multimodal information. Except for the radio setting, which is intended for an external audience that would have access to the speech only, all the other settings rely, at different degrees, on the physical co-location of the participants. It is most obvious in the council meetings where participants often refer to documents they own physical copies of, and in the TV setting where the host seems to be pointing at a person when addressing them. The latter context is quite easily reconstructable from the transcription and common sense but the former is not. The quality of the recording is also far from optimal – the BNC corpus was recorded in the 1990s – and the transcriptions contain numerous <unclear> tags, sometimes hindering the understanding as we further discuss in Section 7.2.4.

The last differences pertain to the content and are probably central for the task discussed in this chapter. Apart from the open conversations, we can imagine that in all the other settings participants had an idea of the topic of discussion before the interaction started. In the interviews, the host and the interviewee

probably agreed on a topic to discuss, sometimes possibly even agreeing on the questions that would be asked. Though some specific pieces of dialogue let us think that the interviewee or guests did not have all the details, we can imagine that these exchanges were somewhat prepared. This aspect goes together with the differences in structure we have already mentioned where the open conversations are completely free and the other settings are more or less structured with the host or secretary guiding the discussion. These features are likely to influence the predictability of the topics in the different settings.

| Parameter | Sub- | Subsub- | Open | Org. | Public |
|----------------------|--------------------------|--------------------------|--|---|--------------------------|
| Participants | Participation structures | Number of participants | 2 / >4 human beings | | |
| | | Functions | All participants involved | Clear sub-groups | Alternate 1 main speaker |
| | Social dynamics | Hierarchy | ∅ | interviewer - interviewee / chairman - council member | host - guest(s) |
| | | (Conversational) History | ++ | ∅ / council history | ∅ |
| | | Common topics | ++ | interview agenda | / broadcast topic |
| Environment & Medium | | | | | |
| | Physical context | | maximal but not for annotators except radio, meant for external understanding | | |
| | Visibility | | maximal but some open conversations involve a task that requires focus, and the setting of the public TV dialogues may hinder the ability to look at some participants | | |
| | Audibility | | maximal but sometimes poor quality for external listeners (and thus transcribers) | | |
| (Verbal) Interaction | | | | | |
| | Co-construction | Coordination signals | maximal but more complex for public TV conversations | | |
| | | Turn-taking | face-to-face | often explicit (interviewer / chair or agenda) | often explicit (host) |
| | | Completion | minimal? but possible | | |
| | Time | Synchrony | fully synchronous exchange except possible delay in radio setting when exchanging with external participants (e.g. envoy) | | |

Table 7.2: Description of the three interaction degrees from our BNC selection

| Parameter | Sub- | Subsub- | Open | Org. | Public |
|------------------------|----------|-------------------|--|--|--|
| Function and Structure | | Planning | ∅ | early knowledge of topic / agenda | early knowledge of topic? |
| | | Ephemerality | clear influence on the topic sometimes | recorded by design / recording does not seem to affect the content | recorded by design |
| | Purpose | | mostly social, daily life | defined by the task | defined by the show |
| | Activity | Success condition | ? | agenda | interview |
| | | Vocabulary | X | quite specific | meant to be understood by general audience |
| | | Structure | no constraint | orderly | orderly |

Table 7.2: Description of the three interaction degrees from our BNC selection (continued)

7.2.3 Statistical description

Table 7.4 reports descriptive statistics for our selection of dialogues. Due to the way the open conversations were sampled, they are generally shorter than the other categories. They also exhibit shorter turns which can be explained by the very different nature of the interactions compared to the other categories in which the participants often contribute with very long explicative or argumentative turns.

All categories except the organised duologues contain a small proportion of turns whose speaker was labelled as unknown in the original transcriptions. During the experiments the annotators also reported that some dialogue turns were attributed to the wrong speaker. While it was clear in some cases, others were more ambiguous which could complicate the annotation. The linguistic content was sometimes unclear for the transcribers, which led to the presence of <unclear> tags in the transcriptions. These tags represent about 2% of all the tokens in our corpus but depending on their position they could hinder the annotation process more or less.

7.2.4 Challenges of naturalistic dialogues

While studying naturalistic data is the best way to produce models and tools usable in real-world situations, it also comes with a number of challenges. In this section we discuss the main issues we encountered when working with the BNC. We first discuss ethical considerations in Section 7.2.4.1 and then present the main challenges related to the nature of naturalistic dialogues (Section 7.2.4.2) and the way it was captured (Section 7.2.4.3).

| Group | Context | # Turns | # Tokens / turn | % Unk. spkr | # Gen. questions | % Specific questions |
|--------------|------------------------------------|---------|-----------------|-------------|------------------|----------------------|
| Open duo | Spontaneous conversations | 47±8 | 8±13 | 7±22 | 66±11 | 19±4 |
| Open multi | Spontaneous conversations | 58±7 | 7±7 | 4±9 | 82±11 | 14±5 |
| Org. duo | Interviews | 107±3 | 17±20 | 0±0 | 155±5 | 17±4 |
| Org. multi | Committee meetings | 97±12 | 18±22 | 25±21 | 141±18 | 19±5 |
| Public duo | ‘Ideas in action’ radio broadcasts | 81±7 | 32±27 | 3±11 | 116±10 | 24±4 |
| Public multi | Radio broadcasts | 106±18 | 23±30 | 6±7 | 154±27 | 18±7 |
| | Television debates / discussions | 105±22 | 14±16 | 6±3 | 152±33 | 21±4 |

Table 7.3: Description of the groups of dialogues in our corpus.

Table 7.4: Statistical description of the dialogues in our corpus per category. *The values are given in mean±std.*

7.2.4.1 Ethical considerations

Working with human produced data requires careful consideration of ethical issues, even more so when the data is collected in naturalistic settings. One of the questioning relates to the intended use of the corpus. While re-using existing corpora has the advantage of avoiding the need to collect new data, it also raises questions about whether the original consent given by participants covers the new use case. In our case, the BNC was collected in the 1990s and according to the Reference Guide, the original intended uses included academic linguistic research, artificial intelligence, and natural processing among others. Semantic and pragmatic information was explicitly mentioned as derivable from the corpus. Our use case is therefore compatible with the original consent.

Nevertheless, the participants were guaranteed ‘complete anonymity’ where ‘all references to full names and addresses have been removed from the corpus and the log’, which we found not to be the case in some of the dialogues.

Pseudonymisation. Personal data protection has become a major concern in recent years. While the BNC was collected before the implementation of the General Data Protection Regulation (GDPR) in 2018, data protection was already discussed in the UK in the 1990s (‘Data Protection – Volume 167: debated on Monday 19 February 1990’, n.d.). Yet, the claim of complete anonymity is clearly not accurate. Data protection law includes a distinction between the concepts of *anonymisation* and *pseudonymisation*. The former refers to the process of irreversibly removing personal identifiers from data, while the latter involves replacing personal identifiers with pseudonyms or codes, allowing for potential re-identification under certain circumstances. In the BNC, some dialogues contain references to specific individuals or locations that could potentially lead to re-identification.

(7.1) shows that the participants expected that their names would be removed from the corpus. Given the explicit mention to ‘Bet’ in the third turn, and the presence of a first name in the description of each file with demographically sampled dialogues, we can assume that it meant that their *family names* would not be mentioned.

(7.1) **PS04F₁** I don't know they said that no names are mentioned

PS04B₂ No

PS04E₃ She takes all them tapes away and Bet's name's not put on there.

Dialogue KBE – 009101 (Open multi) [BNC]

First names are not considered to be personal identifiers, so this does not represent a breach of privacy. However, some dialogues did contain the full names of participants though we only found such occurrences in the contextually sampled component of the corpus. This suggests that the requirements in terms of de-identification during the transcription were not uniform across the two components of the corpus, or even across the different dialogue types in the contextually sampled component. Some participants also expressed their uneagerness to disclose personal information as we can see in (7.2) where PS22H is worried to say where exactly they were born.

(7.2) **PS22G₁** <unclear> about when you worked at er <unclear>

PS22H₂ Yes <unclear> .

PS22H₃ Well where I was born in Needham Market

PS22G₄ Mm.

PS22H₅ need I say where because I, I can't <unclear> day.

PS22H₆ I don't

PS22G₇ Oh, I see.

PS22H₈ <unclear> the people to know

PS22G₉ You was born here then?

PS22H₁₀ I was.

PS22G₁₁ Ah, right well let's say you were born in the house you live in

PS22H₁₂ Well you could say

PS22G₁₃ without without actually saying where it was.

Dialogue HDJ (Org. duo) [BNC]

The recording of the conversation continued anyways and the interviewer found a way to avoid the disclosure of the participant's place of birth. However they later mention that they do not want people to know their age while the different information she provided in the interview would allow to infer it. This illustrates the difficulty of collecting naturalistic dialogues while respecting the privacy of the participants. In particular, (7.2) is a conversation about the life of the interviewee which makes it difficult to avoid the disclosure of personal information, and thus makes full anonymisation impossible (Amblard et al., 2014). Had all named entities been really replaced by pseudonyms or de-identification tokens in the entire corpus, the BNC would have fallen under the pseudonymisation category. However, the presence of real names and locations in some dialogues indicates that the corpus does not fully meet the criteria for either anonymisation nor pseudonymisation. Leveraging advances in NLP since the release of the BNC, a more thorough pseudonymisation process could probably be applied thanks to named entity recognition (NER) systems. This would allow for a more accurate replacement of personal identifiers in the dialogues, reducing the risk of re-identification while preserving the linguistic content of the corpus.

Offensive content. Another ethical consideration is the potential presence of offensive content in the dialogues. The dialogues under the open category are spontaneous conversations with no constraints on the topics discussed. As a result, they may contain offensive language or sensitive topics that could be distressing to some readers or annotators. While the BNC does not provide explicit warnings about

offensive content, it is important to acknowledge this possibility and take appropriate measures to ensure that annotators are aware of the potential for encountering such content. The annotators were informed of this possibility and encouraged to skip any dialogues that they found distressing or offensive. In our case, we encountered only one instance of potentially offensive content where two people exchange insults, seemingly in a joking manner but which could be perceived as offensive by some readers. (7.3) shows the very beginning of this dialogue where PS6NV explicitly states that they are about to use ‘dirty swear’. While we do not have access to the previous context, we can imagine that this ‘topic’ was introduced specifically because the conversation was being recorded. This of course does not change the fact that the content could be distressing to some readers, but it illustrates another difficulty of collecting naturalistic dialogues as the collection itself can influence the content of the conversations.

(7.3) **PS6NV₁** Now , for some fucking dirty swear!

PS6NV₂ Wo oh oh oh!

Continues with 39 turns of various swears, insults, slurs, etc.

Dialogue KSN – 135904 (Open duo) [BNC]

7.2.4.2 Collection setting

Collecting naturally occurring dialogues implies several challenges due to the collection format and setting. First, the presence of a recording device can influence the behaviour of participants, leading to changes in their speech patterns or the topics they choose to discuss. Second, capturing all the aspects of a conversation can be difficult, especially in multi-party face-to-face conversations where a lot of modalities are involved.

Meta-content. The informed consent process for the BNC collection varied depending on the type of interaction. The contextually sampled component was collected through a variety of methods, including direct recordings of meetings, where the participants were informed at the beginning of the meeting and asked for their consent. (7.4) shows the very beginning of such a council meeting where the chairperson explains the recording process and explicitly asks if there are any objections.

(7.4) **J9DPSUNK₁** Just a couple of things by way of introduction.

J9DPSUNK₂ Erm, the first point, relates to a a microphone here, and er and er tape recorder, and I think members of the sub-committee will have received a note from the Chief Executive Department.

J9DPSUNK₃ Longman’s are doing a study into the spoken language, and are looking for, I think, ninety million words that are in common usage.

J9DPSUNK₄ They came to the County Council and asked for our permission, to to record a number of meetings in order to get a variety of settings and words, different words they used.

J9DPSUNK₅ It’s it’s the same procedure that we accepted at the Finance Committee last Friday, and in mentioning it and hoping it will be acceptance that Longman’s can have the benefit of the Personnel’s sub-committee contribution for the next hour and a half or so.

J9DPSUNK₆ There’s, to be clear about it, that it’s completely confidential, that their looking only for particular words that are used in different parts of the country and in different locations.

J9DPSUNK₇ No problems, I presume <pause> Sound.

J9DPSUNK₈ Right.

Dialogue J9D (Org. multi) [BNC]

Moreover, from what we could see in our sub-selection of dialogues, a representative of the BNC team was sometimes present to answer potential questions. Regarding the demographically sampled component, active participants were recruited and asked to record all their conversations for a period ranging from two days to one week. They were told that whenever possible they should not inform their interlocutors of the recording until after the interaction had taken place. After the interaction, they would all be informed and asked for their consent to include the recording in the corpus. However in some cases the participants were clearly all aware of the recording and it could influence their behaviour. In (7.5) several people are watching a TV-show and one of them keeps making comments, where several other participants ask them to be quiet. At the end of this extract, after PS0EH more directly asks PS0EB to be quiet, PS0ED explains that PS0EB's rude behaviour is due to the recording as she wants to record real conversations rather than just the TV-show, which she confirms in the last turn. This illustrates how the presence of a recording device can influence the behaviour of participants and lead to changes in their speech patterns.

- (7.5) **PS0EB**₅₁ Why don't they show us all Britain?
PS0ED₅₂ Sshh <pause> that is Britain, it's just in the winter
PS0EH₅₃ Sshh
PS0EB₅₄ Oh no
PS0EH₅₅ Sshh
PS0EB₅₆ cor this is a bit ugh ah let's all clap for Neil, er, er what's he gonna do go off and have a boxing match?
PS0EH₅₇ Sshh
PS0ED₅₈ Sshh
PS0EH₅₉ why don't you shut up for a moment
PS0EB₆₀ Oh sorry
PS0ED₆₁ cos she's recording and she wants to hear somebody speaking on it <laugh>
PS0EB₆₂ just sounds a bit silly if you can only hear the telly <pause>

Dialogue KCE – 029701 (Open multi) [BNC]

Modalities. The fact that the participants of (7.5) are watching a TV-show becomes quite clear in this extract. However, nothing could indicate it at the beginning of the dialogue and without the visual information, it is even difficult to be sure whether the beginning of the transcription really corresponds to an exchange between the participants or if it is a transcription of the TV-show. This illustrates how working with only the auditive modality can be challenging for understanding the context of the conversation. (7.6) is a piece of conversation from the group *Public TV* which takes place in a TV show where a panel of 100 women are asked questions about a given topic and they can vote in a binary way before some of them explain their answer. Every time the host asks who would like to give their opinion, they seem to explicitly select one participant to take the floor, which is often not obvious in the transcription and we can assume that this selection is done through gestures and eye contact. We can see in turn 17 that the host asks who answered no to the question under discussion and they immediately continue with 'Would you like to say why?' before one person starts to answer. This seems to indicate that she understood that she was the one expected to provide an explanation, though we can only imagine this context and it is not impossible that she was the quickest to answer.

- (7.6) **FLDPS000**₈ Let's find out what one hundred Scottish women think of parents today.
FLDPS000₁₃ Let's start with a question, since you are all children of parents, do you think your parents did a good job?

- FLDPS000₁₄ Button one for yes, button two for no.
 FLDPS000₁₅ Did your parents do a good job?
 FLDPS000₁₆ And, in this particular hundred, eighty five say yes <pause> fifteen say no.
 FLDPS000₁₇ Who said no?
 FLDPS000₁₈ Would you like to say why?
 FLDPS000₁₉ Mhm.
 FLDPS001₂₀ Well I think my mother did a good job but my father was an alcoholic and that's, that made a difference, not just to him but <pause> to the entire family and I think probably to my own children.

Dialogue FLD (Public multi - TV) [BNC]

Similarly, while the transcribers relied on the audio recordings to produce the corpus, they were not asked to produce a *phonetic* transcription of the speech but an *orthographic* one. This means that as users of the corpus, we do not have access to the phonetic information or the prosody which can help disambiguate the meaning of some utterances. (7.7) shows a funny consequence of this limitation. The conversation, an interview for an oral history project, revolved around accents and how certain words were pronounced in different regions of the UK in the interviewee's youth. Without a phonetic transcription or access to the audio recording, discussing the differences between the words 'table' and 'table' becomes rather comical.

- (7.7) PS3F3₃₃ One of the common differences between East and West is the words like table and table in West we'd say table and East they'd say table.

Dialogue HV5 (Org. duo) [BNC]

These examples show that the manner and format data is collected in can create challenges for understanding the conversations and re-constructing the context in which they took place. On top of this aspect, the limitations of the transcription process itself can also hinder the understanding of the conversations.

7.2.4.3 Transcription limitations

Transcribing conversations from their recordings only, often without having been part of the interaction, is a challenging task for several reasons. While as participants in a conversation we can rely on the context and multimodal information to clarify words we may not have heard properly, or access basic information such as 'who said what', the transcribers can only rely on the recording. This means that they rely on its quality and the audibility of the speech and their own ability to distinguish participants – by voice and possibly world knowledge and inferred context – to produce the most accurate transcriptions. This can lead to errors or missing pieces in the transcriptions.

Unclear content. When the transcribers were not able to understand what was said, they used the <unclear> tag to indicate that the content was not clear. This can be due to several reasons such as poor audio quality, overlapping speech, or the use of unfamiliar words or accents. (7.8) is another extract from the same conversation as (7.7) where the interviewee repeats some expressions their grandmother used to say. Due to the accent and possibly the fact that these expressions are region specific, the transcribers were not able to understand what was said and used the <unclear> tag. This is particularly crucial as one of the first motivations for collecting the BNC was to study regional differences in English use. This is not as central for our own task that focuses more on semantic and pragmatics aspects but the presence of <unclear> tags was sometimes hindering for the annotators as they could not understand the content of some turns.

(7.8) **PS3F4**₆₀ I've got a granny who comes from the south isles and she says <unclear> and that's <unclear> .

PS3F4₆₁ If you say something and she's surprised she'll say, oh <unclear> .

Dialogue HV5 (Org. duo) [BNC]

(7.9) is an example where <unclear> tags were particularly problematic, all the more so because they were at the very beginning of the conversation. This extract is one of the open multilogues and the lack of context combined with utterances where most of the content is unclear, as in turn 4, makes it much more difficult to understand what is going on. This also created difficulties for the automatic question generation as it gave room to the model to hallucinate some content as we briefly discuss in Section 7.4.

(7.9) **PS0BA**₁ Have you been?

PS0BG₂ I got slightly lost.

PS0BG₃ Well I sort of went back down the <unclear> Road cos there was a traffic jam <pause> and er I went down <pause> oh sorry mate <pause> went down a road like which I thought was a good short cut, and they've got a lot of these roads and they've put like pavements across the end of them with bollards, and I went back and then there was a traffic jam and I got stuck because they were unloading a lorry <pause> so I've been er basically pissed about.<laugh><pause>

PS0BA₄ Yeah she a bit <unclear> now ain't she?

PS0BC₅ I don't think she has.

PS0BA₆ She has as far as pictures are concerned Rich .

PS0BF₇ She looks like er <unclear>

PS0BA₈ Have you got a number for her?

PS0BF₉ Yeah.

PS0BA₁₀ Different one?

PS0BF₁₁ Yeah.<unclear>

Dialogue KC6 – 055905 (Open multi) [BNC]

Speaker transcription. In addition to the orthographical transcriptions, the speakers' identifiers had to be selected for each utterance. This is challenging based on the recordings only and as a consequence, a small portion of turns in our selection of dialogues have been attributed to an 'unknown' speaker labelled as PSUNK. The exact proportion of such utterances is reported in Table 7.4 for each dialogue group. (7.10) is a piece of conversation where most of the turns are attributed to an unknown speaker which, together with the numerous <unclear> tags, makes it quite hard to understand where the conversation is going. The interaction becomes clearer the more turns we read but our task required to disclose the conversation gradually which means that we could not rely on additional context to improve our understanding.

(7.10) **JJAPSUNK**₁ <unclear> commitments in a recent task.

JJAPSUNK₂ <unclear> .

JJAPSUNK₃ Rights of way are an important part of our heritage <unclear> the road network should not be allowed to degenerate further.

JJAPSUNK₄ All <unclear> is a Labour activity <unclear> by all people <unclear> do things <unclear> cost, unless of course special clothing <unclear> and then <unclear> awfully difficult and making it <unclear> people's education and general health.

JJAPSUNK₅ We feel it's very important that our rights of way network should be maintained and improved where possible which is why we are proposing an issue <unclear> thank you.

PS44T₆ <unclear> I'd like <unclear> .

JJAPSUNK₇ Thank you Mr Chairman I would agree that there was two or three <unclear> on the question of <unclear> .

JJAPSUNK₈ We will <unclear> outside the erm <unclear> the <unclear> supporting it if it wasn't for the fact that we understand <unclear> that the officers would have a great deal of difficulty spending that money.

Dialogue JJA (Org. multi) [BNC]

We also noted some errors in the speaker attribution in some dialogues. In (7.11), PS5T8 is the interviewer and PS5TG is the interviewee who is a scientist who came to discuss how to teach science to children and students. However, turn 15, which is attributed to the interviewer, mentions the 'laboratory' of the speaker, which seems to indicate that the turn was actually produced by the interviewee.

(7.11) **PS5T8₁** Hello.

PS5T8₂ Science is thought of as a subject that is difficult both to teach and to learn.

PS5T8₃ The folklore in school terms is that you have to be a relatively ancient teenager to appreciate physics and chemistry and biology.

PS5T8₄ Is this true?

PS5T8₅ Today I have with me Dr <name>, who's made a study of science teaching in schools.

PS5T8₆ <name>, how early can science be taught to children?

PS5TG₁₂ For most young children in that age group, the world's a magic place and we traditionally like to teach them nature study and flowers and cuddly hamsters and rabbits in school, and that's the nature table syndrome, and that's great and I'm not knocking that .

PS5TG₁₃ There is so much opportunity for children to look at the nature of the physical world around them which isn't taken advantage of, and which could be, and I think that may have something to do with the attitude of teachers as much as the attitude of children.

PS5TG₁₄ But erm they're tremendously curious about the nature of the world around them and they're certainly capable of, if not understanding why, exploring what.

PS5T8₁₅ I took my godson, <name>, round my laboratory the other day, which is a physics laboratory, and he enjoyed it immensely and asked a number of questions, and was absolutely intrigued and fascinated by the various bits of wires and plugs like that, and he asked me the sort of questions that I don't think I would expect sometimes my <laugh>undergraduates to ask.

Dialogue KRH – 27 (Public duo) [BNC]

Such errors in the speaker attribution can be problematic for our task as they can lead to confusion about the roles of the participants in the conversation. In this case, it could lead to a misunderstanding of the context of the conversation and the content of the turns. In particular, (7.11) led to debate between the annotators after the task as they had different interpretations of the conversation, where one of them thought that the turns were well attributed and the other did not. These mistakes and misinterpretations are characteristic of the challenges of working with naturalistic dialogues and in particular from an outside perspective, as we would not have had this issue if we had been part of the conversation.

7.2.4.4 Studying conversations from an outside perspective

Our study requires us to imagine ourselves as participants in the conversations we are annotating. As we have seen in the previous sections, this position is challenging because it forces us to imagine a suitable context for based on the transcriptions alone, which are not always complete or accurate, and are a degraded representation of the original conversations by nature. In some cases, building the context relies on small lexical cues, where we relate a word to a topic or a situation, but it does not always resist when confronted with the subsequent turns. (7.12) is an example of multiparty open conversation, which is arguably the most complex type of dialogue when it comes to coming up with a coherent context, where several lexical fields are mentioned. In the second turn, ‘persil’ is mentioned, followed by the mention of a ‘brush’. With this context, we could imagine that the participants are talking about cooking and that they are preparing a meal, where the brush can be used to apply oil or butter to a dish. However, the subsequent turns mention ‘fil[ing] them down’ before discussing colour and painting, which could relate to nail trimming and nail polish painting. The mention of a ‘sponge’ and ‘dabbl[ing]’ technique can still fit in this context but the mention of ‘spikes’ and a ‘hedgehog’ question this theory. Then in turn 37, the participants talk about ‘paint[ing] the icing’, which could be related to cake decoration. This extract illustrates how each turn can bring new information that can change the context of the conversation and how it is interpreted. The <unclear> part further complicate the interpretation as they prevent us from fully understanding the content of the turns. For instance, turn 36 could mention a ‘hedgehog’ as an element of comparison to describe something else but without the entire content of the turn, there is no way to know for sure whether the participants are talking about an actual hedgehog, a decoration that is supposed to look like one, or no hedgehog at all. The multitude of participants amplify the chaotic energy of this exchange as everyone can contribute with their own needs, intentions, and perceptions. My personal guess at the end of this conversation was that the conversation took place during some kind of sculpting activity and that the mention to ‘persil’ in the second turn was an error of transcription. However, after reading it several times, I could also find ways to explain why this conversation would fit well in a cake decoration lesson or find other explainable contexts. While it is a single example, it illustrates the feeling raised by many other dialogues from the corpus, and in general when studying less organised exchanges one did not take part in.

- (7.12) **PS0KY**₁ what I’m gonna do now.
PS0KY₂ Erm <name₁>, can you pass me the persil please?
PS0L1₃ Do you want this brush?
PS0L1₄ <pause> <name₂>?
KDBPSUNK₅ Yeah that’s fine.
PS0KY₈ I’ll just file them down a bit.<pause>
KDBPSUNK₁₀ Yeah it’s quite runny actually.
KDBPSUNK₁₁ But <unclear>
KDBPSUNK₁₂ What colour you doing it then?
PS0L2₁₃ Purple.
PS0L2₁₄ <pause> But I’m gonna do it over this so it’ll sort of go funny
PS0L6₁₉ Why don’t you kind of dabble it on, <name₂>?
PS0KY₂₀ Yeah.
KDBPSUNK₂₁ <unclear> sponge.
PS0KY₂₂ You mean like that <pause> dabble it on?
KDBPSUNK₂₃ Yeah.

PS0KY₂₃ Or a bit less?
PS0L6₂₄ Yeah.
PS0KY₂₅ Yeah.
PS0L6₃₀ Let me just er
PS0KY₃₁ Yeah.
PS0L6₃₂ Wipe that down there.
KDBPSUNK₃₃ I don't think these spikes belong on there at all.
KDBPSUNK₃₄ Mm?
KDBPSUNK₃₅ He's got a wobbly bottom.
KDBPSUNK₃₆ <unclear> hedgehog.
PS0L6₃₇ You gonna actually paint the icing white?
PS0L6₃₈ Or is it <unclear>
PS0L6₃₉ Oh right.
PS0KY₄₃ That white one there everybody.
PS0KY₄₄ Too big.
KDBPSUNK₄₅ Stick it further down.
PS0KY₄₆ It can be the centrepiece.
KDBPSUNK₄₇ Stick it further down so <unclear>
PS0KY₄₈ Well a lot of the spikes are <unclear>
PS0L6₄₉ Right.
PS0L6₅₀ If you've finished <pause> or if you've finished for today.
PS0L6₅₁ This <pause> paint I want it really really really well washed out of these brushes.
PS0L6₅₂ It's erm very good covering powder, it actually stays in the brushes.
PS0L6₅₆ When you're ready.
PS0KY₅₇ Oh it's hotting up.

Dialogue KDB – 014817 (Open multi) [BNC]

7.3 Method: Automatic question generation and human annotation

We adapt the framework of Y. Wu et al. (2024) to the study of dialogue, as described in Section 7.1.2. Rather than analysing documents composed of sentences, we consider dialogues consisting of utterances $U_1, U_2, \dots, U_n$. Given an anchor utterance U_k and its preceding context $U_1, \dots, U_{k-1}$, we generate potential questions evoked by the dialogue at that point and investigate whether their salience is correlated with the extent to which they are answered in the subsequent interaction $U_{k+1}, \dots, U_n$.

Our methodology, represented in Fig. 7.2, comprises three main steps. First, we generate questions anchored in dialogue excerpts using an LLM. Second, we annotate these questions for salience and answerability using guidelines adapted from Y. Wu et al. (2024) to account for the specific properties of dialogue. Finally, we compute the correlation between the two scores to assess whether more salient questions are more likely to be addressed later in the interaction.

In this section we describe the question generation procedure (Section 7.3.1), the adaptations made to the annotation guidelines (Section 7.3.2), and the annotation process itself (Section 7.3.3).

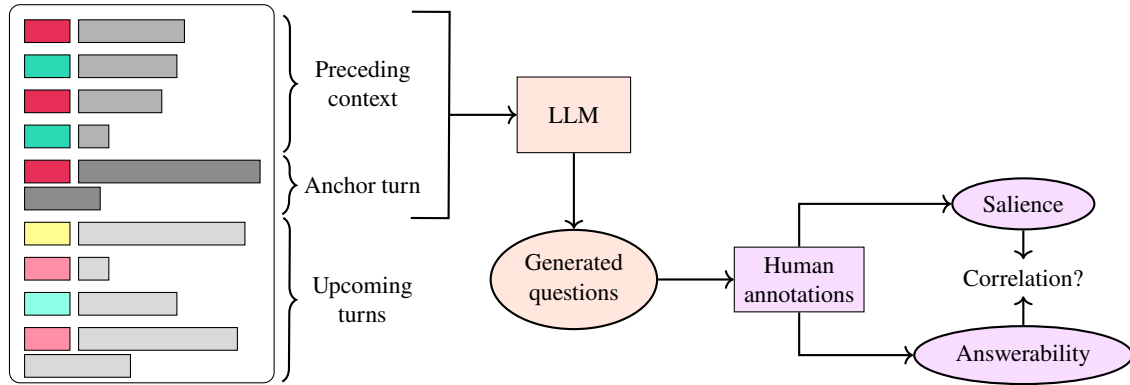


Figure 7.2: Framework for correlating question salience and answerability, adapted from Y. Wu et al. (2024).

7.3.1 Automatic question generation

Following Y. Wu et al. (2024), we generate multiple questions for each anchor position in the dialogue. Starting from the fourth utterance, we select every second utterance as an anchor and generate three questions per anchor using the preceding context and the anchor utterance itself.

To identify a suitable model for question generation, we conducted preliminary experiments with three open-source LLMs: Gemma 3-4B (Team et al., 2024), Llama3-ChatQA-8B (Z. Liu et al., 2024), and Mistral-7B (Jiang et al., 2023). The prompt was adapted from Y. Wu et al. (2024) to account for the dialogic nature of the input. As shown in Fig. 7.3, we explained the format of the utterances and indicated that special tokens should be interpreted as extra information that can help understand the dialogue better, but are not part of the utterance itself. We also instructed the model to generate questions as if it ‘were a participant of the dialogue’. We made this choice to reduce the number of questions the participants clearly already know the answer to. We finished the prompt by specifying the format of the output.

The following is a dialogue where each line is in the format `<speaker>: <utterance>`. All the `<special tokens>` indicate extra linguistic information, such as pauses, vocalizations, etc. You should interpret them as extra information that can help you understand the dialogue better, but they are not part of the utterance itself.

`<conversation up to the anchor turn>`

After reading the previous dialogue, ask 3 questions about a part of the last utterance "`<anchor utterance>`" that you would be curious about if you were a participant of the dialogue and which you don't have an answer for. Generate only the questions, one per line, without any additional text or explanation.

Figure 7.3: Prompt used for question generation.

A pilot evaluation was performed on questions generated by the candidate models. Based on the quality and relevance of the generated questions, we selected Mistral-7B for all subsequent experiments. The final corpus was therefore generated exclusively using this model.

In particular, we dismissed Llama3-ChatQA-8B because of the blatantly sexist questions generated by it as can be seen in (7.13). The conversation is the same as in the previous example (7.9),

where a man mentions getting stuck in a traffic jam before the conversation moves to the topic of a woman in the fourth turn. Due to the numerous unclear parts in the transcription, it is quite difficult to infer the context of this dialogue and the identity of this woman we have no information on. Yet two out of three of the generated questions inquire about the pregnancy of the woman. LLMs are known to be biased, some more than others, and this example illustrates how these biases can manifest in the generated questions. While the model we have chosen in the end, Mistral-7B, is also known for exhibiting biases, we did not observe any obvious similar issues for our task.

(7.13) **PS0BA₁** Have you been?

PS0BG₂ I got slightly lost.

PS0BG₃ Well I sort of went back down the <unclear> Road cos there was a traffic jam <pause> and er I went down <pause> oh sorry mate <pause> went down a road like which I thought was a good short cut, and they've got a lot of these roads and they've put like pavements across the end of them with bollards, and I went back and then there was a traffic jam and I got stuck because they were unloading a lorry <pause> so I've been er basically pissed about.<laugh><pause>

PS0BA₄ Yeah she a bit <unclear> now ain't she?

→ **Generated question:** Why is there uncertainty whether the female in question is pregnant?

→ **Generated question:** How does the speaker know that the female in question isn't pregnant?

Dialogue KC6 – 055905 (Open multi) [BNC]

After a first small batch of annotations (about 150 questions) carried out by one person, we selected Mistral-7B rather than Gemma 3-4B because the generated questions were more salient in average and more often valid. We acknowledge that this sample is too small to draw any strong conclusions but it seemed to be a reasonable choice given the time constraints of the project. We chose to dedicate more time to the final stage of the annotation process in order to compare different types of dialogues on a large enough sample to draw meaningful conclusions. Investigating the impact of the choice of LLM on the quality of the generated questions is an interesting avenue for future work.

7.3.2 Guidelines adaptations for dialogue

The annotation guidelines proposed by Y. Wu et al. (2024) were designed for monologic discourse and required several modifications to accommodate the specific characteristics of dialogue.

Salience Salience measures the extent to which answering a question would contribute to the participants' understanding of the dialogue at a given point. While Y. Wu et al. (2024) employ a five-point scale for valid questions, we found that several dialogue-specific situations were not adequately captured by the original scheme. We therefore introduced three additional categories, shown in italics below. The complete annotation guidelines are provided in the appendix (missing reference: To be added in final version).

- Score = 0: Invalid question, i.e. it contains a grammatical error, contradicts the preceding context or is not anchored in U_k ;
- Score = 1: The question is not related to the preceding context;
- *Score = 2: The question is related to issues in the transcription of the anchor turn;*

- Score = 3: The question is related to the preceding context but answering it is not useful, it may be fully answered by C_p already;
- Score = 4: The question is relevant to the dialogue but the participants clearly already know the answer based on their common ground or shared context;
- Score = 5: The question is related to the preceding context but answering it is not essential;
- Score = 6: The question is related to the preceding context and answering it is somewhat useful;
- Score = 7: The question is a paraphrase of the anchor turn;
- Score = 8: The question is relevant to the preceding context and it should definitely be answered.

The final scheme emerged through several pilot annotation rounds in which alternative dialogue-specific categories were tested and refined.

Answerability Answerability measures the extent to which a question is addressed in the dialogue following the anchor utterance. Unlike Y. Wu et al. (2024), we do not treat questions that are already answered by the preceding context as a separate category. In dialogue, information may be repeated, clarified, or reformulated by participants, meaning that a question can still be answered in subsequent turns even if an answer has previously been mentioned. We therefore employ a three-level scale:

- Score = 0: The question is not answered after the anchor turn;
- Score = 1: The question is partially answered or the participants explicitly state that they do not know the answer after the anchor turn;
- Score = 2: The question is fully answered after the anchor turn.

7.3.3 Annotations

Each question is annotated independently by two annotators for both salience and answerability. In the remainder of the paper, we refer to these as annotator 1 and annotator 2. Annotator 1 corresponds to a single expert annotator, while annotator 2 aggregates annotations from several annotators who worked across different subsets of the data. Although each file was annotated in full by a single individual, different files within annotator 2 were annotated by different people, and in some cases salience and answerability for the same file were produced by different annotators. This design was adopted to make the annotation process more efficient while maintaining two independent annotations per question. The annotators were either core members of the project or researchers from the team who volunteered to annotate some files.

Warning – Ongoing work

Include details on the number of annotators for annotator 2 (8) and the number of files annotated by each of them. Note that one of the annotators covered a large number of files, while the others annotated only a few.

Annotator 1 has formal training in linguistics and experience in dialogue analysis. The annotators contributing to annotator 2 have heterogeneous backgrounds, with varying levels of expertise in linguistics and conversational analysis; the majority of annotations within this group were produced by annotators with limited prior experience in these topics. All annotators were non-native English speakers, which may have introduced additional difficulty in interpreting conversational nuances, particularly in cases involving transcription artefacts or domain-specific vocabulary.

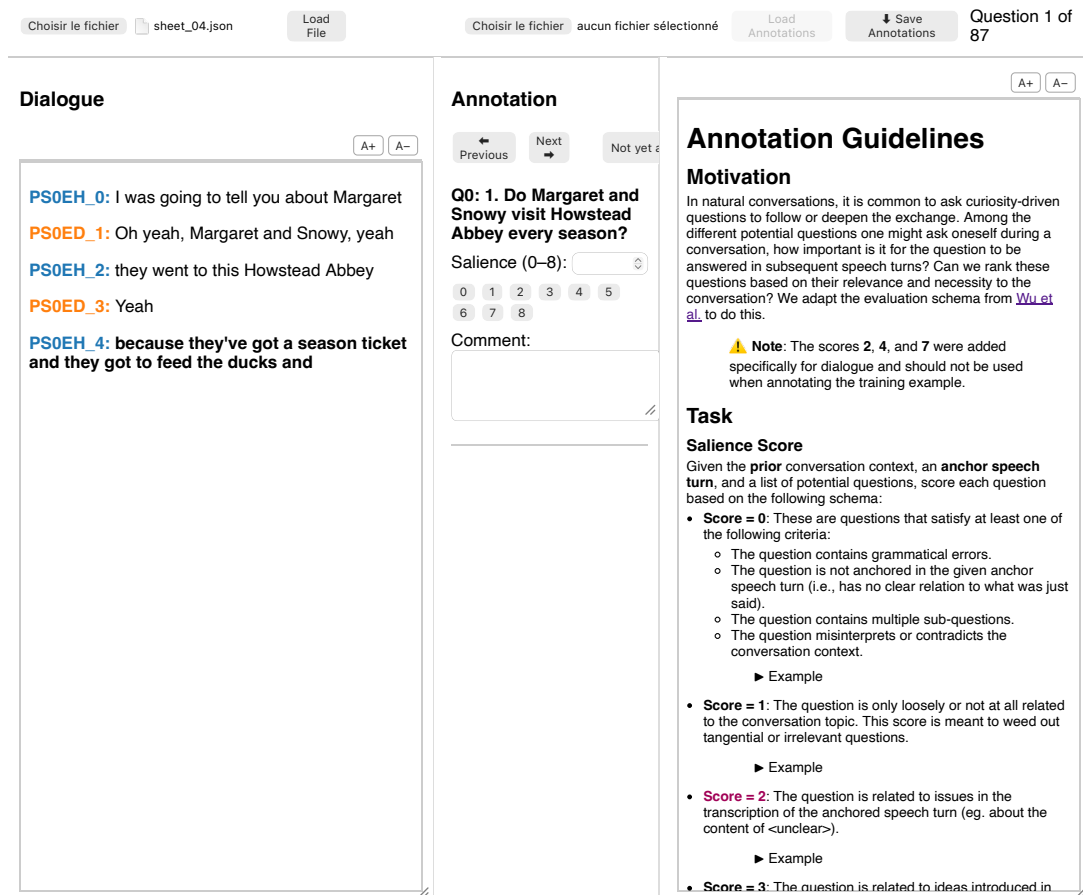


Figure 7.4: Screenshot of the annotation interface for the salience task. The dialogue display is shown on the left, the annotation panel in the centre, and the guidelines panel on the right.

Annotations were collected using a custom-built web-based interface developed for this project²¹ showed in Fig. 7.4. The interface allowed loading and exporting annotations in JSON and CSV formats and supported resuming partially completed annotation sessions. It consisted of three panels: (i) a dialogue display, (ii) a central annotation panel showing the current question, scoring interface, and optional comment field, and (iii) a panel containing the annotation guidelines and examples.

The display of the dialogue differed between the two annotation tasks. For salience annotation, only the context up to the anchor utterance was shown, and subsequent utterances were hidden to avoid contamination from future discourse. For answerability annotation, the full dialogue was displayed. In both tasks, the anchor utterance was visually highlighted to support localisation.

Annotator 1 completed both annotation tasks sequentially, with a one-week interval between them. In contrast, annotators contributing to annotator 2 followed heterogeneous protocols, with variation in task order and temporal spacing between salience and answerability annotation. Annotation time varied substantially across dialogues, ranging from approximately 30 minutes for shorter and more structured interactions to over 90 minutes for longer or less structured ones. Overall annotation effort is estimated at approximately 70 hours for annotator 1 and 100 hours for annotator 2, although these figures are approximate as detailed timing logs were not systematically recorded.

²¹ Salience and Answerability interfaces.

| | $\alpha_{QSalience}$ | α | κ | $w\kappa$ | $w\kappa^2$ |
|-------|----------------------|----------|----------|-----------|-------------|
| Sal. | .63 - .75 | .55 | .36 | .47 | .53 |
| Answ. | .80 | .66 | .55 | .63 | .69 |

Table 7.5: Inter-annotator agreement for salience and answerability annotations. $\alpha_{QSalience}$ refers to the Krippendorff’s alpha reported in Y. Wu et al. (2024) for comparison. κ refers to Cohen’s Kappa, $w\kappa$ to linear weighted Cohen’s Kappa and $w\kappa^2$ to quadratic weighted Cohen’s Kappa.

7.3.3.1 Agreement: a complex task

Given the inherently subjective nature of both salience and answerability annotation, we evaluate inter-annotator agreement using multiple metrics. We report Krippendorff’s Alpha (Krippendorff, 2011)²² for comparability with Y. Wu et al. (2024), as well as Cohen’s Kappa and both linear and quadratic weighted Kappa (J. Cohen, 1968) to account for ordinal structure in the annotation scales. Indeed, given the gradation of the scores, we consider that a disagreement between two adjacent scores is less severe than a disagreement between two scores that are far apart. Weighted κ allows us to take this into account in our agreement analysis, where instead of penalising all disagreements with a weight of 1 as in the case of Cohen’s Kappa, we assign a weight to each disagreement based on the distance between the two scores. Formally, the weight when comparing two scores i and j is defined as $w_{i,j} = |j - i|$ for linear weighted Kappa and $w_{i,j} = (j - i)^2$ for quadratic weighted Kappa.

As shown in Table 7.5, agreement for salience ranges from fair to moderate, while agreement for answerability is moderate to substantial, depending on the metric considered (Artstein & Poesio, 2008). Agreement levels are generally lower for salience than for answerability, consistent with the greater interpretative flexibility of salience judgements. They are also lower than those reported in Y. Wu et al. (2024), which is expected given both the increased complexity of dialogue data and the larger number of annotated questions in the present study. We further compute agreement separately across dialogue categories (Table 7.6) to assess whether conversational structure affects annotation consistency.

Warning – Ongoing work

Include raw percentage agreement.

Include proper statistical comparison for the agreement between the groups. Check this paper?

We also looked into the places of disagreement between the two annotators. Figures 7.5a and 7.5b show the comparison matrices for salience and answerability respectively. The colours of the cells correspond to the portion of questions that received the corresponding scores from the two annotators, while the numbers indicate the raw amount of annotations for each score.

Salience. The salience annotation relied on a large score range. Disagreements are partly attributable to transcription artefacts and the absence of non-linguistic contextual information, both of which are particularly relevant in spontaneous dialogue.

Figure 7.5a shows that agreement is highest for scores 2 and 7 that correspond respectively to questions targeting transcription issues, and cases where the generated question closely paraphrases the anchor utterance. Both categories are relatively constrained and therefore more consistently identifiable across annotators. In particular, score 2 can often be detected using simple heuristics (e.g. the presence of <unclear> markers in the question), which likely contributes to higher consistency. Score 7 is slightly

²²Computed with Castro’s (2017) Python implementation.

| Category | α | κ | $w\kappa$ | $w\kappa^2$ | α | κ | $w\kappa$ | $w\kappa^2$ |
|----------------------|----------|----------|-----------|-------------|----------|----------|-----------|-------------|
| Open duo | .57 | .40 | .50 | .55 | .68 | .56 | .64 | .69 |
| Open multi | .51 | .38 | .45 | .48 | .58 | .47 | .55 | .61 |
| Org. duo | .58 | .38 | .50 | .56 | .69 | .56 | .65 | .72 |
| Org. multi | .50 | .33 | .42 | .48 | .62 | .54 | .61 | .65 |
| Public duo | .53 | .32 | .43 | .50 | .65 | .50 | .59 | .65 |
| Public multi - radio | .55 | .42 | .49 | .52 | .61 | .50 | .58 | .64 |
| Public multi - TV | .44 | .25 | .35 | .40 | .65 | .53 | .61 | .67 |
| ALL | .55 | .36 | .47 | .53 | .65 | .53 | .62 | .68 |

(a) Salience. (b) Answerability.

Table 7.6: Inter-annotator agreement per dialogue group.

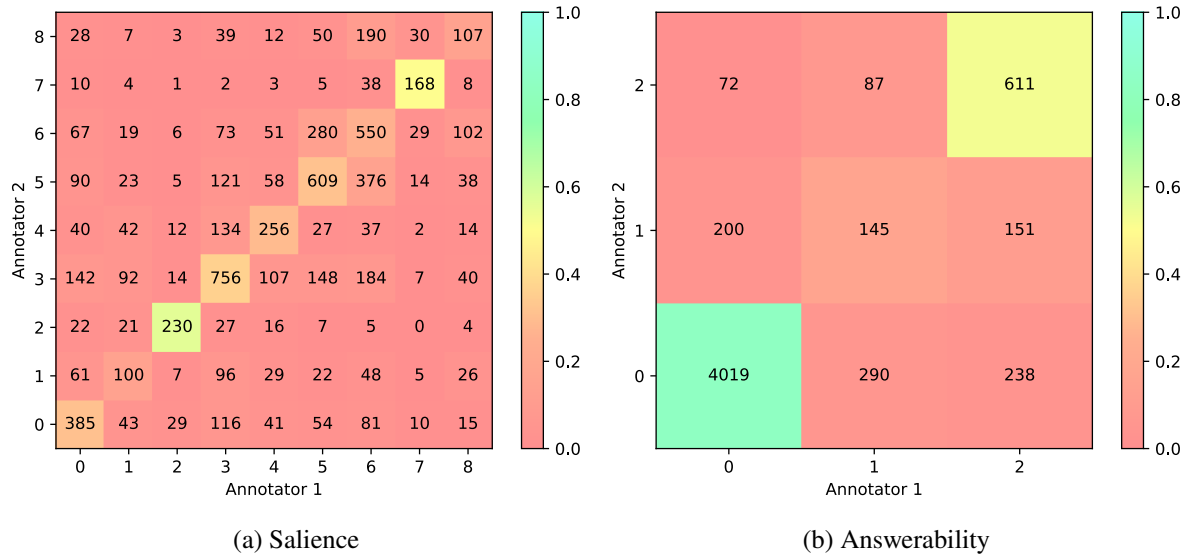


Figure 7.5: Comparison of the annotations for salience and answerability. The colour of the cells corresponds to the portion of questions that received the corresponding scores from the two annotators, while the numbers indicate the actual amount of annotations for each score.

less consensual which can be explained by the fact that the boundary between two sentences being paraphrases of each other, and being very related, is hard to define (Polguère, 1990).

In contrast, lower agreement is observed for scores 1 and 8. Score 1 involves judgements about whether a question is unrelated to the preceding context, a notion that is inherently sensitive to assumptions of discourse coherence. Conversely, score 8 requires assessing whether a question ‘must be answered’, which depends on pragmatic interpretation and access to shared context and participant intentions, both of which are not fully observable unobservable in transcript-based analysis.

Overall, the distribution shows that annotators frequently agree that questions are at least loosely connected to the preceding context, as indicated by the faint orange square spanning scores 3 to 8 with the exception of 7, while differing primarily in the degree of necessity or relevance assigned within that range. This supports the idea that in conversation we tend to assume coherence and relatedness by default, which is a reasonable strategy for joint cooperation (Grice, 1975). However, the extent to which a question is deemed essential or peripheral to the dialogue is more subjective and variable across annotators. In particular it can require a more fine-grained understanding of the dialogue, which can be difficult to achieve without access to the non-linguistic context and to the knowledge of the participants. This source of variability is reflected in the faint orange square spanning scores 3 to 8 with the exception of 7 which shows that the annotators often agree on the question being somewhat related to the preceding context but they can disagree on *how much* it should be answered.

Answerability. Answerability was graded on a small score range. Figure 7.5b shows that agreement is highest for score 0, which corresponds to cases where the question is not answered in the subsequent dialogue. This suggests that identifying absence of resolution is more robust than distinguishing degrees of partial resolution. Disagreements between scores 1 and 2 can be explained by the fact that the boundary between partial and full resolution is ambiguous, particularly for complex or implicit questions. Requesting a confidence score for the answerability annotation could have been useful to capture such cases, but it would also make the annotation process more complex and time-consuming.

7.3.3.2 Aggregation: a perspectivist approach

Given the observed variability in salience annotations, we adopt a *perspectivist* strategy in our analysis (Frenda et al., 2025). Rather than collapsing annotations into a single gold label via averaging, we retain both annotator perspectives. In addition, we construct a restricted ‘gold’ subset consisting only of questions for which both annotators assigned identical salience scores, which corresponds to about 46% of the questions. The repartition between most of the groups is rather balanced, except for the public multilogues where the radio broadcasts provide the highest agreement around 52% and the TV shows the lowest one around 38% (see Table 7.7 for more details). This gold subset allows us to study the general annotation patterns but also to investigate the nature of the disagreements which we consider to be a natural reflection of the complexity of the task and of the nature of dialogue itself, which is perceived differently by its different participants. In such a case, averaging ordinal labels may obscure meaningful differences in interpretation rather than resolving them.

Regarding answerability, we instead compute the mean of the two annotators’ scores to obtain a single continuous estimate. This is justified by the more compositional and less categorically heterogeneous nature of the answerability scale. The resulting aggregation effectively yields a finer-grained scale while preserving ordinal consistency.

7.4 Released dataset: questions and annotations

We release the 7,124 questions generated by the LLM Mistral-7B, together with their salience and answerability annotations, as well as some natural language explanations for some of the questions. Due to the licence of the BNC we cannot include the dialogues themselves but we provide the metadata for each question, which allows to reconstruct the dialogues from the BNC. We also include the code to do so on our GitLab repository.

We release our corpus as a single file where each row corresponds to a question and includes the following information: the dialogue it is anchored in, the anchor turn, the question, the two salience scores, the gold score when the two annotators agreed, comments from the annotators for the salience, the two answerability scores, the gold score, comments from the annotators for the answerability. Figure 7.6 provides an example of a row in our corpus together with an extract of the dialogue it is based on.

PS0EH₀: I was going to tell you about Margaret
PS0ED₁: Oh yeah, Margaret and Snowy, yeah
PS0EH₂: they went to this Howstead Abbey
PS0ED₃: Yeah
PS0EH₄: **because they’ve got a season ticket and they got to feed the ducks and**

| ID _{sheet} | | Type | ID _Q | Cutoff | ID _{ann2} | Question | | | | | |
|---------------------|----------------|----------------|--------------------|--|--------------------|--|----------------|----------------|----------------|--------------------|--------------------|
| 4 | | open multi | 0 | 4 | E | 1. Do Margaret and Snowy visit Howstead Abbey every season? | | | | | |
| S ₁ | S ₂ | S _G | Comm _{S1} | Comm _{S2} | | | A ₁ | A ₂ | A _G | Comm _{A1} | Comm _{A2} |
| 5 | 5 | 5 | / | relevant but not essential to the conversation | | | 0 | 0 | 0 | / | / |

Figure 7.6: Example of a row in our corpus together with an extract of the dialogue it is based on with the anchor turn and the associated question in bold. S_i and A_i indicate the salience and answerability scores for annotator i with G for the gold scores. *Comm* stand for the associated comments. *Beginning of the dialogue KCE – 029701*.

In this section we discuss several limitations of the question generation approach, which are important to consider when interpreting the results of our analysis. We begin by discussing examples that show the lack of world knowledge of the model (Section 7.4.1), then we go through more technical limitations (Section 7.4.2), and finish with some issues most likely created by the prompt (Section 7.4.3).

7.4.1 World knowledge

As discussed in Chapter 3, LLMs are trained on large amounts of text data but do not learn by interacting with the world. This is reflected in several phenomena we observed in the generated questions, which show a lack of world knowledge and common sense reasoning.

Meaning of the words. Some generated questions asked about the meaning of words or phrases in context. While this can be a meaningful question in some cases, as in ‘Does “Mm” imply agreement with the previous statement or a neutral response?’ which indicates a questioning on the intentions of the

speaker, in other cases the meaning of the words is very clear and the question is therefore not relevant. In (7.14), the question concerns the meaning of the word ‘Yeah’, i.e. the only word in the anchor turn, which is very clear in context.

(7.14) **J8DPS003**₆₇ What are those Marian, you don’t get one of those last time?

J8DPS000₆₈ I got <pause> one of these.

J8DPS002₆₉ Yeah.

→ **Generated question:** What does “Yeah” refer to in this context?

Dialogue J8D (Org. multi) [BNC]

This phenomenon also happens with phrases less ambiguous than acknowledgment tokens like ‘Yeah’. In (7.15), the question concerns the meaning of the phrase ‘trim them around the sides’, which requires little explanation in the context of cutting trees.

(7.15) **PS1GE**₂₂ They’re not, they’re not big trees or nothing so I mean there was, it was only a, like one of these small <unclear> power chain saws,<unclear> one hand sort of thing, grab hold of it like that, there was nothing, no really big branches, were only cutting the tops off.

PS1GE₂₃ We, we did trim them around the sides, they were quite big, you know,<unclear> .

→ **Generated question:** What does “trim them around the sides” refer to in this context?

Dialogue KDA – 011605 (Open duo) [BNC]

Epoch anchoring. The collection era of the BNC is also a challenge when generating the questions because pragmatic reasoning and world knowledge are necessary to ask relevant questions. For instance, based on a conversation about the impact of technological progress on scientific publishing, the question ‘What are potential challenges or limitations in using floppy disks for large-scale book publishing due to their capacity limits or outdated technology?’ was generated. This question would clearly be irrelevant at the time of the conversation when floppy disks were no outdated technology and where their capacity limits were not an issue. Similarly, in a conversation where a participant recommends a book, the generated question inquires about the availability of the book in digital format, which was not a relevant question at the time of the conversation: ‘Is “The Honorary Consul” available in digital format as well, or is it only available in paperback?’.

Pragmatic and social reasoning. Conversation is rooted in social and cultural norms, and understanding the pragmatic implications of dialogue often requires knowledge of these norms. In some cases, the generated questions show a lack of such reasoning.

In a council meeting where one point of the agenda was to vote an advance on salary for the council members, the question ‘Why does the speaker hope that the public won’t believe something, as stated in “but certainly I’m not prepared to allow them to think it of me”?’ was generated. As indicated by the context in (7.16) The speaker was worried the public would believe that the council members were voting themselves a raise of salary. The generated question would thus mean ‘Why does the speaker hope that the public won’t believe that the council members are voting themselves a raise of salary?’, which requires little political sense to answer. The question is therefore not relevant in this context and shows the lack of pragmatic reasoning of the model.

(7.16) **PS45W**₈₆ We are elected by the public to do a job and we shouldn’t be ashamed of the amount of money that we’re paid for it, and what I would say is, that no manifesto in May’s election

said we were going to come here and vote for more money for members, nobody put on their leaflets, vote for me and I will raise members allowances by five percent in the coming year, I didn't and I won't support that, and you will argue, perhaps some of you that that's not really what we're doing, but it look's like it, doesn't it?

PS45W₈₇ And I think it's close enough to looking like it that the public may believe it, and I hope the public won't believe it, but certainly I'm not prepared to allow them to think it of me, and therefore I will vote against the motion.

→ **Generated question:** Why does the speaker hope that the public won't believe something, as stated in "but certainly I'm not prepared to allow them to think it of me"?

Dialogue JJG (Org. multi) [BNC]

More socio-cultural aspects were also clearly missing in the generated questions. In the context of a radio broadcast about the Gulf War where several Kuwaiti guests were interviewed, the question 'Does Jossom have any concern about potential civilian casualties on the Iraqi side, or is he only concerned about Kuwaitis?' was generated. The question shows a clear lack of humanity which is inappropriate, all the more so in a conversation about war from which participants and listeners may be direct victims of.

Lastly, some generated questions highlight the importance of conversational structure for assessing the salience of a question. While a question could have been very salient at one point of the exchange, it can become much less relevant later in the conversation due to a change of focus or because the conversation is coming to an end. (7.17) illustrates the latter case as the host asked their guest for a last brief comment while the generated questions inquire about more examples and details. A last 'very brief' comment that would answer one of the generated question at that point of the exchange would most likely be considered a breach of conversational cooperation.

(7.17) **PS5SC₆₂** Ian, I wish we could go on talking endlessly about this because I find the subject quite fascinating and we haven't really talked about the details of your research.

PS5SC₆₃ I wonder if we could just finish with a very brief comment from you in a positive sense as to what would be your advice to somebody that's unemployed?

(5 utterance)

PS5SE₆₉ We found all sorts of different ways that different people were achieving these experiences.

→ **Generated question:** Can you provide examples of the various ways different people are achieving self-esteem during unemployment as per your research?

→ **Generated question:** Were there any common activities among these different methods that showed significant impact on reducing depression levels?

Dialogue KRG – 13 (Public duo) [BNC]

7.4.2 Technical limitations

Leveraging an LLM to automatically generate our questions implies to accept some technical limitations of the model.

Referring to words that are not here. We observed a certain form of 'hallucination' is some generated questions that would refer to entities that were not present in the anchor turn or in the context. In (7.18), the question refers to 'they' and the question explicitly states the scope of the reference which is turn 37. Yet, there is no mention to a plural entity in that turn. The closest valid referent would be in turn 34 but the speaker does not correspond with the one mentioned in the question.

- (7.18) **PS0RE**₃₃ He can't be full time otherwise he wouldn't have enough to do would he?
PS0PN₃₄ I don't know perhaps he does, I don't actually know Ken, I thought the way he was talking though that this chap
PS0RE₃₅ No, it's, it's so many days that they work innit?
PS0PN₃₆ had actually taken him on that he used to know years ago and had finished the
PS0RC₃₇ Oh he's finished the
PS0PN₃₈ I know it's a couple of years ago before he came here
PS0RC₃₉ Oh
→ **Generated question:** What did PS0PN refer to as "they" that PS0RC reacted to with "Oh"?

Dialogue KDM – 030302 (Open multi) [BNC]

A general remark is that the more precise the scope of the question, the more chances there are that it was not logically relevant. In (7.19), the question asks about the content of the '<unclear>' tag, which exists in the anchor turn, but not in the specific phrase quoted in the question.

- (7.19) **PS1GE**₄₉ I don't know whether it's true or not I don't know whether they're fucking <unclear> trouble is it sounds like the sort of thing I do, is we parked in the multi storey car park in Guildon.
→ **Generated question:** What does "<unclear>" refer to in "we parked in the multi storey car park in Guildon"?

Dialogue KDA – 011605 (Open duo) [BNC]

Confusion of substantives and wh-words. A similar issue to that of the too narrow scope of the question is that some questions could be relevant and salient were it not for the use of the wrong sentence topic. (7.20) is an extract of a radio debate about the advantages of the antidepressant Prozac. One of speakers, PS3E6, disagrees with the others and mentions Valium, another antidepressant that was used before Prozac, to explain that while the excitement when it was first introduced was very similar to what is happening with Prozac, Valium was later found to have many side effects. The other participants counter him by stressing that one of the advantages of Prozac is its low amount of side effects so PS3E6 tries to list the known side-effects of the drug. At this point, the generated question inquires about the side-effects of Valium, while we are actually talking about Prozac. The reason for this confusion may be that Prozac had not been mentioned once since turn 8 while Valium is mentioned 6 times between the last mention of Prozac and the anchor turn for that question. Nevertheless, a listener would have needed to pay very little attention to the debate to misunderstand the focus of the last turns of (7.20), showing a disparity between human understanding and LLM processing of the dialogue.

- (7.20) **PS3E4**₉ Oliver what's so wonderful about Prozac as an antidepressant?
(16 utterances)
PS3E6₂₆ Well twenty years ago I was on programmes like this, saying that doctors who said that Valium and the other tranquillizers were perfectly safe were talking rubbish and that there was every possibility that there would be problems found in the future, if we kept prescribing these drugs for vast numbers of people.
(17 utterances)
PS3E5₄₄ <unclear> be six million people in the world are taking this drug saying it's really really helping them and you're telling them it's crap.

PS3E6₄₅ I'm let me just tell you some of the side effects which the company making it admit to .

PS3E5₄₆ Now what do you say to those six million people ?

PS3E6₄₇ Look no let me just tell you what the

PS3E5₄₈ Mm.

PS3E6₄₉ side effects are.

→ **Generated question:** What are some specific side effects associated with Valium?

Dialogue HV1 (Public multi - TV) [BNC]

In the same vein, generated questions often use the wrong wh-word, which technically makes them irrelevant or un-grammatical depending on the situation. ‘Who are “the sports orders” that Adrian is being asked to provide?’ is one such question where the wh-word is clearly not coherent with the subject of the question. Other errors could be more subtle like ‘Who are “them” that will stay in exclusion until they’ve been fixed?’ where ‘them’ actually refers to written reports and not to people.

Providing questions and answers. Another well known limitation of LLMs besides ‘hallucinations’, is that it is complicated to enforce the format of the output. In our case, while we specifically required questions we did not yet have an answer for, some generated questions were followed by an answer in parenthesis like ‘When exactly will we go to the paint shop? (The specific time mentioned was half past four.)’.

7.4.3 Prompt-related issues

LLMs are known to be sensitive to the prompt they are given. In our case, we stressed the importance of generating questions anchored in a specific turn, we required three questions per anchor turn, and the questions were to be generated from the point of view of a participant in the conversation. The following issues likely stem from these constraints.

Short anchors. Following Y. Wu et al. (2024), we generated questions every second turn in the dialogue. Since we were working with naturalistic dialogues, some turns were very short and contain only an acknowledgment token like ‘Mm’ or ‘Yeah’. In such cases, the generated questions were often related to the meaning of the token, which was sometimes obvious as discussed earlier but could also bring pragmatic information, but in other cases the question concerned a much earlier context as we can see in (7.21).

(7.21) J8DPS004₅₄ that is stupid Terry is key stage four!

12 utterances

J8DPS003₆₇ What are those Marian, you don’t get one of those last time?

J8DPS000₆₈ I got <pause> one of these.

J8DPS002₆₉ Yeah.

J8DPSUNK₇₀ Start planning, well you got those are the weeks of the year

J8DPS003₇₁ Mhm.

J8DPSUNK₇₂ so you can actually write in, you know the <pause> at

J8DPS004₇₃ Oh!

→ **Generated question:** What prompted Terry being referred to as “key stage four”?

→ **Generated question:** Could you explain what was unclear about key stage four that was mentioned earlier in the conversation?

This phenomenon may be due to the fact that anchoring a question in a very short turn can be challenging for the model, which may then generate any relevant question from the context, even if it is not directly related to the anchor turn.

Questions answered by design. The prompt required the model to generate a question one would be curious about as a participant in the conversation. This often led to questions that were answered *by design* in the subsequent turns of the dialogue as they roughly amounted to ‘What happens next?’. ‘Are there further discussions or actions to follow after “Yeah, mm”?’ is an example of such a question. While these questions are relevant to the dialogue, in practice they are not very informative for our task as they do not allow us to predict the upcoming topics of the conversation.

First-person questions. The requirement to produce questions one would be curious about *if one were a participant of the conversation* was meant to avoid questions related to entities or events that were clearly part of the common ground of the participants. However, this led to the generation of questions that were phrased in the first person, and in particular from the point of view of an actual participant of the conversation. For instance, in (7.22), the question could have been asked by PS0J1.

- (7.22) PS0J1₁ It doesn’t matter if she’d got a fire guard or not, but it still could easily get onto the
 PS0J3₂ Yeah.
 PS0J1₃ carpet couldn’t it?
 PS0J7₄ What?
 PS0J7₅ What?
 PS0J3₆ They’re on their way squire !
 KD2PSUNK₇ Yeah.
 PS0J1₈ They’re not safe these houses are they dad?
 PS0J3₉ No.
 PS0J4₁₀ Not with, not with that
 PS0J7₁₁ Especially now no one has looked at them.
 PS0J4₁₂ That’s why you’ve gotta go smokeless for.
 PS0J3₁₃ Yeah.

→ **Generated question:** Why do the houses need to be smokeless, Dad?

Dialogue KD2 – 061610 (Open multi) [BNC]

Specific questions. About one fifth of the generated questions contained the word ‘specific’, as reported in Table 7.4, which may be due to the requirement to produce three questions per anchor turn. Indeed, producing several questions about the same turn in the same generation may have led the model to generate questions that are somewhat related to each other. In this case, asking a more ‘specific’ question is a natural way to differentiate the questions. However, this led to questions that were often irrelevant or much too detailed to be considered highly salient. For instance, ‘What specifically does half a day refer to in terms of hours?’ asks for a level of detail that is not necessary to understand the dialogue, and is therefore not a salient question.

7.5 Results and Analysis: Dialogue is less predictable than static text

We begin by analysing salience and answerability separately, focusing on annotation patterns and differences across dialogue types, before examining the relationship between the two measures through correlation analyses.

7.5.1 Salience

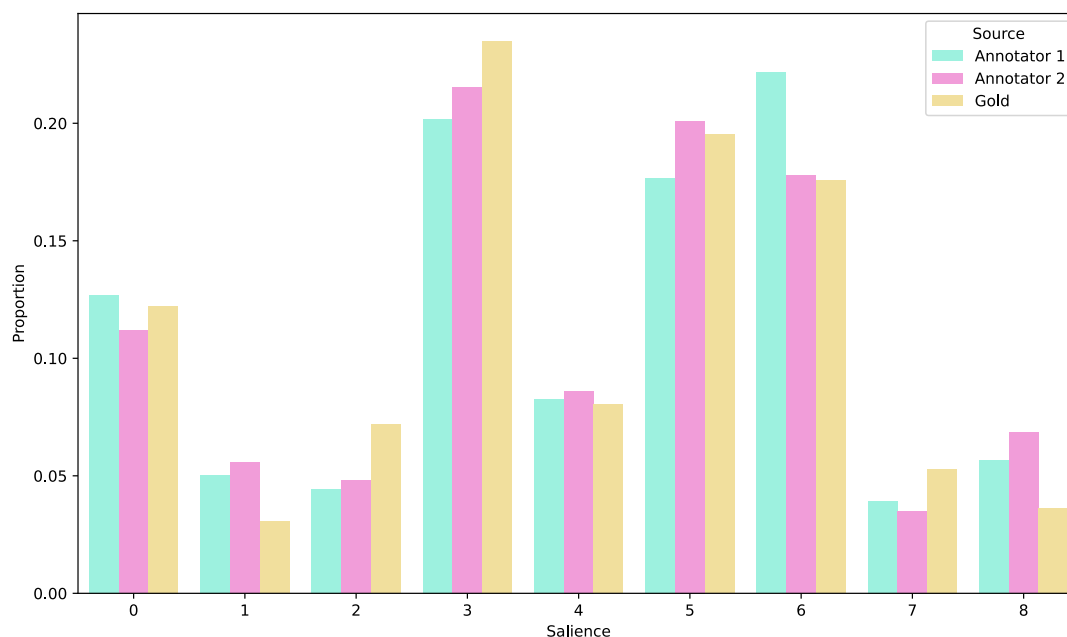


Figure 7.7: Repartition of salience scores across dialogue groups.

Salience scores are not evenly distributed across the range of possible values, as shown in Fig. 7.7. Around 12% of the questions are considered invalid (score 0), often because they are not anchored in the right dialogue turn. Scores 1 and 2, which correspond to questions that are unrelated or due to transcription issues each represent about 5% of the questions, where score 1 reduces to about 3% when considering the gold annotations, and score 2 grows to about 7%. This difference in repartition between the raw annotations and the aggregation is mostly due to the difficulty of annotation for these scores, where the higher agreement on score 2 compared to all the other scores skews the distribution of the gold annotations. Conversely, score 1 is one of the scores with the most disagreements as discussed previously in Section 7.3.3.1. Most questions were considered at least somewhat related to the preceding context, with scores 3, 5 and 6 each representing 18% to 22% of the total. Highly salient questions (scores 7 and 8) are less frequent, with each representing 4% to 6% of the total. This distribution may be influenced by two complementary factors. First, while generating questions that are related to the context can be seen as a task that is relatively easy for an LLM due to the way it is trained, generating questions that are *highly salient* requires more sophisticated reasoning about the dialogue and the participants' intentions. Second, the nature of dialogue itself, and our tendency to look for coherence and relatedness in conversation, may have encouraged the annotators to assign at least a moderate level of salience to most questions, while high scores required more specific types of questions. Finally score 4, which corresponds to questions for which the participants clearly already know the answer even though it does not appear in the transcription, i.e. questions that tap in the common ground of the participants, represents about 8% of the corpus,

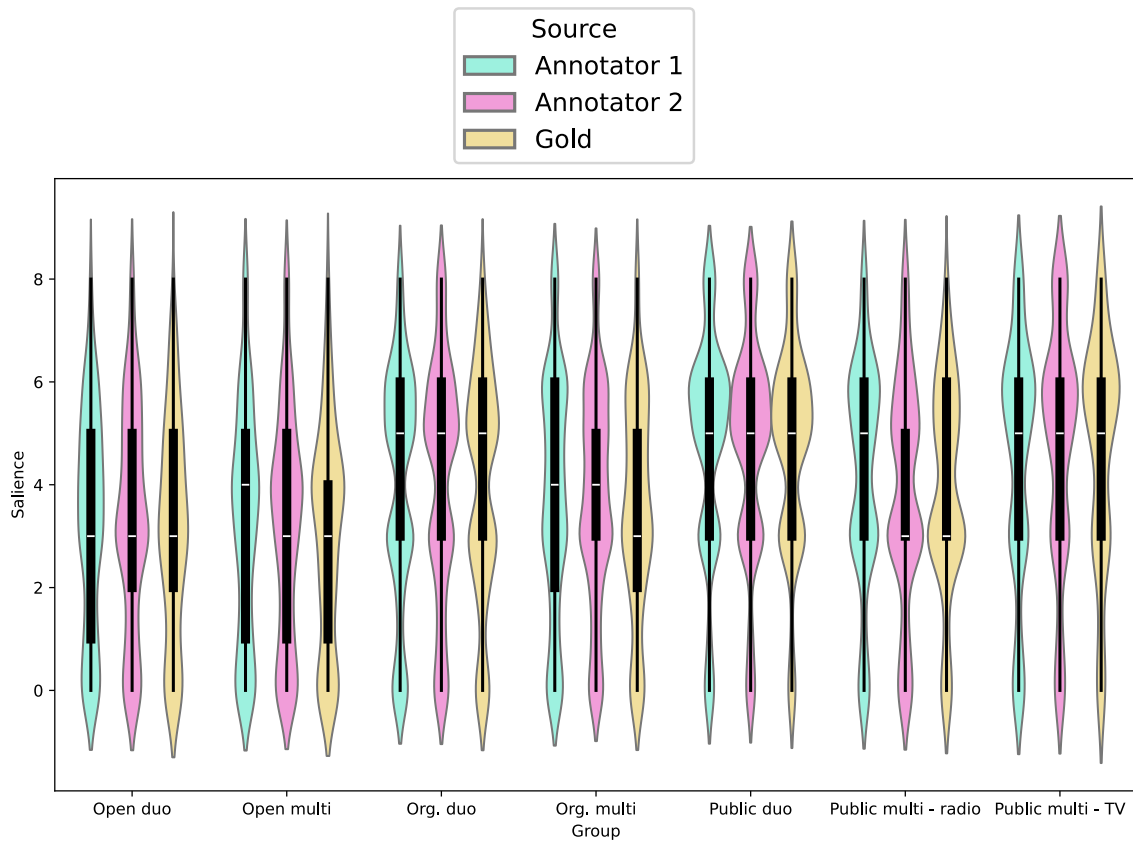


Figure 7.8: Repartition of salience scores across dialogue groups.

showing that this score was indeed necessary to capture the nature of dialogue.

Figure 7.8 shows the distribution of salience scores for each dialogue group, separated by annotator, and Table 7.7 reports the average salience scores for each group, the degree of confidence in the statistical differences between both annotator 1 and 2, as well as the proportion of questions kept for the gold annotation. The distributions are very similar between the two annotators, with only significant differences for the public multilogues, where the radio ones display a higher average salience for annotator 1 than annotator 2, while the opposite is true for the television ones though with a lower degree of confidence. However, the distributions between the different dialogue groups are much more variable. Pairwise comparisons (Fig. 7.9) indicate that only a few pairs of groups display non-significant differences. Open-domain duologues and multilogues are similar together while being different from all the other groups in both annotators' distributions. Public radio multilogues are similar to the organised duologues, while public television multilogues are similar to the public duologues, although the statistical strength of these differences varies by annotator. When looking into the differences for the gold annotations, we observe no significant differences between the dialogue groups, suggesting that disagreements between the annotators highlight more complex interactions, characteristic of the different dialogue types. This last result should be interpreted with caution, as the gold annotations represent a restricted subset of the data which limits the statistical power of the tests. Nevertheless, these results suggest that salience is sensitive to dialogue type and in particular to the number of participants, with duologues displaying more salient questions than multilogues (Table 7.7).

| Group | S_1 | S_2 | S_G | Difference (p-value) | Support Gold # | % |
|----------------------|----------------|----------------|----------------|----------------------|----------------|------|
| Open duo | $3.29 \pm .45$ | $3.32 \pm .64$ | $3.04 \pm .82$ | ns | 321/657 | 48.9 |
| Open multi | $3.26 \pm .46$ | $3.20 \pm .53$ | $3.06 \pm .71$ | ns | 388/822 | 47.2 |
| Org. duo | $4.04 \pm .50$ | $4.12 \pm .69$ | $4.08 \pm .72$ | ns | 742/1,548 | 47.9 |
| Org. multi | $3.80 \pm .29$ | $3.77 \pm .46$ | $3.57 \pm .31$ | ns | 598/1,409 | 42.4 |
| Public duo | $4.79 \pm .52$ | $4.74 \pm .56$ | $4.80 \pm .53$ | ns | 535/1,158 | 46.2 |
| Public multi - radio | $4.22 \pm .56$ | $3.88 \pm .32$ | $3.95 \pm .49$ | *** | 402/768 | 52.3 |
| Public multi - TV | $4.50 \pm .26$ | $4.66 \pm .84$ | $4.81 \pm .60$ | * | 287/762 | 37.7 |
| All | $3.92 \pm .70$ | $3.90 \pm .79$ | $3.82 \pm .91$ | ns | 3,273/7,124 | 45.9 |

Table 7.7: Average salience score per dialogue group (total range: 0-8).

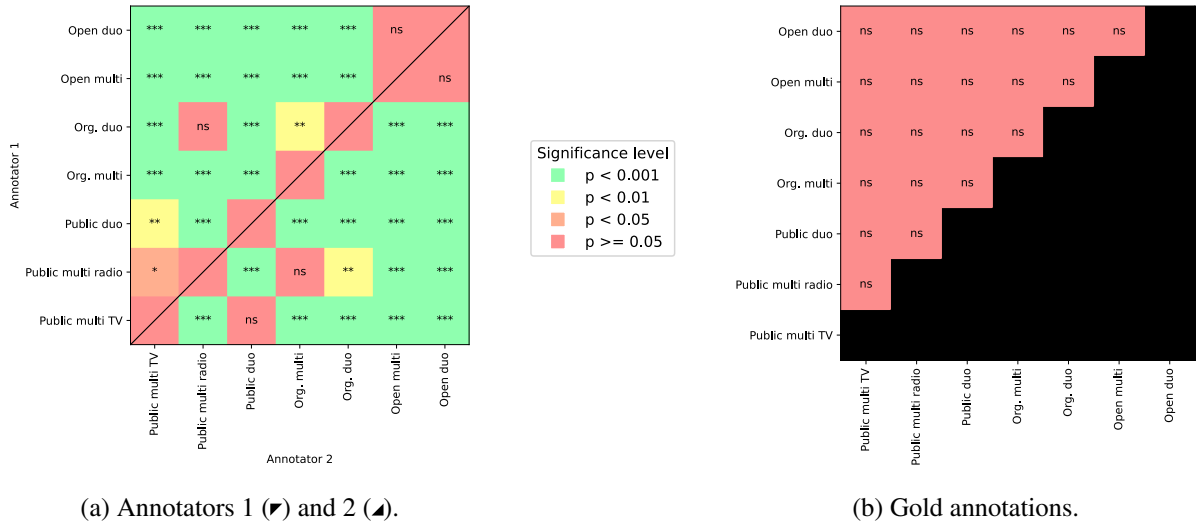


Figure 7.9: Representation of the significance of the difference in salience distributions between dialogue group for a given annotator. Colours indicate the strength of the p-value.

7.5.2 Answerability

As shown in Fig. 7.10, more than 70% of the generated questions are considered not answered in the subsequent dialogue (score 0), about 10% partially answered (score 1), and the rest fully answered (score 2). Annotator 2 assigned slightly more score 0 than annotator 1, and slightly less score 2. In isolation, this repartition could indicate that the generated questions were irrelevant but the salience scores contradict this hypothesis. Instead, it suggests that predicting questions that will actually be answered is complicated which we discuss in Section 7.5.3.

The high proportion of score 0 is reflected in the distributions of answerability scores across dialogue groups (Fig. 7.11), which are skewed towards 0 for all groups. Contrary to salience, the distributions are quite different between the two annotators, here only the open conversations display similar distributions. All the other groups are more or less significantly different (Table 7.8), with annotator 1 consistently assigning higher scores than annotator 2.

Conversely, pairwise statistical tests (Fig. 7.12) indicate fewer significant differences between groups than for salience, although some variation remains. Particularly, open-domain multilogues are signifi-



Figure 7.10: Repartition of the answerability scores for each annotator.

| Group | A_1 | A_2 | A_G | Difference (p-value) |
|----------------------|---------------|---------------|---------------|----------------------|
| Open duo | $.36 \pm .13$ | $.34 \pm .14$ | $.35 \pm .13$ | ns |
| Open multi | $.20 \pm .09$ | $.19 \pm .10$ | $.20 \pm .09$ | ns |
| Org. duo | $.47 \pm .17$ | $.40 \pm .13$ | $.44 \pm .15$ | ** |
| Org. multi | $.41 \pm .09$ | $.30 \pm .07$ | $.35 \pm .07$ | *** |
| Public duo | $.55 \pm .11$ | $.48 \pm .10$ | $.52 \pm .10$ | * |
| Public multi - radio | $.56 \pm .23$ | $.44 \pm .18$ | $.50 \pm .19$ | ** |
| Public multi - TV | $.44 \pm .07$ | $.36 \pm .11$ | $.40 \pm .09$ | * |
| All | $.42 \pm .17$ | $.35 \pm .14$ | $.38 \pm .15$ | *** |

Table 7.8: Average answerability score per dialogue group.

cantly different from all other dialogue types, and organised multilogues are different from all other types except open-domain duologues and public television multilogues. This suggests that answerability is less sensitive to dialogue type than salience, and both the inter-annotator and inter-group differences indicate that open dialogues, and in particular multilogues, are distinct from the other groups. However, the low average answerability scores across *all* groups indicates that most questions are not explicitly answered in the subsequent dialogue (Table 7.8).

7.5.3 Dialogue predictability

The central goal of this study was to determine whether salient questions are more likely to be answered in the subsequent dialogue turns. We therefore examine the relationship between salience and answerability, and how this relationship varies across dialogue types.

7.5.3.1 Average answerability per salience score

Table 7.9 shows the average answerability score for each salience score and Fig. 7.13 supports these results with a representation of the significance of these differences. The average answerability is relatively stable

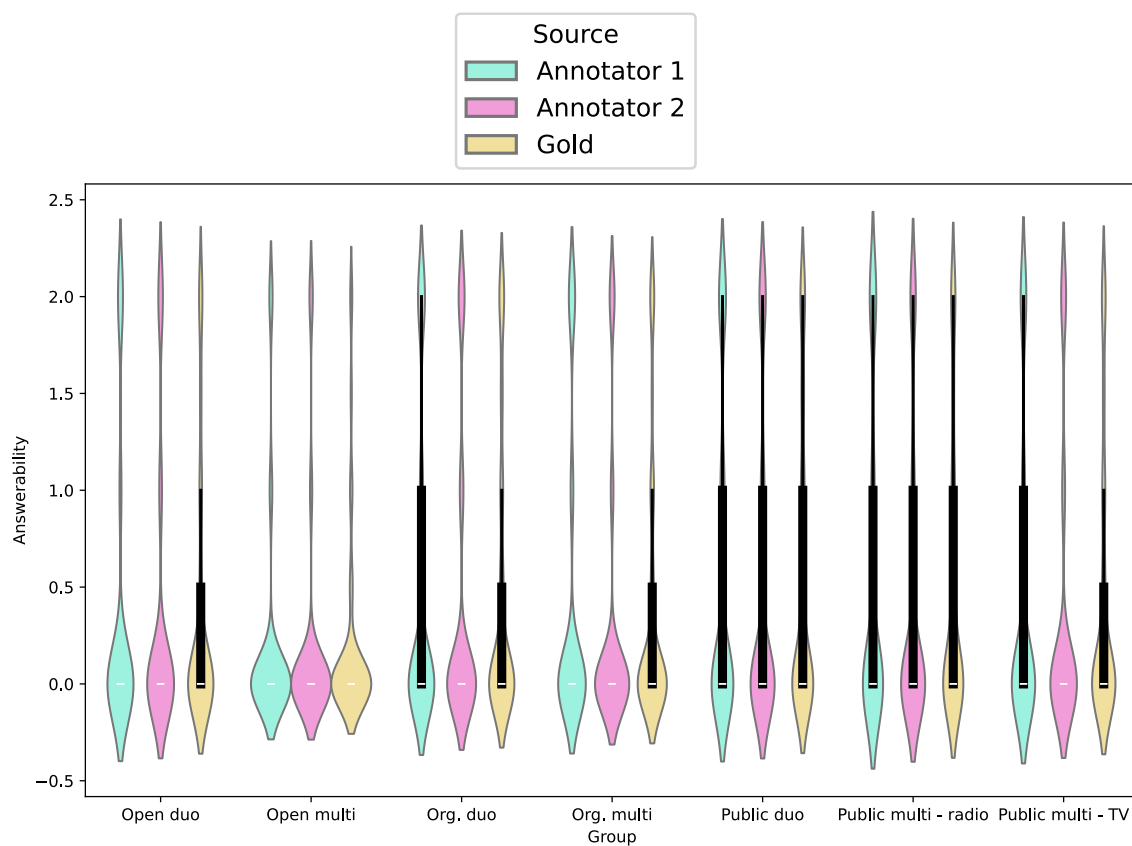


Figure 7.11: Repartition of answerability scores across dialogue groups.

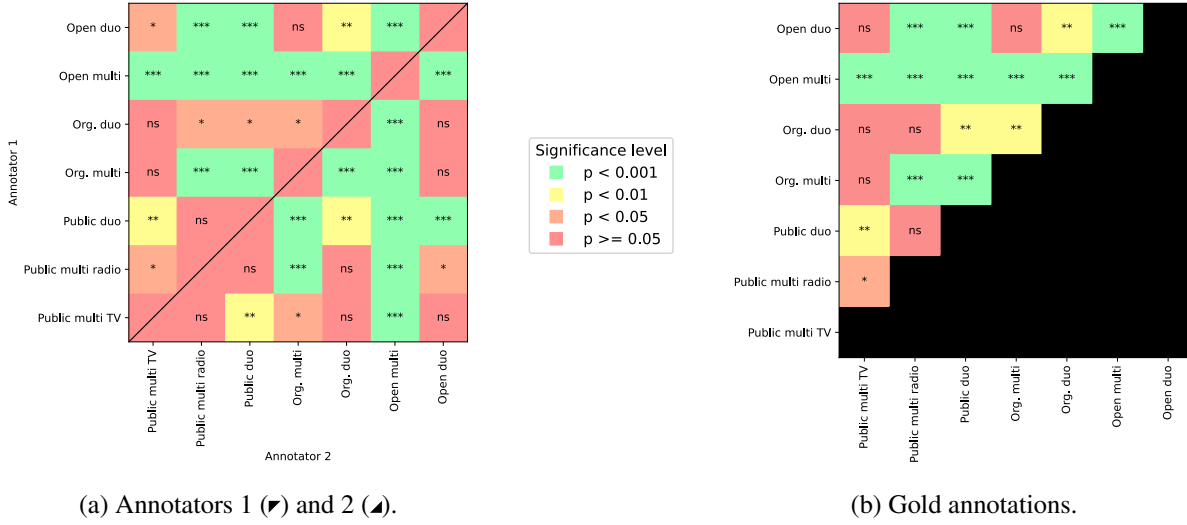


Figure 7.12: Representation of the significance of the difference in answerability distributions between dialogue groups for a given annotator. Colours indicate the strength of the p-value.

| S | A ₁ | A ₂ | A _G |
|---|----------------|----------------|----------------|
| 0 | .23 ± .51 | .25 ± .55 | .18 ± .44 |
| 1 | .29 ± .60 | .33 ± .65 | .32 ± .64 |
| 2 | .17 ± .49 | .17 ± .49 | .13 ± .45 |
| 3 | .30 ± .60 | .33 ± .64 | .30 ± .60 |
| 4 | .30 ± .61 | .32 ± .63 | .33 ± .64 |
| 5 | .27 ± .56 | .28 ± .55 | .22 ± .50 |
| 6 | .52 ± .73 | .52 ± .74 | .53 ± .74 |
| 7 | 1.13 ± .81 | 1.11 ± .83 | 1.21 ± .81 |
| 8 | 1.03 ± .86 | .87 ± .85 | 1.41 ± .76 |

Table 7.9: Average answerability score per salience score. *Results presented as mean ± standard deviation.*

for salience scores below 6, with the exception of score 2, which consistently shows lower answerability across conditions. Higher salience levels (6–8) are associated with significantly higher answerability, all different from each other except for scores 7 and 8 with annotator 1.

7.5.3.2 Correlation between salience and answerability

We compute the correlation between salience and answerability to evaluate whether more salient questions are more likely to be answered in the subsequent dialogue turns. Following Y. Wu et al. (2024), we use Spearman’s rank correlation coefficient to account for the ordinal nature of the annotation scales. We additionally report Kendall’s τ , which is less sensitive to ties and therefore suitable given the discrete label space of answerability. Correlations are computed separately for each annotator as well as for aggregated (‘gold’) annotations. Results are shown in Fig. 7.14, which reports mean answerability per salience level along with regression lines, and the correlation coefficients in the legend.

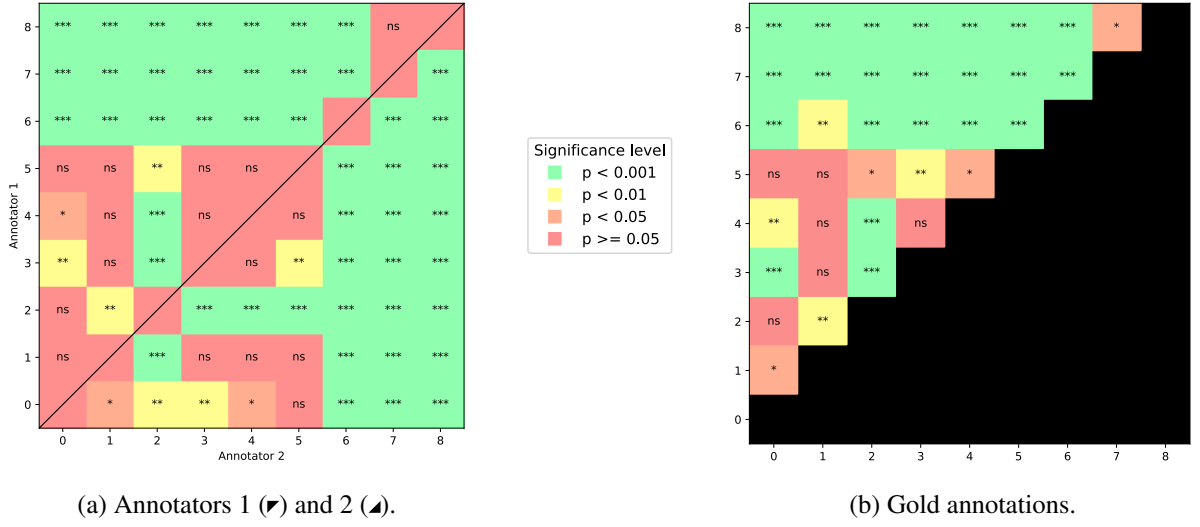


Figure 7.13: Representation of the significance of the difference in average answerability for each salience score. Colours indicate the strength of the p-value.

Warning – Ongoing work

Include correlation scores when excluding questions with salience score 0/2/4/7 for more accurate comparison with Y. Wu et al. (2024) and fully ordinal scale. They actually get even lower:

- Annotator 2: $\rho = 0.19$, $\tau = 0.17$ (p-value around 10^{-37})
- Annotator 1: $\rho = 0.24$, $\tau = 0.21$ (p-value around 10^{-59})
- Gold: $\rho = 0.24$, $\tau = 0.21$ (p-value around 10^{-30})

Across all settings, we observe a positive correlation between salience and answerability. This finding replicates the general trend reported by Y. Wu et al. (2024) in monologic data. However, the magnitude of the correlation is two to three times lower than that reported for news and TED talks ($\rho = 0.59\text{--}0.74$), suggesting weaker alignment between salience and downstream resolution in conversational settings. This reduction in correlation is consistent with the increased variability of dialogue structure, where multiple interacting factors – including turn-taking dynamics, speaker intentions, and shared situational context – may affect whether salient questions are explicitly addressed.

When examining answerability conditioned on salience levels, we observe relatively homogeneous answerability for salience scores below 6, with the exception of score 2. Higher salience levels (6–8) are associated with significantly higher answerability. Score 2 consistently shows lower answerability across conditions, which is expected given that most of these questions ask for clarifications about the transcriptions and are thus unlikely to be answered in the dialogue.

Finally, we examine whether dialogue type affects the salience–answerability relationship (Fig. 7.15). Most differences in correlation are not statistically significant with the exception of open-domain multilogues, which exhibit the lowest ρ coefficient. Public multilogues from television, which have intermediate ρ values for both annotators, also differ significantly from most other groups; only open-domain duologues and organised multilogues show similar correlations. When ranking groups by their ρ coefficients, the two annotators produce nearly identical orderings, with only two groups swapping positions, suggesting that group differences are relatively consistent across annotators. In contrast, the ranking ob-

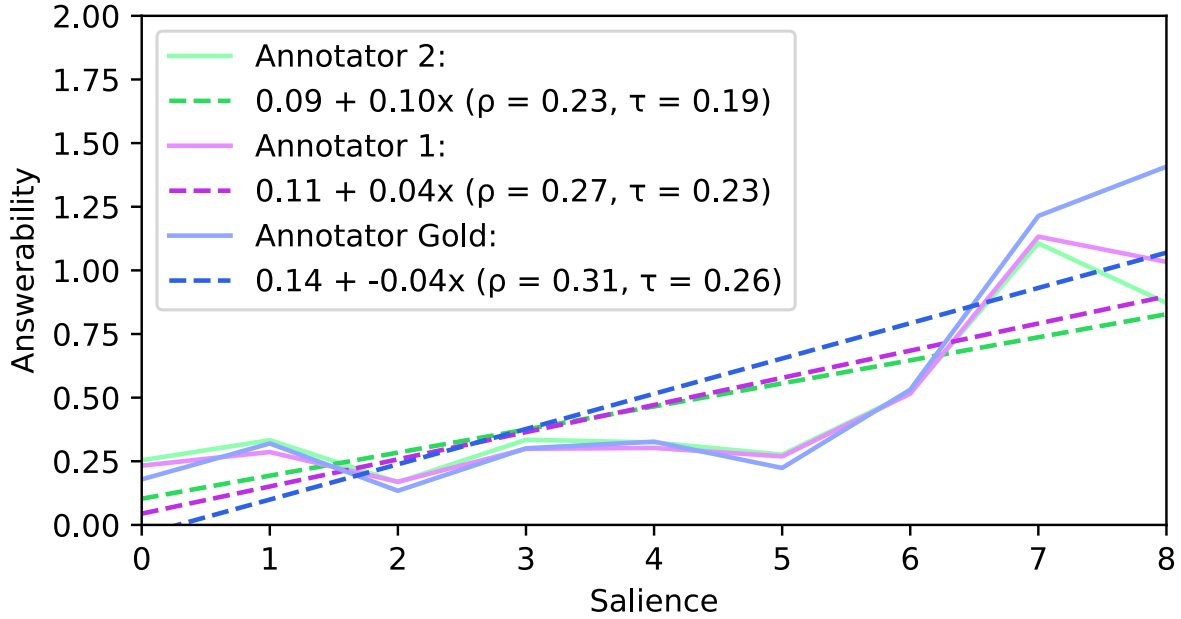


Figure 7.14: Correlation between salience and answerability scores (all p -values $\ll .005$). *Legend in the same order as the plots at 0.*

tained from the gold annotations differs substantially, with only the two highest-ranked groups retaining their positions. This again suggests that annotator disagreements capture meaningful differences in the dialogues.

Table 7.10 shows the correlation coefficient between salience and answerability for each dialogue group. As explained in Section 7.5.3 when ordering the groups by their ρ coefficient, we observe that both annotators have a very similar ranking, with only two groups swapping their position: the open-domain dialogues and the public multilogues from the television that are ranked second and fifth based on annotator 1 in increasing order, and fifth and second based on annotator 2. This suggests that the differences between groups are relatively consistent across annotators but that these specific two groups may be more or less complex to annotate depending on the annotator. However, this ranking is very different from the one obtained on the gold annotations, where only the two groups with highest ρ coefficients – the organised dialogues and the public dialogues – keep the same ranking.

Nevertheless, Fig. 7.15 shows that most group differences in correlation are not statistically significant with the exception of open-domain multilogues that have the lowest ρ coefficient. Public multilogues from the television, with the intermediate ρ coefficient for both annotators, also show significant differences with most other groups, open-domain dialogues and organised multilogues being the only groups similar to them. Moreover, there are even less differences between groups when looking at the gold annotations. The homogenisation of the results in the gold annotations suggests that the disagreements between the annotators reveal some meaningful differences in the dialogues.

7.6 Conclusion

In this chapter, we investigated whether the relationship between question salience and answerability observed in monologic discourse also holds in dialogue. Adapting the framework of Y. Wu et al. (2024), we applied it to 60 dialogues from the British National Corpus representing a range of interactional settings.

| Group | ρ_1 | ρ_2 | ρ_G |
|--------------------|----------|----------|----------|
| Open duo | .159 | .202 | .236 |
| Open multi | .134 | .149 | .244 |
| Org. duo | .331 | .290 | .418 |
| Org. multi | .219 | .174 | .211 |
| Public duo | .303 | .239 | .361 |
| Public multi radio | .253 | .158 | .198 |
| Public multi TV | .289 | .190 | .332 |

Table 7.10: Correlation between salience and answerability by dialogue group. ρ_i and p_i refer to the Spearman’s rank correlation coefficient and associated p-value for annotator i , while ρ_G and p_G refer to the same metrics for the gold annotations. The p-value for each correlation was below 0.001, indicating statistical significance.

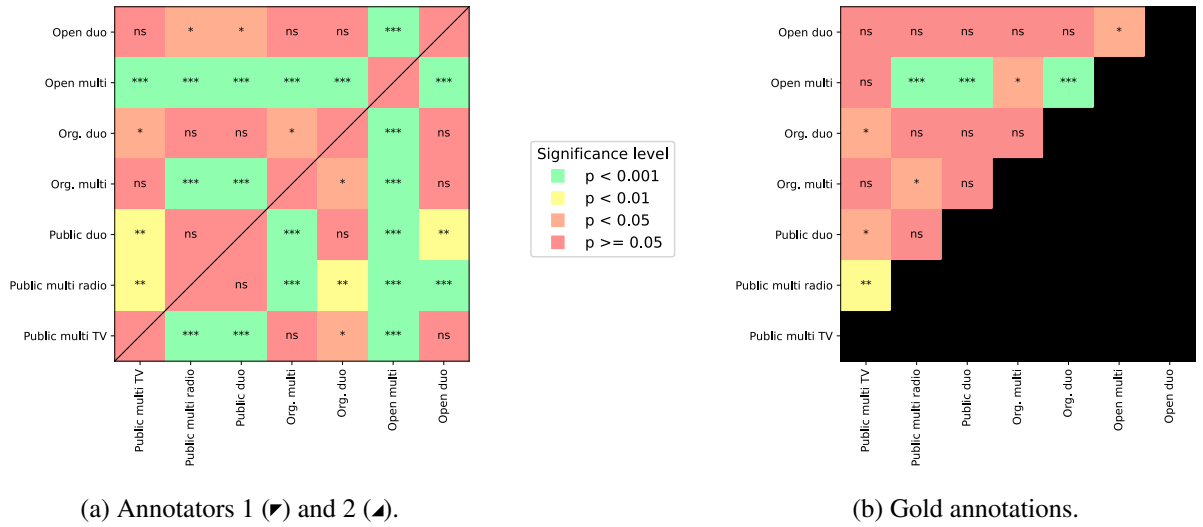


Figure 7.15: Representation of the significance of the difference in correlation between dialogue group. Colours indicate the strength of the p-value.

Our results show a positive correlation between salience and answerability: questions judged to be more salient are more likely to be addressed later in the conversation. However, this relationship is substantially weaker than that reported for news articles and TED talks, suggesting that the structure of dialogue is less predictable than that of monologic discourse. We also show that this relationship is much weaker in open-domain multilogues. Moreover, the more uniform values we get for the annotations and the correlation across dialogue types when considering only the gold annotations suggest that the disagreements between the annotators reveal meaningful differences.

Beyond these empirical findings, our work provides one of the first large-scale operationalisations of QUD-driven approaches for modelling ND. The moderate agreement obtained for salience annotation further highlights both the complexity of the task and the importance of accounting for annotator perspective when studying dialogue structure. At the level of this thesis, this study clearly shows that data characterisation is crucial, as we found differences between dialogue groups despite the relatively small sample size.

Future work should investigate the temporal dynamics of question resolution, including when ques-

tions are answered and by whom. The dynamic nature of conversation creates complex situations that may not be captured by the simple salience–answerability model used in this experiment. For instance, the salience of a question may vary throughout the dialogue, while remaining relevant, depending on the current focus of the participants. If such a question was generated at several points in the dialogue, we may end up with several different salience scores bound to the same answerability score, which would dilute the correlation between the two measures. Similarly, the distance between the anchor that triggers a question and its actual answer should be taken into account when considering the ‘predictability’ of a conversation. One lead to address this issue would be to consider a question answered only if the answer is provided in a turn accessible from the anchor turn in terms of dialogue structure. This leads us back to the tension between linear and hierarchical models of topical structure, as accessibility would be defined very differently depending on the model. This would also raise questions on the relevance of such structures depending on the type of dialogue, where open multilogues may be less constrained by structural expectations than other dialogue types. A somewhat circular reasoning emerges from these considerations, as the structure of dialogue, and the degree to which it really constrains conversation production, is precisely what we are trying to understand through the analysis of salience and answerability.

Another promising direction is the development of automatic dialogue salience and answerability prediction models, enabling large-scale analysis of interactional structure. While we tried to work with a sample of dialogues large enough to support statistically significant analyses, and varied enough to compare a range of interactional settings, the sample size remains relatively small due to the time and effort required for manual annotation. The corpus developed in this study could be used to train and evaluate an automatic salience prediction model, and an extension of our annotations could be leveraged to produce an answerability prediction model. Y. Wu et al. (2024) have shown that a salience prediction model can be developed for monologic discourse, and a project carried out by first-year Master students in the NLP program at the University of Lorraine gives hope that it can be adapted for dialogue as well. Predicting answerability is likely to be more challenging, particularly because useful models should identify the portions of a dialogue that answer a question rather than merely assigning a score. Such models would nevertheless have direct applications to information retrieval and dialogue systems, two very active areas, providing a strong motivation for further research.

The automation of the annotation step would allow us to compare the results obtained when generating the questions with different models, and possibly compare them with human generations. While we focused more on what the questions could teach us in the present study, which was somewhat model agnostic as long as our final sample contained a variety of salience scores, it would be interesting to determine whether the correlation value is stable across different models. Showing that it is the case would support the idea that the salience–answerability relationship can be used to discriminate between dialogue types, and maybe even further discourse types. A limitation of this approach is that our corpus does not represent all possible dialogue types, and building automatic salience and answerability prediction models based on it may not generalise well to all types of language use. Nevertheless, this could be verified by validating the predictions through a smaller set of annotations, allowing us to select which types of dialogues require additional annotations to improve the model’s generalisation.

Summary and Conclusion

8

Epilogue

Warning – Ongoing work

Summary:

- Conversation is something we observe through its realisation and not through the underlying processes that produce it. Yet there seem to exist very different forms of realisations so most likely different processes as well or at least different use configurations.
- We thus tried to propose a descriptive framework grounded in theory but still quite agnostic and flexible in its operationalisation so that its practical use can allow to position data and task constraints in a multidimensional space to select relevant data.
- We have demonstrated how this framework can help address widespread shortcomings within the NLP community, particularly in terms of defining objectives and describing data. These shortcomings obscure the quality and the work carried out beyond these aspects, because a lack of grounding in the data leads to a loss of meaning and a disconnect from the linguistic aspect, which is, after all, the main focus and motivation of NLP/CL.
- We then focused more specifically on the structural dimension of dialogue. First of all in a very upfront way through the linear topic segmentation of transcriptions of the Friends TV show. Even though these exchanges are not naturalistic, they were scripted by professionals and involve multiple participants in various open-ended settings. This leads to a complex conversational format and indeed recent segmentation techniques based on ML or LLMs in a zero-shot setting do not perform so well while rule-based approaches, though still unsatisfying, are better because they do not over-generate boundaries.
- An issue highlighted by this experiment besides the limitations of the methods when confronted to such complex conversations, is that the task itself is not flexible enough. We thus interested ourselves to the idea of topical similarity in a more continuous way, i.e. to the degree of topical similarity *throughout* a conversation. We did this on Reddit interactions because they allow participants to explicitly position themselves as topically different from one another through the structure of the platform. We thus worked on *real* conversations (as opposed to the scripted ones from Friends) but in a written format – initially, not because of a transcription – and asynchronous.
- *Summary of the results with the new version of the model.*

- This reveals an idea of latent structure: the phenomenon seems to exist but we needed a way to explicit it or at least quantify it. We thus finished by leveraging a QUD-driven approach, one of the formal theories to model topics, to analyse the predictability of topics in dialogue. This time we worked with transcriptions of naturally occurring face-to-face (for the most part) conversations from the BNC. We established that conversation is overall less predictable than monologic forms of text such as news or TED talks. Besides, there seems to be differences in the predictability depending on the number of participants and the interactional setting, validating the need to clearly define and characterise the data we are working with.

Discussion:

- Describing and characterising the data we work with is crucial as it is the basis of everything we do in NLP. It is also a way to make explicit the assumptions we make about the data and the task we are working on. The idea behind this claim is not to point fingers at a certain dataset or type thereof but to question the motivations we may have when working with a given corpus. Do we use it because it is the only one available despite its unsuitability? In such a case, is there a way to rethink the task within the limitations of the data? Or is it possible to invest the required time and resources in the collection and annotation of data?
- LLMs have become a big topic in this data production question where they are used for the creation of language data but also of annotations. While the practical aspect is undeniable, this practice requires questioning regarding the naturality of such data. The advantages seem clear and central for some domains like the medical domain where the sensitive nature of data makes sharing and large scale exploitation very hard. It is also a domain where using make-believe data was already entrenched through exercises of training to clinical interview. This seems to argue in favour of using indirect methods, but we must carefully consider the stages of the process where the use of LLMs makes sense, and ensure that qualified personnel – both medically and in terms of the more technical aspects – are involved in these protocols. In particular, we need facilitators to ensure mutual understanding amongst the various parties involved, so that everyone understands the issues at stake for the others.
- The increasingly massive use of LLMs raises other questions of a more practical nature, in particular regarding our own discourse in the NLP community. One of the advantages that is often highlighted is the comparatively low cost of using LLMs for data production and annotation. However, this is a very relative notion that is usually based solely on the price billed by the provider of the LLM, which amounts to discuss who got the best deal with the huge companies that we depend on for these services. This is however a very narrow and misleading perspective that overshadows a large part of the real cost of these tools which, despite not being paid upfront by researchers still exists somewhere. The huge cost of building and sustaining data centers these tools rely on must be absorbed by some entity. It seems that the companies that provide these services currently take it on themselves but the lack of sustainability of this model raises questions on their motivations. Moreover, the will to become more sovereign in terms of data and technologies in Europe leads to massive investments of public funds in the construction of data centers and more local LLMs. These have a cost that is directly observable in terms of fall of other fundings to compensate. There is also a clear ecological impact due to the water and energy consumption of these tools, and the very fast-paced renewal of the models but also of the hardware they rely on raises

questions about material shortage and waste management. All of this means that reducing the cost of LLMs to the one billed for using the API is very misleading and that it is one of our responsibilities as scientists specialists of these technologies to try to formulate a more complete and accurate discourse on the matter.

- The relative ease of access to LLMs and in particular conversational agents creates an influx of works mechanically related to the notion of dialogue. However, the authors of these studies often have a limited interest in the conversation itself and use it as a tool to reach a goal without much theoretical grounding. This leads to the transfer of existing evaluation metrics to tasks where they are not relevant, and to the use of data that is not suitable for the task at hand. Our work here is to try to propose tools and evaluation metrics that are more directly applicable and relevant so that the interdisciplinary nature of dialogue research is a motor rather than the source of studies with a weak theoretical frame. The framework we proposed in part 1 is directly grounded in this idea.

Perspectives:

Data characterisation:

- Extend the QUD operationalisation to other types of data / larger corpora through the production of automatic annotation models informed by the annotations we release with this thesis.
- Go further in this idea and leverage generated QUDs to model the topical structure of conversation through some kind of clustering. We can imagine that it would be helpful in disentangling parallel topical threads in complex conversations.

Pathological data / psychological angle:

- Reproduce the QUD experiment on pathological vs control data to see if the predictability of topics is different in these two types of data.
- Same on data used to train psychological counselling models to see if we can point a difference with actual counselling practices.

Meta science:

- Automate (at least partially) the annotation of research papers as in Chapter 3 to be able to analyse the evolution of the field and its practices over time.

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Appendices

Warning – Ongoing work

The appendices will be added in the final version of the thesis. Here is a tentative list of the appendices that will be included:

- Full text of the OintMind news article mentioned in Chapter 4;
- Extended results for the experiments presented in Chapter 5;
- Complete guidelines for the annotation of QUDs in dialogue, as mentioned in Chapter 7;
- Details on the dialogue collection experiment carried out during this thesis, briefly mentioned in Chapter 3.